LIOUVILLE ENERGY LOCATES THE RIGHTMOST ZETA ZERO

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ABSTRACT. The sliding-window energy $Q(N)$ of the Liouville function λ satisfies $Q(N) = N^{2\Theta+o(1)}$, where Θ is the supremum of the real parts of the zeros of ζ, and a bound along one sequence of N already locates Θ. Unconditionally $Q(N) > 2.81N$ infinitely often, and $Q(N)/N$ is bounded exactly when the summatory function of λ is $O(\sqrt{x})$. Changing λ on x^β primes can break the exponent when $\beta > 1/2$ and never when $\beta \le 1/2$.

1. INTRODUCTION

Let λ be the Liouville function, $\lambda(n) = (-1)^{\Omega(n)}$, and let $L(x) = \sum_{n\le x}\lambda(n)$. For a sequence $a = (a_n)$ and integers $N, H \ge 1$ we measure the size of the sums of a over all windows of length H that meet $[1, N]$:

$$\mathcal{Q}_a(N,H) = \frac{1}{H}\sum_{t\in\mathbb{Z}}\Big|\sum_{t<n\le t+H} a_n \mathbf{1}_{[1,N]}(n)\Big|^2, \qquad Q_a(N) = \mathcal{Q}_a(N,N).$$

We call $Q_\lambda(N)$ the *Liouville energy* and write $Q(N)$ for it. It is a single positive quantity, and Fejér's identity expresses it through the shifted correlations of λ with the triangular weights $1 - h/N$.[1] Let

$$\Theta = \sup\{\operatorname{Re}\rho : \ \zeta(\rho) = 0\} \in [\tfrac{1}{2}, 1].$$

Since the nontrivial zeros are symmetric about the critical line, the Riemann hypothesis is the statement $\Theta = 1/2$.

The guiding fact of this paper is that the Liouville energy sees exactly Θ. Its growth exponent is 2Θ, with a genuine limit; a bound for it along one infinite sequence of N already bounds Θ; and the exponent does not move under a change of λ on a set of primes of size $x^{1/2+o(1)}$, while for every $\beta > 1/2$ some change on a set of size x^β breaks it. Every bound for Q strong enough to prove the Riemann hypothesis is therefore equivalent to a zero-free half-plane. The paper does not prove the Riemann hypothesis. It identifies what any proof through the energy must supply, and it proves unconditionally what can be proved along the way.

1.1. Main results.

The first theorem collects the equivalence and the exact exponent (Theorem 3.1 and Corollaries 3.2 and 4.3).

Theorem A (Energy and the rightmost zero)**.** *Let $1/2 \le \theta \le 1$. The following are equivalent: $\zeta(s) \ne 0$ for $\operatorname{Re} s > \theta$; $L(x) \ll_\varepsilon x^{\theta+\varepsilon}$ for every $\varepsilon > 0$; and $Q(N) \ll_\varepsilon N^{2\theta+\varepsilon}$ for every $\varepsilon > 0$. Moreover*

$$\lim_{N\to\infty}\frac{\log Q(N)}{\log N} = 2\Theta.$$

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[1]The energy is the mean square of the window sums, so it records cancellation inside windows of every position at once. For independent random signs its expectation is N; for the constant sequence it is $(2N^2+1)/3$; for alternating signs it equals 1 (Table 2).

The second theorem is a zero-one law for the exceptional set of the energy (Theorem 4.2). Its proof uses the kernel of Pintz [31, 32], which isolates one zero of ζ inside a mean value of the summatory function.

Theorem B (Zero-one law)**.** *If $\sigma < \Theta$, then $Q(N) > N^{2\sigma}$ for all sufficiently large N; if $\sigma > \Theta$, then $Q(N) \le N^{2\sigma}$ for all sufficiently large N. Consequently, if $Q(N_k) \le N_k^{2\sigma}$ along a single infinite sequence $N_k \to \infty$, then $\Theta \le \sigma$; and the Riemann hypothesis holds if and only if, for each $\varepsilon > 0$, $Q(N) \le N^{1+\varepsilon}$ for infinitely many N.*

Thus there is no intermediate regime away from $\sigma = \Theta$: the set of N at which the energy exceeds $N^{2\sigma}$ is finite or cofinite. A statement of the form "the energy is small for almost all N", or for a sequence of positive density, or along a sparse sequence, is exactly as strong as the bound for all N.

The third theorem measures how much freedom the exponent leaves in the signs (Theorem 6.3 and Corollary 6.5). For a set S of primes let λ_S be the completely multiplicative function with $\lambda_S(p) = +1$ for $p \in S$ and $\lambda_S(p) = -1$ otherwise; these are the modified Liouville functions of Borwein, Choi and Coons [11].[2]

Theorem C (Sensitivity of the exponent)**.** *If $\sum_{p\in S} p^{-\sigma} < \infty$ for every $\sigma > 1/2$, in particular if $\#\{p \le x : p \in S\} \le x^{1/2+o(1)}$, then Q_{λ_S} has the same growth exponent 2Θ as Q_λ. For every $\beta \in (1/2, 1)$ there is a set S with $\#\{p \le x : p \in S\} \asymp x^\beta$ for which the growth exponent of Q_{λ_S} lies between 2β and $2\max(\beta, \Theta)$; it therefore exceeds 2Θ whenever $\beta > \Theta$.*

The lower bound in the second statement is a form of a theorem of Aymone [2] (Remark 6.4). The new point is the two-sided threshold: for every $\beta > 1/2$ some S of size x^β breaks the exponent, and no S of size $x^{1/2+o(1)}$ does. The functions λ_S keep every property of λ listed in Table 6, including Tao's logarithmically averaged two-point Chowla theorem [35] and Davenport's saving in additive twists. No argument for the critical bound can therefore rest only on properties of λ that survive a change of $x^{1/2+\delta}$ prime values.

At the critical scale $Q(N) \asymp N$ the energy is controlled by the oscillation of $L(x)/\sqrt{x}$ (Theorems 5.6, 5.13, Proposition 5.8 and Corollary 5.10).

Theorem D (Size at the critical scale)**.** *Unconditionally $Q(N) > 2.8139\,N$ for infinitely many N. The ratio $Q(N)/N$ is bounded if and only if $L(x) \ll \sqrt{x}$, and if it is bounded, then the Riemann hypothesis holds, every zero is simple, and every tail of the sequence of positive ordinates satisfies an integer linear relation. Under the Riemann hypothesis, if $\sum_{0<\gamma\le T} |\zeta'(\rho)|^{-2} \ll T$, then*

$$\lim_{X\to\infty} \frac{1}{\log X} \sum_{N\le X} \frac{Q(N)}{N^2} = \frac{2}{3\zeta(\frac12)^2} + \sum_{\gamma} K(\tfrac12 + i\gamma)\Big|\frac{\zeta(2\rho)}{\rho\,\zeta'(\rho)}\Big|^2 > 0.4579,$$

where γ runs over the ordinates of both signs and $K(\frac12 + i\gamma) = 2 - 3/(\frac94 + \gamma^2)$ is the weight of Section 3.2.

The first statement transfers an oscillation theorem of Mossinghoff and Trudgian [29], the second rests on Ingham's theorem [20], and the third on a mean-value lemma of Humphries [19]. In particular the set of N with $Q(N) > N$ is infinite, which under the Riemann hypothesis is the set B_Θ on which Theorem B is silent.

The last group concerns approximation and smoothing. Beurling [7] proved that the dilates of the fractional part are complete in $L^p(0,1)$ exactly when ζ has no zero in $\operatorname{Re} s > 1/p$, and Báez-Duarte [3] showed that at $p = 2$ the natural dilates $\{1/(nx)\}$ suffice. The endpoint theorem (Theorem 13.5) carries Báez-Duarte's form to every exponent: in the weighted norms $\|\cdot\|_\sigma$ the distance $d_{M,\sigma}$ from 1 to the span of $g_n(x) = \{1/(nx)\} - \frac1n\{1/x\}$, $n \le M$, tends to 0 if and only if $\zeta(s) \ne 0$ for $\operatorname{Re} s > \sigma$, it does so at a power rate inside $(\Theta, 1)$, at the

[2] In Section 6 the same function is written f_S.

rate $(\log M)^{-1/4+o(1)}$ at $\sigma = 1/2$ under the Riemann hypothesis, and at an unconditional rate for $\sigma \geq 1$. For the exponentially smoothed sum $B(X) = \sum_n \lambda(n)\mathrm{e}^{-n/X}$ we prove the following.

Theorem E (Unconditional bounds)**.** *There are absolute constants $c > 0$ and X_0 such that*

$$|B(X)| \leq X \exp\Big(-\frac{c \log X}{(\log\log X)^{2/3}(\log\log\log X)^{1/3}}\Big) \qquad (X \geq X_0),$$

and $\liminf_{X\to\infty} B(X)/\sqrt{X} \leq -\sqrt{\pi}/(2|\zeta(1/2)|) = -0.60685\ldots$. *For every $x \geq 1$ and every integer $N \geq 1$,*

$$|L(x)| \leq \frac{x}{2641} + 8.3\sqrt{x}, \qquad Q(N) \leq 5\Big(\frac{N}{2641} + 8.3\sqrt{N}\Big)^2.$$

The first bound is Theorem 12.4: the smoothing replaces the height X in the Vinogradov–Korobov region by the height $\log X$, which is why the saving is larger than the one known for $L(x)$ itself. The second is Proposition 12.8, and comes from the pole of $\Gamma(s)\zeta(2s)/\zeta(s)$ at $s = 1/2$. The explicit bounds are Theorem 15.1 and rest on the estimate of Cohen, Dress and El Marraki [14] for the Mertens function.

Table 1 lists the main statements with their hypotheses, and Figure 1 shows how the four layers of the paper are joined.

TABLE 1. The main results. "None" means that the statement is unconditional. Every statement that would give the Riemann hypothesis is an equivalence, and its other side is a zero-free half-plane.

Hypothesis	Conclusion	Where
$\zeta \neq 0$ on $\mathrm{Re}\, s > \theta$	$L \ll x^{\theta+\varepsilon}$ and $Q \ll N^{2\theta+\varepsilon}$, and conversely	Thm. 3.1
none	$\log Q(N)/\log N \to 2\Theta$	Cor. 4.3
$\sigma < \Theta$	$Q(N) > N^{2\sigma}$ for all large N	Thm. 4.2
$Q(N_k) \leq N_k^{1+\varepsilon}$ on one sequence, each ε	Riemann hypothesis, and conversely	Thm. 4.2
$\#S \cap [1,x] \leq x^{1/2+o(1)}$	exponent of Q_{λ_S} equals 2Θ	Cor. 6.5
$\beta \in (\frac{1}{2}, 1)$	some S of size x^β has exponent $\geq 2\beta$	Thm. 6.3
none	$Q(N) > 2.8139\,N$ infinitely often	Thm. 5.6
none	$Q(N)/N$ bounded $\Leftrightarrow$ $L(x) \ll \sqrt{x}$	Prop. 5.8
$\zeta \neq 0$ on $\mathrm{Re}\, s > \sigma$	$d_{M,\sigma} \to 0$, and conversely	Thm. 13.5
Riemann hypothesis	$d_{M,1/2}^2 \ll (\log M)^{-1/2+o(1)}$	Thm. 13.9
none	$\lvert B(X)\rvert \leq X\exp(-c\log X/\ell(\log X))$	Thm. 12.4
none	$\liminf B(X)/\sqrt{X} \leq -0.60685$	Prop. 12.8
none	$\lvert L(x)\rvert \leq x/2641 + 8.3\sqrt{x}$	Thm. 15.1

1.2. **What a proof of the critical bound must contain.** The critical bound is $Q(N) \ll N^{1+\varepsilon}$. By Theorems A and B it is equivalent to the Riemann hypothesis, and it cannot be weakened to a statement about a subsequence, a density or an average over N. By Theorem C it cannot follow from any property of λ that is stable under changing $x^{1/2+\delta}$ prime values, and in particular not from the correlation and exponential-sum estimates of Table 6. Short windows behave differently: the theorem of Matomäki and Radziwiłł [25] gives $\mathcal{Q}(N,H) = o(NH)$ as $H \to \infty$ (Theorem 15.2), with a saving that is only a power of $\log\log H/\log H$. A conditional input "ζ has no zeros in $\mathrm{Re}\, s > \theta$" gives the exponent 2θ and nothing better; with $\theta < 1$ fixed it would already be a new theorem. The unconditional exponent is 2, with the saving of Theorem 15.1.[3]

[3]Large-value estimates for Dirichlet polynomials, in particular those of Guth and Maynard [18], improve zero-density bounds but give no zero-free half-plane, and therefore no power saving in Q.

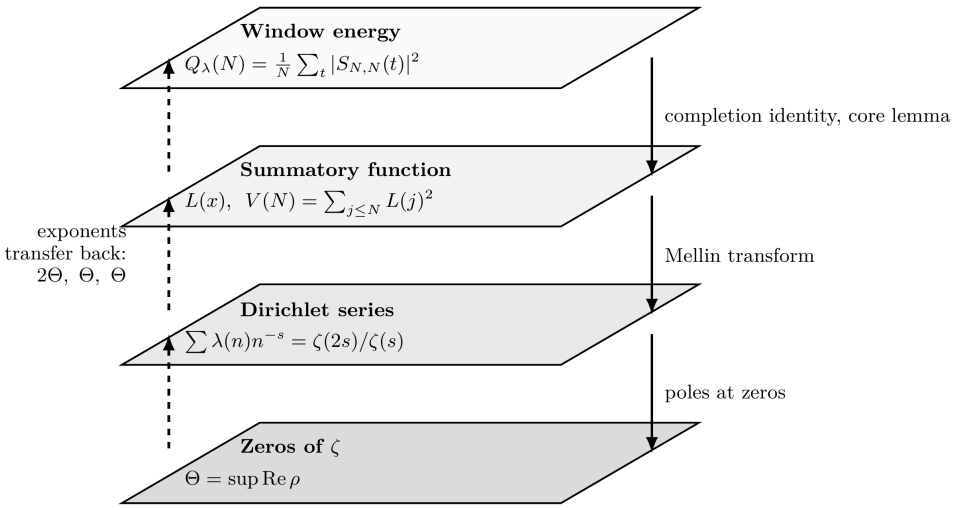


FIGURE 1. The four layers of the paper. The energy Q is bounded above and below by the square of the summatory function L (Corollary 2.2); L and the Dirichlet series $\zeta(2s)/\zeta(s)$ are related by the Mellin transform; the poles of that series in $\operatorname{Re} s > 1/2$ are the zeros of ζ. Each solid arrow is a passage from one layer to the next, and each dashed arrow returns the exponent: 2Θ for the energy, Θ for L and for the abscissa of the series. The top plane therefore carries exactly the information of the bottom one.

1.3. **Organisation and verification.** Section 2 proves the energy identities, and Sections 3 and 4 prove Theorems A and B. The sensitivity theorem and the obstructions it implies occupy Sections 6 to 11. Sections 12 to 15 contain the smoothed sum, the reciprocal dilates, Dirichlet characters and the unconditional bounds. Section 16 returns to the critical bound. The appendices contain the full calculations. The finite identities and the numerical constants are checked by a Lean 4 project and by Julia programs, described in Appendix K; no proof depends on them.

2. The Liouville energy

2.1. **Windows and the Fejér identity.** Let $a = (a_n)_{n\ge 1}$ be a real sequence, and write

$$L_a(x) = \sum_{n\le x} a_n, \qquad C_{a,h}(M) = \sum_{n\le M} a_n a_{n+h}, \qquad A_N(\alpha) = \sum_{n\le N} a_n \mathrm{e}(n\alpha),$$

where $\mathrm{e}(\alpha) = e^{2\pi i\alpha}$. Fix integers $N \ge 1$ and $H \ge 1$. The truncation $a_1, \dots, a_N$ is extended by zero to all of $\mathbb{Z}$, and for $t \in \mathbb{Z}$ the window sum is

$$S_{N,H}(t) = \sum_{t<n\le t+H} a_n \mathbf{1}_{[1,N]}(n).$$

Only the $N + H - 1$ windows with $1 - H \le t \le N - 1$ meet $[1, N]$. The *window energy* is

$$\mathcal{Q}_a(N,H) = \frac{1}{H}\sum_{t\in\mathbb{Z}} S_{N,H}(t)^2, \qquad Q_a(N) = \mathcal{Q}_a(N,N). \tag{2.1}$$

The factor $1/H$ normalizes the diagonal: if the products $a_m a_n$ with $m \ne n$ cancel, then $\mathcal{Q}_a(N,H) = \sum_{n\le N} a_n^2$. For the Liouville function we write $L = L_\lambda$, $\mathcal{Q}_\lambda$ and Q_λ.

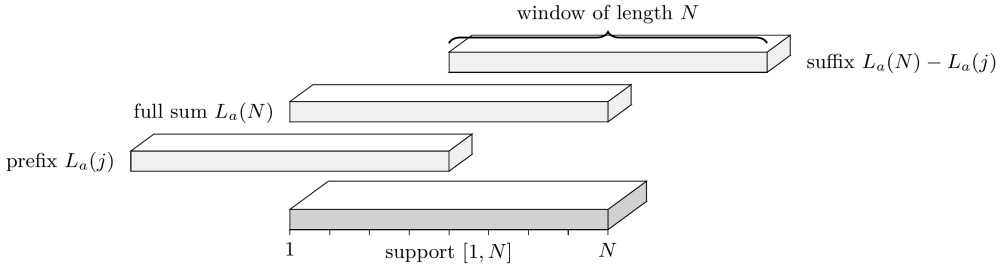


FIGURE 2. Windows of length N in the zero-extended sequence $a_n \mathbf{1}_{[1,N]}(n)$. The bottom slab is the support $[1, N]$. A window that starts to the left of 1 sees a prefix sum $L_a(j)$; the window $(0, N]$ sees the full sum $L_a(N)$; a window that ends to the right of N sees a suffix sum $L_a(N) - L_a(j)$. Each proper prefix and each proper suffix of $[1, N]$ occurs in exactly one window, which is the content of (2.5).

Proposition 2.1 (Fejér identity)**.** *Let $F_H(\alpha) = H^{-1}\big|\sum_{j=1}^{H} \mathrm{e}(j\alpha)\big|^2$ be the Fejér kernel. For every real sequence a and all integers $N, H \geq 1$,*

$$\mathcal{Q}_a(N, H) = \int_0^1 |A_N(\alpha)|^2 F_H(\alpha)\, \mathrm{d}\alpha \tag{2.2}$$

$$= \sum_{n \leq N} a_n^2 + 2 \sum_{h=1}^{\min(H,N)-1} \Big(1 - \frac{h}{H}\Big) C_{a,h}(N - h). \tag{2.3}$$

Moreover

$$\mathcal{Q}_a(N, H) \geq \frac{H}{N + H - 1}\, L_a(N)^2, \tag{2.4}$$

and if $H \geq N$, then

$$\mathcal{Q}_a(N, H) = \frac{H - N + 1}{H}\, L_a(N)^2 + \frac{1}{H} \sum_{j=1}^{N-1} \Big(L_a(j)^2 + \big(L_a(N) - L_a(j)\big)^2\Big). \tag{2.5}$$

Proof. Expanding the square in (2.1) gives

$$\mathcal{Q}_a(N, H) = \frac{1}{H} \sum_{m,n \leq N} a_m a_n\, c(m, n),$$

where $c(m, n)$ is the number of $t \in \mathbb{Z}$ with $m, n \in (t, t + H]$. By Lemma A.1(a), $c(m, n) = (H - |m - n|)_+$, with $u_+ = \max(u, 0)$. Grouping the pairs by $h = |m - n|$ gives (2.3). The Fejér kernel has the expansion $F_H(\alpha) = \sum_{|h|<H} (1 - |h|/H)\mathrm{e}(h\alpha)$, so that

$$\int_0^1 |A_N(\alpha)|^2 F_H(\alpha)\, \mathrm{d}\alpha = \sum_{m,n \leq N} a_m a_n \Big(1 - \frac{|m - n|}{H}\Big)_+,$$

which is the same sum. This proves (2.2).

Each $n \in [1, N]$ lies in exactly H windows, so $\sum_t S_{N,H}(t) = H\, L_a(N)$. The sum has at most $N + H - 1$ nonzero terms, and the Cauchy–Schwarz inequality gives $H^2 L_a(N)^2 \leq (N + H - 1) \sum_t S_{N,H}(t)^2 = (N + H - 1) H\, \mathcal{Q}_a(N, H)$. This is (2.4).

Let $H \geq N$. By Lemma A.1(b), exactly $H - N + 1$ windows contain $[1, N]$ and have sum $L_a(N)$. For each $1 \leq j \leq N - 1$, exactly one window meets $[1, N]$ in the prefix $[1, j]$, with sum $L_a(j)$, and exactly one meets it in the suffix $[j + 1, N]$, with sum $L_a(N) - L_a(j)$. There are no other windows with a nonempty intersection. Summing the squares gives (2.5). □

At $H = N$ the identity $x^2 + (y-x)^2 = 2(x - \frac{1}{2}y)^2 + \frac{1}{2}y^2$, applied in (2.5) with $x = L_a(j)$ and $y = L_a(N)$, gives the *completion identity*

$$Q_a(N) = \frac{N+1}{2N}\, L_a(N)^2 + \frac{2}{N} \sum_{j=1}^{N-1} \Big(L_a(j) - \tfrac{1}{2} L_a(N) \Big)^2. \tag{2.6}$$

Corollary 2.2. *For every real sequence a and every $N \geq 1$,*

$$\tfrac{1}{2}\, L_a(N)^2 \leq Q_a(N) \leq 5 \max_{j \leq N} L_a(j)^2. \tag{2.7}$$

Proof. The lower bound is the first term of (2.6), since $(N+1)/(2N) \geq \frac{1}{2}$. For the upper bound put $m = \max_{j \leq N} |L_a(j)|$. In (2.5) with $H = N$, the first term is at most m^2/N, and each of the $N-1$ summands in the bracket is at most $m^2 + (2m)^2 = 5m^2$; so $Q_a(N) \leq m^2/N + 5m^2(N-1)/N \leq 5m^2$. □

The energy is therefore bounded below by the square of the sum at its own endpoint, and above by the square of the largest prefix sum. Both bounds lose only constants, which is what makes $Q_\lambda(N)$ an exact gauge for the zeros of ζ in Section 3.

Shorter windows are less informative. If $H \leq N$, then (2.4) gives $L_a(N)^2 \leq (2N/H)\, \mathcal{Q}_a(N, H)$, so a bound for $\mathcal{Q}_a(N, H)$ controls $L_a(N)$ with a loss of the factor N/H. This loss is recorded in Corollary 3.4.

2.2. **Three model sequences.** Three sequences with $|a_n| = 1$ show the range of the energy.

Alternating signs. Let $a_n = (-1)^n$. Then $L_a(j) = -1$ for odd j and $L_a(j) = 0$ for even j. If N is even, then $L_a(N) = 0$ and the sum in (2.5) has $N/2$ odd indices, each contributing 2. If N is odd, then $L_a(N) = -1$, the first term contributes 1, and each of the $N-1$ indices j contributes exactly 1: from $L_a(j)^2$ when j is odd and from $(L_a(N) - L_a(j))^2$ when j is even. In both cases $Q_a(N) = 1$.

Fair signs. Let ϵ_n be independent random signs with $\mathbf{P}(\epsilon_n = 1) = \mathbf{P}(\epsilon_n = -1) = \frac{1}{2}$. Then $\mathbf{E}\, C_{\epsilon,h}(M) = 0$ for $h \geq 1$, and (2.3) gives $\mathbf{E}\, Q_\epsilon(N) = N$.

Constant signs. Let $a_n = 1$. Then $L_a(j) = j$, and (2.5) gives

$$Q_{\mathbf{1}}(N) = \frac{1}{N}\Big(N^2 + 2 \sum_{j=1}^{N-1} j^2 \Big) = N + \frac{(N-1)(2N-1)}{3} = \frac{2N^2+1}{3}.$$

TABLE 2. The energy $Q_a(N)$ for three model sequences and for the Liouville function. The first three rows are exact. The last row is Corollary 3.2 together with Corollary 4.3; here Θ is the supremum of the real parts of the zeros of ζ.

sequence a_n	size of L_a on $[1, N]$	$Q_a(N)$	exponent of N
$(-1)^n$	1	1	0
fair signs (means)	$\big(\mathbf{E}\, L_\epsilon(N)^2\big)^{1/2} = N^{1/2}$	N	1
1	N	$(2N^2+1)/3$	2
$\lambda(n)$	$N^{\Theta + o(1)}$	$N^{2\Theta + o(1)}$	$2\Theta \in [1, 2]$

All three sequences have the same absolute values, and their energies have three different orders of magnitude. The Riemann hypothesis is the statement that λ belongs in the second row, up to factors N^ε.

2.3. **Positivity.** The identity (2.5) writes the energy as a sum of squares. This positivity holds for every sequence, and for that reason it carries no information about zeros.

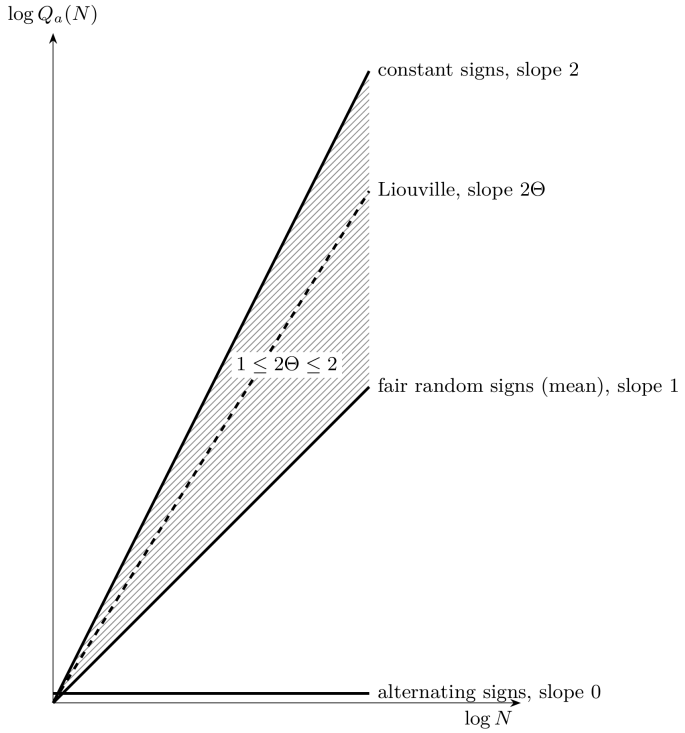


FIGURE 3. Growth exponents of the energy, on a logarithmic scale. Alternating signs give $Q_a(N) = 1$, fair random signs give $\mathbf{E}\, Q_\epsilon(N) = N$, and constant signs give $Q_{\mathbf{1}}(N) = (2N^2+1)/3$ (Table 2). The Liouville energy grows like $N^{2\Theta+o(1)}$ (Corollary 4.3), so its line lies in the hatched sector $1 \le 2\Theta \le 2$; the Riemann hypothesis places it on the lower edge.

Proposition 2.3 (The energy is a positive definite Toeplitz form)**.** *For every real sequence a and every $N \ge 1$,*

$$Q_a(N) = \frac{1}{N}\sum_{j=0}^{N-1} \Big(L_a(j)^2 + \big(L_a(N) - L_a(j)\big)^2 \Big) = a^{\mathsf{T}} T_N a, \tag{2.8}$$

where $L_a(0) = 0$ and $T_N = (1 - |m-n|/N)_{1\le m,n\le N}$. The matrix T_N is positive definite, and $a \mapsto Q_a(N)$ is a convex function on $\mathbb{R}^N$. On the cube $\{\pm 1\}^N$,

$$1 \le Q_a(N) \le \frac{2N^2+1}{3}, \tag{2.9}$$

both bounds are attained, and the mean of $Q_a(N)$ over the cube is N.

Proof. The first equality in (2.8) is (2.5) at $H = N$, with the term $j = 0$ equal to $L_a(N)^2$. The second is (2.3) at $H = N$. The first expression is a sum of squares of linear forms in a. If it vanishes, then $L_a(j) = 0$ for all $0 \le j \le N$, hence $a_j = L_a(j) - L_a(j-1) = 0$ for all j. So T_N is positive definite, and a positive definite quadratic form is convex.

Let $a \in \{\pm 1\}^N$. Then $L_a(j) \equiv j \pmod 2$. In the first expression of (2.8), an index j for which $L_a(j)$ is odd contributes at least 1 through $L_a(j)^2$. An index j for which $L_a(j)$ is even and $L_a(N)$ is odd contributes at least 1 through $(L_a(N) - L_a(j))^2$. If N is odd, every j is of one of these two kinds, and $Q_a(N) \ge N/N = 1$. If N is even, then $L_a(N)$ is even, each of the $N/2$ odd indices j contributes at least 2, and again $Q_a(N) \ge 1$. The alternating sequence attains the value 1. Since the entries of T_N are positive, $a^{\mathsf{T}} T_N a \le \mathbf{1}^{\mathsf{T}} T_N \mathbf{1} = (2N^2+1)/3$. The mean over the cube is the trace of T_N, which is N. □

Every sum-of-squares, Gram or Cauchy–Schwarz argument applied to the form $a^{\mathsf{T}}T_N a$ bounds it from below. The inequality (2.4), the lower bound in (2.7) and the pole bounds of Section 5 are of this kind. The estimate that matters for the zeros of ζ is an upper bound at the single vector $a = \lambda$, and by (2.9) the cube contains vectors at which the form is as large as N^2. An upper bound for $Q_\lambda(N)$ must therefore use arithmetic information about λ; positivity of the form gives none.[4]

2.4. **The monotone core.** The energy $Q_a(N)$ is not monotone in N. It is comparable, however, to a monotone quantity. Put

$$V_a(N) = \sum_{j \le N} L_a(j)^2, \qquad V(N) = V_\lambda(N). \tag{2.10}$$

Lemma 2.4 (Monotone core)**.** *For every real sequence a and every $N \ge 1$,*

$$V_a(N) \le (2N+1)\, Q_a(N). \tag{2.11}$$

If in addition $|a_n| \le 1$ for all n, then

$$Q_a(N) \le 11\, V_a(N)^{2/3}. \tag{2.12}$$

In particular $V(N) \le (2N+1)Q_\lambda(N)$ and $Q_\lambda(N) \le 11\, V(N)^{2/3}$ for all $N \ge 1$.

Proof. Write $L = L_a$, $Q = Q_a(N)$ and $V = V_a(N)$. By (2.6), $L(N)^2 \le 2Q$ and $\sum_{j<N}(L(j) - \frac{1}{2}L(N))^2 \le \frac{1}{2}NQ$. Since $L(j)^2 \le 2(L(j) - \frac{1}{2}L(N))^2 + \frac{1}{2}L(N)^2$, summation over $j < N$ and addition of the term $j = N$ give $V \le NQ + (N-1)Q + 2Q$, which is (2.11).

For (2.12), the inequality $(L(N) - L(j))^2 \le 2L(N)^2 + 2L(j)^2$ in (2.5) gives $Q \le 2L(N)^2 + 3V/N$. Let $m = |L(N)|$. Since $|a_n| \le 1$, the sum L changes by at most 1 at each step, so $|L(j)| \ge m/2$ for the integers $j \in [N - m/2, N]$. There are at least $m/2$ of them, and all are positive since $m \le N$. Hence $V \ge m^3/8$, that is, $L(N)^2 \le 4V^{2/3}$. Finally $V \le \sum_{j \le N} j^2 \le N^3$ gives $V/N \le V^{2/3}$. Together, $Q \le 8V^{2/3} + 3V^{2/3}$. The individual steps are written out in Appendix A.2. □

The lemma is used as follows. Since V is nondecreasing, a bound for Q_λ at one point N bounds the mean square of L on the whole interval $[1, N]$:

$$\int_1^{N+1} L(x)^2 \, \mathrm{d}x = V(N) \le (2N+1)\, Q_\lambda(N). \tag{2.13}$$

This is the bridge between the energy at a single point and the mean-value method of Section 4.

3. The exact exponent

Throughout, $\rho = \beta + i\gamma$ denotes a nontrivial zero of ζ, and

$$\Theta = \sup_\rho \beta.$$

The functional equation and the existence of nontrivial zeros give $\frac{1}{2} \le \Theta \le 1$, and the Riemann hypothesis is $\Theta = \frac{1}{2}$. No value $\Theta < 1$ is known. For $\operatorname{Re} s > 1$ the Euler product gives

$$\sum_{n \ge 1} \frac{\lambda(n)}{n^s} = \prod_p \frac{1}{1 + p^{-s}} = \frac{\zeta(2s)}{\zeta(s)}, \qquad M(s) := \frac{\zeta(2s)}{s\,\zeta(s)} = \int_1^\infty L(x) x^{-s-1} \, \mathrm{d}x, \tag{3.1}$$

the second identity by partial summation [36, Section 1.2]. The function M is meromorphic on $\mathbb{C}$. In the half-plane $\operatorname{Re} s \ge \frac{1}{2}$ its poles are the simple pole at $s = \frac{1}{2}$, which comes from

[4] This is the point of contrast with Weil's criterion [38], in which the positivity of a certain quadratic form is itself equivalent to the Riemann hypothesis. In the notation of Section 3.2, the kernel $\langle w_s, w_z \rangle = 1 - \frac{1}{1+s} - \frac{1}{1+\bar z} + \frac{2}{1+s+\bar z}$ is a Gram kernel, hence positive definite on $\operatorname{Re} s > -\frac{1}{2}$, whatever the zeros are. The zeros enter through the size of the vector U_X of Section 3.2, and no sign condition is involved.

$\zeta(2s)$ and has residue $1/\zeta(\frac{1}{2})$, and the zeros ρ of ζ with $\beta \geq \frac{1}{2}$; at such a zero $\zeta(2\rho) \neq 0$, because ζ does not vanish on $\operatorname{Re} s \geq 1$ and $2\rho \neq 1$. The zero of $1/\zeta$ at $s = 1$ makes M regular there.

3.1. **Zero-free half-planes.**

Theorem 3.1 (Equivalent zero-free bounds)**.** *Let $\frac{1}{2} \leq \theta \leq 1$. The following conditions are equivalent.*

(i) $\zeta(s) \neq 0$ *for* $\operatorname{Re} s > \theta$.
(ii) $L(x) \ll_{\theta,\varepsilon} x^{\theta+\varepsilon}$ *for every* $\varepsilon > 0$ *and all* $x \geq 1$.
(iii) $Q_\lambda(N) \ll_{\theta,\varepsilon} N^{2\theta+\varepsilon}$ *for every* $\varepsilon > 0$ *and all* $N \geq 1$.

In particular, the Riemann hypothesis is equivalent to $Q_\lambda(N) \ll_\varepsilon N^{1+\varepsilon}$.

Proof. (ii)⇔(iii). This is Corollary 2.2: $L(N)^2 \leq 2Q_\lambda(N)$ and $Q_\lambda(N) \leq 5 \max_{j \leq N} L(j)^2$, with $L(x) = L(\lfloor x \rfloor)$.

(ii)⇒(i). Under (ii) the integral in (3.1) converges absolutely and locally uniformly on $\operatorname{Re} s > \theta$, so M is holomorphic there. The identity $\zeta(s)\, sM(s) = \zeta(2s)$ holds on $\operatorname{Re} s > 1$, hence on $\operatorname{Re} s > \theta$. If $\zeta(\rho) = 0$ with $\operatorname{Re} \rho > \theta \geq \frac{1}{2}$, the left side vanishes at ρ, so $\zeta(2\rho) = 0$ with $\operatorname{Re}(2\rho) > 1$, which is impossible.

(i)⇒(ii). For $\theta = 1$ the bound is trivial. For $\theta < 1$ it is Proposition A.3 of Appendix A.3, which combines a bound $1/\zeta(s) \ll (1 + |t|)^\eta$ to the right of the line $\operatorname{Re} s = \theta$ (Lemma A.2) with the truncated Perron formula.

For the last assertion take $\theta = \frac{1}{2}$; a zero with $\beta < \frac{1}{2}$ would be reflected by the functional equation to a zero with real part $1 - \beta > \frac{1}{2}$. □

Corollary 3.2 (Exact growth exponent)**.**

$$\limsup_{N \to \infty} \frac{\log Q_\lambda(N)}{\log N} = 2\Theta.$$

Proof. By (2.9), $Q_\lambda(N) \geq 1$, so the logarithm is nonnegative. Theorem 3.1 with $\theta = \Theta$ gives the upper bound 2Θ. Suppose the limit superior is $2\kappa < 2\Theta$. If $\Theta > \frac{1}{2}$, choose θ with $\max(\kappa, \frac{1}{2}) < \theta < \Theta$; then (iii) holds for θ, so ζ has no zero in $\operatorname{Re} s > \theta$, contrary to the definition of Θ. If $\Theta = \frac{1}{2}$, then $\kappa < \frac{1}{2}$ and (2.7) gives $L(x) \ll x^{\kappa'}$ for some $\kappa' < \frac{1}{2}$. Then M is holomorphic on $\operatorname{Re} s > \kappa'$, whereas it has a pole at $s = \frac{1}{2}$. □

Corollary 4.3 below shows that the limit superior is a limit.

Corollary 3.3 (A constraint at the square-root scale)**.** *Suppose that $Q_\lambda(N) \ll N(\log 2N)^B$ for some $B \geq 0$. Then the Riemann hypothesis holds and every nontrivial zero of ζ has multiplicity at most $1 + B/2$. In particular, if $B < 2$, all nontrivial zeros are simple.*

Proof. Theorem 3.1 gives the Riemann hypothesis, and (2.7) gives $L(x) \ll x^{1/2}(\log 2x)^{B/2}$. Let $\rho = \frac{1}{2} + i\gamma$ be a zero of multiplicity m. For $0 < u \leq 1$, the substitution $v = \log x$ in (3.1) gives

$$\begin{aligned} |M(\rho + u)| &\ll \int_1^\infty x^{-1-u} (\log 2x)^{B/2}\, \mathrm{d}x \\ &= \int_0^\infty e^{-uv} (v + \log 2)^{B/2}\, \mathrm{d}v \ll_B u^{-1-B/2}. \end{aligned}$$

Since $\zeta(2\rho) \neq 0$, the function M has a pole of order exactly m at ρ, so $|M(\rho + u)| \gg u^{-m}$ as $u \downarrow 0$. Hence $m \leq 1 + B/2$. □

Corollary 3.4 (Short windows)**.** *Let $1 \leq H(N) \leq N$ be integers and let $0 < \delta \leq 1$. If $\mathcal{Q}_\lambda(N, H(N)) \ll N^{1-\delta} H(N)$ for all N, then $\zeta(s) \neq 0$ for $\operatorname{Re} s > 1 - \delta/2$.*

Proof. By (2.4), $L(N)^2 \leq (2N/H)\, \mathcal{Q}_\lambda(N, H) \ll N^{2-\delta}$. Apply Theorem 3.1 with $\theta = 1 - \delta/2$. □

Since $0 \le F_H \le H$ and $\int_0^1 |A_N|^2 \,\mathrm{d}\alpha = N$, the identity (2.2) gives the trivial bound $\mathcal{Q}_\lambda(N,H) \le NH$. The hypothesis of Corollary 3.4 is thus a power saving over the trivial bound, at an arbitrary window length. Every such saving is a zero-free half-plane.

3.2. **A mean-square identity.** The energy has an exact description as the squared norm of a vector, and its Mellin transform is a multiple of $M(s)$. Let

$$\mathcal{H} = L^2\big((0,1);\mathbb{C}^2\big),$$

with inner product $\langle f,g\rangle = \int_0^1 (f_1\overline{g_1} + f_2\overline{g_2})\,\mathrm{d}u$. Set $L(x) = 0$ for $0 < x < 1$, and for $X > 0$ and $\operatorname{Re} s > -\frac12$ define

$$U_X(u) = \big(L(uX),\, L(X) - L(uX)\big), \qquad w_s(u) = \big(u^s,\, 1-u^s\big) \qquad (0<u<1). \tag{3.2}$$

Lemma 3.5. *(a) For an integer $N \ge 1$ and $0 \le a < 1$,*

$$\|U_{N+a}\|^2 = \frac{N\,Q_\lambda(N) + a\,L(N)^2}{N+a}. \tag{3.3}$$

In particular $\|U_N\|^2 = Q_\lambda(N)$, and for $N \le X < N+1$,

$$N\,Q_\lambda(N) \le X\,\|U_X\|^2 \le (N+2)\,Q_\lambda(N). \tag{3.4}$$

(b) The map $s \mapsto w_s$ is holomorphic from $\operatorname{Re} s > -\frac12$ to $\mathcal{H}$, and for $s = \sigma + i\tau$,

$$K(s) := \|w_s\|^2 = 1 + \frac{2}{1+2\sigma} - \frac{2(1+\sigma)}{(1+\sigma)^2+\tau^2}. \tag{3.5}$$

For $\sigma \ge \frac12$,

$$4\sqrt{2} - 5 \le K(s) < 2, \qquad 4\sqrt{2}-5 = 1 - \frac{2}{3+2\sqrt{2}} = 0.65685\ldots, \tag{3.6}$$

with equality on the left only at $s = 1/\sqrt{2}$. Moreover $K(\frac12) = \frac23$ and $K(\frac12 + i\tau) = 2 - 2\operatorname{Re}\frac{1}{3/2+i\tau}$.

(c) For $\operatorname{Re} s > 1$, with an absolutely convergent $\mathcal{H}$-valued integral,

$$\int_1^\infty U_X\, X^{-s-1}\,\mathrm{d}X = M(s)\,w_s. \tag{3.7}$$

Proof. (a) With $X = N+a$ and $v = uX$, $\|U_X\|^2 = X^{-1}\int_0^X \big(L(v)^2 + (L(N)-L(v))^2\big)\,\mathrm{d}v$. On $[j, j+1)$ with $0 \le j < N$ the integrand is $L(j)^2 + (L(N)-L(j))^2$, and on $[N, X)$ it is $L(N)^2$. By (2.8) the integral equals $N\,Q_\lambda(N) + a\,L(N)^2$. The inequalities (3.4) follow from $0 \le a < 1$ and $L(N)^2 \le 2Q_\lambda(N)$.

(b) Direct integration gives $\int_0^1 u^{2\sigma}\,\mathrm{d}u = 1/(1+2\sigma)$ and $\int_0^1 |1-u^s|^2\,\mathrm{d}u = 1 - 2\operatorname{Re}\frac{1}{1+s} + \frac{1}{1+2\sigma}$, which is (3.5). Holomorphy follows because, on compact subsets of $\operatorname{Re} s > -\frac12$, the derivative $u^s \log u$ is dominated in $L^2(0,1)$. The bounds (3.6) are proved in Appendix A.4.

(c) Since $|L(x)| \le x$, we have $\|U_X\| \le 2X$, and the integral converges absolutely for $\operatorname{Re} s > 1$. Its value may be computed pointwise in u.[5] For the first coordinate, the substitution $v = uX$ and $L(v) = 0$ for $v < 1$ give $\int_1^\infty L(uX) X^{-s-1}\,\mathrm{d}X = u^s \int_u^\infty L(v) v^{-s-1}\,\mathrm{d}v = u^s M(s)$. The second coordinate is $M(s) - u^s M(s)$. □

Theorem 3.6 (Mean-square identity)**.** *Let $\sigma > \frac12$, and suppose that*

$$\mathcal{I}(\sigma) := \int_1^\infty \|U_X\|^2 X^{-2\sigma-1}\,\mathrm{d}X < \infty. \tag{3.8}$$

Then M has no pole in the closed half-plane $\operatorname{Re} s \ge \sigma$, and

$$\int_1^\infty \|U_X\|^2 X^{-2\sigma-1}\,\mathrm{d}X = \frac{1}{2\pi}\int_{-\infty}^\infty \big|M(\sigma+i\tau)\big|^2 K(\sigma+i\tau)\,\mathrm{d}\tau. \tag{3.9}$$

[5]For $g \in \mathcal{H}$, Fubini's theorem gives $\langle \int_1^\infty U_X X^{-s-1}\,\mathrm{d}X, g\rangle = \int_0^1 \big(\int_1^\infty U_X(u) X^{-s-1}\,\mathrm{d}X\big)\cdot\overline{g(u)}\,\mathrm{d}u$, so the Bochner integral is represented by the pointwise integral.

The condition (3.8) *holds if and only if* $\sum_{N\ge1} Q_\lambda(N)N^{-1-2\sigma}<\infty$, *and it holds for every* $\sigma>\max(\Theta,\frac12)$. *Consequently, for* $\sigma>\frac12$,

$$\sigma>\Theta \implies \sum_{N\ge1}\frac{Q_\lambda(N)}{N^{1+2\sigma}}<\infty \implies \sigma\ge\Theta, \tag{3.10}$$

and the Riemann hypothesis holds if and only if the series converges for every $\sigma>\frac12$.

Proof. Put $F(t)=e^{-\sigma t}U_{e^t}$ for $t\ge0$ and $F(t)=0$ for $t<0$. Then $\mathcal{I}(\sigma)=\int_0^\infty\|F(t)\|^2\,\mathrm{d}t$, so $F\in L^2(\mathbb{R};\mathcal{H})$. For $\eta>0$ the function $F_\eta(t)=e^{-\eta t}F(t)$ lies in $L^1\cap L^2$ by the Cauchy–Schwarz inequality. Hence

$$\Phi(s)=\int_0^\infty F(t)e^{-(s-\sigma)t}\,\mathrm{d}t=\int_1^\infty U_X X^{-s-1}\,\mathrm{d}X$$

is a holomorphic $\mathcal{H}$-valued function on $\operatorname{Re}s>\sigma$. By (3.7), $\Phi(s)=M(s)w_s$ for $\operatorname{Re}s>1$. For each fixed $g\in\mathcal{H}$, both $\langle\Phi(s),g\rangle$ and $M(s)\langle w_s,g\rangle$ are meromorphic on $\operatorname{Re}s>\sigma$, so they agree there. Taking $g=w_{s_0}$ at a point s_0 with $\operatorname{Re}s_0>\sigma$ shows that $M(s)\langle w_s,w_{s_0}\rangle$ is bounded near s_0; since $\langle w_{s_0},w_{s_0}\rangle=K(s_0)>0$, the function M has no pole at s_0. Thus $\Phi(s)=M(s)w_s$ on $\operatorname{Re}s>\sigma$.

The Fourier transform of F_η is $\widehat{F_\eta}(\tau)=\Phi(\sigma+\eta+i\tau)=M(\sigma+\eta+i\tau)\,w_{\sigma+\eta+i\tau}$. As $\eta\downarrow0$, $F_\eta\to F$ in $L^2(\mathbb{R};\mathcal{H})$ by dominated convergence, so by the Plancherel theorem[6] $\widehat{F_\eta}\to\widehat{F}$ in L^2. On the other hand $\widehat{F_\eta}(\tau)\to M(\sigma+i\tau)w_{\sigma+i\tau}$ for every τ such that $\sigma+i\tau$ is not a pole of M, and the poles form a discrete set. A sequence that converges in L^2 has a subsequence converging almost everywhere to the same limit. Hence $\widehat{F}(\tau)=M(\sigma+i\tau)w_{\sigma+i\tau}$ for almost every τ, and the Plancherel identity $\int\|F\|^2\,\mathrm{d}t=(2\pi)^{-1}\int\|\widehat{F}\|^2\,\mathrm{d}\tau$ is (3.9).

If M had a pole at $\sigma+i\tau_0$, then $|M(\sigma+i\tau)|^2\ge c|\tau-\tau_0|^{-2}$ near τ_0, and since $K>\frac12$ by (3.6), the right side of (3.9) would diverge. So M has no pole on $\operatorname{Re}s=\sigma$.

Put $\iota_N=\int_N^{N+1}X^{-2\sigma-2}\,\mathrm{d}X$. By (3.4),

$$N\,Q_\lambda(N)\,\iota_N\le\int_N^{N+1}\|U_X\|^2X^{-2\sigma-1}\,\mathrm{d}X\le(N+2)\,Q_\lambda(N)\,\iota_N,$$

and both outer terms are of exact order $Q_\lambda(N)N^{-1-2\sigma}$, with constants depending only on σ. This gives the equivalence with the series. If $\sigma>\Theta$, Theorem 3.1 with $\varepsilon=\sigma-\Theta$ gives $Q_\lambda(N)\ll N^{2\sigma-\varepsilon}$ and the series converges. If the series converges, M has no pole in $\operatorname{Re}s\ge\sigma$; every zero of ζ with real part at least $\frac12$ is a pole of M, so $\Theta\le\sigma$. □

By (3.6) the weight K in (3.9) lies between two positive constants. The mean square of the energy on the scale $X^{2\sigma}$ is therefore comparable, with explicit constants, to $\int|\zeta(2s)/(s\zeta(s))|^2\,\mathrm{d}\tau$ on the line $\operatorname{Re}s=\sigma$. Averaging over N does not separate the energy from the zeros.

4. No exceptional set

Theorem 3.1 concerns bounds that hold for all N. One may ask for less: a bound $Q_\lambda(N)\le N^{2\sigma}$ for most N, or along a sequence. For $\sigma\ge0$ let

$$B_\sigma=\{N\ge1:\ Q_\lambda(N)>N^{2\sigma}\}.$$

The result of this section is that B_σ is finite or cofinite, according to the position of σ relative to Θ. A weaker bound of this kind is therefore not easier than the zero-free half-plane itself.

The proof uses a kernel that removes every singularity of M except one chosen pole. This device is due to Pintz, who used it to prove that a zero $\beta_0+i\gamma_0$ of ζ forces $\frac1Y\int_1^Y|M_\mu(x)|\,\mathrm{d}x\gg Y^{\beta_0}$ for all large Y, where M_μ is the summatory function of the Möbius function [31], and

[6]For functions with values in a separable Hilbert space the Plancherel theorem follows from the scalar case applied to the coordinates in an orthonormal basis.

more generally for mean values of arithmetic error terms [32].[7] Lemma 2.4 transfers the mean value to the energy.

Proposition 4.1 (Lower bound from one pole)**.** *Let $\rho_0 = \beta_0 + i\gamma_0$ be either a zero of ζ with $\beta_0 > \frac{1}{2}$, or the point $\rho_0 = \frac{1}{2}$. For every $\varepsilon > 0$ there is a constant $c = c(\rho_0, \varepsilon) > 0$ such that, for all $N \geq 1$,*

$$V(N) \geq c\, N^{1+2\beta_0-\varepsilon} \qquad \text{and} \qquad Q_\lambda(N) \geq c\, N^{2\beta_0-\varepsilon}. \tag{4.1}$$

In particular $Q_\lambda(N) \gg_\varepsilon N^{1-\varepsilon}$ for all N, unconditionally.

Proof. Let ν be the order of the zero ρ_0 in the first case, and put

$$W(s) = \begin{cases} \dfrac{s(s-1)\zeta(s)}{(s-\rho_0)^\nu}\, e^{(s-\rho_0)^2} & \text{if } \zeta(\rho_0) = 0, \\ s(s-1)\zeta(s)\, e^{(s-1/2)^2} & \text{if } \rho_0 = \frac{1}{2}. \end{cases} \tag{4.2}$$

In both cases W is entire. Define, for $u > 0$ and any real κ,

$$w(u) = \frac{1}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} W(s) u^{-s}\, \mathrm{d}s.$$

Lemma A.4 of Appendix A.5 establishes the following four facts.

(K1) The integral converges absolutely and does not depend on κ. For every $A \geq 1$ there is C_A with $|w(u)| \leq C_A \min(u^A, u^{-A})$ for all $u > 0$. In particular $C_w := \int_0^\infty |w(u)|^2 u^{-2}\, \mathrm{d}u < \infty$.

(K2) For $Y > 0$, with $G(s) = W(s)M(s)$,

$$I(Y) := \int_1^\infty L(x)\, w(x/Y) \frac{\mathrm{d}x}{x} = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} G(s) Y^s\, \mathrm{d}s.$$

(K3) The function G is meromorphic on $\mathbb{C}$. If $\zeta(\rho_0) = 0$, then $G(s) = (s-1)\zeta(2s) e^{(s-\rho_0)^2} (s-\rho_0)^{-\nu}$, and its only pole in $\operatorname{Re} s > \frac{1}{2}$ is at ρ_0. If $\rho_0 = \frac{1}{2}$, then $G(s) = (s-1)\zeta(2s) e^{(s-1/2)^2}$, and its only pole in $\mathbb{C}$ is the simple pole at $\frac{1}{2}$.

(K4) There are $\kappa_0 > 0$ and $Y_0 \geq 1$ such that $|I(Y)| \geq \kappa_0 Y^{\beta_0}$ for all $Y \geq Y_0$.

The essential point is (K3): the factor $\zeta(s)$ in W cancels the denominator of M, so that every zero of ζ other than ρ_0 disappears, and (K4) is the residue at the one remaining pole.

Let $N \geq 1$, let $0 < \varepsilon < 1$, and put $Y = N^{1-\varepsilon}$ and $A = 2/\varepsilon$. Split $I(Y)$ at $x = N+1$. On $[1, N+1)$ the Cauchy–Schwarz inequality, (2.13) and (K1) give

$$\begin{aligned} \Big| \int_1^{N+1} L(x)\, w(x/Y) \frac{\mathrm{d}x}{x} \Big| &\leq \Big(\int_1^{N+1} L(x)^2\, \mathrm{d}x \Big)^{1/2} \Big(\int_0^\infty |w(x/Y)|^2 \frac{\mathrm{d}x}{x^2} \Big)^{1/2} \\ &= \Big(\frac{C_w V(N)}{Y} \Big)^{1/2}. \end{aligned}$$

On $[N+1, \infty)$ we use $|L(x)| \leq x$ and $|w(x/Y)| \leq C_A (Y/x)^A$:

$$\begin{aligned} \Big| \int_{N+1}^\infty L(x)\, w(x/Y) \frac{\mathrm{d}x}{x} \Big| &\leq C_A Y^A \int_N^\infty x^{-A}\, \mathrm{d}x \\ &\leq C_A Y^A N^{1-A} = C_A N^{1-A\varepsilon} = C_A N^{-1}. \end{aligned}$$

If $Y \geq Y_0$ and $\kappa_0 Y^{\beta_0} \geq 2C_A$, then (K4) gives $\frac{1}{2}\kappa_0 Y^{\beta_0} \leq (C_w V(N)/Y)^{1/2}$, that is,

$$V(N) \geq \frac{\kappa_0^2}{4C_w} Y^{1+2\beta_0} = \frac{\kappa_0^2}{4C_w} N^{(1-\varepsilon)(1+2\beta_0)} \geq \frac{\kappa_0^2}{4C_w} N^{1+2\beta_0-3\varepsilon}.$$

[7] The classical argument of Landau, which uses a Dirichlet integral with a coefficient of constant sign, gives such lower bounds only for arbitrarily large values of the variable. Pintz's kernel replaces positivity by a contour shift past a single pole, and gives the bound for all large values.

This holds for all $N \geq N_0(\rho_0, \varepsilon)$. For $N < N_0$ we have $V(N) \geq 1$, and the first bound in (4.1) follows, after replacing 3ε by ε and reducing the constant. The second bound follows from (2.11): $Q_\lambda(N) \geq V(N)/(2N+1) \geq V(N)/(3N)$. □

Theorem 4.2 (Zero-one law)**.** *Let $\sigma \geq 0$.*

(a) If $\sigma > \Theta$, then B_σ is finite.

(b) If $\sigma < \Theta$, then B_σ is cofinite: $Q_\lambda(N) > N^{2\sigma}$ for all sufficiently large N.

In particular, if $\frac{1}{2} \leq \sigma < 1$ and $Q_\lambda(N_k) \leq N_k^{2\sigma}$ along some infinite sequence $N_1 < N_2 < \cdots$, then $\zeta(s) \neq 0$ for $\operatorname{Re} s > \sigma$.

Proof. (a) Theorem 3.1 with $\theta = \Theta$ and $\varepsilon = \sigma - \Theta$ gives $Q_\lambda(N) \leq CN^{2\sigma-\varepsilon}$, which is smaller than $N^{2\sigma}$ for large N.

(b) Suppose first that $\Theta > \frac{1}{2}$ and $\sigma < \Theta$. By the definition of Θ there is a zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 > \max(\sigma, \frac{1}{2})$. Proposition 4.1 with $\varepsilon = \beta_0 - \sigma$ gives $Q_\lambda(N) \geq cN^{2\sigma+\varepsilon} > N^{2\sigma}$ for large N. If $\Theta = \frac{1}{2}$ and $\sigma < \frac{1}{2}$, apply Proposition 4.1 with $\rho_0 = \frac{1}{2}$ and $\varepsilon = \frac{1}{2} - \sigma$.

For the last assertion, the complement of B_σ is infinite, so $\sigma < \Theta$ is excluded by (b). Hence $\Theta \leq \sigma$. □

At $\sigma = \Theta$ the theorem says nothing, and the answer depends on finer information. For instance, if $\Theta = \frac{1}{2}$, then $B_{1/2}$ is the set of N with $Q_\lambda(N) > N$, and Theorem 5.6 below shows that this set is infinite; whether its complement is infinite is not known.

Corollary 4.3 (The limit exists)**.**

$$\lim_{N\to\infty} \frac{\log Q_\lambda(N)}{\log N} = 2\Theta \qquad \textit{and} \qquad \lim_{N\to\infty} \frac{\log V(N)}{\log N} = 1 + 2\Theta.$$

Proof. Theorem 4.2 gives $N^{2\sigma_1} < Q_\lambda(N) \leq N^{2\sigma_2}$ for all large N, whenever $\sigma_1 < \Theta < \sigma_2$. For V, Theorem 3.1 gives $V(N) \leq N \max_{j\leq N} L(j)^2 \ll_\varepsilon N^{1+2\Theta+\varepsilon}$. In the other direction, Proposition 4.1, applied to a zero with $\beta_0 > \max(\Theta - \varepsilon/2, \frac{1}{2})$ if $\Theta > \frac{1}{2}$ and to $\rho_0 = \frac{1}{2}$ if $\Theta = \frac{1}{2}$, gives $V(N) \gg_\varepsilon N^{1+2\Theta-2\varepsilon}$. □

Corollary 4.4 (One sequence suffices)**.** *The following statements are equivalent.*

(i) The Riemann hypothesis.

(ii) $Q_\lambda(N) \ll_\varepsilon N^{1+\varepsilon}$ for every $\varepsilon > 0$ and all N.

(iii) For each $\varepsilon > 0$ there is an infinite sequence $N_1 < N_2 < \cdots$, which may depend on ε, such that $Q_\lambda(N_k) \leq N_k^{1+\varepsilon}$ for all k.

(iv) For each $\varepsilon > 0$ there is an infinite sequence $N_1 < N_2 < \cdots$ such that $\int_1^{N_k} L(x)^2\,\mathrm{d}x \leq N_k^{2+\varepsilon}$ for all k.

Proof. (i)⇔(ii) is Theorem 3.1. (ii)⇒(iii): apply (ii) with $\varepsilon/2$ and take all sufficiently large N. (iii)⇒(iv): by (2.13), a sequence as in (iii) for $\varepsilon/2$ satisfies $\int_1^{N_k} L(x)^2\,\mathrm{d}x \leq 3N_k\, Q_\lambda(N_k) \leq 3N_k^{2+\varepsilon/2} \leq N_k^{2+\varepsilon}$ for large k. (iv)⇒(i): suppose $\Theta > \frac{1}{2}$, and choose a zero with $\beta_0 = \frac{1}{2} + \varepsilon_0$, $\varepsilon_0 > 0$. Since $\int_1^N L(x)^2\,\mathrm{d}x = V(N-1)$, Proposition 4.1 with $\varepsilon = \varepsilon_0$ gives $\int_1^N L(x)^2\,\mathrm{d}x \geq c(N-1)^{2+\varepsilon_0}$ for all $N \geq 2$. This contradicts (iv) with $\varepsilon = \varepsilon_0/2$ for large k. □

Corollary 4.5 (No almost-all bound below the zero-free exponent)**.** *Let $\frac{1}{2} \leq \sigma < 1$, and let $E \subset \mathbb{N}$ be any set whose complement is infinite. If $Q_\lambda(N) \leq N^{2\sigma}$ for all $N \notin E$, then $\zeta(s) \neq 0$ for $\operatorname{Re} s > \sigma$. The same conclusion holds if $V(N) \leq N^{1+2\sigma}$ for all $N \notin E$.*

Proof. The first statement is the last assertion of Theorem 4.2. For the second, if $\Theta > \sigma$, Proposition 4.1 applied to a zero with $\beta_0 > \sigma$ and $\varepsilon = \beta_0 - \sigma$ gives $V(N) > N^{1+2\sigma}$ for all large N. □

Table 3 lists the hypotheses on the energy and what each gives.

TABLE 3. Bounds for the energy and their consequences for Θ. In each row $\frac{1}{2} \le \sigma < 1$, and the bound is $Q_\lambda(N) \le N^{2\sigma}$ or the stated variant.

hypothesis	conclusion	reference
$Q_\lambda(N) \ll_\varepsilon N^{2\sigma+\varepsilon}$, all N	equivalent to $\Theta \le \sigma$	Thm. 3.1
$\sum_N Q_\lambda(N) N^{-1-2\sigma} < \infty$	$\Theta \le \sigma$; no zero on $\operatorname{Re} s = \sigma$	Thm. 3.6
bound outside a set of density 0	$\Theta \le \sigma$	Cor. 4.5
bound along one infinite sequence	$\Theta \le \sigma$	Thm. 4.2
$\#\{j \le N : \lvert L(j)\rvert > N^\sigma\} \le N^{1-\delta}$, one sequence	$\Theta \le \sigma$	Prop. 4.7
$V(N_k) \le N_k^{1+2\sigma}$ along one sequence	$\Theta \le \sigma$	Cor. 4.5
$\mathcal{Q}_\lambda(N, H) \ll N^{2\sigma-1}H$, $H = H(N)$	$\Theta \le \sigma$	Cor. 3.4

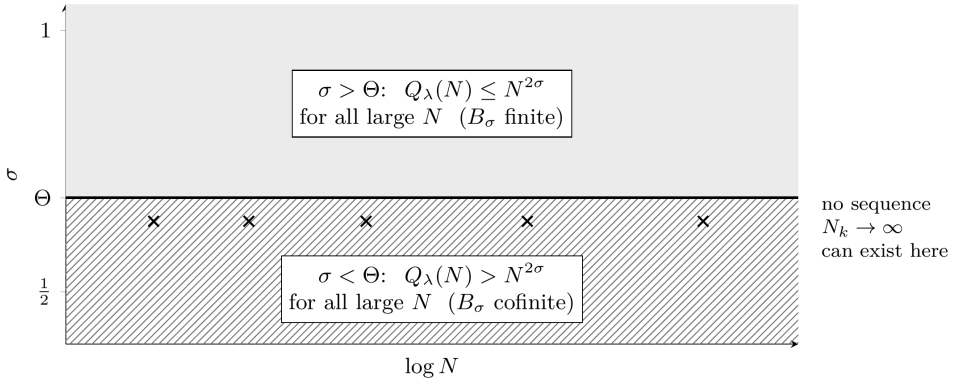


FIGURE 4. The zero-one law (Theorem 4.2). Above the line $\sigma = \Theta$ the inequality $Q_\lambda(N) \le N^{2\sigma}$ holds for all large N; below it the reverse inequality holds for all large N. No infinite sequence N_k can satisfy $Q_\lambda(N_k) \le N_k^{2\sigma}$ with $\sigma < \Theta$ (crosses), so a single such sequence already gives the zero-free half-plane $\operatorname{Re} s > \sigma$. At $\sigma = \Theta$ itself the theorem makes no claim.

Remark 4.6 (Zero-density estimates). Let $N(\sigma, T)$ be the number of zeros with $\beta \ge \sigma$ and $|\gamma| \le T$. Zero-density estimates bound $N(\sigma, T)$ by a power $T^{A(\sigma)(1-\sigma)+o(1)}$; the estimate of Guth and Maynard [18] gives $A(\sigma) \le 30/13$ uniformly. In many problems on primes such estimates replace the Riemann hypothesis for almost all values of a parameter, and one might hope in the same way for a bound $Q_\lambda(N) \le N^{2\sigma}$, with some fixed $\sigma < 1$, for all N outside a sparse set. Corollary 4.5 shows that this is not possible without proving $\Theta \le \sigma$: any derivation of such an almost-all bound from a zero-density estimate would be a derivation of a zero-free half-plane from that estimate. The reason is visible in the proof of Proposition 4.1. A zero-density estimate limits the number of zeros to the right of a line. A single zero $\beta_0 + i\gamma_0$ already forces $V(N) \gg N^{1+2\beta_0-\varepsilon}$ for every N, and V is monotone, so neither averaging over N nor discarding a set of N can hide it. The unconditional bounds for $Q_\lambda(N)$ that are available come from the zero-free region and save less than any power of N.[8]

The same mechanism controls the number of positions at which L is large.

Proposition 4.7 (Almost all positions)**.** *Let $\frac{1}{2} \le \sigma < 1$ and $0 < \delta \le 1$, and put $E_N = \{1 \le j \le N : |L(j)| > N^\sigma\}$. If $\#E_N \le N^{1-\delta}$ for infinitely many N, then $\Theta \le \sigma$. Conversely, if $\Theta < \sigma$, then $E_N = \varnothing$ for all large N. In particular, the Riemann hypothesis holds if and*

[8]From the Vinogradov–Korobov zero-free region one has $\sum_{n \le y} \mu(n) \ll y \exp(-c\,\eta(y))$ with $\eta(y) = (\log y)^{3/5}(\log\log y)^{-1/5}$ [37]. Since $L(x) = \sum_{d \le \sqrt{x}} \sum_{n \le x/d^2} \mu(n)$, the same bound holds for $L(x)$ with a smaller constant c, and (2.7) gives $Q_\lambda(N) \ll N^2 \exp(-2c\,\eta(N))$.

only if for every $\varepsilon > 0$ there are $\delta > 0$ and infinitely many N with $\#\{j \le N : |L(j)| > N^{1/2+\varepsilon}\} \le N^{1-\delta}$.

Proof. If $\Theta < \sigma$, Theorem 3.1 gives $|L(j)| \ll j^{(\Theta+\sigma)/2} \le N^{(\Theta+\sigma)/2} < N^{\sigma}$ for $j \le N$ and N large.

Suppose $\Theta > \sigma$ and $\#E_N \le N^{1-\delta}$ for infinitely many N. Choose a zero $\beta_0 + i\gamma_0$ with $\beta_0 > \sigma$ and $\Theta - \beta_0 < \delta/4$, and put $\varepsilon = \min(\delta/8, (\beta_0 - \sigma)/2)$. Proposition 4.1 gives $V(N) \ge cN^{1+2\beta_0-\varepsilon}$ for all N, and Theorem 3.1 gives $|L(j)| \le C' j^{\Theta+\varepsilon}$. For N in the sequence,

$$V(N) = \sum_{j \in E_N} L(j)^2 + \sum_{j \le N,\ j \notin E_N} L(j)^2 \le C'^2 N^{1-\delta+2\Theta+2\varepsilon} + N^{1+2\sigma}.$$

Both exponents are smaller than $1 + 2\beta_0 - \varepsilon$, because $2(\Theta - \beta_0) < \delta/2 < \delta - 3\varepsilon$ and $\varepsilon < 2(\beta_0 - \sigma)$. This contradicts the lower bound for large N. The last assertion follows from the first two with $\sigma = \frac{1}{2} + \varepsilon$. □

Remark 4.8. Proposition 4.7 is a direct consequence of Pintz's mean-value principle (Proposition 4.1, [31]). Kaczorowski and Pintz [23, 24] studied the measure of the sets on which oscillation of arithmetic error terms is large; statements of this type in their work may contain Proposition 4.7, and no priority is claimed.

5. Pole energies

Section 3 determines the exponent of $Q_\lambda(N)$. The vector identity (3.7) also gives lower bounds for the constant in front of the power, with one contribution from each pole of M on the line $\operatorname{Re} s = \Theta$. We keep the notation $\mathcal{H}$, U_X, w_s, $K(s)$ of Section 3.2.

5.1. **Poles on one vertical line.**

Proposition 5.1 (A sum of pole contributions)**.** *Fix $\beta \ge \frac{1}{2}$ and an integer $r \ge 1$. Let $\mathcal{P}$ be a finite set of distinct poles $\rho = \beta + i\gamma$ of M, each of order exactly r, and write $A_\rho = \lim_{s\to\rho}(s-\rho)^r M(s)$. Then*

$$C := \limsup_{N\to\infty} \frac{Q_\lambda(N)}{N^{2\beta}(\log N)^{2r-2}} \ \ge\ \sum_{\rho\in\mathcal{P}} K(\rho) \left| \frac{A_\rho}{(r-1)!} \right|^2 . \tag{5.1}$$

If $C < \infty$, then M has no pole in $\operatorname{Re} s > \beta$, and every pole of M on $\operatorname{Re} s = \beta$ has order at most r.

Proof. If $C = \infty$ there is nothing to prove. Let $C < \infty$. By (3.4), the limit superior of $\|U_X\|^2 X^{-2\beta}(\log X)^{-(2r-2)}$ over real $X \to \infty$ is also C. Put $F(t) = e^{-\beta t} U_{e^t}$ for $t \ge 0$. For every $\varepsilon > 0$ there is t_ε with

$$\|F(t)\|^2 \le (C + \varepsilon)\, t^{2r-2} \qquad (t \ge t_\varepsilon), \tag{5.2}$$

and $\|F\|$ is bounded on $[0, t_\varepsilon]$. Hence $\Phi(s) = \int_0^\infty F(t) e^{-(s-\beta)t}\,\mathrm{d}t$ is holomorphic on $\operatorname{Re} s > \beta$, and, as in the proof of Theorem 3.6, $\Phi(s) = M(s) w_s$ there and M has no pole in $\operatorname{Re} s > \beta$. At a point ρ of the line $\operatorname{Re} s = \beta$, (5.2) gives $\|M(\rho+\delta)w_{\rho+\delta}\| \le \int_0^\infty \|F(t)\| e^{-\delta t}\,\mathrm{d}t = O(\delta^{-r})$ as $\delta \downarrow 0$. Since $w_{\rho+\delta} \to w_\rho \ne 0$, a pole at ρ has order at most r.

For $\delta > 0$ let $\langle \cdot, \cdot \rangle_\delta$ be the inner product of $L^2([0,\infty), e^{-\delta t}\,\mathrm{d}t; \mathcal{H})$, and put $G_r = \Gamma(2r-1) = (2r-2)!$. By (5.2) and Euler's integral $\int_0^\infty t^{2r-2} e^{-\delta t}\,\mathrm{d}t = G_r \delta^{-(2r-1)}$,

$$\limsup_{\delta\downarrow 0}\ \delta^{2r-1} \|F\|_\delta^2 \le C\, G_r. \tag{5.3}$$

For $\rho = \beta + i\gamma \in \mathcal{P}$ define the test function

$$g_{\rho,\delta}(t) = \frac{\delta^{r-1/2}}{\sqrt{G_r}}\, t^{r-1} e^{i\gamma t}\, v_\rho, \qquad v_\rho = \frac{w_\rho}{\|w_\rho\|}.$$

Then $\|g_{\rho,\delta}\|_\delta = 1$, and for $\rho \neq \rho'$ in $\mathcal{P}$,

$$|\langle g_{\rho,\delta}, g_{\rho',\delta}\rangle_\delta| = \Big|\frac{\delta}{\delta - i(\gamma-\gamma')}\Big|^{2r-1} |\langle v_\rho, v_{\rho'}\rangle| \;\longrightarrow\; 0 \qquad (\delta \downarrow 0).$$

So the Gram matrix of the family $(g_{\rho,\delta})_{\rho\in\mathcal{P}}$ tends to the identity, and its largest eigenvalue Λ_δ tends to 1. For any vector F and any finite family (g_ρ) with Gram matrix of largest eigenvalue Λ,

$$\sum_\rho |\langle F, g_\rho\rangle|^2 \le \Lambda \, \|F\|^2, \tag{5.4}$$

because the synthesis map $(c_\rho) \mapsto \sum_\rho c_\rho g_\rho$ has squared norm Λ, and $F \mapsto (\langle F, g_\rho\rangle)_\rho$ is its adjoint.

It remains to evaluate $\langle F, g_{\rho,\delta}\rangle_\delta$. Differentiating $\Phi(s) = M(s)w_s$ under the integral sign $r-1$ times,

$$\begin{aligned}\langle F, g_{\rho,\delta}\rangle_\delta &= \frac{\delta^{r-1/2}}{\sqrt{G_r}}\Big\langle \int_0^\infty t^{r-1} F(t) e^{-(\rho+\delta-\beta)t}\,\mathrm{d}t,\; v_\rho\Big\rangle \\ &= \frac{\delta^{r-1/2}}{\sqrt{G_r}}\Big\langle (-1)^{r-1}\frac{\mathrm{d}^{\,r-1}}{\mathrm{d}s^{r-1}}\big[M(s)w_s\big]_{s=\rho+\delta},\; v_\rho\Big\rangle.\end{aligned}$$

Near ρ, $M(s)w_s = A_\rho w_\rho (s-\rho)^{-r} + R(s)$, where R is an $\mathcal{H}$-valued function with a pole of order at most $r-1$ at ρ. Since

$$(-1)^{r-1}\frac{\mathrm{d}^{\,r-1}}{\mathrm{d}s^{r-1}}(s-\rho)^{-r} = \frac{(2r-2)!}{(r-1)!}\,(s-\rho)^{-(2r-1)},$$

and the corresponding derivative of R is $O(|s-\rho|^{-(2r-2)})$, we obtain

$$\lim_{\delta\downarrow 0}\; \delta^{r-1/2}\langle F, g_{\rho,\delta}\rangle_\delta = \sqrt{G_r}\,\frac{A_\rho}{(r-1)!}\,\|w_\rho\|.$$

Apply (5.4) in the space with inner product $\langle\cdot,\cdot\rangle_\delta$, multiply by δ^{2r-1}, let $\delta\downarrow 0$, and use (5.3) and $\Lambda_\delta \to 1$:

$$G_r \sum_{\rho\in\mathcal{P}} \Big|\frac{A_\rho}{(r-1)!}\Big|^2 \|w_\rho\|^2 \le C\, G_r.$$

Since $\|w_\rho\|^2 = K(\rho)$, this is (5.1). □

Corollary 5.2 (A critical-line lower bound)**.** *For every finite set $\mathcal{Z}$ of simple zeros of ζ on the critical line,*

$$\limsup_{N\to\infty}\frac{Q_\lambda(N)}{N} \;\ge\; \frac{2}{3\,\zeta(1/2)^2} + \sum_{\rho\in\mathcal{Z}}\Big(2 - 2\,\mathrm{Re}\,\frac{1}{1+\rho}\Big)\,\Big|\frac{\zeta(2\rho)}{\rho\,\zeta'(\rho)}\Big|^2. \tag{5.5}$$

The left side is infinite if the Riemann hypothesis fails, or if some zero on the critical line is multiple. Unconditionally,

$$\limsup_{N\to\infty}\frac{Q_\lambda(N)}{N} \;\ge\; \frac{2}{3\,\zeta(1/2)^2} > \frac{2(3-2\sqrt{2})}{3} > 0.114. \tag{5.6}$$

Proof. At $s = \frac12$ the function M has a simple pole with residue $1/\zeta(\frac12)$, and $K(\frac12) = \frac23$. At a simple zero $\rho = \frac12 + i\gamma$, M has a simple pole with residue $\zeta(2\rho)/(\rho\zeta'(\rho)) \neq 0$, and $K(\rho) = 2 - 2\,\mathrm{Re}\,\frac{1}{1+\rho}$ by (3.5). Proposition 5.1 with $\beta = \frac12$, $r = 1$ and $\mathcal{P} = \{\frac12\}\cup\mathcal{Z}$ gives (5.5). If the left side is finite, the same proposition excludes poles of M in $\mathrm{Re}\,s > \frac12$ and poles of order greater than 1 on the line; by the functional equation a zero off the line produces a zero to its right. For (5.6), the alternating series $(1-\sqrt{2})\zeta(\frac12) = \sum_{n\ge1}(-1)^{n-1}n^{-1/2}$ has a sum in $(0,1)$, so $0 < -\zeta(\frac12) < 1/(\sqrt{2}-1) = 1+\sqrt{2}$, and $2/(3(1+\sqrt{2})^2) = 2(3-2\sqrt{2})/3$. □

The zeros with $|\gamma| \le 3 \cdot 10^{12}$ are known to lie on the critical line [33]. Each simple zero among them may be used in (5.5), and each adds a positive term. Values of the right side are given in Table 4.

5.2. **Positive weights.** The bound (5.5) adds the squares of the pole contributions. A positive weight in the variable $t = \log X$ adds the contributions themselves, as vectors, and gives larger constants when their phases are aligned.

Proposition 5.3 (Positive-weight bound)**.** *Let $C = \limsup_{N\to\infty} Q_\lambda(N)/N$, and suppose $C < \infty$. For real ω let $R(\omega)$ be the residue of M at $\frac12 + i\omega$, so that $R(\omega) = 0$ when M is regular there. Let*

$$P(t) = \sum_{\omega\in\Omega} p_\omega e^{-i\omega t}$$

be a trigonometric polynomial with a finite set Ω of distinct real frequencies, such that $P(t) \ge 0$ for all real t and $p_0 = 1$. Then

$$C \;\ge\; \Big\| \sum_{\omega\in\Omega} p_\omega\, R(\omega)\, w_{1/2+i\omega} \Big\|^2. \tag{5.7}$$

Proof. By Proposition 5.1 with $\beta = \frac12$ and $r = 1$, M has no pole in $\operatorname{Re} s > \frac12$ and only simple poles on the line. Put $F(t) = e^{-t/2} U_{e^t}$. As in that proof, $\limsup_{t\to\infty} \|F(t)\|^2 = C$ and $\int_0^\infty F(t) e^{-(s-1/2)t}\,\mathrm{d}t = M(s) w_s$ for $\operatorname{Re} s > \frac12$. For $\delta > 0$ consider

$$J_\delta = \delta \int_0^\infty e^{-\delta t} P(t) F(t)\,\mathrm{d}t = \sum_{\omega\in\Omega} p_\omega\, \delta M(\tfrac12 + \delta + i\omega)\, w_{1/2+\delta+i\omega}.$$

Since the poles on the line are at most simple, $\delta M(\frac12 + \delta + i\omega) \to R(\omega)$ as $\delta \downarrow 0$, and $w_{1/2+\delta+i\omega} \to w_{1/2+i\omega}$ in $\mathcal{H}$. So J_δ converges to the vector in (5.7). On the other hand, $P \ge 0$ gives

$$\|J_\delta\| \le \delta \int_0^\infty e^{-\delta t} P(t) \|F(t)\|\,\mathrm{d}t.$$

Fix $\varepsilon > 0$ and t_ε with $\|F(t)\| \le \sqrt{C} + \varepsilon$ for $t \ge t_\varepsilon$. The part of the integral over $[0, t_\varepsilon]$ is bounded, and vanishes after multiplication by δ. Since $\delta \int_0^\infty e^{-\delta t} P(t)\,\mathrm{d}t = \sum_\omega p_\omega \delta/(\delta + i\omega) \to p_0 = 1$, we get $\limsup_{\delta\downarrow 0} \|J_\delta\| \le \sqrt{C} + \varepsilon$. □

When $C = \infty$ the lower bounds below hold trivially, so they are unconditional.

Corollary 5.4 (Fejér weights)**.** *Let $\rho_j = \frac12 + i\gamma_j$, $1 \le j \le J$, be simple zeros of ζ with $0 < \gamma_1 < \cdots < \gamma_J$, and let $m \ge 1$. Put*

$$a = -\frac{1}{\zeta(1/2)} > 0, \qquad A_j = \frac{\zeta(2\rho_j)}{\rho_j\, \zeta'(\rho_j)}.$$

Assume that the $(2m+1)^J$ numbers $n_1\gamma_1 + \cdots + n_J\gamma_J$ with integers $|n_j| \le m$ are pairwise distinct, and that $\zeta(\frac12 + i\omega) \ne 0$ for each of them other than $\pm\gamma_1, \dots, \pm\gamma_J$. Then

$$\limsup_{N\to\infty} \frac{Q_\lambda(N)}{N} \;\ge\; \Big\| a\, w_{1/2} + \frac{2m}{m+1} \sum_{j=1}^{J} |A_j|\, \operatorname{Re} w_{\rho_j} \Big\|^2, \tag{5.8}$$

where $\operatorname{Re} w_\rho = \frac12 (w_\rho + w_{\bar\rho})$. The norm is computed from

$$\langle \operatorname{Re} w_z, \operatorname{Re} w_{z'} \rangle = 1 - \operatorname{Re}\frac{1}{1+z} - \operatorname{Re}\frac{1}{1+z'} + \operatorname{Re}\frac{1}{1+z+z'} + \operatorname{Re}\frac{1}{1+z+\overline{z'}}, \tag{5.9}$$

valid for $\operatorname{Re} z, \operatorname{Re} z' > -\frac12$; for real z the left side is $\langle w_z, \operatorname{Re} w_{z'} \rangle$.

Proof. We may assume $C < \infty$. Choose real θ_j with $e^{i\theta_j} A_j = -|A_j|$, and let

$$P(t) = \prod_{j=1}^{J} \frac{1}{m+1} \Big| \sum_{\ell=0}^{m} e^{i\ell(\gamma_j t - \theta_j)} \Big|^2 = \prod_{j=1}^{J} \sum_{|n| \le m} \Big(1 - \frac{|n|}{m+1}\Big) e^{in(\gamma_j t - \theta_j)}.$$

Then $P \ge 0$. Its frequencies are the numbers $-(n_1\gamma_1 + \cdots + n_J\gamma_J)$, which are pairwise distinct by assumption, so the coefficient of each exponential is the product of the corresponding coefficients of the factors. In particular $p_0 = 1$, and the coefficient at $\omega = \mp\gamma_j$ is $\frac{m}{m+1} e^{\mp i\theta_j}$. By assumption $R(\omega) = 0$ at every other nonzero frequency. Further $R(0) = 1/\zeta(\frac{1}{2}) = -a$, $R(\gamma_j) = A_j$, and $R(-\gamma_j) = \overline{A_j}$ because $M(\bar{s}) = \overline{M(s)}$. The vector in (5.7) is therefore

$$-a\, w_{1/2} + \frac{m}{m+1} \sum_{j=1}^{J} \big(e^{i\theta_j} A_j\, w_{\rho_j} + e^{-i\theta_j} \overline{A_j}\, w_{\bar{\rho}_j} \big)$$

$$= -\Big(a\, w_{1/2} + \frac{2m}{m+1} \sum_{j=1}^{J} |A_j|\ \mathrm{Re}\, w_{\rho_j} \Big).$$

The formula (5.9) is verified in Appendix A.4. □

The hypothesis of Corollary 5.4 concerns finitely many explicit numbers. For $J = 1$ it says that no multiple $k\gamma_1$ with $2 \le k \le m$ is the ordinate of a zero. For the first zero and $m = 100$, and for the first two zeros and $m = 30$, it has been confirmed by interval arithmetic,[9] and the right side of (5.8) then has the computed values 0.65142 and 0.72446 of Table 4.

Remark 5.5 (Status of the constants). The inequality (5.6), with the constant $2(3-2\sqrt{2})/3 > 0.114$, is proved without computation. The larger constants follow from Corollaries 5.2 and 5.4 once the zeros involved have been located and the finitely many nonvanishing conditions have been checked numerically. With the first two zeros and $m = 30$ this gives $\limsup_{N\to\infty} Q_\lambda(N)/N > 0.7244$. Nothing else in the paper uses these values.

Under the Riemann hypothesis and the simplicity of the zeros, (5.5) bounds $\limsup Q_\lambda(N)/N$ below by the full sum $\frac{2}{3}\zeta(\frac{1}{2})^{-2} + \sum_\rho K(\rho)|A_\rho|^2$, finite or infinite, and Table 4 shows that the Fejér weights exceed the first thousand terms of that sum. Larger lower bounds require weights with more frequencies, and then the hypothesis of Corollary 5.4 involves more integer combinations of ordinates. Subsection 5.3 removes that hypothesis and gives a larger constant.

5.3. **The size of $Q_\lambda(N)/N$.** At the critical scale the energy is governed by the oscillation of $L(x)/\sqrt{x}$. The first statement transfers a published oscillation theorem; the second shows that nothing is lost in the transfer.

Theorem 5.6 (Oscillation of the energy)**.**

$$\limsup_{N\to\infty} \frac{Q_\lambda(N)}{N} > 2.8139.$$

Consequently $B_{1/2} = \{N : Q_\lambda(N) > N\}$ is infinite, and under the Riemann hypothesis the set $B_\Theta = B_{1/2}$ is infinite.

Proof. Mossinghoff and Trudgian [29] proved, unconditionally, that $\liminf_{x\to\infty} L(x)/\sqrt{x} < -2.3723$ (see also [30, (1.2)]). The proof rests on the classical dichotomy: if the Riemann hypothesis fails or some zero is multiple, then $L(x)/\sqrt{x}$ is unbounded in both directions [20, §3], [28, Ch. 15]; otherwise their weak-independence form of Ingham's method applies.

[9] In the second case there are $61^2 = 3721$ frequencies; one checks that they are distinct and that $\zeta(\frac{1}{2} + i\omega) \ne 0$ at the 3716 frequencies other than $0, \pm\gamma_1, \pm\gamma_2$. These verifications, and the numerical values in Table 4, are computations and are not used in any proof in this paper.

TABLE 4. Constants of Section 5. Rows marked "exact" are proved in the text, the last from the theorem of Mossinghoff and Trudgian. Rows marked "computed" are numerical values, rounded to the digits shown; they illustrate the theorems and no proof depends on them. In the rows for (5.5), $\mathcal{Z}$ consists of the first n zeros above the real axis and their conjugates.

quantity	value	status
$K(\frac{1}{2})$	$2/3$	exact
$\min_{\mathrm{Re}\, s\geq 1/2} K(s)$, attained at $s=1/\sqrt{2}$	$4\sqrt{2}-5=0.65685\ldots$	exact
$\sup_{\mathrm{Re}\, s\geq 1/2} K(s)$	2, not attained	exact
analytic lower bound (5.6)	$2(3-2\sqrt{2})/3=0.11438\ldots$	exact
$\zeta(\frac{1}{2})$	-1.46035	computed
$2/(3\zeta(\frac{1}{2})^2)$, the pole at $\frac{1}{2}$ alone	0.31260	computed
$\lvert A_1\rvert$, $\gamma_1=14.13472\ldots$	0.17372	computed
$\lvert A_2\rvert$, $\gamma_2=21.02203\ldots$	0.03476	computed
right side of (5.5), $n=1$	0.43241	computed
right side of (5.5), $n=10$	0.44722	computed
right side of (5.5), $n=100$	0.45578	computed
right side of (5.5), $n=1000$	0.45770	computed
right side of (5.8), $J=1$, $m=100$	0.65142	computed
right side of (5.5), $n=3000$	0.45794	computed
right side of (5.8), $J=2$, $m=30$	0.72446	computed
$\lVert Z(y_0)\rVert^2$ of Theorem 5.11, $m=1400$	1.98545	computed
$2.3723^2/2$, Theorem 5.6	$2.81390\ldots$	exact
$\pi/(4\zeta(\frac{1}{2})^2)$, the boundary cost in (5.16)	0.36828	computed

Take x large with $L(x)<-2.3723\sqrt{x}$ and put $N=\lfloor x\rfloor$. Then $L(N)=L(x)$ and $\sqrt{N}\leq\sqrt{x}$, so $L(N)^2>2.3723^2N$. By (2.6), $Q_\lambda(N)\geq\frac{N+1}{2N}L(N)^2>\frac{2.3723^2}{2}N$, and $2.3723^2/2=2.81390\ldots>2.8139$. Such N are arbitrarily large.

For the second assertion, if $\Theta>\frac{1}{2}$ then $B_{1/2}$ is cofinite by Theorem 4.2(b). If $\Theta=\frac{1}{2}$, the first assertion gives $Q_\lambda(N)>2.8N>N$ for infinitely many N. Under the Riemann hypothesis $\Theta=\frac{1}{2}$ and $B_\Theta=B_{1/2}$. □

Remark 5.7. The value $\liminf L(x)/\sqrt{x}\leq-1.389278$ of Borwein, Ferguson and Mossinghoff [12] gives only $\limsup Q_\lambda(N)/N\geq 0.9650$. Any constant larger than 1 obtained in this way needs $\lvert L(x)\rvert>\sqrt{2x}$ for arbitrarily large x.

Proposition 5.8 (Boundedness of the normalized energy)**.** *Let* $\ell=\limsup_{x\to\infty}\lvert L(x)\rvert/\sqrt{x}\in[0,\infty]$. *Then*

$$\tfrac{1}{2}\,\ell^2\ \leq\ \limsup_{N\to\infty}\frac{Q_\lambda(N)}{N}\ \leq\ \tfrac{10}{3}\,\ell^2.$$

In particular $\limsup Q_\lambda(N)/N<\infty$ *if and only if* $L(x)\ll\sqrt{x}$.

Proof. Write C for the middle term. For the lower bound take $x_k\to\infty$ with $\lvert L(x_k)\rvert/\sqrt{x_k}\to\ell$, and put $N_k=\lfloor x_k\rfloor$. By (2.6), $Q_\lambda(N_k)/N_k\geq L(N_k)^2/(2N_k)\geq L(x_k)^2/(2x_k)$, which tends to $\ell^2/2$.

For the upper bound assume $\ell<\infty$, and fix $\varepsilon>0$. There is J with $\lvert L(j)\rvert\leq(\ell+\varepsilon)\sqrt{j}$ for $j\geq J$, and $B_0=\max_{j<J}\lvert L(j)\rvert$ is finite. Let $N\geq J$. By the definition (2.1) with $H=N$, equivalently (2.5),

$$N\,Q_\lambda(N)=\sum_{j=0}^{N-1}\Big(L(j)^2+\big(L(N)-L(j)\big)^2\Big).$$

For $J \le j < N$ the summand is at most $(\ell+\varepsilon)^2(j+(\sqrt{N}+\sqrt{j})^2) = (\ell+\varepsilon)^2(2j+N+2\sqrt{Nj})$. Since $\sum_{j<N}\sqrt{j} \le \int_0^N \sqrt{t}\,\mathrm{d}t = \frac{2}{3}N^{3/2}$,

$$\sum_{j=0}^{N-1}\bigl(2j+N+2\sqrt{Nj}\bigr) \le N(N-1)+N^2+\tfrac{4}{3}N^2 \le \tfrac{10}{3}N^2.$$

For $j < J$ the summand is at most $B_0^2 + 2L(N)^2 + 2B_0^2 \le 3B_0^2 + 2(\ell+\varepsilon)^2 N$. Hence

$$\frac{Q_\lambda(N)}{N} \le \tfrac{10}{3}(\ell+\varepsilon)^2 + \frac{J\bigl(3B_0^2+2(\ell+\varepsilon)^2N\bigr)}{N^2},$$

and the last term tends to 0. Let $N \to \infty$ and then $\varepsilon \to 0$. The final assertion follows, because $L(x) \ll \sqrt{x}$ is equivalent to $\ell < \infty$. □

Whether $L(x) \ll \sqrt{x}$ is not known, even under the Riemann hypothesis. Ingham showed that it fails under an independence hypothesis on the ordinates.

Theorem 5.9 (Ingham [20, Theorem A and §3])**.** *Let $0 < \gamma_1 < \gamma_2 < \cdots$ be the distinct ordinates of the zeros of ζ above the real axis. Suppose that the Riemann hypothesis holds, that every zero is simple, and that for some m the numbers γ_n with $n > m$ are linearly independent over $\mathbb{Q}$. Then*

$$\limsup_{x\to\infty}\frac{L(x)}{\sqrt{x}} = +\infty, \qquad \liminf_{x\to\infty}\frac{L(x)}{\sqrt{x}} = -\infty.$$

If the Riemann hypothesis fails, or some zero is multiple, the same conclusion holds without any independence hypothesis.

Ingham states the hypothesis as "no relation $\sum c_n\gamma_n = 0$ with integers c_n not all 0, or only a finite number of such relations". Since a relation has infinitely many integer multiples, "finitely many" must be read as "the relations span a finite-dimensional space"; this holds exactly when some tail $\{\gamma_n : n > m\}$ is independent, which is the form used in his proof (Kronecker's theorem applied to the γ_n with $\gamma_m < \gamma_n < T$). The proof rests on two results from the same paper: a Tauberian comparison with the Fejér-weighted trigonometric polynomials [20, Theorem 1], and the divergence

$$\sum_{\gamma>0}|A_\gamma| = \infty \qquad \text{(Riemann hypothesis, simple zeros)}, \tag{5.10}$$

which is [20, Theorem 2(ii)]: the residue of $\zeta(1+2s)/((\frac{1}{2}+s)\zeta(\frac{1}{2}+s))$ at $s = i\gamma$ is A_γ.

Corollary 5.10 (Bounded energy forces relations)**.** *Suppose that $\limsup_{N\to\infty} Q_\lambda(N)/N < \infty$. Then the Riemann hypothesis holds, every zero of ζ is simple, and for every m the ordinates γ_n with $n > m$ satisfy a nontrivial linear relation with integer coefficients; equivalently, the integer relations among the positive ordinates span an infinite-dimensional space. In particular, if the positive ordinates are linearly independent over $\mathbb{Q}$, then $\limsup Q_\lambda(N)/N = \infty$.*

Proof. Write C for the upper limit. By (2.6), $L(N)^2 \le 2Q_\lambda(N) \le 2(C+1)N$ for large N, and $L(x) = L(\lfloor x\rfloor)$. So $L(x)/\sqrt{x}$ is bounded, and Theorem 5.9 excludes each alternative. □

The positive weights of Proposition 5.3 have a continuous form, which needs no condition on integer combinations of ordinates.

Theorem 5.11 (Continuous positive-weight bound)**.** *Let $m > 0$, and let k be a continuous even function on $\mathbb{R}$ with support in $[-m,m]$ and $k(0) = 1$, such that*

$$P(t) = \frac{1}{2\pi}\int_{-m}^{m} k(\omega)e^{i\omega t}\,\mathrm{d}\omega$$

satisfies $P \ge 0$ and $P \in L^1(\mathbb{R})$. Let Ω be the finite set of $\omega \in (-m, m)$ at which M has a pole at $\frac12 + i\omega$, and put

$$Z(y) = \sum_{\omega\in\Omega} k(\omega)R(\omega)e^{i\omega y}\, w_{1/2+i\omega} \in \mathcal{H} \qquad (y \in \mathbb{R}).$$

Then $\limsup_{N\to\infty} Q_\lambda(N)/N \ge \|Z(y_0)\|^2$ for every $y_0 \in \mathbb{R}$. Explicitly, with $a_\omega = k(\omega)R(\omega)e^{i\omega y_0}$,

$$\|Z(y_0)\|^2 = \sum_{\omega,\omega'\in\Omega} a_\omega\overline{a_{\omega'}}\,\langle w_{1/2+i\omega}, w_{1/2+i\omega'}\rangle, \qquad \langle w_s, w_z\rangle = 1 - \frac{1}{1+s} - \frac{1}{1+\bar z} + \frac{2}{1+s+\bar z}.$$

Proof. Let $C = \limsup Q_\lambda(N)/N$ and assume $C < \infty$. By Proposition 5.1 with $\beta = \frac12$ and $r = 1$, M has no pole in $\operatorname{Re} s > \frac12$ and only simple poles on the line $\operatorname{Re} s = \frac12$; by Theorem 3.1 the Riemann hypothesis holds. Put $F(t) = e^{-t/2}U_{e^t}$ for $t \ge 0$ and $F(t) = 0$ for $t < 0$. By (3.4), $\|F(t)\|^2 = \|U_X\|^2/X \le (N+2)Q_\lambda(N)/N^2$ for $N \le X = e^t < N+1$, so $\limsup_{t\to\infty}\|F(t)\|^2 \le C$. Since F is also bounded on bounded intervals, $\|F\| \le B_0$ on $\mathbb{R}$ for some B_0. By Lemma 3.5(c) and analytic continuation, $\Phi(s) = \int_0^\infty F(t)e^{-(s-1/2)t}\,\mathrm{d}t = M(s)w_s$ for $\operatorname{Re} s > \frac12$.

For $\delta > 0$ the function $e^{-\delta t}F(t)$ is integrable, and Fubini's theorem gives

$$\int_{\mathbb{R}} P(y-t)e^{-\delta t}F(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-m}^{m} k(\omega)e^{i\omega y}\,\Phi(\tfrac12+\delta+i\omega)\,\mathrm{d}\omega.$$

The function $M(s)w_s$ is meromorphic on $\operatorname{Re} s > -\frac12$, with finitely many poles, all simple and on the line, in a neighbourhood of the rectangle $[\frac12, \frac12+\delta_0]\times[-m-1, m+1]$. Hence

$$\Phi(s) = \sum_{|\gamma|\le m+1} \frac{R(\gamma)w_{1/2+i\gamma}}{s - \frac12 - i\gamma} + \Psi(s),$$

where the sum runs over the poles $\frac12 + i\gamma$ and Ψ is continuous on the rectangle. Using $1/(\delta + i(\omega-\gamma)) = \int_0^\infty e^{-(\delta+i(\omega-\gamma))u}\,\mathrm{d}u$, the contribution of the pole at $\frac12+i\gamma$ is

$$R(\gamma)w_{1/2+i\gamma}\int_0^\infty e^{-\delta u}e^{i\gamma u}P(y-u)\,\mathrm{d}u \;\xrightarrow[\delta\to0]{}\; R(\gamma)w_{1/2+i\gamma}\,e^{i\gamma y}\Big(k(\gamma) - \int_y^\infty e^{-i\gamma v}P(v)\,\mathrm{d}v\Big),$$

by dominated convergence and the inversion formula $\int_{\mathbb{R}} P(v)e^{-i\gamma v}\,\mathrm{d}v = k(\gamma)$. Letting $\delta \to 0$ on both sides, with dominated convergence on the left ($\|F\| \le B_0$, $P \in L^1$) and uniform continuity of Ψ on the right,

$$\begin{aligned}(P*F)(y) = Z(y) &+ \frac{1}{2\pi}\int_{-m}^{m} k(\omega)e^{i\omega y}\Psi(\tfrac12+i\omega)\,\mathrm{d}\omega \\ &- \sum_{|\gamma|\le m+1} R(\gamma)w_{1/2+i\gamma}e^{i\gamma y}\int_y^\infty e^{-i\gamma v}P(v)\,\mathrm{d}v,\end{aligned}$$

since $k(\gamma) = 0$ for $|\gamma| \ge m$. As $y\to\infty$ the middle term tends to 0 by the Riemann–Lebesgue lemma and the last sum tends to 0 because $P \in L^1$. So $\|Z(y) - (P*F)(y)\| \to 0$.

Fix $\varepsilon > 0$ and t_ε with $\|F(t)\| \le \sqrt{C} + \varepsilon$ for $t \ge t_\varepsilon$. Since $P \ge 0$ and $\int P = k(0) = 1$,

$$\|(P*F)(y)\| \le \int_{\mathbb{R}} P(y-t)\|F(t)\|\,\mathrm{d}t \le \sqrt{C} + \varepsilon + B_0\int_{y-t_\varepsilon}^\infty P(v)\,\mathrm{d}v,$$

and the last term tends to 0. Hence $\limsup_{y\to\infty}\|Z(y)\| \le \sqrt{C}$.

Finally Z is a finite exponential sum. Given $\eta > 0$ and $Y > 0$, Dirichlet's theorem on simultaneous approximation, applied to the numbers $Y\omega/2\pi$ ($\omega \in \Omega$), gives a positive integer q with $|e^{i\omega qY} - 1| < \eta$ for all $\omega \in \Omega$. With $h = qY \ge Y$, $\|Z(y_0+h) - Z(y_0)\| \le \eta\sum_\omega |a_\omega|\,\|w_{1/2+i\omega}\|$. So $\|Z(y_0)\| \le \limsup_{y\to\infty}\|Z(y)\| \le \sqrt{C}$. The Gram formula is the integral $\int_0^1 (u^s\overline{u^z} + (1-u^s)(1-\overline{u^z}))\,\mathrm{d}u$. ∎

Remark 5.12. Theorem 5.11 is a vector form of Ingham's Tauberian theorem [20, Theorem 1], and it contains Proposition 5.3. It needs no hypothesis on integer combinations of ordinates, unlike Corollary 5.4. Pairing $Z(y)$ with $e_2 = (0,1)$, using $\langle w_{1/2+i\omega}, e_2\rangle = \frac{1/2+i\omega}{3/2+i\omega}$, recovers the scalar bound. A convenient kernel is that of Jurkat and Peyerimhoff, $k(\omega) = (1-\tau)\cos\pi\tau + \frac{1}{\pi}\sin\pi\tau$ for $\tau = |\omega|/m \le 1$: it is the normalized autocorrelation of $\cos(\pi\omega/m)\mathbf{1}_{|\omega|\le m/2}$, so P is a positive multiple of a squared modulus and $P(t) = O(t^{-4})$. With $m = 1400$ (the 983 zeros with $0 < \gamma < 1400$) and $y_0 = 561488971144772334329867737844 7$, a point at which the phases γy_0 of the first 110 zeros were aligned by lattice reduction, the right side has the computed value $\|Z(y_0)\|^2 = 1.98545$. This value is a computation, like those of Table 4; it gives $B_{1/2}$ infinite independently of [29], and it is smaller than the published constant of Theorem 5.6.

Upper limits are not the only information available. On a logarithmic scale the mean of $Q_\lambda(N)/N$ is exactly the full pole sum of (5.5).

Theorem 5.13 (Logarithmic mean)**.** *Let*

$$S = \frac{2}{3\zeta(\frac12)^2} + \sum_{\gamma} K(\tfrac12 + i\gamma)\,|A_\gamma|^2,$$

where γ runs over the ordinates of both signs of the zeros on the critical line, with $S = \infty$ if the Riemann hypothesis fails or some zero is multiple.

(a) Unconditionally, $\limsup_{X\to\infty} \dfrac{1}{\log X} \displaystyle\sum_{N\le X} \frac{Q_\lambda(N)}{N^2} \ge S.$

(b) If the Riemann hypothesis holds, every zero is simple, and $J_{-1}(T) = \sum_{0<\gamma\le T} |\zeta'(\rho)|^{-2} \ll T$*, then the limit exists and equals S.*

Proof. Put $G(y) = e^{-y}\|U_{e^y}\|^2$ and $E(y) = e^{-y/2}L(e^y)$ for $y \ge 0$, with $E(y) = 0$ for $y < 0$. The substitution $u = e^{-t}$ in the definition of $\|U_X\|^2$ gives, exactly,

$$G(y) = \int_0^\infty \Big(e^{-t}E(y-t)^2 + \big(E(y) - e^{-t/2}E(y-t)\big)^2 \Big) e^{-t}\,\mathrm{d}t. \tag{5.11}$$

By (3.4), $\int_N^{N+1} \|U_X\|^2 X^{-2}\,\mathrm{d}X = Q_\lambda(N)N^{-2}\big(1 + O(1/N)\big)$, and $\int_1^X \|U_v\|^2 v^{-2}\,\mathrm{d}v = \int_0^{\log X} G$. Under the Riemann hypothesis $Q_\lambda(N) \ll N^{1+\varepsilon}$ (Theorem 3.1), so

$$\sum_{N\le X} \frac{Q_\lambda(N)}{N^2} = \int_0^{\log X} G(y)\,\mathrm{d}y + O(1). \tag{5.12}$$

(a) If $\Theta > \frac12$, choose σ with $\frac12 < \sigma < \Theta$; by Theorem 4.2, $Q_\lambda(N)/N^2 > N^{2\sigma-2}$ for all large N, so the sum in (a) grows like a positive power of X and the upper limit is infinite. Assume $\Theta = \frac12$. For $\sigma = \frac12 + \delta$, Theorem 3.6 in the variable $y = \log X$ reads

$$2\delta \int_0^\infty G(y) e^{-2\delta y}\,\mathrm{d}y = \frac{\delta}{\pi} \int_{-\infty}^{\infty} \big|M(\tfrac12 + \delta + i\tau)\big|^2 K(\tfrac12 + \delta + i\tau)\,\mathrm{d}\tau.$$

Let $\frac12 + i\gamma$ be a pole of M of order r, and $\eta > 0$ small. If $r = 1$, then $M(\frac12 + \delta + i\tau) = R(\gamma)/(\delta + ix) + O(1)$ for $x = \tau - \gamma$, $|x| < \eta$, and

$$\frac{\delta}{\pi}\int_{-\eta}^{\eta} \frac{|R(\gamma)|^2}{\delta^2 + x^2}\,\mathrm{d}x \to |R(\gamma)|^2, \qquad \frac{\delta}{\pi}\int_{-\eta}^{\eta} \frac{\mathrm{d}x}{\sqrt{\delta^2 + x^2}} = O\big(\delta\log(1/\delta)\big).$$

Since K is continuous and positive, the contribution of $|\tau - \gamma| < \eta$ has $\liminf_{\delta\to0}$ at least $|R(\gamma)|^2 K(\frac12 + i\gamma)$. If $r \ge 2$, the same contribution is $\gg \delta^{2-2r} \to \infty$. Since $K > 0$ everywhere, summing over any finite set of disjoint intervals shows that $\liminf_{\delta\to0} 2\delta\int_0^\infty G e^{-2\delta y}\,\mathrm{d}y$ is at least every partial sum of S, hence at least S, with $R(0) = 1/\zeta(\frac12)$, $K(\frac12) = \frac23$ and

$R(\gamma) = A_\gamma$. Now put $H(Y) = \int_0^Y G$. If $\limsup H(Y)/Y = c < \infty$, then for each $\varepsilon > 0$ there is B_ε with $H(y) \le (c+\varepsilon)y + B_\varepsilon$, and

$$s\int_0^\infty e^{-sy}G(y)\,\mathrm{d}y = s^2\int_0^\infty e^{-sy}H(y)\,\mathrm{d}y \le c + \varepsilon + sB_\varepsilon,$$

so, with $s = 2\delta \to 0$, $c \ge S$. Together with (5.12) this proves (a).

(b) Put $P_T(y) = 1/\zeta(\frac12) + \sum_{|\gamma|<T} A_\gamma e^{i\gamma y}$, a real trigonometric polynomial. Under the hypotheses of (b), Humphries [19, §5] proves

$$\limsup_{Y\to\infty} \frac{1}{Y}\int_0^Y |E(y) - P_T(y)|^2\,\mathrm{d}y \le \eta(T), \qquad \eta(T) \ll \frac{\log T(\log\log T)^2}{T^{1/4}}.$$

For a locally bounded function D vanishing on $(-\infty, 0)$, let G_D be defined by (5.11) with D in place of E. Then $G_D(y)$ is the squared norm of a linear image of D, and $(u-v)^2 \le 2u^2 + 2v^2$ gives $G_D(y) \le 2D(y)^2 + 3\int_0^\infty e^{-2t}D(y-t)^2\,\mathrm{d}t$. Hence

$$\frac{1}{Y}\int_0^Y G_D \le \frac{7}{2}\cdot\frac{1}{Y}\int_0^Y D^2.$$

With $\widetilde{P}_T = P_T\mathbf{1}_{[0,\infty)}$, the inequality $|G_E - G_{\widetilde{P}_T}| \le G_{E-\widetilde{P}_T} + 2\sqrt{G_{E-\widetilde{P}_T}G_{\widetilde{P}_T}}$ and the Cauchy–Schwarz inequality show that the upper and lower limits of the means of G_E and of $G_{\widetilde{P}_T}$ differ by $O\big(\eta(T) + \sqrt{\eta(T)\Sigma}\big)$, where $\Sigma = \zeta(\frac12)^{-2} + \sum_\gamma |A_\gamma|^2 < \infty$ under $J_{-1}(T) \ll T$ and $|\zeta(1+2i\gamma)| \ll \log|\gamma|$. For a trigonometric polynomial $P = \sum_\omega c_\omega e^{i\omega y}$, the truncation at 0 changes $G_P(y)$ by $O(e^{-y})$, and the mean value of G_P is

$$\begin{aligned}\sum_\omega |c_\omega|^2 \int_0^\infty \Big(e^{-t} + \big|1 - e^{-(1/2+i\omega)t}\big|^2\Big)e^{-t}\,\mathrm{d}t &= \sum_\omega |c_\omega|^2\Big(2 - 2\,\mathrm{Re}\,\frac{1}{3/2+i\omega}\Big)\\ &= \sum_\omega |c_\omega|^2 K(\tfrac12 + i\omega).\end{aligned}$$

For P_T this is $\frac{2}{3\zeta(1/2)^2} + \sum_{|\gamma|<T} K(\frac12+i\gamma)|A_\gamma|^2$. Letting $T\to\infty$, the mean of G_E over $[0,Y]$ converges to S, and (5.12) gives (b). □

Remark 5.14. The terms of S are positive, and the sum over the first 3000 zeros above the real axis and their conjugates gives the computed value 0.45794; hence $S > 0.45794$. Over $10^4 \le N \le 3\cdot 10^7$ the logarithmic mean $\sum Q_\lambda(N)/N^2 / \sum 1/N$ is 0.4540 (Section K). Under the hypotheses of (b), $\sum_{N\le X,\ Q_\lambda(N)\ge N} 1/N \le \sum_{N\le X} Q_\lambda(N)/N^2$, so the upper logarithmic density of $B_{1/2}$ is at most S.

5.4. **A weighted Mellin energy.** The same mechanism applies to the smoothed sum

$$B(X) = \sum_{n\ge1} \lambda(n)e^{-n/X} \qquad (X > 0),$$

whose Mellin transform carries a Gamma factor. For $\mathrm{Re}\,s > 1$, termwise integration with $\int_0^\infty e^{-n/X}X^{-s-1}\,\mathrm{d}X = \Gamma(s)n^{-s}$ gives

$$\int_0^\infty B(X)X^{-s-1}\,\mathrm{d}X = \Gamma(s)\frac{\zeta(2s)}{\zeta(s)}. \tag{5.13}$$

Define

$$J(\sigma) = \int_1^\infty |B(X)|^2 X^{-2\sigma-1}\,\mathrm{d}X \in [0,\infty], \qquad H(s) = \int_0^1 B(X)X^{-s-1}\,\mathrm{d}X.$$

Since $|B(X)| \le \sum_n e^{-n/X} \le 2e^{-1/X}$ for $0 < X \le 1$, the function H is entire.

Theorem 5.15 (Finite weighted energy excludes zeros)**.** *Let $\sigma > \frac12$. If $J(\sigma) < \infty$, then $\zeta(s) \ne 0$ for $\mathrm{Re}\,s > \sigma$, and on that half-plane*

$$\Big|\Gamma(s)\frac{\zeta(2s)}{\zeta(s)} - H(s)\Big|^2 \le \frac{J(\sigma)}{2(\mathrm{Re}\,s - \sigma)}. \tag{5.14}$$

Conversely $J(\sigma) < \infty$ for every $\sigma > \max(\Theta, \frac{1}{2})$. Consequently the Riemann hypothesis is equivalent to $J(\sigma) < \infty$ for every $\sigma > \frac{1}{2}$.

Proof. Let $T(s) = \int_1^\infty B(X) X^{-s-1}\,\mathrm{d}X$. For $\operatorname{Re} s > \sigma$ the Cauchy–Schwarz inequality gives

$$|T(s)|^2 \le J(\sigma) \int_1^\infty X^{-2(\operatorname{Re} s - \sigma)-1}\,\mathrm{d}X = \frac{J(\sigma)}{2(\operatorname{Re} s - \sigma)},$$

and the same estimate, applied locally uniformly, shows that T is holomorphic on $\operatorname{Re} s > \sigma$. By (5.13), $\zeta(s)\big(T(s) + H(s)\big) = \Gamma(s)\zeta(2s)$ for $\operatorname{Re} s > \max(1, \sigma)$, and the identity extends to $\operatorname{Re} s > \sigma$. The right side is holomorphic and has no zero on $\operatorname{Re} s > \sigma \ge \frac{1}{2}$, so ζ has no zero there, and $T = \Gamma\zeta(2\cdot)/\zeta - H$. This gives (5.14).

For the converse, partial summation gives $B(X) = X^{-1}\int_1^\infty L(x) e^{-x/X}\,\mathrm{d}x$. If $\theta > \Theta$, then $L(x) \ll x^\theta$ by Theorem 3.1, so

$$|B(X)| \ll \frac{1}{X}\int_0^\infty x^\theta e^{-x/X}\,\mathrm{d}x = \Gamma(1+\theta)\, X^\theta,$$

and $J(\sigma) < \infty$ for $\sigma > \theta$.[10] If $J(\sigma) < \infty$ for every $\sigma > \frac{1}{2}$, then ζ has no zero in $\operatorname{Re} s > \frac{1}{2}$, and the functional equation gives the Riemann hypothesis. □

Proposition 5.16 (The cost of a pole)**.** *Let ρ be a zero of ζ of order m with $\beta = \operatorname{Re}\rho > \frac{1}{2}$, and put*

$$a_\rho = \frac{m!\,\Gamma(\rho)\,\zeta(2\rho)}{\zeta^{(m)}(\rho)} \neq 0.$$

Then, with the value $+\infty$ allowed for J,

$$\liminf_{u \downarrow 0}\; u^{2m-1} J(\beta + u/2) \ge |a_\rho|^2. \tag{5.15}$$

Independently of the Riemann hypothesis,

$$\liminf_{u \downarrow 0}\; u\, J(\tfrac{1}{2} + u/2) \ge \Big|\frac{\Gamma(1/2)}{2\zeta(1/2)}\Big|^2 = \frac{\pi}{4\,\zeta(1/2)^2} > 0. \tag{5.16}$$

Proof. Let $u > 0$. If $J(\beta + u/2) = \infty$ the inequality is trivial. Otherwise apply (5.14) with $\sigma = \beta + u/2$ at the point $s = \rho + u$, where $2(\operatorname{Re} s - \sigma) = u$:

$$u^{2m-1} J(\beta + u/2) \ge \Big| u^m \Big(\Gamma(s)\frac{\zeta(2s)}{\zeta(s)} - H(s)\Big)\Big|^2_{s=\rho+u}.$$

Near ρ, $\Gamma(s)\zeta(2s)/\zeta(s) = a_\rho (s-\rho)^{-m} + O(|s-\rho|^{-m+1})$, with $a_\rho \neq 0$ because $\operatorname{Re}(2\rho) > 1$, and H is entire. So the right side tends to $|a_\rho|^2$. If $\beta < \Theta$, then $J(\beta + u/2) = \infty$ for all small u by Theorem 5.15; the inequality has content when $\beta = \Theta$. For (5.16) the same argument applies at $s = \frac{1}{2} + u$ with $\sigma = \frac{1}{2} + u/2$, using the simple pole of $\zeta(2s)$ at $s = \frac{1}{2}$, where $\Gamma(s)\zeta(2s)/\zeta(s)$ has residue $\Gamma(\frac{1}{2})/(2\zeta(\frac{1}{2}))$. □

The pole at $s = \frac{1}{2}$ is present whether or not the Riemann hypothesis holds. By (5.16) the energy $J(\sigma)$ grows at least like $0.368/(2\sigma - 1)$ as $\sigma \downarrow \frac{1}{2}$, so a proof of the Riemann hypothesis through Theorem 5.15 cannot bound $J(\sigma)$ uniformly in $\sigma > \frac{1}{2}$. Table 5 compares the three energies of this part.

[10] Bounds for $B(X)$ under a zero-free hypothesis are treated in Proposition 12.1.

TABLE 5. Three energy criteria. In each row the finiteness or growth condition, for every $\sigma > \frac{1}{2}$, is equivalent to the Riemann hypothesis; for a fixed σ it lies between $\Theta < \sigma$ and $\Theta \le \sigma$.

energy	condition at σ	transform	reference
$Q_\lambda(N)$	$Q_\lambda(N) \ll N^{2\sigma}$	$M(s)\, w_s$	Theorem 3.1
$\sum_N Q_\lambda(N) N^{-1-2\sigma}$	finite	$\frac{1}{2\pi}\int \lvert M\rvert^2 K \,\mathrm{d}\tau$	Theorem 3.6
$J(\sigma)$	finite	$\Gamma(s)\zeta(2s)/\zeta(s) - H(s)$	Theorem 5.15

6. Sensitivity to sparse sets of primes

Theorem 3.1 identifies the growth exponent of $Q_\lambda(N)$ with 2Θ. A proof that this exponent is 1 must therefore use some property of λ that fails for every sequence whose energy grows faster. This section measures how fine that property has to be. We change the sign of λ on a set of primes and ask how thin the set can be while the exponent still moves. The answer is a set with about x^β elements up to x, for any $\beta > 1/2$, and no thinner.

For a real sequence a write $L_a(x) = \sum_{n \le x} a_n$, so that $L = L_\lambda$, and recall from Proposition 2.1 and (2.6) that

$$\tfrac{1}{2}|L_a(N)|^2 \le Q_a(N) \le 5 \max_{0 \le j \le N} |L_a(j)|^2 \qquad (N \ge 1). \tag{6.1}$$

Definition 6.1. Let S be a set of primes. The *modified Liouville function* f_S is the completely multiplicative function with $f_S(p) = +1$ for $p \in S$ and $f_S(p) = -1$ for every other prime.[11] We write $\pi_S(x) = \#\{p \in S : p \le x\}$ and

$$G_S(s) = \prod_{p \in S} \frac{1 + p^{-s}}{1 - p^{-s}} = \sum_{d \ge 1} \frac{g_S(d)}{d^s},$$

where g_S is the multiplicative function with $g_S(p^j) = 2$ for $p \in S$, $j \ge 1$, and $g_S(p^j) = 0$ at the other prime powers.

Thus $f_\emptyset = \lambda$, and $g_S(d) = 2^{\omega(d)}$ when every prime factor of d lies in S.

Lemma 6.2 (Two convolutions)**.** *Let S be a set of primes with $\sum_{p \in S} p^{-\sigma_0} < \infty$ for some $\sigma_0 > 0$. The Dirichlet inverse g_S^{-1} is the multiplicative function with $g_S^{-1}(p^j) = 2(-1)^j$ for $p \in S$, $j \ge 1$, and zero at the other prime powers. Both series $\sum_d g_S(d) d^{-s}$ and $\sum_d g_S^{-1}(d) d^{-s}$ converge absolutely for $\operatorname{Re} s \ge \sigma_0$, with $\sum_d |g_S^{\pm 1}(d)| d^{-\sigma} = G_S(\sigma)$ for real $\sigma \ge \sigma_0$. Moreover $f_S = \lambda * g_S$ and $\lambda = f_S * g_S^{-1}$, so that for every $x \ge 1$*

$$L_{f_S}(x) = \sum_{d \le x} g_S(d) L(x/d), \qquad L(x) = \sum_{d \le x} g_S^{-1}(d) L_{f_S}(x/d). \tag{6.2}$$

Proof. For $|u| < 1$ one has $(1+u)/(1-u) = 1 + 2\sum_{j \ge 1} u^j$ and $(1-u)/(1+u) = 1 + 2\sum_{j \ge 1}(-1)^j u^j$. These are the local factors of g_S and of its inverse at $p \in S$, with $u = p^{-s}$. Since $p^{-\sigma_0} \le 2^{-\sigma_0} < 1$ and $\sum_{p \in S} p^{-\sigma_0} < \infty$, the product $G_S(\sigma)$ converges for $\sigma \ge \sigma_0$, and it equals the sum of the absolute values of either series. Both sides of $f_S = \lambda * g_S$ are multiplicative. At p^j with $p \notin S$ only $d = 1$ contributes, giving $\lambda(p^j) = f_S(p^j)$. At p^j with $p \in S$ the convolution equals $(-1)^j + 2\sum_{i=1}^{j}(-1)^{j-i} = 1$. Convolving with g_S^{-1} gives the second identity, and summing over $n \le x$ gives (6.2). □

Theorem 6.3 (Sensitivity to a sparse set of primes)**.** *Let $1/2 < \beta < 1$.*

[11] This is the class studied by Borwein, Choi and Coons [11], who call its members Liouville-like functions and determine, among other things, which of them have small partial sums when S is defined by a congruence condition.

(i) *Let S be a set of primes for which $\sum_{p\in S} p^{-\sigma}$ is finite for every $\sigma > \beta$ and infinite at $\sigma = \beta$. Then, for every $\alpha < \beta$,*

$$L_{f_S}(x) \neq O(x^{\alpha}), \qquad Q_{f_S}(N) \neq O(N^{2\alpha}).$$

(ii) *For the same S and every $\varepsilon > 0$,*

$$L_{f_S}(x) \ll_{\varepsilon,S} x^{\max(\beta,\Theta)+\varepsilon}, \qquad Q_{f_S}(N) \ll_{\varepsilon,S} N^{2\max(\beta,\Theta)+\varepsilon}.$$

(iii) *Such sets exist with $\pi_S(x) \asymp x^{\beta}$ for $x \geq x_0$. In particular $\sum_{p\in S} 1/p < \infty$.*

Proof. (iii) By the prime number theorem [28] the interval $(2^j, 2^{j+1}]$ contains more than $2^{j\beta}$ primes once $j \geq j_0$. For each such j let S_j consist of exactly $\lfloor 2^{j\beta} \rfloor$ of them, and put $S = \bigcup_{j\geq j_0} S_j$. For $2^J \leq x < 2^{J+1}$ with $J > j_0$,

$$\lfloor 2^{(J-1)\beta} \rfloor \leq \pi_S(x) \leq \sum_{j\leq J} 2^{j\beta} \leq \frac{2^{J\beta}}{1-2^{-\beta}},$$

so $\pi_S(x) \asymp x^{\beta}$. Moreover

$$\sum_{j\geq j_0} \lfloor 2^{j\beta} \rfloor 2^{-(j+1)\sigma} \leq \sum_{p\in S} p^{-\sigma} \leq \sum_{j\geq j_0} 2^{j(\beta-\sigma)},$$

and both outer series converge exactly when $\sigma > \beta$.

(i) Write $f = f_S$ and suppose $L_f(x) \ll x^{\alpha}$ with $\alpha < \beta$; we may take $\alpha > 1/2$. By partial summation

$$F(s) = s\int_1^{\infty} L_f(x) x^{-s-1}\,\mathrm{d}x$$

is holomorphic on $\operatorname{Re} s > \alpha$ and equals $\sum_n f(n)n^{-s}$ for $\operatorname{Re} s > 1$. Comparing Euler factors, $(1-p^{-s})^{-1}$ for $p \in S$ against $(1+p^{-s})^{-1}$, gives

$$F(s) = \frac{\zeta(2s)}{\zeta(s)}\, G_S(s) \qquad (\operatorname{Re} s > 1). \tag{6.3}$$

By Lemma 6.2 the function G_S is holomorphic on $\operatorname{Re} s > \beta$, where $\zeta(2s)/\zeta(s)$ is meromorphic. Hence (6.3) persists on $\operatorname{Re} s > \beta$ as an identity of meromorphic functions. On the real segment $\beta < \sigma < 1$ the factor $\zeta(2\sigma)/\zeta(\sigma)$ is finite and nonzero: $2\sigma > 1$, and $\zeta(\sigma) < 0$ on $(0,1)$ because $(1-2^{1-\sigma})\zeta(\sigma) = \sum_{n\geq 1}(-1)^{n-1}n^{-\sigma} > 0$. As $\sigma \downarrow \beta$ the left side of (6.3) tends to the finite value $F(\beta)$ and $\zeta(2\sigma)/\zeta(\sigma)$ tends to the finite nonzero value $\zeta(2\beta)/\zeta(\beta)$. So $G_S(\sigma)$ stays bounded. But $g_S(p) = 2$ and $g_S \geq 0$ give

$$G_S(\sigma) \geq 1 + 2\sum_{p\in S} p^{-\sigma} \longrightarrow \infty \qquad (\sigma \downarrow \beta)$$

by monotone convergence. This contradiction proves the first assertion. If $Q_f(N) \ll N^{2\alpha}$, then (6.1) gives $L_f(N) \ll N^{\alpha}$ at the integers, hence at all real x, which has just been excluded.

(ii) If $\Theta = 1$ the bounds are trivial. Otherwise let $\theta' = \max(\beta,\Theta) + \varepsilon < 1$. Theorem 3.1 gives $|L(y)| \leq Cy^{\theta'}$ for $y \geq 1$, and $G_S(\theta') < \infty$. By (6.2),

$$|L_f(x)| \leq C\sum_{d\leq x} g_S(d)\Big(\frac{x}{d}\Big)^{\theta'} \leq C\, G_S(\theta')\, x^{\theta'}.$$

The bound for Q_f follows from the upper inequality in (6.1). □

Remark 6.4 (Attribution). Part (i) is not new. Aymone [2] proved that if a completely multiplicative f with values in $[-1,1]$ has $\sum_{n\leq x} f(n) \ll x^{1-\delta}$ and its Dirichlet series vanishes at $s = 1$, then $\sum_{p\leq x}(1+f(p))\log p \ll x^{1-\delta+\varepsilon}$. For $f = f_S$ the last sum is $2\sum_{p\leq x,\, p\in S}\log p$, so a bound $L_{f_S}(x) \ll x^{\alpha}$ forces $\pi_S(x) \ll x^{\alpha+\varepsilon}$, and part (i) is the contrapositive within the class of Definition 6.1. The argument above is a short self-contained proof of this case. Section 10 gives a refinement in which the Dirichlet series of f_S is computed in closed form up to a nonvanishing factor.

The content of the theorem for the energy problem is the following two-sided threshold.

Corollary 6.5 (The threshold 1/2)**.** *Let* $1/2 \le \theta < 1$.

(a) *For every* β *with* $\theta < \beta < 1$, *some set* S *of primes with* $\pi_S(x) \asymp x^\beta$ *breaks the exponent: there is a completely multiplicative* $f : \mathbb{N} \to \{\pm 1\}$ *that agrees with* λ *at every prime outside* S *and for which* $Q_f(N) \ll_\varepsilon N^{2\theta+\varepsilon}$ *fails for every* $\varepsilon < 2(\beta - \theta)$. *This holds whether or not* ζ *has zeros in* $\operatorname{Re} s > \theta$.

(b) *No set of size* $x^{1/2+o(1)}$ *does. If* $\sum_{p\in S} p^{-\sigma} < \infty$ *for every* $\sigma > 1/2$, *in particular if* $\pi_S(x) \ll x^{1/2+o(1)}$, *then for every* $\theta \in [1/2, 1]$ *the three statements*

$$Q_{f_S}(N) \ll_\varepsilon N^{2\theta+\varepsilon} \ \text{ for every } \varepsilon > 0, \qquad Q_\lambda(N) \ll_\varepsilon N^{2\theta+\varepsilon} \ \text{ for every } \varepsilon > 0,$$
$$\zeta(s) \neq 0 \ \text{ for } \operatorname{Re} s > \theta$$

are equivalent, and $\limsup_N \log(1 + Q_{f_S}(N))/\log N = 2\Theta$.

Consequently, any argument that proves $Q_a(N) \ll_\varepsilon N^{2\theta+\varepsilon}$ *from hypotheses on* a *which hold for every completely multiplicative sign sequence agreeing with* λ *outside a set of* $O(x^\beta)$ *primes, for some* $\beta > \theta$, *is invalid. A proof of the exponent* 1 *must detect a change of* λ *on* $O(x^{1/2+\delta})$ *primes up to* x, *for every* $\delta > 0$.

Proof. (a) Take S from Theorem 6.3(iii) and $f = f_S$. If $Q_f(N) \ll N^{2\theta+\varepsilon}$ with $\theta + \varepsilon/2 < \beta$, then part (i) with $\alpha = \theta + \varepsilon/2$ is contradicted.

(b) Partial summation shows that $\pi_S(x) \ll x^{1/2+o(1)}$ implies the convergence of $\sum_{p\in S} p^{-\sigma}$ for $\sigma > 1/2$. Under that convergence Lemma 6.2 applies with every $\sigma_0 > 1/2$. Let $\theta \ge 1/2$ and $\varepsilon > 0$. If $|L(y)| \le Cy^{\theta+\varepsilon}$, the first identity in (6.2) gives $|L_{f_S}(x)| \le CG_S(\theta+\varepsilon)x^{\theta+\varepsilon}$. If $|L_{f_S}(y)| \le Cy^{\theta+\varepsilon}$, the second gives the same bound for L, because $\sum_d |g_S^{-1}(d)|d^{-\theta-\varepsilon} = G_S(\theta+\varepsilon)$. By (6.1), bounds $O(x^{\theta+\varepsilon})$ for the partial sums for all ε are equivalent to bounds $O(N^{2\theta+\varepsilon})$ for the energy for all ε, for either function. Theorem 3.1 supplies the third statement.

For the exponent, write $\kappa = \limsup_N \log(1 + Q_{f_S}(N))/\log N$. Since $\Theta \ge 1/2$ and $\zeta \neq 0$ on $\operatorname{Re} s > \Theta$, the equivalence with $\theta = \Theta$ gives $\kappa \le 2\Theta$. If $\kappa < 2\Theta$ and $\Theta > 1/2$, choose $\theta \in [1/2, \Theta)$ with $2\theta > \kappa$; then $Q_{f_S}(N) \ll N^{2\theta+\varepsilon}$ for every $\varepsilon > 0$, and the equivalence gives $\zeta \neq 0$ on $\operatorname{Re} s > \theta$, contradicting the definition of Θ. If $\kappa < 1 = 2\Theta$, choose $\kappa < 2\kappa' < 1$. By (6.1), $L_{f_S}(x)^2 \le 2Q_{f_S}(\lfloor x \rfloor) \ll x^{2\kappa'}$, so the Dirichlet series $F_S(s) = \sum_n f_S(n)n^{-s}$ converges and is holomorphic on $\operatorname{Re} s > \kappa'$, in particular bounded near $s = 1/2$. For real $\sigma > 1/2$, however, comparing Euler factors gives

$$F_S(\sigma) = \frac{\zeta(2\sigma)}{\zeta(\sigma)}\, G_S(\sigma), \qquad G_S(\sigma) = \prod_{p\in S} \frac{1 + p^{-\sigma}}{1 - p^{-\sigma}} \ge 1,$$

and as $\sigma \downarrow 1/2$ one has $\zeta(2\sigma) \to +\infty$ and $\zeta(\sigma) \to \zeta(1/2) < 0$, so $F_S(\sigma) \to -\infty$. This contradiction gives $\kappa = 2\Theta$. □

The threshold is sharp in both directions: above 1/2 some sparse set changes the exponent, and at 1/2 none does.

Remark 6.6 (Every set with abscissa β). The divergence at $\sigma = \beta$ in Theorem 6.3(i) can be replaced by the weaker hypothesis that $\beta \in (1/2, 1)$ is the abscissa of convergence of $\sum_{p\in S} p^{-s}$, which holds for instance whenever $\pi_S(x) \asymp x^\beta$. Indeed G_S has nonnegative coefficients and the same abscissa of convergence, so by Landau's theorem [1, Ch. 11] the point $s = \beta$ is a singularity of G_S. If $L_{f_S}(x) \ll x^\alpha$ with $1/2 < \alpha < \beta$, then $G_S(s) = F(s)\zeta(s)/\zeta(2s)$ would be holomorphic near every real point of $(\alpha, 1)$, since $\zeta(2\sigma) > 1$ there. This agrees with Aymone's theorem quoted in Remark 6.4.

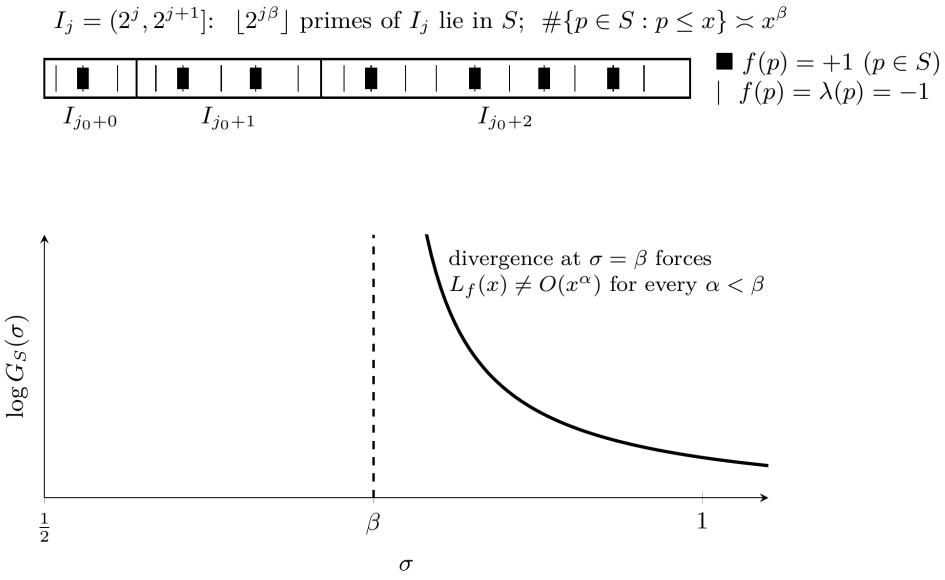


FIGURE 5. The sparse flip of Corollary 6.5. Top: in each dyadic interval $I_j = (2^j, 2^{j+1}]$ about $2^{j\beta}$ primes are placed in S (filled), and f is changed from $\lambda(p) = -1$ to $+1$ on them; all other primes keep $f(p) = \lambda(p)$. Bottom: the factor $G_S(\sigma)$ of the Dirichlet series diverges as $\sigma \downarrow \beta$, so the series of f has a singularity at β, and $L_f(x)$ cannot be $O(x^\alpha)$ for any $\alpha < \beta$.

The next proposition lists what the functions f_S share with λ. For multiplicative f, g with values in the closed unit disc, the pretentious distance up to x is

$$\mathbb{D}(f, g; x) = \Big(\sum_{p \le x} \frac{1 - \operatorname{Re} f(p)\overline{g(p)}}{p} \Big)^{1/2}.$$

Proposition 6.7 (Hypotheses invariant under the flip)**.** *Let S be a set of primes with $\sum_{p \in S} 1/p < \infty$ and $f = f_S$.*

(a) *f is completely multiplicative with values ± 1, and the primes with $f(p) \neq \lambda(p)$ have relative logarithmic density zero among the primes. If $\pi_S(x) \ll x^\beta$ with $\beta < 1$, they have relative natural density zero.*

(b) *$\mathbb{D}(f, \lambda; x)^2 \le 2 \sum_{p \in S} 1/p$ for every x. For every Dirichlet character χ, real t and $x \ge 2$,*

$$|\mathbb{D}(f, \chi n^{it}; x) - \mathbb{D}(\lambda, \chi n^{it}; x)| \le \Big(2 \sum_{p \in S} \frac{1}{p} \Big)^{1/2}.$$

(c) *f satisfies the hypothesis of Tao's theorem* [35]*, and therefore, for each fixed $d \ge 1$,*

$$\sum_{t \le T} \frac{f(t) f(t+d)}{t} = o_{d,S}(\log T).$$

(d) *If moreover $\sum_{p \in S} p^{-\sigma_0} < \infty$ for some $\sigma_0 < 1$, then for every $A > 0$, uniformly in real α,*

$$\sum_{n \le x} f(n) \mathrm{e}(n\alpha) \ll_{A,S} \frac{x}{(\log x)^A} \qquad (x \ge 2).$$

(e) *For any $P \ge 2$, removing from S the primes up to P preserves the hypotheses and conclusions of Theorem 6.3 and gives $f(n) = \lambda(n)$ for all $n \le P$.*

Proof. (a) This is the definition, and $\sum_{p \in S,\, p \le x} 1/p = O(1)$ while $\sum_{p \le x} 1/p \to \infty$.

(b) At a prime $p \notin S$ the two functions agree, and at $p \in S$ one has $1 - f(p)\lambda(p) = 2$. For the second claim, let q be the modulus of χ and $g(n) = \chi(n)n^{it}$. At primes $p \nmid q$ all three functions are unimodular and $1 - \operatorname{Re} u\bar{v} = \frac{1}{2}|u - v|^2$. Hence

$$\mathbb{D}(f, g; x)^2 = \sum_{\substack{p \le x \\ p \nmid q}} \frac{|f(p) - g(p)|^2}{2p} + \sum_{\substack{p \le x \\ p \mid q}} \frac{1}{p},$$

and likewise for λ. The first sums are squared ℓ^2 norms, to which Minkowski's inequality applies, and the second sums coincide. The claim follows from the inequality $|\sqrt{a^2 + c^2} - \sqrt{b^2 + c^2}| \le |a - b|$.

(c) Tao's theorem applies to a multiplicative function g_1 bounded by 1 such that, for every A, one has $\inf \mathbb{D}(g_1, \chi n^{it}; x)^2 \ge A$ for all large x, the infimum being over characters of period at most A and real $|t| \le Ax$. It is shown in [35], using [26], that λ has this property. By (b) it passes to f, with A replaced by $(\sqrt{A} + (2\sum_{p \in S} 1/p)^{1/2})^2$. The conclusion of the theorem with $g_1 = g_2 = f$ is the displayed estimate.

(d) By Lemma 6.2, $\sum_{n \le x} f(n)\mathrm{e}(n\alpha) = \sum_{d \le x} g_S(d) \sum_{m \le x/d} \lambda(m)\mathrm{e}(md\alpha)$. Davenport's bound $\sup_\alpha |\sum_{m \le y} \lambda(m)\mathrm{e}(m\alpha)| \le C_A y(\log 2y)^{-A}$, [22, Ch. 13], [26], applies to the inner sum for $d \le \sqrt{x}$, where $\log(2x/d) \ge \frac{1}{2}\log x$. For $d > \sqrt{x}$ use the trivial bound x/d and $\sum_{d > \sqrt{x}} g_S(d)/d \le x^{(\sigma_0 - 1)/2} G_S(\sigma_0)$. Altogether

$$\Big|\sum_{n \le x} f(n)\mathrm{e}(n\alpha)\Big| \le 2^A C_A G_S(1) \frac{x}{(\log x)^A} + G_S(\sigma_0) x^{(1+\sigma_0)/2}.$$

(e) Removing finitely many primes changes neither the convergence nor the divergence of $\sum_{p \in S} p^{-\sigma}$, and an integer $n \le P$ has all its prime factors at most P. □

Table 6 summarizes the comparison. Every hypothesis in its upper block holds for λ and for each f_S of Theorem 6.3, so none of them, alone or in combination, can yield an energy exponent below 2β; and β may be taken arbitrarily close to 1. In particular the logarithmically averaged Chowla theorem cannot give any exponent smaller than 2. What separates λ from these functions is the location of the singularities of $\sum_n f(n)n^{-s}$ in the strip $1/2 < \operatorname{Re} s < 1$, which for λ are exactly the zeros of ζ.

TABLE 6. Properties of λ and of a modified Liouville function f_S with $\sum_{p \in S} p^{-\sigma}$ convergent exactly for $\sigma > \beta$, where $1/2 < \beta < 1$. The upper block is invariant under the flip; the lower block is not.

Property	λ	f_S	Reference
completely multiplicative, values ± 1	yes	yes	Def. 6.1
value -1 at a density-one set of primes	yes	yes	Prop. 6.7(a)
bounded pretentious distance from λ	yes	yes	Prop. 6.7(b)
non-pretentious in the sense of [35]	yes	yes	Prop. 6.7(c)
log-averaged two-point Chowla	yes	yes	Prop. 6.7(c)
saving $(\log x)^{-A}$ in every additive twist	yes	yes	Prop. 6.7(d)
agreement with λ on $[1, P]$, any fixed P	yes	can be imposed	Prop. 6.7(e)
simple zero of $\sum f(n)n^{-s}$ at $s = 1$	yes	yes	Prop. 10.2
derivative of $\sum f(n)n^{-s}$ at $s = 1$ equal to $\zeta(2)$	yes	no	Prop. 10.2
singularities in $1/2 < \operatorname{Re} s < 1$	zeros of ζ	also $s = \beta$	Thm. 6.3
exponent of $Q_f(N)$	2Θ	$2\max(\beta, \Theta)$	Thm. 10.1

7. Shift correlations and the maximal collar

The expansion (7.2) below writes $|\sum_{n\le N} a_n \mathrm{e}(n\alpha)|^2$ as a sum over shifts of two-point correlations. A natural way to bound it is to estimate each correlation separately, in absolute value and uniformly in its endpoint, and to add the results. This section examines that approach. The hypothesis it needs would imply the generalized Riemann hypothesis for every Dirichlet character; no estimate that is uniform in the coefficients can supply it; the general maximal inequality that is available is a full power of N away; and Sections 8 and 9 show that the hypothesis fails for natural model sequences and, at its endpoint, for λ itself.

7.1. **Definitions and two identities.** Write $\mathrm{e}(t) = e^{2\pi i t}$. A sequence $a = (a_n)_{n\ge 1}$ is real with $|a_n| \le 1$ unless stated otherwise. Put

$$A_N(\alpha) = \sum_{n=1}^{N} a_n \mathrm{e}(n\alpha), \qquad C_{a,h}(Y) = \sum_{n=1}^{Y} a_n a_{n+h} \qquad (h \ge 1,\ Y \ge 0),$$

so that $C_{a,h}(0) = 0$. For $1 \le h < N$ let

$$C^*_{a,h}(N) = \max_{0\le Y\le N-h} |C_{a,h}(Y)|,$$

and put $C^*_{a,h}(N) = 0$ for $h \ge N$. The *collar energy* on a dyadic range of shifts is

$$\mathbb{E}_a(N,H) = \sum_{H<h\le 2H} C^*_{a,h}(N)^2 \qquad (H = 1, 2, 4, \dots).$$

The dyadic ranges cover every shift $h \ge 2$ once. The *maximal collar hypothesis* for a is the family of bounds

$$\mathbb{E}_a(N,H) \ll_\varepsilon \frac{N^{2+\varepsilon}}{H} \qquad (1 \le H < N,\ H \text{ dyadic},\ \varepsilon > 0). \tag{7.1}$$

It asserts square-root cancellation, on average over $h \in (H, 2H]$, in every prefix of every shift correlation.

Lemma 7.1 (Square and twist identities)**.** *For every real α,*

$$|A_N(\alpha)|^2 = \sum_{n\le N} a_n^2 + 2\,\mathrm{Re} \sum_{h=1}^{N-1} \mathrm{e}(h\alpha)\, C_{a,h}(N-h). \tag{7.2}$$

For every finite set J of shifts,

$$\Big(\sum_{h\in J} C^*_{a,h}(N)^2\Big)^{1/2} = \max_{(Y_h)} \sup_{\sum_{h\in J}|\beta_h|^2\le 1} \Big|\sum_{h\in J} \beta_h C_{a,h}(Y_h)\Big|, \tag{7.3}$$

where $0 \le Y_h \le N-h$ for $h < N$ and the terms with $h \ge N$ are omitted.

Proof. Group the terms of $A_N(\alpha)\overline{A_N(\alpha)}$ by the difference of their indices; the pairs $(n+h, n)$ and $(n, n+h)$ with $1 \le n \le N-h$ contribute $\mathrm{e}(h\alpha)a_n a_{n+h}$ and its conjugate. For fixed endpoints (Y_h), the Cauchy–Schwarz inequality gives the dual norm in (7.3), with equality at $\beta_h = \overline{C_{a,h}(Y_h)}/\|C\|_2$. The maxima over the Y_h are attained and are independent of one another. □

The second identity explains the name: $\mathbb{E}_a(N,H)^{1/2}$ is the largest value of a linear form in the correlations in which the coefficients β_h and the endpoints Y_h may be chosen separately for every shift.

Lemma 7.2 (Square divisors)**.** *For every $n \ge 1$, $\lambda(n) = \sum_{r^2\mid n} \mu(n/r^2)$. Consequently, for $Y, h \ge 1$,*

$$C_{\lambda,h}(Y) = \sum_{r\le\sqrt{Y}} \sum_{s\le\sqrt{Y+h}} \sum_{\substack{1\le n\le Y \\ r^2\mid n,\ s^2\mid n+h}} \mu(n/r^2)\,\mu((n+h)/s^2),$$

and the inner sum is empty unless $\gcd(r,s)^2 \mid h$.

Proof. Both sides of the first identity are multiplicative, and at p^e only the term with $e-2j \in \{0,1\}$ survives in $\sum_{2j\le e}\mu(p^{e-2j})$, giving $(-1)^e$. Insert the identity in both factors of the correlation. The congruences $n \equiv 0 \pmod{r^2}$ and $n \equiv -h \pmod{s^2}$ are compatible exactly when $\gcd(r^2,s^2) = \gcd(r,s)^2$ divides h. □

The stratum $r = s = 1$ is the Möbius correlation $\sum_{n\le Y}\mu(n)\mu(n+h)$. The identity keeps all signs. Replacing the Möbius values by their absolute values gives a quantity of order Y, that is, the trivial bound behind the coefficient-uniform maximum of Proposition 7.5.

7.2. **What the hypothesis would give.**

Theorem 7.3 (Consequence of the full maximal collar)**.** *Suppose* (7.1) *holds for $a = \lambda$. Then*

$$\sup_{\alpha\in\mathbb{R}}\Big|\sum_{n\le N}\lambda(n)\mathrm{e}(n\alpha)\Big| \ll_\varepsilon N^{1/2+\varepsilon}$$

for every $\varepsilon > 0$, and every Dirichlet L-function $L(s,\chi)$, as well as ζ, is free of zeros in $\operatorname{Re} s > 1/2$.

Proof. By (7.2), $|A_N(\alpha)|^2 \le N + 2\sum_{h<N} C^*_{\lambda,h}(N)$. The shift $h = 1$ contributes at most N. On a dyadic block the Cauchy–Schwarz inequality and (7.1) give

$$\sum_{H<h\le 2H} C^*_{\lambda,h}(N) \le \sqrt{H}\,\mathbb{E}_\lambda(N,H)^{1/2} \ll_\varepsilon N^{1+\varepsilon/2},$$

and there are at most $1+\log_2 N$ blocks. This proves the first assertion after renaming ε.

Let χ be a character modulo q and $\widehat{\chi}(r) = q^{-1}\sum_{b \bmod q}\chi(b)\mathrm{e}(-rb/q)$, so that $\chi(n) = \sum_{r \bmod q}\widehat{\chi}(r)\mathrm{e}(rn/q)$ and $\sum_r |\widehat{\chi}(r)|^2 \le 1$ by Parseval's identity. Then $\sum_r |\widehat{\chi}(r)| \le \sqrt{q}$ and

$$\sum_{n\le x}\lambda(n)\chi(n) \ll_\varepsilon \sqrt{q}\,x^{1/2+\varepsilon}.$$

By partial summation, $F_\chi(s) = s\int_1^\infty\big(\sum_{n\le x}\lambda(n)\chi(n)\big)x^{-s-1}\,\mathrm{d}x$ is holomorphic on $\operatorname{Re} s > 1/2$. For $\operatorname{Re} s > 1$ the Euler factors $(1+\chi(p)p^{-s})^{-1} = (1-\chi(p)p^{-s})/(1-\chi(p)^2p^{-2s})$ give

$$L(s,\chi)\,F_\chi(s) = L(2s,\chi^2).$$

The identity extends to $\operatorname{Re} s > 1/2$, where the right side is a convergent Euler product without zeros. A zero of $L(s,\chi)$ there would make the left side vanish. The trivial character modulo 1 gives the statement for ζ. □

For primitive χ the functional equation then confines the nontrivial zeros of $L(s,\chi)$ to the critical line. The hypothesis (7.1) for λ is thus at least as strong as the Riemann hypothesis for all Dirichlet L-functions simultaneously, with the uniformity in q of the displayed character-sum bound. Only the short and middle ranges of shifts need the collar form:

Proposition 7.4 (Terminal-range form)**.** *Fix $0 < \eta < 1$. Suppose that, for every $\delta > 0$,*

$$\mathbb{E}_a(N,H) \ll_{\eta,\delta} \frac{N^{2+\delta}}{H} \quad (H \le N^{1-\eta},\ H \text{ dyadic}),$$
$$\sum_{N^{1-\eta}<h<N} C^*_{a,h}(N) \ll_{\eta,\delta} N^{1+\delta}.$$

Then $\sup_\alpha |A_N(\alpha)| \ll_{\eta,\varepsilon} N^{1/2+\varepsilon}$ for every $\varepsilon > 0$. For $a = \lambda$ this implies RH.

Proof. The term $h = 1$ is at most N. Each dyadic block beginning at $H \le N^{1-\eta}$ contributes $O_{\eta,\delta}(N^{1+\delta/2})$ to $\sum_h C^*_{a,h}(N)$, as in the preceding proof, and there are $O(\log 2N)$ of them. Every shift not covered by these blocks or by $h = 1$ exceeds $N^{1-\eta}$, and the sum over those is bounded by hypothesis. Now use (7.2). For $a = \lambda$ take $\alpha = 0$ and apply Theorem 3.1. □

7.3. **The coefficient-uniform maximum.**

Proposition 7.5 (Exact coefficient-uniform maximum)**.** *For all N and all dyadic H,*

$$\sup_{|a_n|\le 1} \mathbb{E}_a(N,H) = \sum_{H<h\le \min(2H,N-1)} (N-h)^2.$$

The supremum is attained by $a_n = 1$ and by $a_n = (-1)^n$. If $2H \le N/2$ it lies between $HN^2/4$ and HN^2.

Proof. One has $|C_{a,h}(Y)| \le Y \le N-h$. For the two stated sequences $a_n a_{n+h}$ is constant in n and of modulus one, so $|C_{a,h}(N-h)| = N-h$. If $2H \le N/2$, each of the H terms lies between $N^2/4$ and N^2. □

The two extremal sequences behave in opposite ways in the signed energy of Proposition 2.1: $Q_{\mathbf{1}}(N) = (2N^2+1)/3$ while $Q_a(N) = 1$ for $a_n = (-1)^n$. The collar energy cannot tell them apart, because the absolute value in C^* removes the sign $(-1)^h$ that produces the cancellation between shifts. Table 7 compares the uniform maximum with the scale in (7.1).

TABLE 7. Exponents of N at $H \asymp N^u$, $0 < u < 1$. The last column is the ratio of the uniform maximum to the collar scale with $\varepsilon = 0$.

	general u	$u = 1/4$	$u = 1/2$	$u = 3/4$
$\sup_{\lvert a_n\rvert\le 1} \mathbb{E}_a(N,H)$	$2+u$	$9/4$	$5/2$	$11/4$
collar scale N^2/H	$2-u$	$7/4$	$3/2$	$5/4$
ratio	$2u$	$1/2$	1	$3/2$

Corollary 7.6 (No coefficient-uniform proof)**.** *Let $0 < u < 1$ and let H be the largest dyadic integer not exceeding N^u. For all sufficiently large N,*

$$\mathbb{E}_{\mathbf{1}}(N,H) \ge \tfrac{1}{8} N^{2+u}, \qquad \frac{N^{2+\varepsilon}}{H} \le 2N^{2-u+\varepsilon}.$$

Hence no bound of the form (7.1) with $\varepsilon < 2u$ holds with a constant independent of the bounded sequence a. In particular no argument that uses only $|a_n| \le 1$ proves (7.1) for any $\varepsilon < 2$.

Proof. Here $N^u/2 < H \le N^u$, and $2H \le N/2$ for large N. Proposition 7.5 gives $\mathbb{E}_{\mathbf{1}}(N,H) \ge HN^2/4 > N^{2+u}/8$. □

7.4. **The maximal Bessel bound.** A maximal inequality over prefixes is available whenever a Bessel inequality holds on a common index set. It is the Rademacher–Menshov device, which costs only a squared logarithm.[12]

Theorem 7.7 (Maximal Bessel bound)**.** *Let (Ω,μ) be a measure space and let $\rho_1,\dots,\rho_M \in L^2(\Omega,\mu)$ satisfy*

$$\int_\Omega \Big|\sum_{n=1}^{M} b_n \rho_n\Big|^2 \,\mathrm{d}\mu \le B \sum_{n=1}^{M} |b_n|^2 \qquad \text{for every } b \in \mathbb{C}^M.$$

Then, with $K = \lceil \log_2 M \rceil$,

$$\int_\Omega \max_{0\le Y\le M} \Big|\sum_{n\le Y} b_n \rho_n\Big|^2 \,\mathrm{d}\mu \le (K+1)^2 B \sum_{n=1}^{M} |b_n|^2.$$

[12]The dyadic decomposition of a prefix into aligned blocks goes back to Rademacher and Menshov in their work on orthogonal series. The form below is the one that combines with spectral large sieve inequalities of Deshouillers–Iwaniec type [16]: if the fixed-endpoint inequality holds with constant $B \ll_\varepsilon (T^2+N)(TN)^\varepsilon$ for coefficients supported on $N < n \le 2N$, then the maximal version holds with the same exponents.

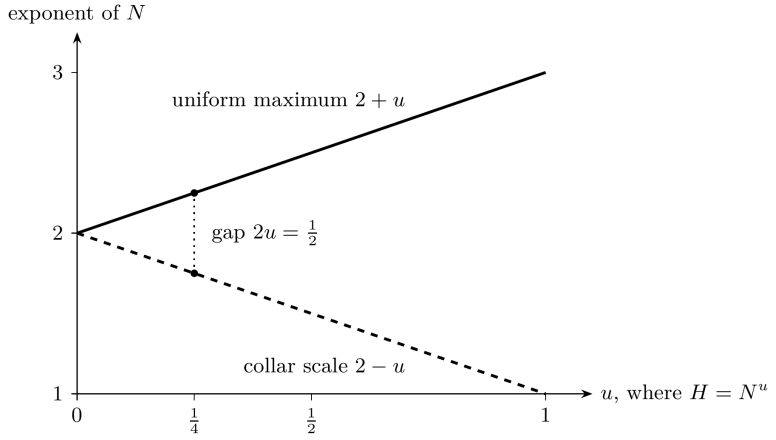


FIGURE 6. The exact coefficient-uniform maximum of the collar energy, of size N^{2+u} when $H = N^u$, against the collar scale N^{2-u}. The two lines separate at once; at $u = \frac{1}{4}$ the gap is already $N^{1/2}$. No bound that treats the shift correlations uniformly in the coefficients can close this gap.

Proof. Extend b by zero to length 2^K. Call an interval $(i2^k, (i+1)2^k]$ with $0 \le k \le K$ aligned. Reading the binary expansion of Y from the top shows that $[1, Y]$ is a disjoint union of at most one aligned interval of each length, hence of at most $K+1$ aligned intervals. By the Cauchy–Schwarz inequality the squared prefix sum is at most $K+1$ times the sum of the squares of the corresponding block sums, and therefore at most $K+1$ times the sum over all aligned intervals I of $|\sum_{n\in I} b_n\rho_n|^2$, which no longer depends on Y. Integrate. For each k the aligned intervals of length 2^k partition $[1, 2^K]$, and the hypothesis applied to each restricted vector bounds the integral of that level by $B\sum_n |b_n|^2$. There are $K+1$ levels. □

Proposition 7.8 (The additive benchmark)**.** *For every real sequence a with $|a_n| \le 1$, every $N \ge 2$ and every dyadic H,*

$$\mathbb{E}_a(N, H) \le (1 + \lceil \log_2 N \rceil)^2 \, \|A_N\|^2_{L^\infty(\mathbb{R}/\mathbb{Z})} \sum_{n \le N} a_n^2.$$

In particular $\mathbb{E}_\lambda(N, H) \ll_A N^3 (\log N)^{-A}$ for every $A > 0$, uniformly in H.

Proof. Let $a^{(N)} = a\mathbf{1}_{[1,N]}$, extended by zero to $\mathbb{Z}$, and $K = \lceil \log_2 N \rceil$. For an aligned interval $I \subset [1, 2^K]$ put

$$c_I(h) = \sum_{n \in I} a_n^{(N)} \, a_{n+h}^{(N)} \qquad (h \in \mathbb{Z}).$$

If $0 \le Y \le N - h$, then $n + h \le N$ for every $n \le Y$, so $C_{a,h}(Y)$ is the sum of $c_I(h)$ over the at most $K+1$ aligned intervals that make up $[1, Y]$. As in the proof of Theorem 7.7,

$$C^*_{a,h}(N)^2 \le (K+1) \sum_I |c_I(h)|^2.$$

Let $A_I(\alpha) = \sum_{n\in I} a_n^{(N)} \mathrm{e}(n\alpha)$. The Fourier series of $h \mapsto c_I(h)$ is $\overline{A_I(\alpha)} A_N(\alpha)$, and by Parseval's identity

$$\sum_{h \in \mathbb{Z}} |c_I(h)|^2 = \int_0^1 |A_I(\alpha)|^2 |A_N(\alpha)|^2 \, \mathrm{d}\alpha \le \|A_N\|^2_\infty \sum_{n \in I} |a_n^{(N)}|^2.$$

Summing over the intervals of one length gives at most $\|A_N\|^2_\infty \sum_{n \le N} a_n^2$, and there are $K+1$ lengths. For λ insert Davenport's bound $\|A_N\|_\infty \ll_A N(\log N)^{-A}$, in the Liouville form quoted in [26]; see also [22, Ch. 13]. □

Proposition 7.8 is essentially all that the maximal method yields from additive information: a bound $\|A_N\|_\infty \ll N^{1/2+\varepsilon}$, which is the conclusion of Theorem 7.3, returns only $\mathbb{E}_a(N,H) \ll N^{2+\varepsilon}$, without the factor $1/H$ of (7.1). The loss is real. Theorem 8.2 exhibits a sign sequence with $\|A_N\|_\infty \ll \sqrt{N}$ and $\mathbb{E}(3H,H) \geq H^2$. Table 8 collects the scales.

TABLE 8. The collar energy $\mathbb{E}_a(N,H)$ at $N = 3H$: hypothesis, benchmarks and lower bounds.

Statement	Size at $N = 3H$	Source
maximal collar hypothesis (7.1)	$\ll H^{1+\varepsilon}$	would imply GRH
uniform maximum over $\lvert a_n\rvert \leq 1$	$\asymp H^3$	Prop. 7.5
additive benchmark for λ	$\ll_A H^3(\log H)^{-A}$	Prop. 7.8
Rudin–Shapiro signs	$\geq H^2$	Thm. 8.2
independent signs, almost surely	$\geq H^2/2$	Prop. 8.3
random multiplicative signs, infinitely often	$\gg H^2/\log H$	Prop. 8.4
every sign sequence	$\mathbb{E}/H \to \infty$; $\gg H\log H$ for a positive proportion of scales	Thm. 8.5
λ, fixed endpoint M	mean of $\mathbb{E}/H$ at least $M\log 2$	Thm. 9.1
λ, under power-saving Chowla	$\gg H^{1+\kappa}$	Prop. 9.2

8. Model sequences

Five models show that (7.1) fails for sequences that share with λ the properties one might hope to use. The first keeps the arithmetic data at any finite set of primes. The second has square-root cancellation in every additive twist, which is the conclusion of Theorem 7.3. The third and fourth are random. The last is a statement about every sign sequence.

8.1. **Finite prime data.**

Theorem 8.1 (Finite prime data)**.** *Let P be a finite set of primes and $(\epsilon_p)_{p\in P}$ signs. There is a completely multiplicative $f:\mathbb{N}\to\{\pm1\}$ with $f(p)=\epsilon_p$ for $p\in P$ and*

$$\Big|\sum_{n\leq N} f(n)\Big| \ll_P (1+\log N)^{|P\cup\{3\}|}.$$

There is an integer $D\geq 1$ depending only on P such that $\mathbb{E}_f(N,H)\geq HN^2/(32D)$ whenever $2D\leq H\leq N/4$.

Proof. Let $S = P\cup\{3\}$, $W=\prod_{p\in S}p$, and let χ_3 be the nonprincipal character modulo 3. Put $f(p)=\epsilon_p$ on P, $f(3)=1$ if $3\notin P$, and $f(p)=\chi_3(p)$ for $p\notin S$. Write $n=bm$ with b composed of primes in S and $(m,W)=1$; then $f(n)=f(b)\chi_3(m)$. The function $m\mapsto \chi_3(m)\mathbf{1}_{(m,W)=1}$ has period W, and its sum over a period vanishes: by the Chinese remainder theorem the reduced residues modulo W that are 1 modulo 3 and those that are 2 modulo 3 are equinumerous. Each inner sum over $m\leq N/b$ is therefore at most W in absolute value, and the number of admissible $b\leq N$ is at most $\prod_{p\in S}(1+\log N/\log p)$.

Choose k with $\sum_{p\in S}p^{-k}\leq 1/4$ and put $D=\prod_{p\in S}p^k$. Let $D\mid h$ and let n satisfy $p^k\nmid n$ for every $p\in S$. Then $v_p(n+h)=v_p(n)$ for $p\in S$, so n and $n+h$ have the same factor b; moreover $3\mid h/b$, so the cofactors agree modulo 3. Hence $f(n+h)=f(n)$. At most $Y/4$ integers $n\leq Y$ are exceptional, so $C_{f,h}(Y)\geq Y/2$ whenever $D\mid h$. For $2D\leq H\leq N/4$ the range $(H,2H]$ contains at least $H/(2D)$ multiples of D, and for each of them $Y=N-h\geq N/2$ is admissible. Summing $(N/4)^2$ over these shifts gives the bound. □

So the collar energy of f is within a constant of the uniform maximum, although f agrees with λ, or with any other prescription, on P, and its partial sums are bounded by a power of $\log N$.

8.2. **Rudin–Shapiro signs.** The Rudin–Shapiro polynomials are defined by $P_0 = Q_0 = 1$ and

$$P_{m+1}(z) = P_m(z) + z^{2^m} Q_m(z), \qquad Q_{m+1}(z) = P_m(z) - z^{2^m} Q_m(z). \tag{8.1}$$

The coefficients of P_{m+1} begin with those of P_m, so there is a sign sequence $r = (r_n)_{n\ge1}$ with $r_n = [z^{n-1}]P_m$ for $n \le 2^m$.

Theorem 8.2 (Additive cancellation without the collar)**.** *The Rudin–Shapiro sequence satisfies*

$$\sup_{\alpha} \Big| \sum_{n\le N} r_n \mathrm{e}(n\alpha) \Big| \le (2+2\sqrt{2})\sqrt{N}, \qquad |C_{r,h}(Y)| \le 5\sqrt{hY} \quad (h, Y \ge 1).$$

Nevertheless $C_{r,2H}(H) = H$ and hence $\mathbb{E}_r(3H, H) \ge H^2$ for every dyadic H.

Proof. The parallelogram law applied to (8.1) gives $|P_{m+1}|^2 + |Q_{m+1}|^2 = 2(|P_m|^2 + |Q_m|^2)$ on $|z| = 1$, hence

$$|P_m(z)|^2 + |Q_m(z)|^2 = 2^{m+1} \qquad (|z| = 1). \tag{8.2}$$

By (8.1) the coefficient vector of P_{m+1} is that of P_m followed by that of Q_m, and the coefficient vector of Q_{m+1} is that of P_m followed by that of $-Q_m$. By induction every aligned block $(i2^j, (i+1)2^j]$ of r is the coefficient vector of one of $\pm P_j, \pm Q_j$. The three assertions follow from this block structure; the computations are carried out in Appendix B.4. In outline: a prefix of length N is a union of at most one aligned block of each length $2^j \le N$, each of sup norm at most $2^{(j+1)/2}$ by (8.2), which gives the first bound. For the second, if $h > Y$ the trivial bound suffices. Otherwise cut $[1, Y]$ into aligned blocks of length 2^{k+1}, where 2^k is the power of two in $[\sqrt{hY}, 2\sqrt{hY})$. Inside a block the pairs $(n, n+h)$ lying in the same half contribute the lag-h autocorrelations of P_k and of Q_k, whose sum is the h-th Fourier coefficient of (8.2) and vanishes. The remaining pairs number at most $2h$ per block. Finally P_{m+2} consists of the four blocks P_m, Q_m, P_m, $-Q_m$, so for $H = 2^m$ the first and third blocks agree term by term and $C_{r,2H}(H) = \sum_{n\le H} r_n^2 = H$. The endpoint $Y = H$ is admissible for $N = 3H$ and $h = 2H$. □

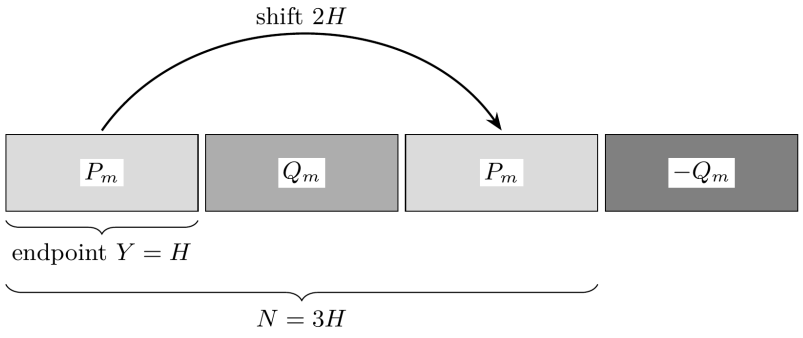


FIGURE 7. Rudin–Shapiro blocks of length H. The first and third blocks agree entry by entry, so the correlation at shift $2H$ is as large as possible. The endpoint $Y = H$ sees this agreement before the fourth block, with the opposite sign, contributes.

The prefix maximum in C^* is essential here. With the full endpoint $Y = N - h$ the fourth block would enter and the second estimate of the theorem would apply. The sequence r satisfies the conclusion of Theorem 7.3 with $\varepsilon = 0$, and every one of its correlations has square-root size uniformly in h and Y; the collar hypothesis at $N = 3H$ would require $\mathbb{E}_r(3H, H) \ll H^{1+\varepsilon}$.

8.3. **Random signs.**

Proposition 8.3 (Independent signs)**.** *Let (ϵ_n) be independent fair signs. Almost surely $\mathbb{E}_\epsilon(3H,H) \geq H^2/2$ for all sufficiently large dyadic H.*

Proof. Put $T_H = \sum_{H<h\leq 2H} |C_{\epsilon,h}(H)|^2$, so that $\mathbb{E}_\epsilon(3H,H) \geq T_H$. Expanding the squares, the diagonal contributes H for each h, in total H^2. The other terms are $2\epsilon_n\epsilon_m\epsilon_{n+h}\epsilon_{m+h}$ with $1 \leq n < m \leq H < n+h < m+h$. The four indices are distinct and determine (n,m,h), so these monomials are orthonormal. Hence $\mathbf{E}\, T_H = H^2$ and $\operatorname{Var} T_H = 4H\binom{H}{2} \leq 2H^3$. Chebyshev's inequality gives $\mathbf{P}(T_H < H^2/2) \leq 8/H$, which is summable over dyadic H. Apply the Borel–Cantelli lemma. □

Proposition 8.4 (Random completely multiplicative signs)**.** *Give the primes independent fair signs and extend completely multiplicatively to f. Almost surely $\mathbb{E}_f(3H,H) \gg H^2/\log H$ for infinitely many dyadic H, with an absolute implied constant.*

Proof. Fix a dyadic $H \geq 2$ and condition on the signs of all primes up to $2H$. A composite integer at most $3H$ has all its prime factors at most $3H/2$. So in the vector $(C_{f,h}(H))_{H<h\leq 2H}$ the only signs not yet determined are $\epsilon_p = f(p)$ for primes $p \in (2H, 3H]$ occurring as $n+h$. Write the vector as $u + \sum_p \epsilon_p A_p$, where u is determined by the conditioning and $A_p(h) = f(p-h)$ if $1 \leq p-h \leq H$, and $A_p(h) = 0$ otherwise. Let $T = \sum_p \|A_p\|_2^2$. A prime $p \in (2H, 5H/2]$ has $\|A_p\|_2^2 = 3H - p + 1 \geq H/2$, and the prime number theorem gives $T \gg H^2/\log H$.

Let $Z = \|u + \sum_p \epsilon_p A_p\|_2^2$. Its conditional mean is $m = \|u\|_2^2 + T$, and by orthogonality of the monomials ϵ_p and $\epsilon_p\epsilon_q$,

$$\operatorname{Var}(Z) = 4\sum_p \langle u, A_p\rangle^2 + 4\sum_{p<q} \langle A_p, A_q\rangle^2 \leq 4\|u\|_2^2 T + 2T^2 \leq 2m^2.$$

Thus $\mathbf{E}[Z^2] \leq 3m^2$. Since $\mathbf{E}[Z\mathbf{1}_{Z\geq m/2}] \geq m/2$, the Cauchy–Schwarz inequality gives $\mathbf{P}(Z \geq m/2) \geq 1/12$ conditionally. The events $\{Z \geq T/2\}$ at smaller dyadic scales depend only on primes up to $3H/2$ and are measurable with respect to the conditioning at scale H. Iterating, the probability that none of k given dyadic scales succeeds is at most $(11/12)^k$. Let $k \to \infty$ and take a countable union over starting scales. Finally $\mathbb{E}_f(3H,H) \geq Z$. □

8.4. **A deterministic logarithmic obstruction.**

Theorem 8.5 (Gram matrices)**.** *There is an absolute constant C_0 with the following property. For every sign sequence a and every $B \geq 0$, the number of dyadic H with*

$$\mathbb{E}_a(3H,H) \leq BH$$

is at most $C_0(B+1)$. Consequently $\mathbb{E}_a(3H,H)/H \to \infty$ along the dyadic integers, and for all large J at least a quarter of the scales $H = 2^j$, $J/2 \leq j \leq J$, satisfy $\mathbb{E}_a(3H,H) \geq c_0 H\log H$ with an absolute $c_0 > 0$.

Proof. Restrict to exponents j in one residue class modulo 3, so that distinct scales differ by a factor at least 8. Let $M = \lceil 64(B+1)\rceil$ and suppose that there are $K = 2M$ scales $H_1 < \cdots < H_K$ in the class, all at least M, with $\mathbb{E}_a(3H_i,H_i) \leq BH_i$. Since $M \geq 64$, each H_i is divisible by 4; let $I_i = (3H_i/2, 7H_i/4] \cap \mathbb{Z}$, a set of $H_i/4$ integers. For $n \geq 1$ let $v_n = (a_{n+1},\dots,a_{n+M})^{\mathsf{T}}$ and

$$B_i = \frac{1}{|I_i|}\sum_{n\in I_i} v_n v_n^{\mathsf{T}}.$$

Each B_i is a positive semidefinite $M\times M$ matrix of trace M, so $\operatorname{tr}(B_i^2) \leq M^2$. Let $i<j$, $n \in I_i$, $m \in I_j$ and $h = m-n$. Since $H_i \leq H_j/8$, one has $H_j < h < 2H_j$ and $m+M \leq 3H_j$. Therefore both n and $n+M$ are admissible endpoints at $N = 3H_j$, and

$$|v_n^{\mathsf{T}} v_m| = |C_{a,h}(n+M) - C_{a,h}(n)| \leq 2C^*_{a,h}(3H_j).$$

For a fixed h there are at most $|I_i|$ pairs (n,m) with $m-n=h$. Hence

$$\operatorname{tr}(B_iB_j)=\frac{1}{|I_i||I_j|}\sum_{n\in I_i}\sum_{m\in I_j}(v_n^{\mathsf T}v_m)^2\le\frac{4\,\mathbb{E}_a(3H_j,H_j)}{|I_j|}\le 16B.$$

On the other hand $\sum_i B_i$ is positive semidefinite with trace KM and M eigenvalues, so $\operatorname{tr}((\sum_i B_i)^2)\ge K^2M$ by the Cauchy–Schwarz inequality. Combining,

$$K^2M\le KM^2+16BK(K-1).$$

With $K=2M$ this gives $2M^3<64BM^2$, that is $M<32B$, contrary to the choice of M.

So each residue class contains fewer than $2M$ such scales above M, and there are at most $1+\log_2 M$ dyadic scales below M. This proves the first claim. For the last, apply it with $B=c_0J$: for c_0 small and J large the exceptional scales number at most $J/4$ among the $J/2+O(1)$ scales of the range, and on the others $\mathbb{E}_a(3H,H)>c_0JH\ge c_0H\log_2 H$. □

The theorem applies to λ without any information on its correlations. It shows that (7.1) with $\varepsilon=0$ fails for every sign sequence, along every infinite set of dyadic scales. The mechanism is linear algebra: a sign sequence has too many windows of length M for them to be pairwise almost orthogonal in $\mathbb{R}^M$.

Table 9 lists the values of $\mathbb{E}_a(3H,H)/H$ for three sequences, as a numerical illustration. For λ the ratio grows linearly in H in this range, as it does for random signs. The theorems of this paper do not prove that growth for λ; Section 9 proves what the available correlation estimates give.

TABLE 9. Computed values of $\mathbb{E}_a(3H,H)/H$. The last column is the uniform maximum of Proposition 7.5 divided by H. The random column is one sample.

H	$a=\lambda$	Rudin–Shapiro	random signs	maximum
16	35.4	41.0	34.1	573.5
64	155.4	162.8	159.7	9461.5
256	638.4	673.4	672.0	152533.5
1024	2630.6	2684.0	2622.6	2445141.5
2048	5399.2	5366.1	5656.7	9783637.5

9. The endpoint for Liouville itself

The models of Section 8 are other sequences, and Theorem 8.5 gives for λ only a logarithmic lower bound at most scales. Tao's logarithmically averaged two-point Chowla theorem [35] gives an asymptotic for a second moment of the Liouville correlations themselves. We use it in the form

$$\sum_{t\le T}\frac{\lambda(t)\lambda(t+d)}{t}=o_d(\log T)\qquad(d\ge 1\text{ fixed}).\tag{9.1}$$

Theorem 9.1 (The endpoint for Liouville)**.** *For every fixed integer $M\ge 1$,*

$$\sum_{h\le T}\frac{|C_{\lambda,h}(M)|^2}{h}=M\log T+o_M(\log T)\qquad(T\to\infty).\tag{9.2}$$

Consequently

$$\liminf_{J\to\infty}\frac{1}{J}\sum_{j=0}^{J-1}\frac{\mathbb{E}_\lambda(3\cdot 2^j,2^j)}{2^j}\ge M\log 2\quad\textit{for every } M,\tag{9.3}$$

so the left side is $+\infty$, and the bound (7.1) *with $\varepsilon=0$ fails for λ at $N=3H$.*

Proof. Expand

$$|C_{\lambda,h}(M)|^2 = \sum_{1\le n,m\le M} \lambda(n)\lambda(m)\lambda(n+h)\lambda(m+h).$$

The diagonal $n = m$ contributes M, hence $M \log T + O(M)$ to (9.2). Fix $n < m$ and put $d = m - n$, $t = n + h$. Then

$$\begin{aligned}\sum_{h\le T} \frac{\lambda(n+h)\lambda(m+h)}{h} &= \sum_{n<t\le T+n} \frac{\lambda(t)\lambda(t+d)}{t} + O\Big(\sum_{t>n} \frac{n}{t(t-n)}\Big)\\ &= o_d(\log T) + O_M(1)\end{aligned}$$

by (9.1); the change of the range of summation costs $O_M(1)$. The pairs with $m < n$ are treated in the same way. There are fewer than M^2 pairs, which proves (9.2).

Let $H \ge M$ be dyadic and $H < h \le 2H$. The endpoint M is admissible at $N = 3H$, so $|C_{\lambda,h}(M)| \le C^*_{\lambda,h}(3H)$, and $1/h < 1/H$. Hence the part of the sum (9.2) with $H < h \le 2H$ is at most $\mathbb{E}_\lambda(3H, H)/H$. Summing over the dyadic H with $M \le H < 2^J$ and using (9.2) with $T = 2^J$ gives

$$\sum_{j<J} \frac{\mathbb{E}_\lambda(3\cdot 2^j, 2^j)}{2^j} \ge MJ\log 2 - o_M(J) - O_M(1),$$

where the last term accounts for the shifts $h \le 2M$. Divide by J. □

The divergence in (9.3) alone already follows from Theorem 8.5, for every sign sequence. What the correlation theorem adds is the exact constant in (9.2): on logarithmic average over the shift, the correlations of λ with a fixed endpoint M have the mean square M of independent signs. By Proposition 6.7(c) the same asymptotic holds for every modified Liouville function f_S with $\sum_{p\in S} 1/p < \infty$. The argument takes M fixed before $T \to \infty$ and gives no positive power of H. The next proposition shows what uniformity in M would give.

Proposition 9.2 (A power saving in ordinary correlations contradicts the collar)**.** *Suppose there are $\delta, \eta > 0$ and $K < \infty$ such that, for all sufficiently large dyadic H and all $1 \le n < m \le H^\eta$,*

$$\Big|\sum_{H<h\le 2H} \lambda(h+n)\lambda(h+m)\Big| \le KH^{1-\delta}.$$

Then $\mathbb{E}_\lambda(3H, H) \gg H^{1+\kappa}$ for every $0 < \kappa < \min(\delta, \eta, 1)$ and all large dyadic H. In particular (7.1) *fails for λ for every $\varepsilon < \kappa$.*

Proof. Let $M = \lfloor H^\kappa \rfloor \le H$ and expand $\sum_{H<h\le 2H} |C_{\lambda,h}(M)|^2$. The diagonal equals HM. The off-diagonal terms sum to at most $M(M-1)KH^{1-\delta}$ in absolute value, and the ratio of this to HM is at most $KH^{\kappa-\delta} \to 0$. Since M is an admissible endpoint at $N = 3H$, the collar energy is at least this second moment. □

The hypothesis of Proposition 9.2 is a uniform form of the two-point Chowla conjecture with a power saving, which is the expected truth. Under it the maximal collar hypothesis is false, and Theorem 7.3 would be an implication with a false premise. The reason is a matter of scale. If every correlation $C_{\lambda,h}(Y)$ had square-root size and no smaller, then $C^*_{\lambda,h}(N)^2$ would be of order N and $\mathbb{E}_\lambda(N, H)$ of order HN, which exceeds the collar scale N^2/H as soon as $H > \sqrt{N}$. The best cancellation one expects in each term is therefore incompatible with (7.1) in the upper half of the range of shifts, and this is the precise sense in which the termwise approach cannot work. For comparison, the two-point Chowla conjecture at each fixed shift would give, for fixed M,

$$\frac{1}{H}\sum_{H<h\le 2H} |C_{\lambda,h}(M)|^2 \longrightarrow M \qquad (H \to \infty),$$

by the same expansion, and hence $\mathbb{E}_\lambda(3H,H)/H \to \infty$ with explicit dependence on M and without averaging over scales. Only the logarithmically averaged statement (9.1) is known, and Theorem 9.1 uses nothing more.

The signed energy $Q_\lambda(N)$ of Proposition 2.1 avoids this defect. In its shift expansion the correlations enter with the Fejér weights $1-h/N$, at the single endpoint $N-h$, and without absolute values.

10. Sparse primes and Abel data

Theorem 6.3 produces a singularity at $s=\beta$ without describing it. With a more careful choice of the set of primes the Dirichlet series can be computed up to a nonvanishing factor, and the resulting functions can be made to imitate λ to any finite precision in a neighbourhood of $s=1$. The local behaviour at $s=1$ is what Abel summation $\sum_n f(n)n^{-1-\delta}$, $\delta \downarrow 0$, detects, and we refer to the Taylor coefficients there as Abel data.

10.1. **A rounded construction with an explicit factorization.** Fix $1/2<\beta<1$ and an integer $P\ge 2$. For primes $p>P$ put

$$q_p=\tfrac12 p^{\beta-1}, \qquad S(t)=\sum_{P<p\le t} q_p, \qquad I_p=\lfloor S(p)\rfloor-\lfloor S(p)-q_p\rfloor\in\{0,1\}. \tag{10.1}$$

Let $\mathcal{S}_P=\{p>P: I_p=1\}$, let $f_P=f_{\mathcal{S}_P}$ be the corresponding modified Liouville function, and $G_P=G_{\mathcal{S}_P}$. Thus $f_P(p)=2I_p-1$ for $p>P$ and $f_P(n)=\lambda(n)$ for all $n\le P$. The set $\mathcal{S}_P$ contains a prime each time the sum S passes an integer. Telescoping the integer parts gives

$$E(t):=\sum_{P<p\le t}(I_p-q_p)=\lfloor S(t)\rfloor-S(t)\in(-1,0] \qquad (t\ge P). \tag{10.2}$$

Theorem 10.1 (Explicit factorization and exact exponent)**.** *The function $f=f_P$ has the following properties.*

(i) *One has $\pi_{\mathcal{S}_P}(x)=\frac12\sum_{P<p\le x}p^{\beta-1}+O(1)$, which is asymptotic to $x^\beta/(2\beta\log x)$. The series $\sum_{p\in\mathcal{S}_P}p^{-\sigma}$ converges for $\sigma>\beta$ and diverges at $\sigma=\beta$.*

(ii) *The Dirichlet series $F_f(s)=\sum_n f(n)n^{-s}$ extends meromorphically to $\operatorname{Re}s>1/2$ as*

$$F_f(s)=\frac{\zeta(2s)\,\zeta(s+1-\beta)}{\zeta(s)}\,R(s), \tag{10.3}$$

where R is holomorphic and zero-free on $\operatorname{Re}s>1/2$ and positive on the real axis. It has a simple zero at $s=1$ and a simple pole at $s=\beta$ with residue $\zeta(2\beta)R(\beta)/\zeta(\beta)\neq 0$.

(iii) *$L_f(x)\ll_{A,f} x(\log x)^{-A}$ and $Q_f(N)\ll_{A,f} N^2(\log N)^{-A}$ for every $A>0$.*

(iv) *The energy exponent is exact:*

$$\limsup_{N\to\infty}\frac{\log(1+Q_f(N))}{\log N}=2\max(\Theta,\beta).$$

Proof. (i) By (10.2), $\pi_{\mathcal{S}_P}(x)=S(x)+E(x)$, and the asymptotic follows from the prime number theorem by partial summation. The centred series $A(s)=\sum_{p>P}(I_p-q_p)p^{-s}$ satisfies, by partial summation and the boundedness of E,

$$A(s)=s\int_P^\infty E(t)\,t^{-s-1}\,\mathrm{d}t \qquad (\operatorname{Re}s>0), \tag{10.4}$$

and is holomorphic there. For real $\sigma>\beta$,

$$\sum_{p\in\mathcal{S}_P}p^{-\sigma}=\frac12\sum_{p>P}p^{-(\sigma+1-\beta)}+A(\sigma)<\infty.$$

As $\sigma\downarrow\beta$ the first term tends to $\frac12\sum_{p>P}1/p=\infty$ while $A(\sigma)$ stays bounded, so the series diverges at β by monotone convergence.

(ii) For $\operatorname{Re} s > 1$, $F_f = \zeta(2s)G_P(s)/\zeta(s)$ as in (6.3). Subtracting the logarithm of the Euler product of $\zeta(s+1-\beta)$ from that of G_P gives, for $\operatorname{Re} s > 1$,

$$\log G_P(s) - \log\zeta(s+1-\beta) = 2A(s) - \sum_{p\le P} p^{-(s+1-\beta)} + 2\sum_{p\in\mathcal{S}_P}\sum_{\substack{k\ge 3\\ k\ \text{odd}}} \frac{p^{-ks}}{k} - \sum_{p}\sum_{k\ge 2}\frac{p^{-k(s+1-\beta)}}{k}. \tag{10.5}$$

The details are in Appendix B.1. The first line is holomorphic on $\operatorname{Re} s > 0$. The two double series converge absolutely and locally uniformly on $\operatorname{Re} s > 1/2$, being dominated there by $\sum_p p^{-3\sigma}$ and $\sum_p p^{-2(\sigma+1-\beta)}$, whose exponents exceed 1. Let $R(s)$ be the exponential of the right side of (10.5). It is holomorphic and zero-free on $\operatorname{Re} s > 1/2$, positive at real points, and $G_P(s) = \zeta(s+1-\beta)R(s)$; this proves (10.3). At $s = 1$ the factors $\zeta(2s)$, $\zeta(s+1-\beta)$ and R are finite and positive and $1/\zeta$ has a simple zero. At $s = \beta$ the factor $\zeta(s+1-\beta)$ has a simple pole with residue 1, while $\zeta(2\beta)$ is finite and positive, $\zeta(\beta) < 0$ and $R(\beta) > 0$.

(iii) This is Proposition 6.7(d) at $\alpha = 0$, applicable by (i) with any $\sigma_0 \in (\beta, 1)$. For the energy use (6.1), bounding $|L_f(j)|$ by $\sqrt{N}$ for $j \le \sqrt{N}$.

(iv) The upper bound is Theorem 6.3(ii). For the lower bound let ρ be a pole of F_f in $\operatorname{Re} s > 1/2$. If the limit superior were less than $2\operatorname{Re}\rho$, choose α with $\alpha < \operatorname{Re}\rho$ and $Q_f(N) \ll N^{2\alpha}$. Then $L_f(x) \ll x^{\alpha}$ by (6.1), and $s\int_1^\infty L_f(x)x^{-s-1}\,dx$ would continue F_f holomorphically to $\operatorname{Re} s > \alpha$, a contradiction. The point β is a pole. So is every zero ρ of ζ with $\operatorname{Re}\rho > \beta$: there $\operatorname{Re} 2\rho > 1$ and $\operatorname{Re}(\rho+1-\beta) > 1$, so neither zeta factor in the numerator of (10.3) vanishes, and $R(\rho) \ne 0$. If $\Theta > \beta$ such zeros have real parts arbitrarily close to Θ. □

The same function f_P serves for all A in (iii); it is one completely multiplicative function, fixed once β and P are chosen. Table 21 in Appendix B.1 lists the first steps of the rounding for $\beta = 3/4$, $P = 2$.

10.2. The zero at $s = 1$ and its leading coefficient.

Proposition 10.2 (Rigidity of the leading coefficient)**.** *Let $g : \mathbb{N} \to \{\pm 1\}$ be completely multiplicative, and suppose its Dirichlet series F_g continues holomorphically to a neighbourhood of $s = 1$ with $F_g(1) = 0$. Then the zero is simple, $F_g'(1) \ge \zeta(2)$, and equality holds if and only if $g = \lambda$.*

Proof. For real $\sigma > 1$ the Euler products give

$$\frac{F_g(\sigma)}{F_\lambda(\sigma)} = \prod_{p:\, g(p)=1}\frac{1+p^{-\sigma}}{1-p^{-\sigma}} \ge 1, \qquad F_\lambda(1+\delta) = \frac{\zeta(2+2\delta)}{\zeta(1+\delta)} \sim \zeta(2)\,\delta \quad (\delta \downarrow 0).$$

Hence $F_g(1+\delta) \ge (1+o(1))\zeta(2)\delta$. A zero of order at least two would give $F_g(1+\delta) = O(\delta^2)$. Dividing by δ and letting $\delta \downarrow 0$ gives $F_g'(1) \ge \zeta(2)$. If $g(p_0) = 1$ for some prime p_0, the ratio is at least $(1+p_0^{-\sigma})/(1-p_0^{-\sigma})$, which tends to $(p_0+1)/(p_0-1) > 1$, so $F_g'(1) > \zeta(2)$. Otherwise $g(p) = -1$ for all p and $g = \lambda$. □

For $g = f_P$ the proposition applies, and $F_{f_P}'(1) = \zeta(2)G_P(1) > \zeta(2)$ because $\mathcal{S}_P$ is nonempty: $S(t) \to \infty$ since $\sum_p p^{\beta-1}$ diverges. Thus complete multiplicativity, agreement with λ up to P, every logarithmic saving and a simple zero at $s = 1$ together still allow energy growth $N^{2\beta}$. Within the class of completely multiplicative sign functions the exact value $F'(1) = \zeta(2)$ singles out λ. The next subsection shows that no approximate form of this equality does.

10.3. **Approximation of finite Abel jets.** Normalize the zero at $s=1$ by putting

$$\mathcal{J}_\lambda(s)=\frac{\zeta(2s)}{\big((s-1)\zeta(s)\big)^2},\qquad \mathcal{J}_P(s)=\mathcal{J}_\lambda(s)\,G_P(s), \tag{10.6}$$

so that $\mathcal{J}_f(s)=F_f(s)/((s-1)^2\zeta(s))$ for $f=\lambda$ and $f=f_P$, with the removable value at $s=1$ assigned. The derivatives $\mathcal{J}^{(k)}(1)$, $0\le k\le K$, form the Abel jet of order K; $\mathcal{J}_f(1)=F_f'(1)$.

Lemma 10.3 (Euler-factor bound)**.** *For real $\sigma>\beta$,*

$$\sup_{\operatorname{Re}s\ge\sigma}|\log G_P(s)|\le\Delta(P,\sigma):=\frac{P^{\beta-\sigma}}{(\sigma-\beta)(1-P^{-2\sigma})},$$

where $\log G_P$ is defined by its Euler series. Consequently $|G_P(s)-1|\le e^{\Delta(P,\sigma)}-1$ for $\operatorname{Re}s\ge\sigma$.

Proof. For $T>P$, partial summation and $E\le 0$ give

$$\sum_{P<p\le T}(I_p-q_p)p^{-\sigma}=E(T)T^{-\sigma}+\sigma\int_P^T E(t)t^{-\sigma-1}\,\mathrm{d}t\le 0.$$

Therefore

$$\sum_{p\in\mathcal{S}_P}p^{-\sigma}\le\frac12\sum_{p>P}p^{-(1+\sigma-\beta)}\le\frac12\sum_{n>P}n^{-(1+\sigma-\beta)}\le\frac{P^{\beta-\sigma}}{2(\sigma-\beta)},$$

the last step comparing the decreasing sum with $\int_P^\infty t^{-(1+\sigma-\beta)}\,\mathrm{d}t$. For $\operatorname{Re}s\ge\sigma$,

$$|\log G_P(s)|=\Big|2\sum_{p\in\mathcal{S}_P}\sum_{j\ \text{odd}}\frac{p^{-js}}{j}\Big|\le 2\sum_{p\in\mathcal{S}_P}\frac{p^{-\sigma}}{1-p^{-2\sigma}}\le\frac{2}{1-P^{-2\sigma}}\sum_{p\in\mathcal{S}_P}p^{-\sigma}.$$

The last assertion follows from $|e^w-1|\le e^{|w|}-1$. □

Theorem 10.4 (Approximation of every finite Abel jet)**.** *Let $0<r<1-\beta$ and $C_r=\zeta(2-2r)/(1-2r)^2$. The functions* (10.6) *are holomorphic on a neighbourhood of $|s-1|\le r$, and for every integer $k\ge 0$,*

$$|\mathcal{J}_P^{(k)}(1)-\mathcal{J}_\lambda^{(k)}(1)|\le k!\,r^{-k}\,C_r(e^{\Delta(P,1-r)}-1). \tag{10.7}$$

For fixed r and K the right side tends to zero as $P\to\infty$, uniformly in $0\le k\le K$. Explicitly, if $\eta>0$, $d=1-r-\beta$, $M=C_r\max_{k\le K}k!\,r^{-k}$ and

$$P\ge\max\Big\{2,\ \Big[\frac{2}{d\,\log(1+\eta/M)}\Big]^{1/d}\Big\}, \tag{10.8}$$

then $|\mathcal{J}_P^{(k)}(1)-\mathcal{J}_\lambda^{(k)}(1)|\le\eta$ for all $0\le k\le K$.

Proof. Euler summation gives, for $\operatorname{Re}s>0$,

$$\zeta(s)=\frac{s}{s-1}-s\int_1^\infty\{x\}x^{-s-1}\,\mathrm{d}x,$$

and the integral has modulus at most $1/\operatorname{Re}s$ [36, §2.1]. Since $r<1/2$, on $|s-1|\le r$ this yields

$$|(s-1)\zeta(s)|\ge|s|\Big(1-\frac{|s-1|}{\operatorname{Re}s}\Big)\ge(1-r)\Big(1-\frac{r}{1-r}\Big)=1-2r>0.$$

Also $|\zeta(2s)|\le\zeta(2-2r)$ there. Hence $|\mathcal{J}_\lambda|\le C_r$ on the disc. Since $1-r>\beta$, Lemma 10.3 bounds $|G_P-1|$ by $e^{\Delta(P,1-r)}-1$ on the disc, and Cauchy's inequalities for $\mathcal{J}_P-\mathcal{J}_\lambda=\mathcal{J}_\lambda(G_P-1)$ on $|s-1|=r$ give (10.7). For $P\ge 2$ and $\sigma=1-r>1/2$ one has $1-P^{-2\sigma}\ge 1/2$, so $\Delta(P,1-r)\le 2P^{-d}/d$, and (10.8) makes this at most $\log(1+\eta/M)$. □

Theorem 10.5 (The local energy still fails)**.** *Let* $B_P(X) = \sum_{n\ge1} f_P(n)e^{-n/X}$ *and* $\mathcal{H}_{B_P}(X) = \int_{X/2}^{2X} |B_P(Y)|^2\,\mathrm{d}Y/Y$*. For every* P *and every* $0 < \varepsilon < 2\beta - 1$*,*

$$\mathcal{H}_{B_P}(X) \neq O(X^{1+\varepsilon}), \qquad B_P(X) \neq O(X^{(1+\varepsilon)/2}).$$

Proof. The second statement follows from the first. Suppose $\mathcal{H}_{B_P}(X) \ll X^{1+\varepsilon}$. For real $\sigma > (1+\varepsilon)/2$ the Cauchy–Schwarz inequality on $[2^j, 2^{j+1}] \subset [2^j/2, 2\cdot 2^j]$ gives

$$\int_{2^j}^{2^{j+1}} |B_P(Y)|\, Y^{-\sigma}\,\frac{\mathrm{d}Y}{Y} \le (\log 2)^{1/2} 2^{-j\sigma}\mathcal{H}_{B_P}(2^j)^{1/2} \ll 2^{j((1+\varepsilon)/2-\sigma)}.$$

The sum over $j \ge 0$ converges, locally uniformly in σ. On $0 < Y \le 1$ one has $|B_P(Y)| \le (e^{1/Y}-1)^{-1}$, which decays faster than any power. Hence $\int_0^\infty B_P(Y)Y^{-s}\,\mathrm{d}Y/Y$ is holomorphic on $\operatorname{Re} s > (1+\varepsilon)/2$. For $\operatorname{Re} s > 1$ it equals $\Gamma(s)F_{f_P}(s)$, by absolute convergence and $\int_0^\infty e^{-n/Y}Y^{-s}\,\mathrm{d}Y/Y = \Gamma(s)n^{-s}$. This would continue $\Gamma(s)F_{f_P}(s)$ holomorphically across $s = \beta > (1+\varepsilon)/2$, where it has a pole by Theorem 10.1(ii) and $\Gamma(\beta) \neq 0$. □

Theorems 10.4 and 10.5 concern the same function. Table 10 gives certified instances of (10.7); the arithmetic is in Appendix B.2. In the first row every value $f_P(n)$ with $n \le 2^{320}$ equals $\lambda(n)$, the first four Abel derivatives agree with those of λ to within $2\cdot10^{-7}$, and nevertheless $\mathcal{H}_{B_P}(X) \neq O(X^{1+\varepsilon})$ for every $\varepsilon < 1/2$. Exact agreement is different: by Proposition 10.2, $\mathcal{J}_P(1) = \zeta(2)G_P(1) > \zeta(2) = \mathcal{J}_\lambda(1)$ for every P, although the difference tends to zero.

TABLE 10. Upper bounds for $\max_{k\le K}|\mathcal{J}_P^{(k)}(1) - \mathcal{J}_\lambda^{(k)}(1)|$ from (10.7), with C_r replaced by the rational bound $(2-2r)/(1-2r)^3$.

β	r	K	P	$\Delta(P, 1-r) \le$	bound
3/4	1/8	3	2^{320}	2^{-36}	$1.9\cdot10^{-7}$
3/4	1/8	0	2^{160}	2^{-16}	$6.4\cdot10^{-5}$
3/4	1/8	6	2^{480}	2^{-56}	$1.1\cdot10^{-8}$
5/8	1/4	3	2^{320}	2^{-36}	$6.8\cdot10^{-8}$
7/8	1/16	3	2^{960}	2^{-55}	$2.0\cdot10^{-12}$

10.4. **Bounded coefficients with prescribed vanishing moments.** The preceding results stay within completely multiplicative sign functions and match Abel data only approximately. If multiplicativity is dropped, any finite number of Taylor coefficients at $s = 1$ can be made to vanish exactly while the smoothed sum is as large as X^β.

Theorem 10.6 (Finitely many vanishing moments)**.** *Let* $k \ge 1$*,* $1/2 < \beta < 1$ *and* $\gamma > 0$*. There are a real sequence* (a_n) *with* $|a_n| \le 1$ *and a number* $\varepsilon_0 > 0$ *such that:*

(i) $D_a(s) = \sum_n a_n n^{-s}$ *converges absolutely for* $\operatorname{Re} s > \beta$ *and* $D_a^{(j)}(1) = 0$ *for* $0 \le j < k$*;*
(ii) D_a *is meromorphic on* $\mathbb{C}$*, with simple poles at* $\beta \pm i\gamma$ *of residue* $\varepsilon_0/2$ *and no others;*
(iii) *with* $z = \beta + i\gamma$*, for* $X \ge 1$*,*

$$\sum_{n\ge1} a_n e^{-n/X} = \varepsilon_0\,\operatorname{Re}\bigl(\Gamma(z)X^z\bigr) + O_{k,\beta,\gamma}(1).$$

In particular $\limsup_{X\to\infty} X^{-\beta}\sum_n a_n e^{-n/X} = \varepsilon_0|\Gamma(z)|$*, the limit inferior is* $-\varepsilon_0|\Gamma(z)|$*, and the smoothed sum is not* $O(X^{1/2+\eta})$ *for any* $\eta < \beta - 1/2$*.*

Proof. Let $r_n = n^{\beta-1}\cos(\gamma\log n) = \operatorname{Re} n^{z-1}$ and $\mathcal{T} = \{2^m : 0 \le m < k\}$. For $0 \le j < k$ the moments

$$t_j = \sum_{n\notin\mathcal{T}} \frac{r_n(-\log n)^j}{n}$$

converge absolutely, since $|r_n| \le n^{\beta-1}$. Choose real numbers $b_0, \dots, b_{k-1}$ with

$$\sum_{m=0}^{k-1} \frac{b_m(-m\log 2)^j}{2^m} = -t_j \qquad (0 \le j < k). \tag{10.9}$$

After scaling rows and columns the matrix of this system is the Vandermonde matrix $(m^j)_{0\le j,m<k}$, which is invertible; Appendix B.3 gives the solution in closed form. Put

$$\varepsilon_0 = \frac{1}{1+\max_m |b_m|}, \qquad a_n = \varepsilon_0 b_m \ \ (n = 2^m \in \mathcal{T}), \qquad a_n = \varepsilon_0 r_n \ \ (n \notin \mathcal{T}).$$

Then $|a_n| \le 1$, and (10.9) is the statement $D_a^{(j)}(1) = 0$, the series being differentiable term by term. For $\operatorname{Re} s > \beta$,

$$D_a(s) = \frac{\varepsilon_0}{2}\bigl(\zeta(s+1-z) + \zeta(s+1-\bar z)\bigr) + \varepsilon_0 \sum_{m=0}^{k-1} (b_m - r_{2^m}) 2^{-ms}. \tag{10.10}$$

The finite sum is entire, and the two shifted zeta functions have their only poles at $s = z$ and $s = \bar z$. This proves (i) and (ii).

For (iii) let $\phi_X(t) = t^{z-1}e^{-t/X}$. For a continuously differentiable integrable ϕ on $[1,\infty)$ with integrable derivative, $|\sum_{n\ge1}\phi(n) - \int_1^\infty \phi| \le \int_1^\infty |\phi'|$. Here, for $X \ge 1$,

$$\int_1^\infty |\phi_X'(t)|\,\mathrm{d}t \le |z-1|\int_1^\infty t^{\beta-2}\,\mathrm{d}t + \frac{1}{X}\int_0^\infty t^{\beta-1}e^{-t/X}\,\mathrm{d}t \le \frac{|z-1|}{1-\beta} + \Gamma(\beta),$$

and $\int_0^1 |\phi_X| \le 1/\beta$. Since $\int_0^\infty t^{z-1}e^{-t/X}\,\mathrm{d}t = \Gamma(z)X^z$,

$$\sum_{n\ge1} n^{z-1}e^{-n/X} = \Gamma(z)X^z + O_{\beta,\gamma}(1).$$

Take real parts, multiply by ε_0, and note that changing the k coefficients on $\mathcal{T}$ alters the sum by $O_k(1)$. The limits follow by choosing $\gamma\log X \equiv -\arg\Gamma(z)$ and $\gamma\log X \equiv \pi - \arg\Gamma(z)$ modulo 2π. □

The poles in (10.10) are built into the coefficients; they are not zeros of ζ. The sequence depends on k, and ε_0 may tend to zero as k grows, so no single sequence with all moments vanishing is obtained. The theorem excludes a general inference from boundedness and finitely many exact cancellations at $s = 1$ to a square-root bound for the smoothed sum. Theorem 10.4 shows that the multiplicative structure does not restore that inference when the cancellations are matched only to finite precision.

11. Local energy and the prime tail

By Proposition 12.1, RH is equivalent to $B(X) \ll_\varepsilon X^{1/2+\varepsilon}$ for the smoothed sum $B(X) = \sum_n \lambda(n)e^{-n/X}$. This section records four exact reformulations of that bound: a mean square over neighbouring scales, a one-sided bound for a primitive correlation, a signed dilation identity over prime powers, and a cancellation in the tail over large primes. In each case the reformulation is proved and the place where an arithmetic estimate is missing is identified. Table 13 at the end of the section summarizes them.

11.1. **Reconstruction of a point value from a local mean square.** For a complex sequence with $|a_n| \le 1$ put

$$F(X) = \sum_{n\ge1} a_n e^{-n/X}, \qquad \mathcal{H}_F(X) = \int_{X/2}^{2X} |F(Y)|^2 \frac{\mathrm{d}Y}{Y} \qquad (X > 0).$$

Theorem 11.1 (Analytic localization)**.** *Let $X \ge 1$ and $m = \lceil 8\log_2(2X)\rceil$. Then*

$$|F(X)|^2 \le 10(m+2)\,\mathcal{H}_F(X) + X^{-13}. \tag{11.1}$$

Moreover there is a polynomial $K_m(0,u)$ of degree at most m, independent of the sequence, with

$$F(X) = \int_{-1/4}^{1/4} K_m(0,u)\, F\Big(\frac{X}{1+u}\Big)\,\mathrm{d}u + \mathcal{R}_X, \qquad |\mathcal{R}_X| \le \frac{\sqrt{2(m+2)}}{128X^7}.$$

Proof. The function $G_X(z) = \sum_n a_n e^{-n(1+z)/X}$ is holomorphic on $\operatorname{Re} z > -1$, $G_X(u) = F(X/(1+u))$ for real u, and on $|z| \le 1/2$

$$|G_X(z)| \le \sum_{n\ge1} e^{-n/(2X)} = \frac{1}{e^{1/(2X)}-1} \le 2X.$$

Let p_m be its Taylor polynomial of degree m at 0. Cauchy's inequalities on $|z| = 1/2$ give

$$\sup_{|u|\le1/4} |G_X(u) - p_m(u)| \le 2X \sum_{j>m} 2^j 4^{-j} = 2X\, 2^{-m} =: \delta \le \frac{1}{128X^7},$$

and $p_m(0) = F(X)$. Let Λ_j be the Legendre polynomial of degree j, normalized by $\Lambda_j(1) = 1$, so that $\int_{-1}^{1} \Lambda_i\Lambda_j = 2\delta_{ij}/(2j+1)$. The polynomials $\sqrt{2(2j+1)}\,\Lambda_j(4u)$, $0 \le j \le m$, are orthonormal on $[-1/4, 1/4]$, and

$$K_m(v,u) = 2\sum_{j=0}^{m} (2j+1)\Lambda_j(4v)\Lambda_j(4u)$$

is the reproducing kernel of the space of polynomials of degree at most m in $L^2[-1/4,1/4]$. Hence, for such a polynomial p,

$$p(0) = \int_{-1/4}^{1/4} K_m(0,u)p(u)\,\mathrm{d}u, \qquad |p(0)|^2 \le K_m(0,0)\int_{-1/4}^{1/4} |p(u)|^2\,\mathrm{d}u. \tag{11.2}$$

Appendix B.5 proves the facts about Λ_j used here and the bound

$$K_m(0,0) = 2\sum_{i\le m/2} (4i+1)\Big(\binom{2i}{i}4^{-i}\Big)^2 \le 4(m+2). \tag{11.3}$$

Apply (11.2) to p_m with $|p_m|^2 \le 2|G_X|^2 + 2\delta^2$:

$$|F(X)|^2 \le 8(m+2)\int_{-1/4}^{1/4} |G_X(u)|^2\,\mathrm{d}u + 4(m+2)\delta^2.$$

The substitution $Y = X/(1+u)$ gives

$$\int_{-1/4}^{1/4} |G_X(u)|^2\,\mathrm{d}u = \int_{4X/5}^{4X/3} |F(Y)|^2 \frac{X}{Y}\frac{\mathrm{d}Y}{Y} \le \frac{5}{4}\,\mathcal{H}_F(X).$$

Since $m+2 \le 8\log_2 X + 11 \le 16X$, the last term is at most $64X/(128X^7)^2 \le X^{-13}$. This proves (11.1). Finally, by (11.2) for p_m,

$$\mathcal{R}_X = \int_{-1/4}^{1/4} K_m(0,u)\big(p_m(u) - G_X(u)\big)\,\mathrm{d}u,$$

$$|\mathcal{R}_X| \le \delta\Big(\tfrac12\int_{-1/4}^{1/4} K_m(0,u)^2\,\mathrm{d}u\Big)^{1/2} = \delta\sqrt{K_m(0,0)/2}.$$

□

The polynomial approximation acts on the analytic dependence on the scale, not on the coefficients, and uses no cancellation. For $F = B$ it shows that

$$\mathcal{H}_B(X) \ll_\varepsilon X^{1+\varepsilon} \qquad (\varepsilon > 0,\ X \ge 1) \tag{11.4}$$

implies $B(X) \ll_\varepsilon X^{1/2+\varepsilon}$, and the converse follows by integration. So (11.4) is equivalent to RH. The proof of Theorem 10.5 gives the forward implication in another way: (11.4) makes the Mellin transform of B, which is $\Gamma(s)\zeta(2s)/\zeta(s)$, holomorphic on $\operatorname{Re} s > 1/2$.

11.2. **Diagonal and signed parts.** Let $K(v) = \int_{1/2}^{2} e^{-v/t}\,\mathrm{d}t/t$ for $v > 0$. Expanding $|B(Y)|^2$ and integrating term by term, which is justified by absolute convergence, gives

$$\mathcal{H}_B(X) = \mathcal{D}(X) + \mathcal{N}(X), \qquad \mathcal{D}(X) = \int_{X/2}^{2X} \frac{1}{e^{2/Y}-1}\,\frac{\mathrm{d}Y}{Y},$$
$$\mathcal{N}(X) = 2\sum_{h\ge 1}\sum_{n\ge 1} \lambda(n)\lambda(n+h)K\Big(\frac{2n+h}{X}\Big). \tag{11.5}$$

Proposition 11.2 (The diagonal)**.** *For every $X > 0$,*

$$0 < \mathcal{D}(X) - \Big(\frac{3X}{4} - \log 2\Big) < \frac{1}{4X}.$$

Proof. For $t > 0$,

$$0 < \frac{1}{e^t-1} - \frac{1}{t} + \frac{1}{2} < \frac{t}{12}. \tag{11.6}$$

The lower inequality is equivalent to $\varphi(t) = (t-2)e^t + t + 2 > 0$, which holds because $\varphi(0) = \varphi'(0) = 0$ and $\varphi''(t) = te^t > 0$. After multiplication by $12t(e^t-1)$ the upper inequality becomes

$$(t^2 - 6t + 12)(e^t - 1) - 12t = \sum_{j\ge 5} \frac{(j-3)(j-4)}{j!}\,t^j > 0.$$

Put $t = 2/Y$ and integrate against $\mathrm{d}Y/Y$ over $[X/2, 2X]$. The terms $Y/2 - 1/2$ give $3X/4 - \log 2$, and the error is less than $\frac{1}{6}\int_{X/2}^{2X} Y^{-2}\,\mathrm{d}Y = 1/(4X)$. □

Since $\mathcal{D}(X) \asymp X$ for $X \ge 1$, the bound (11.4), and with it RH, is equivalent to the upper bound $\mathcal{N}(X) \le K_\varepsilon X^{1+\varepsilon}$ for the signed part. Bounding the terms of $\mathcal{N}$ in absolute value gives order X^2. The positivity of $\mathcal{H}_B$ gives the lower bound $\mathcal{N}(X) \ge -\mathcal{D}(X)$ and nothing from above. Table 11 checks the proposition numerically.

Table 11. Numerical values of the diagonal $\mathcal{D}(X)$ against Proposition 11.2.

X	$\mathcal{D}(X)$	$\mathcal{D}(X) - (3X/4 - \log 2)$	$1/(4X)$
1	0.282975	0.226122	0.250000
10	6.831824	0.024971	0.025000
100	74.309353	0.002500	0.002500
1000	749.307103	0.000250	0.000250

11.3. **The primitive correlation.** Complete multiplicativity removes common factors from the pairs in $|B|^2$. Put

$$c(k) = \sum_{\substack{1\le a<k\\(a,k)=1}} \lambda(a)\lambda(k-a),$$
$$\mathcal{C}_\lambda(X) = \sum_{\substack{a,b\ge 1\\(a,b)=1}} \lambda(a)\lambda(b)e^{-(a+b)/X} = \sum_{k\ge 2} c(k)e^{-k/X}.$$

Proposition 11.3 (Primitive decomposition)**.** *For every $X > 0$, with absolutely convergent series,*

$$B(X)^2 = \sum_{d\ge 1} \mathcal{C}_\lambda(X/d), \qquad \mathcal{C}_\lambda(X) = \sum_{d\ge 1} \mu(d)B(X/d)^2, \tag{11.7}$$

and

$$\mathcal{H}_B(X) = \sum_{k\ge 2} c(k) \int_{X/2}^{2X} \frac{1}{e^{k/Y}-1}\,\frac{\mathrm{d}Y}{Y}.$$

The term $k = 2$, where $c(2) = 1$, is the diagonal $\mathcal{D}(X)$.

Proof. Sort the pairs (m, n) in $B(X)^2$ by $d = \gcd(m, n)$ and write $m = da$, $n = db$ with $(a, b) = 1$. Since $\lambda(d)^2 = 1$, $\lambda(da)\lambda(db) = \lambda(a)\lambda(b)$; this gives the first identity. For the second insert $\mathbf{1}_{(m,n)=1} = \sum_{d|m,\, d|n} \mu(d)$ in the definition of $\mathcal{C}_\lambda$. The rearrangements are justified by the majorant $\sum_d (\sum_a e^{-da/X})^2 = \sum_d (e^{d/X} - 1)^{-2} < \infty$. For the third, sum $e^{-d(a+b)/Y}$ over d first, put $k = a + b$, note $(a, b) = 1$ if and only if $(a, k) = 1$, and integrate. The pair $a = b = 1$ collects exactly the pairs $m = n$. □

Theorem 11.4 (A one-sided primitive criterion)**.** *The following are equivalent.*

(i) RH.
(ii) $B(X) \ll_\varepsilon X^{1/2+\varepsilon}$ *for every* $\varepsilon > 0$ *and* $X \geq 1$.
(iii) *For every* $\varepsilon \in (0, 1]$ *there is* K_ε *with* $\mathcal{C}_\lambda(X) \leq K_\varepsilon X^{1+\varepsilon}$ *for all* $X \geq 1$.

In (iii) *only an upper bound is required, and* $\mathcal{C}_\lambda$ *is not assumed to have a sign.*

Proof. The equivalence of (i) and (ii) is Proposition 12.1. Assume (iii). For $0 < X < 1$, $|\mathcal{C}_\lambda(X)| \leq (e^{1/X} - 1)^{-2} \leq X^2 \leq X^{1+\varepsilon}$, so after enlarging K_ε to at least 1 the bound holds for all $X > 0$. The first identity in (11.7) gives

$$0 \leq B(X)^2 \leq K_\varepsilon \sum_{d \geq 1} (X/d)^{1+\varepsilon} = K_\varepsilon\, \zeta(1 + \varepsilon)\, X^{1+\varepsilon},$$

which is (ii). Conversely, (ii) and $|B(X)| \leq (e^{1/X} - 1)^{-1} \leq X$ for $X < 1$ give $B(X)^2 \leq A_\varepsilon X^{1+\varepsilon}$ for all $X > 0$, and the second identity in (11.7) gives

$$|\mathcal{C}_\lambda(X)| \leq A_\varepsilon X^{1+\varepsilon} \sum_{d \geq 1} \frac{\mu(d)^2}{d^{1+\varepsilon}} = A_\varepsilon \frac{\zeta(1 + \varepsilon)}{\zeta(2 + 2\varepsilon)} X^{1+\varepsilon}.$$

□

The criterion involves one weighted sum of the coefficients $c(k)$ and no bound for an individual $c(k)$ or for a fixed additive shift. The upper bound in (iii) is not proved here. The identities (11.7) do not supply it, and $B(X)^2 \geq 0$ bounds a sum of values of $\mathcal{C}_\lambda$ from below only. At the level of coefficients the same proof gives $A(k) = \sum_{d|k} c(k/d)$ and $c(k) = \sum_{d|k} \mu(d) A(k/d)$, where $A(k) = \sum_{a=1}^{k-1} \lambda(a)\lambda(k - a)$.

11.4. **Prime dilations.** For a completely multiplicative f one has $f \log = f * (f\Lambda)$, and $\lambda(p^k)\Lambda(p^k) = (-1)^k \log p$. Smoothing gives the exact identity

$$\sum_{n \geq 1} \lambda(n) \log n\, e^{-n/X} = \sum_{p^k} (-1)^k \log p\, B(X/p^k), \tag{11.8}$$

with absolute convergence for each X. It expresses a logarithmic derivative of B through signed dilations of B itself. One might hope to treat the dilations as nearly orthogonal, or the right side as a contraction. Neither is possible in general. For distinct primes p, q,

$$\sum_{n \leq N} \lambda(pn)\lambda(qn) = N,$$

so prime dilates of λ are perfectly correlated.[13] The following theorem computes the norm of the dilation operator exactly. Let

$$\mathcal{K}_\sigma = L^2((0, \infty), X^{-2\sigma}\, \mathrm{d}X/X), \qquad T_R g(X) = \sum_{p^k \leq R} (-1)^k \log p\, g(X/p^k).$$

[13]Results on multiplicative functions in almost all short intervals [25] compare short averages with long ones. They do not give the square-root size of the long average, which is what is needed here.

Theorem 11.5 (Exact dilation norm)**.** *For every real σ and $R \geq 2$,*

$$\|T_R\|_{\mathcal{K}_\sigma \to \mathcal{K}_\sigma} = \sum_{p^k \leq R} (\log p)\, p^{-k\sigma}. \tag{11.9}$$

For fixed $1/2 < \sigma < 1$, $\|T_R\| \sim R^{1-\sigma}/(1-\sigma)$ as $R \to \infty$.

Proof. The map $Ug(t) = e^{-\sigma t} g(e^t)$ is an isometry of $\mathcal{K}_\sigma$ onto $L^2(\mathbb{R})$, and it carries $g(X/d)$ to $d^{-\sigma}(Ug)(t - \log d)$. So UT_RU^{-1} is a finite combination of translations, with Fourier multiplier

$$m_{R,\sigma}(\tau) = \sum_{p^k \leq R} (-1)^k (\log p)\, p^{-k\sigma} e^{-ik\tau \log p}.$$

By Plancherel's theorem the operator norm is $\sup_\tau |m_{R,\sigma}(\tau)|$, which is at most the right side of (11.9). For the reverse inequality it suffices to find τ with $e^{-i\tau \log p}$ arbitrarily close to -1 for all primes $p \leq R$ simultaneously, since then every term $(-1)^k e^{-ik\tau \log p}$ is close to 1. The numbers $\log p$, $p \leq R$, are linearly independent over $\mathbb{Q}$ by unique factorization, so this is Kronecker's theorem [36, Ch. VIII]. Directly: for integers (n_p) not all zero the frequency $\omega = \sum_p n_p \log p$ is nonzero and $(2T)^{-1} \int_{-T}^{T} e^{-i\tau\omega}\, \mathrm{d}\tau \to 0$; by approximation with trigonometric polynomials, the averages over $\tau \in [-T, T]$ of any continuous function on the torus evaluated along $(e^{-i\tau \log p})_{p \leq R}$ converge to its Haar integral, and a nonnegative bump supported near $(-1, \dots, -1)$ has positive integral.

For the asymptotic, the prime number theorem $\sum_{p \leq u} \log p \sim u$ [28] and partial summation give $\sum_{p \leq R} (\log p) p^{-\sigma} \sim R^{1-\sigma}/(1-\sigma)$. The terms with $k \geq 2$ contribute at most $(1 - 2^{-\sigma})^{-1} \sum_p (\log p) p^{-2\sigma} < \infty$. □

Proposition 11.6 (A sign change at low frequency)**.** *Fix $1/2 < \sigma < 1$ and let $P_R = \sum_{p \leq R} (\log p) p^{-\sigma}$, $\tau_R = \pi/\log R$. Then*

$$m_{R,\sigma}(0) = -P_R + O_\sigma(1), \qquad m_{R,\sigma}(\tau_R) = P_R + O_\sigma\Big(\frac{R^{1-\sigma}}{\log R}\Big).$$

Proof. The terms with $k \geq 2$ are $O_\sigma(1)$ at every frequency, which gives the first formula. The terms with $k = 1$ at τ_R are

$$-\sum_{p \leq R} (\log p) p^{-\sigma} e^{-i\pi \log p/\log R} = \sum_{p \leq R} (\log p) p^{-\sigma} e^{i\pi \log(R/p)/\log R},$$

and $|e^{i\vartheta} - 1| \leq |\vartheta|$ bounds the difference from P_R by

$$\frac{\pi}{\log R} \sum_{p \leq R} (\log p) p^{-\sigma} \log \frac{R}{p} = \frac{\pi}{\log R} \int_2^R P_u \frac{\mathrm{d}u}{u} \ll_\sigma \frac{R^{1-\sigma}}{\log R},$$

using $P_u \ll_\sigma u^{1-\sigma}$. □

So the operator in (11.8), truncated at R, has norm of size $R^{1-\sigma}$ on every space $\mathcal{K}_\sigma$ with $\sigma < 1$, and its multiplier changes sign between frequency 0 and frequency $\pi/\log R$ at nearly full size. No contraction estimate and no fixed sign are available, and smoothing in frequency near zero does not help. These are statements about the operator; they do not assert that B itself is extremal, nor that $B \in \mathcal{K}_\sigma$, which for every $\sigma > 1/2$ is again equivalent to RH. Table 12 illustrates (11.9).

11.5. **A finite-prime model and the prime tail.** For $y \geq 2$ let f_y be the completely multiplicative function with $f_y(p) = -1$ for $p \leq y$ and $f_y(p) = +1$ for $p > y$, and put

$$B_y(X) = \sum_{n \geq 1} f_y(n) e^{-n/X}, \qquad c_y(s) = \prod_{p \leq y} \frac{1 - p^{-s}}{1 + p^{-s}}, \qquad c_y = c_y(1).$$

Thus $c_y = \prod_{p \leq y} (p-1)/(p+1)$.

TABLE 12. The norm (11.9) at $\sigma = 3/4$, compared with its asymptotic value and with $\log R$.

R	$\|T_R\|$	$R^{1/4}/(1/4)$	$\|T_R\|/\log R$
10^2	7.89	12.65	1.71
10^3	17.86	22.49	2.58
10^4	35.39	40.00	3.84
10^5	66.51	71.13	5.78
10^6	121.85	126.49	8.82

Theorem 11.7 (Finite-prime main term)**.** *For $X > 0$, $y \geq 2$ and $0 < \sigma < 1$,*

$$|B_y(X) - c_y X| \leq \frac{X^\sigma}{c_y(\sigma)}. \tag{11.10}$$

For $y = (\log X)^2$ and $X \to \infty$,

$$B_y(X) = c_y X + O(X^{1/2+o(1)}), \qquad c_y \geq (\log X)^{-4},$$

and hence $B_y(X) \geq X/(2(\log X)^4)$ for $X \geq X_0$.

Proof. Let g_y be multiplicative with $g_y(p^j) = 2(-1)^j$ for $p \leq y$, $j \geq 1$, and zero at powers of larger primes. Comparing Euler factors, $f_y = \mathbf{1} * g_y$, $\sum_d g_y(d) d^{-s} = c_y(s)$ and $\sum_d |g_y(d)| d^{-\sigma} = 1/c_y(\sigma)$, all series being finite products of geometric series for $\sigma > 0$. Hence

$$B_y(X) = \sum_{d \geq 1} \frac{g_y(d)}{e^{d/X} - 1}, \qquad c_y X = \sum_{d \geq 1} g_y(d) \frac{X}{d}.$$

By (11.6), $0 < 1/t - 1/(e^t - 1) < \min(1/2, 1/t) \leq t^{-\sigma}$ for $t > 0$. Applying this with $t = d/X$ term by term gives (11.10).

For $u = p^{-1/2} \leq 2^{-1/2}$ one has $\log((1+u)/(1-u)) \leq 2u/(1-u) < 7u$, so $-\log c_y(1/2) \leq 7 \sum_{p \leq y} p^{-1/2} \ll \sqrt{y}/\log y$ by Chebyshev's bound; the computation is in Appendix B.6. For $y = (\log X)^2$ this is $O(\log X/\log\log X)$, so $1/c_y(1/2) = X^{o(1)}$. With $r = \lfloor y \rfloor$,

$$c_y \geq \prod_{n=2}^{r} \frac{n-1}{n+1} = \frac{2}{r(r+1)} \geq y^{-2}.$$

□

So complete multiplicativity together with agreement with λ at all primes up to $(\log X)^2$ is compatible with a smoothed sum of size $X/(\log X)^4$ at scale X. To pass from f_y to λ, let $h_y(d) = 2^{\omega(d)}\lambda(d)$ if every prime factor of d exceeds y and $h_y(d) = 0$ otherwise, with $h_y(1) = 1$. Its local factor at $p > y$ is $(1 - p^{-s})/(1 + p^{-s})$, so $\lambda = f_y * h_y$ and

$$B(X) = B_y(X) + T_y(X), \qquad T_y(X) = \sum_{\substack{d > 1 \\ P^-(d) > y}} h_y(d)\, B_y(X/d), \tag{11.11}$$

where $P^-(d)$ is the least prime factor. The series converges absolutely for each X.

Proposition 11.8 (The prime tail)**.** *Let $y = (\log X)^2$.*

(i) *For every $A > 0$, $T_y(X) = -c_y X + O_A(X(\log X)^{-A})$. In particular $T_y(X)/(c_y X) \to -1$.*

(ii) *The estimate $T_y(X) = -c_y X + O_\varepsilon(X^{1/2+\varepsilon})$ for every $\varepsilon > 0$ is equivalent to* RH.

Proof. By (11.11) and Theorem 11.7,

$$T_y(X) + c_y X = B(X) - (B_y(X) - c_y X) = B(X) + O(X^{1/2+o(1)}).$$

Part (ii) follows from Proposition 12.1. For (i), partial summation gives $B(X) = X^{-1}\int_1^\infty L(t)e^{-t/X}\,\mathrm{d}t$. The prime number theorem with the classical error term gives $|L(t)| \le C_A t(\log 2t)^{-A}$ for $t \ge 1$ and every $A > 0$ [28]. For $X \ge 4$ the range $t \le \sqrt{X}$ contributes at most $1/2$, and on $t \ge \sqrt{X}$ one has $\log 2t \ge \frac{1}{2}\log X$, so

$$|B(X)| \le \frac{1}{2} + \frac{2^A C_A}{X(\log X)^A}\int_0^\infty te^{-t/X}\,\mathrm{d}t = \frac{1}{2} + \frac{2^A C_A X}{(\log X)^A}.$$

Since $c_y \ge (\log X)^{-4}$, taking $A > 4$ gives the last statement. □

Part (i) is a genuine cancellation over all large primes in the exponential kernel: the tail removes the main term $c_y X$ of the model. It follows from classical additive cancellation and is far from the square-root error of part (ii). There is a companion cancellation in the Abel weight, where it is exact. For real $s > 1$,

$$H_y(s) := \sum_{d\ge 1}\frac{h_y(d)}{d^s} = \frac{\zeta(2s)}{\zeta(s)^2\,c_y(s)}, \tag{11.12}$$

and, uniformly for $0 < \delta \le 1$ with $\delta\log y \le 1$,

$$H_y(1+\delta) = \frac{\zeta(2)\,\delta^2}{c_y}\bigl(1 + O(\delta(1+\log y))\bigr). \tag{11.13}$$

Indeed $\zeta(2+2\delta)/\zeta(1+\delta)^2 = \zeta(2)\delta^2(1+O(\delta))$, and

$$\frac{\mathrm{d}}{\mathrm{d}s}\log c_y(s) = 2\sum_{p\le y}\frac{(\log p)\,p^{-s}}{1-p^{-2s}} \ll 1 + \log y \qquad (s \ge 1)$$

by Mertens' estimate $\sum_{p\le y}(\log p)/p \ll 1+\log y$ [1]. So $\sum_{d>1} h_y(d)d^{-1-\delta} \to -1$ as $\delta \downarrow 0$ for fixed y: the tail cancels the term $d = 1$ to second order. The two cancellations are not interchangeable. The series of absolute values is $\sum_d |h_y(d)|d^{-s} = \zeta(s)^2 c_y(s)/\zeta(2s)$, with a double pole at $s = 1$, so no comparison in absolute value between the weight $d^{-1-\delta}$ of (11.13) and the kernel $B_y(X/d)$ of (11.11) preserves the small signed value. Nor does the double zero at $s = 1$ constrain zeros of ζ elsewhere: if ρ is a zero of order r with $\mathrm{Re}\,\rho > 1/2$, then by (11.12)

$$H_y(s) = \frac{(r!)^2\,\zeta(2\rho)}{c_y(\rho)\,\zeta^{(r)}(\rho)^2}\,(s-\rho)^{-2r} + O\bigl(|s-\rho|^{1-2r}\bigr),$$

with a nonzero leading coefficient. Theorems 10.4 and 10.6 show in two different classes that local information at $s = 1$ of this kind cannot be converted into the bound of part (ii).

TABLE 13. Reformulations of the square-root bound for $B(X)$. In each row the stated estimate, for every $\varepsilon > 0$, is equivalent to RH; the last column records what is proved unconditionally.

Object	Estimate equivalent to RH	Proved here
local mean square	$\mathcal{H}_B(X) \ll X^{1+\varepsilon}$	point value bounded by $\mathcal{H}_B$ with loss $O(\log X)$ (Thm. 11.1)
signed part	$\mathcal{N}(X) \le K_\varepsilon X^{1+\varepsilon}$	$\mathcal{D}(X) = \frac{3}{4}X - \log 2 + O(1/X)$ (Prop. 11.2)
primitive correlation	$\mathcal{C}_\lambda(X) \le K_\varepsilon X^{1+\varepsilon}$, one-sided	exact decomposition (Prop. 11.3)
prime dilations	$B \in \mathcal{K}_\sigma$ for all $\sigma > 1/2$	exact norm $\|T_R\| \asymp R^{1-\sigma}$ (Thm. 11.5)
prime tail, $y = (\log X)^2$	$T_y(X) = -c_y X + O(X^{1/2+\varepsilon})$	error $O_A(X(\log X)^{-A})$ (Prop. 11.8)

12. The smoothed sum

The sharp cut-off in $L(x)$ can be replaced by an exponential weight without changing what the sum detects. For $X > 0$ put

$$B(X) = \sum_{n\geq 1} \lambda(n)\mathrm{e}^{-n/X}.$$

The series converges absolutely, and

$$|B(X)| \leq \sum_{n\geq 1} \mathrm{e}^{-n/X} = \frac{1}{\mathrm{e}^{1/X}-1} \leq \min(X,\, 2\mathrm{e}^{-1/X}), \tag{12.1}$$

where the second bound is used for $X \leq 1$. Since $\int_0^\infty \mathrm{e}^{-n/X} X^{-s}\,\mathrm{d}X/X = \Gamma(s)n^{-s}$ for $\operatorname{Re} s > 0$, summation over n gives, for $\operatorname{Re} s > 1$,

$$\int_0^\infty B(X)X^{-s}\frac{\mathrm{d}X}{X} = \Phi(s), \qquad \Phi(s) = \Gamma(s)\frac{\zeta(2s)}{\zeta(s)}. \tag{12.2}$$

The part of the integral over $0 < X \leq 1$ is an entire function of s by (12.1). The function Φ is meromorphic in the plane. In the half-plane $\operatorname{Re} s > 1/2$ the factors $\Gamma(s)$ and $\zeta(2s)$ are holomorphic and have no zeros, so the poles of Φ there are exactly the zeros of ζ, with the same orders; at $s = 1$ the pole of ζ makes Φ vanish. At $s = 1/2$ the factor $\zeta(2s)$ has a simple pole with residue $1/2$, and $\zeta(1/2) < 0$,[14] so Φ has a simple pole there with residue

$$\operatorname*{Res}_{s=1/2} \Phi(s) = \frac{\sqrt{\pi}}{2\zeta(1/2)} = -c_0, \qquad c_0 = -\frac{\sqrt{\pi}}{2\zeta(1/2)} = 0.60685\ldots \tag{12.3}$$

12.1. **The equivalence.**

Proposition 12.1 (Smoothed form of the criterion)**.** *Let $1/2 \leq \theta \leq 1$. The following are equivalent.*

(1) $\zeta(s) \neq 0$ for $\operatorname{Re} s > \theta$.
(2) $B(X) \ll_{\theta,\varepsilon} X^{\theta+\varepsilon}$ for $X \geq 1$, for every $\varepsilon > 0$.

Consequently

$$\limsup_{X\to\infty} \frac{\log(1+|B(X)|)}{\log X} = \Theta.$$

Proof. Assume the second condition. With (12.1) for $X \leq 1$, the integral in (12.2) converges absolutely and locally uniformly in $\operatorname{Re} s > \theta$, so it is holomorphic there. It agrees with Φ for $\operatorname{Re} s > 1$, hence Φ is holomorphic in $\operatorname{Re} s > \theta$, and ζ has no zero in that half-plane.

Assume the first condition. Theorem 3.1 gives $L(u) \ll_{\theta,\varepsilon} u^{\theta+\varepsilon}$ for $u \geq 1$. Partial summation gives

$$B(X) = \int_{1^-}^\infty \mathrm{e}^{-u/X}\,\mathrm{d}L(u) = \frac{1}{X}\int_1^\infty L(u)\mathrm{e}^{-u/X}\,\mathrm{d}u,$$

and therefore

$$|B(X)| \ll_{\theta,\varepsilon} \frac{1}{X}\int_0^\infty u^{\theta+\varepsilon}\mathrm{e}^{-u/X}\,\mathrm{d}u = \Gamma(1+\theta+\varepsilon)X^{\theta+\varepsilon}.$$

Let r denote the limit superior. The equivalence at $\theta = \Theta$ gives $r \leq \Theta$. If $r < \Theta$, the Mellin integral is holomorphic in $\operatorname{Re} s > r+\varepsilon$ for every $\varepsilon > 0$. When $\Theta > 1/2$ this half-plane contains a zero of ζ with real part greater than $1/2$, which is a pole of Φ; when $\Theta = 1/2$ it contains the pole of Φ at $s = 1/2$. Both are impossible. □

The proof uses only that the Mellin transform $\Gamma(s)$ of the weight has no zeros. This is a real restriction on the weight, as the next remark shows.

[14] For $0 < s < 1$ one has $(1-2^{1-s})\zeta(s) = \sum_{n\geq 1}(-1)^{n-1}n^{-s} > 0$ and $1-2^{1-s} < 0$, so $\zeta(s) < 0$ on $(0,1)$ [36, §2.2].

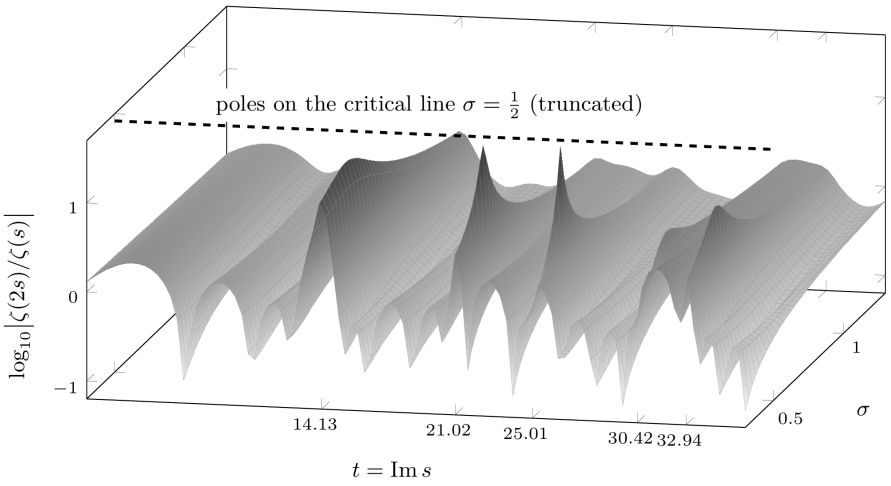


FIGURE 8. The surface $\log_{10}|\zeta(2s)/\zeta(s)|$ over $0.3 \le \sigma \le 1.3$, $2 \le t \le 36$. The peaks on the line $\sigma = \frac{1}{2}$ are the poles at the zeros of ζ (truncated for display). Any zero of ζ with $\sigma > \frac{1}{2}$ would appear as a peak away from that line, and by Proposition 12.1 it would force $|B(X)|$ to exceed $X^{\beta-\varepsilon}$ infinitely often.

Remark 12.2 (Weights that cancel the poles). Let V be smooth and compactly supported in $(0,\infty)$, write $\widehat{V}(s) = \int_0^\infty V(x)x^{s-1}\,\mathrm{d}x$, which is entire, and form the divisor-convolution weight $W(x) = \sum_{m\ge1} V(mx)$. For each $X > 0$ the sums below are finite, and

$$\sum_{n\ge1} \lambda(n)W(n/X) = \sum_{r\ge1} V(r/X) \sum_{n\mid r} \lambda(n) = \sum_{j\ge1} V(j^2/X),$$

because $\sum_{n\mid r}\lambda(n)$ equals 1 when r is a square and 0 otherwise: at a prime power p^ℓ the sum is $\sum_{k=0}^{\ell}(-1)^k$. If V is supported in $[a,b]$, the last sum has at most $\sqrt{bX}$ terms, so the weighted sum is $O_V(\sqrt{X})$ for $X \ge 1$, with no hypothesis on the zeros. There is no contradiction with Proposition 12.1. For $\operatorname{Re} s > 1$ the Mellin transform of W is $\zeta(s)\widehat{V}(s)$, and so

$$\int_0^\infty \Big(\sum_{n\ge1} \lambda(n)W(n/X)\Big) X^{-s}\,\frac{\mathrm{d}X}{X} = \frac{\zeta(2s)}{\zeta(s)} \cdot \zeta(s)\widehat{V}(s) = \zeta(2s)\widehat{V}(s).$$

The factor $\zeta(s)$ contributed by the weight cancels every pole that a zero of ζ would produce; only the pole at $s = 1/2$ survives, and it accounts for the size $\sqrt{X}$. The argument of Proposition 12.1 works for any weight whose Mellin transform has no zeros in $\operatorname{Re} s > 1/2$; the exponential weight, with transform $\Gamma(s)$, is one such weight, and W is not.

12.2. **The unconditional bound.** Write

$$\ell(t) = (\log t)^{2/3}(\log\log t)^{1/3} \qquad (t \ge 3),$$

an increasing function. The strongest known zero-free region is that of Vinogradov and Korobov.

Lemma 12.3 (Vinogradov–Korobov)**.** *There are absolute constants $c_1 > 0$ and C_1 such that, for $|t| \ge 3$ and $\sigma \ge 1 - c_1/\ell(|t|)$,*

$$\zeta(\sigma + it) \ne 0 \qquad \textit{and} \qquad \left|\frac{1}{\zeta(\sigma+it)}\right| \le C_1\,\ell(|t|).$$

This is classical: see [36, Notes to Ch. VI] and [21, Ch. 6]; Ford [17] gives the zero-free region with the explicit constant $c_1 = 1/57.54$.[15]

Theorem 12.4 (The smoothed sum, unconditionally)**.** *There are absolute constants $c > 0$ and X_0 such that, for $X \geq X_0$,*

$$|B(X)| \leq X \exp\left(-\frac{c \log X}{(\log\log X)^{2/3}(\log\log\log X)^{1/3}}\right).$$

For comparison, partial summation from the unconditional bound for L in Theorem 15.1 gives only

$$B(X) \ll X \exp(-c\,\eta(X)), \qquad \eta(X) = (\log X)^{3/5}(\log\log X)^{-1/5}.$$

The smoothing gains because $\Gamma(s)$ decays exponentially on vertical lines: at scale X the sum $B(X)$ is sensitive to the zeros of height at most a constant multiple of $\log X$, where the Vinogradov–Korobov region is much wider than at height X.

Proof. For $y > 0$, Mellin inversion gives $\mathrm{e}^{-y} = \frac{1}{2\pi i}\int_{\operatorname{Re} s=2} \Gamma(s) y^{-s}\,\mathrm{d}s$ [28, §5.1]. Taking $y = n/X$, multiplying by $\lambda(n)$ and summing, which is permitted by absolute convergence, gives

$$B(X) = \frac{1}{2\pi i}\int_{\operatorname{Re} s=2} \Phi(s) X^s\,\mathrm{d}s \qquad (X > 0). \tag{12.4}$$

Choose $\omega_0 \in (0, 1/4]$ such that ζ has no zero in the rectangle $1 - \omega_0 \leq \sigma \leq 2$, $|t| \leq 3$. This is possible because ζ has finitely many zeros in $1/2 \leq \sigma \leq 1$, $|t| \leq 3$, and none of them has real part 1.[16] Let X_0 be so large that $T = \log X \geq 3$ and

$$\omega = \frac{c_1}{\ell(T)} \leq \omega_0 \qquad \text{for } X \geq X_0.$$

Let $\mathcal{C}_X$ be the path made of the half-line $\operatorname{Re} s = 2$, $\operatorname{Im} s \leq -T$; the segment from $2 - iT$ to $1 - \omega - iT$; the segment from $1 - \omega - iT$ to $1 - \omega + iT$; the segment from $1 - \omega + iT$ to $2 + iT$; and the half-line $\operatorname{Re} s = 2$, $\operatorname{Im} s \geq T$.

The closed rectangle $R_X = \{1 - \omega \leq \sigma \leq 2,\ |t| \leq T\}$ lies between the line $\operatorname{Re} s = 2$ and $\mathcal{C}_X$. The function Φ is holomorphic on a neighbourhood of R_X: $\Gamma(s)$ and $\zeta(2s)$ are holomorphic in $\operatorname{Re} s > 1/2$; the reciprocal $1/\zeta$ is holomorphic wherever ζ has no zero, with a zero at $s = 1$; for $|t| \leq 3$ there is no zero in R_X by the choice of ω_0; and for $3 \leq |t| \leq T$ the monotonicity of ℓ gives $1 - \omega \geq 1 - c_1/\ell(|t|)$, so Lemma 12.3 applies. Cauchy's theorem on R_X therefore replaces the segment of $\operatorname{Re} s = 2$ with $|t| \leq T$ by the other three sides, and

$$B(X) = \frac{1}{2\pi i}\int_{\mathcal{C}_X} \Phi(s) X^s\,\mathrm{d}s.$$

The three kinds of piece are estimated in Appendix D, with absolute implied constants:

$$\int_{\operatorname{Re} s=2,\ |t|\geq T} |\Phi(s)X^s|\,|\mathrm{d}s| \ll X^{2-\pi/2}(\log X)^{3/2}, \tag{12.5}$$

$$\int_{1-\omega}^{2} |\Phi(\sigma \pm iT)| X^{\sigma}\,\mathrm{d}\sigma \ll X^{2-\pi/2}(\log X)^{2}, \tag{12.6}$$

$$\int_{-T}^{T} |\Phi(1 - \omega + it)| X^{1-\omega}\,\mathrm{d}t \ll X^{1-\omega}. \tag{12.7}$$

[15] The sources state these facts for $|t| \geq t_0$. On $3 \leq |t| \leq t_0$ the function ζ has finitely many zeros in $1/2 \leq \sigma \leq 1$ and none on $\sigma = 1$, so after decreasing c_1 and increasing C_1 the statement holds from $|t| = 3$ on.

[16] In fact ζ has no zero at all with $|t| < 14.13$, so any $\omega_0 \leq 1/4$ would do. The argument does not need this numerical fact.

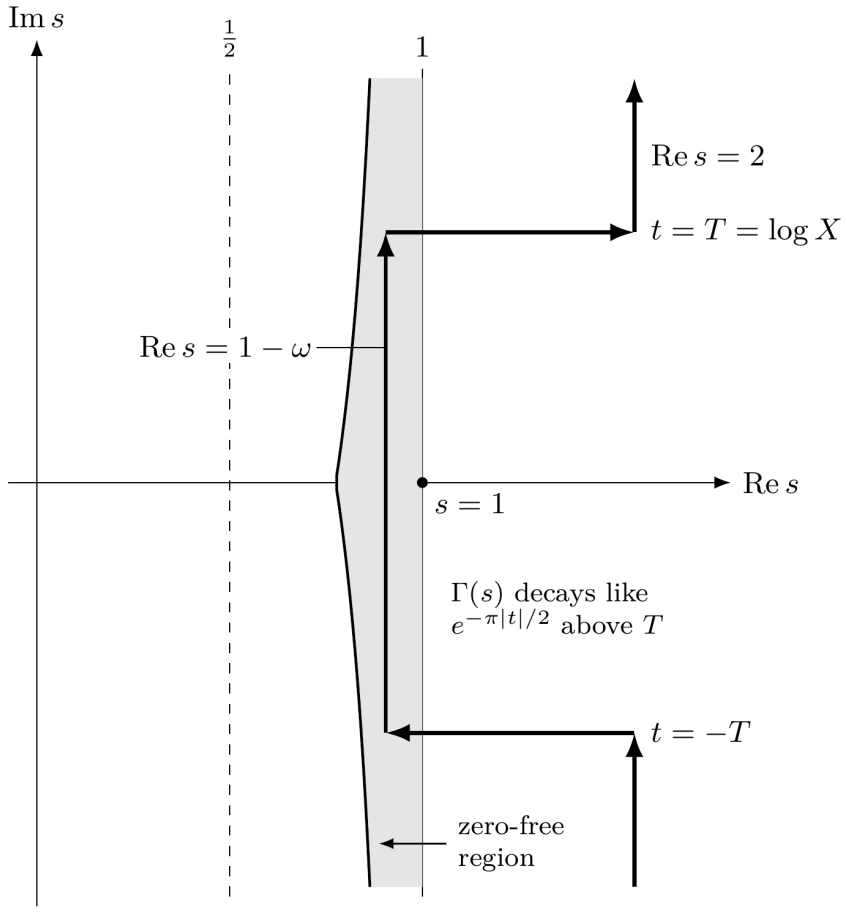


FIGURE 9. The contour $\mathcal{C}_X$ in the proof of Theorem 12.4, with $T = \log X$ and $\omega = c_1/\ell(T)$. The integral starts on the line $\operatorname{Re} s = 2$ (dashed) and is moved left to the solid path. On the vertical segment $\operatorname{Re} s = 1-\omega$ the integrand is $O(X^{1-\omega})$ times logarithmic factors; on the horizontal segments and on the half-lines the factor $\Gamma(s)$ gives $O(X^2 \mathrm{e}^{-\pi T/2}) = O(X^{2-\pi/2})$. The shaded region is the Vinogradov–Korobov zero-free region.

The first two come from $|\Gamma(\sigma+it)| \ll (1+|t|)^{3/2}\mathrm{e}^{-\pi|t|/2}$ for $1/2 \le \sigma \le 2$ and from $\mathrm{e}^{-\pi T/2} = X^{-\pi/2}$; the third from the same bound, which makes the integral over t converge, and from Lemma 12.3. Since $2 - \pi/2 = 0.4292\ldots < 1/2$, we obtain

$$B(X) \ll X^{1-\omega} + X^{0.43} \ll X \exp\left(-\frac{c_1 \log X}{(\log\log X)^{2/3}(\log\log\log X)^{1/3}}\right).$$

Replacing c_1 by $c = c_1/2$ and enlarging X_0 absorbs the implied constant. □

12.3. **Why no power saving is possible.** By Proposition 12.1, a bound $B(X) \ll X^{1-\delta}$ with a fixed $\delta > 0$ is the statement that ζ has no zero in $\operatorname{Re} s > 1-\delta$. No such half-plane is known, so Theorem 12.4 cannot be improved to a power saving by any argument that does not also produce a zero-free half-plane.

The proof shows the mechanism. A zero $\rho = \beta + i\gamma$ enters $B(X)$ with the weight $X^{\beta}|\Gamma(\rho)|$, and $|\Gamma(\rho)|$ is of order $|\gamma|^{\beta-1/2}\mathrm{e}^{-\pi|\gamma|/2}$. Against a target $X^{1-\delta}$, the zero is negligible when $\frac{\pi}{2}|\gamma|$ exceeds $(\beta - 1 + \delta)\log X$ by a margin, and not otherwise. A saving of $X^{-\delta}$ at the single scale X therefore needs a rectangle

$$\operatorname{Re} s > 1-\delta, \qquad |\operatorname{Im} s| \le \frac{2\delta}{\pi}\log X$$

free of zeros, together with a bound for $1/\zeta$ inside it; as $X \to \infty$ these rectangles exhaust the half-plane $\operatorname{Re} s > 1 - \delta$. The Vinogradov–Korobov region has width $c_1/\ell(\log X)$ at height $\log X$, and this width is what appears in the exponent of Theorem 12.4. Conversely, a single zero to the right of the critical line forces oscillations of exact order X^β in both directions; this is Theorem 12.7 below.

Remark 12.5 (A finite range). Platt and Trudgian [33] verified that every zero of ζ with $0 < \operatorname{Im} s \le 3 \cdot 10^{12}$ lies on the critical line. For $1 \le X \le \exp(3 \cdot 10^{12})$ and any fixed $\sigma_1 \in (1/2, 1)$, the contour above may therefore be drawn with left side $\operatorname{Re} s = \sigma_1$, and the same estimates give

$$|B(X)| \le K(\sigma_1) X^{\sigma_1} \qquad (1 \le X \le \exp(3 \cdot 10^{12})),$$

where $K(\sigma_1)$ is a finite constant depending on the maximum of $|1/\zeta|$ on the compact set $\sigma_1 \le \sigma \le 2$, $|t| \le 3 \cdot 10^{12}$. We have not evaluated $K(\sigma_1)$. The statement concerns a bounded range of X and carries no information about Θ.

Table 14 records values of $B(X)/\sqrt{X}$. They are consistent with the negative bias $-c_0$ produced by the pole at $s = 1/2$ and prove nothing about large X.

TABLE 14. The smoothed sum at small scales, computed from the definition. The last column is the constant of (12.3).

X	1	3	10^2	10^3	10^4	10^5	$-c_0$
$B(X)/\sqrt{X}$	0.1957	−0.0457	−0.5069	−0.5752	−0.5969	−0.6037	−0.6069

12.4. **The positivity principle.** The pole at $s = 1/2$ suggests that $B(X)$ might eventually be negative. The following classical principle, due to Landau for Dirichlet series [1, Ch. 11], turns a sign condition into information about singularities. The integral form is [28, Lemma 15.1]; we include the short proof.

Lemma 12.6 (Positivity principle). *Let $f : [0, \infty) \to [0, \infty)$ be locally integrable, and suppose that*

$$\mathcal{L}(s) = \int_0^\infty f(t) \mathrm{e}^{-st} \, \mathrm{d}t$$

converges for some real s. Let a be the infimum of the real s for which it converges. Then the integral converges absolutely and is holomorphic in $\operatorname{Re} s > a$. If $a > -\infty$, then $\mathcal{L}$ has no holomorphic continuation to any neighbourhood of the point $s = a$.

Proof. Since $f \ge 0$, convergence at a real point s_1 gives absolute and locally uniform convergence in $\operatorname{Re} s > s_1$; this proves the first assertion. Suppose that a is finite and that $\mathcal{L}$ continues holomorphically to a disc of radius r about a. Put $b = a + r/4$. The disc of radius $r/2$ about b lies in the disc of radius r about a, so the Taylor series of $\mathcal{L}$ at b converges at $b - z$ for every real $z \in (r/4, r/2)$. Differentiating under the integral sign, which is justified by convergence at a real point between a and b, gives

$$\frac{(-1)^j \mathcal{L}^{(j)}(b)}{j!} = \frac{1}{j!} \int_0^\infty t^j f(t) \mathrm{e}^{-bt} \, \mathrm{d}t \ge 0.$$

Hence, by Tonelli's theorem,

$$\mathcal{L}(b - z) = \sum_{j \ge 0} \frac{z^j}{j!} \int_0^\infty t^j f(t) \mathrm{e}^{-bt} \, \mathrm{d}t = \int_0^\infty f(t) \mathrm{e}^{-(b-z)t} \, \mathrm{d}t < \infty.$$

But $b - z < a$, which contradicts the definition of a. □

With $X = X_0 \mathrm{e}^t$ the lemma applies to integrals $\int_{X_0}^\infty A(X) X^{-s} \, \mathrm{d}X/X$ with $A \ge 0$ on $[X_0, \infty)$. We call a the abscissa of the integral.

Theorem 12.7 (Both signs forced by a right-hand zero)**.** *Let $\rho = \beta + i\gamma$ be a zero of ζ of order r with $\beta > 1/2$, and put*

$$C_\rho = \frac{r\,|\Gamma(\rho)\zeta(2\rho)|}{|\zeta^{(r)}(\rho)|} > 0.$$

Then

$$\limsup_{X\to\infty} \frac{B(X)}{X^\beta (\log X)^{r-1}} \geq C_\rho, \qquad \liminf_{X\to\infty} \frac{B(X)}{X^\beta (\log X)^{r-1}} \leq -C_\rho.$$

Proof. Near ρ we have $\Phi(s) = A_\rho (s-\rho)^{-r} + O(|s-\rho|^{1-r})$ with

$$A_\rho = \frac{r!\,\Gamma(\rho)\zeta(2\rho)}{\zeta^{(r)}(\rho)}, \qquad |A_\rho| = (r-1)!\,C_\rho,$$

and $A_\rho \neq 0$ because $\mathrm{Re}(2\rho) > 1$. Also $\gamma \neq 0$ and $\beta < 1$, and Φ is holomorphic at every real point of $(1/2, \infty)$, since ζ has no zero on $(1/2, 1)$ and its pole at 1 is a zero of Φ.

Suppose the first inequality fails. Then there are $c \in (0, C_\rho)$ and $X_1 > 1$ with

$$A(X) = cX^\beta (\log X)^{r-1} - B(X) \geq 0 \qquad (X \geq X_1).$$

Let $I(s) = \int_{X_1}^\infty A(X) X^{-s}\,\mathrm{d}X/X$. It converges for real $s > 1$ by (12.1). Using $\int_1^\infty X^{\beta-s}(\log X)^{r-1}\,\mathrm{d}X/X = (r-1)!/(s-\beta)^r$ and (12.2), we find for $\mathrm{Re}\, s > 1$

$$I(s) = J(s), \qquad J(s) = \frac{c\,(r-1)!}{(s-\beta)^r} - \Phi(s) + E(s), \tag{12.8}$$

where E is entire and accounts for the integrals over the bounded ranges $[1, X_1]$ and $(0, X_1]$. The function J is meromorphic in the plane.

Let $a \leq 1$ be the abscissa of I. By Lemma 12.6, I is holomorphic in $\mathrm{Re}\, s > a$, and by the identity theorem (12.8) holds there; in particular J is holomorphic in $\mathrm{Re}\, s > a$. Now J has a pole at the real point β, because Φ and E are holomorphic there. Hence $a \geq \beta$, and a is finite. If $a > \beta$, then J would be holomorphic at the real point $a \in (\beta, 1]$ and would continue I to a neighbourhood of a, contrary to Lemma 12.6. Thus $a = \beta$.

Since $A \geq 0$, for every $\delta > 0$

$$|J(\beta + \delta + i\gamma)| = |I(\rho + \delta)| \leq I(\beta + \delta) = J(\beta + \delta).$$

Multiply by δ^r and let $\delta \downarrow 0$. On the left, the first term of J is bounded because $\gamma \neq 0$, the term E is bounded, and $\delta^r \Phi(\rho + \delta) \to A_\rho$; the limit is $|A_\rho|$. On the right, Φ and E are bounded near β and the limit is $c\,(r-1)!$. Therefore $c \geq |A_\rho|/(r-1)! = C_\rho$, a contradiction.

For the second inequality apply the same argument to the function

$$cX^\beta (\log X)^{r-1} + B(X),$$

for which

$$J(s) = \frac{c\,(r-1)!}{(s-\beta)^r} + \Phi(s) - E(s).$$

□

In particular $\limsup_{X\to\infty} |B(X)| X^{-\beta} (\log X)^{1-r} \geq C_\rho$. The theorem gives lower bounds only. At a zero of large height the constant C_ρ carries the factor $|\Gamma(\rho)| \asymp |\gamma|^{\beta - 1/2} \mathrm{e}^{-\pi|\gamma|/2}$ and is extremely small, but the exponent β is unaffected.

Proposition 12.8 (The real pole and an eventual sign)**.** *With c_0 as in* (12.3)*,*

$$\liminf_{X\to\infty} \frac{B(X)}{\sqrt{X}} \leq -c_0,$$

unconditionally. If moreover $B(X) \leq 0$ for all sufficiently large X, then the Riemann hypothesis holds, every nontrivial zero ρ is simple, and

$$\left| \frac{\Gamma(\rho)\zeta(2\rho)}{\zeta'(\rho)} \right| \leq c_0.$$

Proof. Suppose the limit inferior exceeds $-c_0$. Then there are $c \in (0, c_0)$ and $X_1 > 1$ with $B(X) + c\sqrt{X} \geq 0$ for $X \geq X_1$. As in the proof of Theorem 12.7, the integral $I(s) = \int_{X_1}^{\infty} (B(X) + c\sqrt{X}) X^{-s} \, \mathrm{d}X/X$ equals, for $\operatorname{Re} s > 1$,

$$J(s) = \Phi(s) + \frac{c}{s - 1/2} + E(s)$$

with E entire. The function J has a simple pole at $s = 1/2$ with residue $c - c_0 < 0$ and is holomorphic at every real point of $(1/2, \infty)$. As before, the pole gives $a \geq 1/2$ for the abscissa of I, and Lemma 12.6 excludes $a > 1/2$. So $I(s) = J(s)$ for real $s > 1/2$. But $I(s) \geq 0$ there, while $J(s) \to -\infty$ as $s \downarrow 1/2$.

Assume that $B(X) \leq 0$ for $X \geq X_1$ and apply the same reasoning to $I(s) = \int_{X_1}^{\infty} (-B(X)) X^{-s} \, \mathrm{d}X/X$, which equals $J(s) = -\Phi(s) + E(s)$ for $\operatorname{Re} s > 1$. The pole of J at $1/2$ and its regularity on $(1/2, \infty)$ give abscissa $1/2$. Hence Φ is holomorphic in $\operatorname{Re} s > 1/2$, so ζ has no zero there, and the functional equation gives the Riemann hypothesis. Let $\rho = 1/2 + i\gamma$ be a zero of order r. Since $\zeta(2\rho) = \zeta(1 + 2i\gamma) \neq 0$, the function Φ has a pole of order exactly r at ρ. Positivity gives

$$|J(\rho + \delta)| \leq J(1/2 + \delta) = \frac{c_0}{\delta} + O(1) \qquad (\delta \downarrow 0).$$

The left side is of exact order δ^{-r}, so $r = 1$, and comparison of the coefficients of δ^{-1} gives the stated inequality. □

An eventual sign would thus give more than the square-root bound with an X^{ε} loss: it would give simplicity of all zeros and a uniform bound for the residues. Neither the sign nor the upper bound is proved here. The first assertion bounds B from one side only and does not control positive excursions.[17]

12.5. **A lower bound for the local energy.** For a locally square-integrable F on $(0, \infty)$ define the local energy over one octave on each side,

$$\mathcal{H}_F(X) = \int_{X/2}^{2X} |F(Y)|^2 \frac{\mathrm{d}Y}{Y}.$$

For the model $F_0(X) = -c_0\sqrt{X}$ one has $\mathcal{H}_{F_0}(X) = \frac{3}{2} c_0^2 X$ exactly.

Proposition 12.9 (Local-energy lower bound)**.** *Unconditionally,*

$$\limsup_{X \to \infty} \frac{\mathcal{H}_B(X)}{X} \geq \frac{3}{2} c_0^2 = \frac{3\pi}{8\,\zeta(1/2)^2} = 0.5524\ldots$$

The proof is in Appendix E: a bound $\mathcal{H}_B(X) \leq CX$ gives a bound for the Mellin transform of $|B|^2$ near its abscissa, and the Cauchy–Schwarz inequality compares it with the residue (12.3).

The constant is the one dictated by the real residue, as the model F_0 shows. The proposition constrains any proposed upper bound for $\mathcal{H}_B$ from below; it gives no upper bound. Table 15 collects the statements of this section.

13. Reciprocal dilates in weighted norms

The criterion of Nyman and Beurling expresses the Riemann hypothesis as an approximation problem in $L^2(0, 1)$ for dilates of the fractional part of $1/x$, and Báez-Duarte showed that integer dilates suffice [3]; see also [7, 4, 13]. This section places the problem in a one-parameter family of weighted norms. In that family the set of exponents at which approximation succeeds is exactly $[\Theta, 1)$, the approximation succeeds unconditionally at the exponent 1 with an explicit rate, and at the critical exponent every fixed taper of the natural coefficients fails.

[17] The sign of $B(X)$ is not constant: $B(1) > 0.195$ and $B(3) < -0.079$, by the elementary evaluation in Appendix J.

TABLE 15. What is known about $B(X) = \sum_n \lambda(n)\mathrm{e}^{-n/X}$. Here $\rho = \beta + i\gamma$ is a zero of order r with $\beta > 1/2$.

Hypothesis	Conclusion	Reference
none	$\lvert B(X)\rvert \le X\exp(-c\log X/\ell(\log X))$	Theorem 12.4
none	$\liminf B(X)/\sqrt{X} \le -c_0$	Proposition 12.8
none	$\limsup \mathcal{H}_B(X)/X \ge \frac{3}{2}c_0^2$	Proposition 12.9
$\zeta \ne 0$ in $\operatorname{Re} s > \theta$	$B(X) \ll X^{\theta+\varepsilon}$, and conversely	Proposition 12.1
a zero ρ exists	$B(X) = \Omega_\pm\big(X^\beta(\log X)^{r-1}\big)$	Theorem 12.7
$B(X) \le 0$ eventually	RH, simple zeros, bounded residues	Proposition 12.8

13.1. **Balanced dilates and their Mellin transforms.** For $n \ge 1$ define

$$g_n(x) = \Big\{\frac{1}{nx}\Big\} - \frac{1}{n}\Big\{\frac{1}{x}\Big\} \qquad (x > 0).$$

Then $|g_n| \le 1$, $g_1 = 0$, and $g_n(x) = 0$ for $x > 1$, since both fractional parts are then equal to their arguments. Writing $y = 1/x$,

$$g_n(1/y) = \frac{\lfloor y\rfloor}{n} - \Big\lfloor\frac{y}{n}\Big\rfloor \qquad (y > 0). \tag{13.1}$$

For $\sigma \ge 1/2$ and a measurable u on $(0,1)$ set

$$\|u\|_\sigma^2 = \int_0^1 |u(x)|^2 x^{2\sigma-1}\,\mathrm{d}x, \qquad d_{M,\sigma} = \inf_{c_1,\dots,c_M\in\mathbb{C}} \Big\|1 - \sum_{n\le M} c_n g_n\Big\|_\sigma.$$

At $\sigma = 1/2$ this is the norm of $L^2(0,1)$. The norms decrease as σ increases, and $d_{M,\sigma}$ is nonincreasing in M.

Lemma 13.1 (Exact Mellin transform)**.** *For* $\operatorname{Re} s > 0$*,*

$$\int_0^1 g_n(x)x^{s-1}\,\mathrm{d}x = -\big(n^{-s} - n^{-1}\big)\frac{\zeta(s)}{s},$$

and both sides are holomorphic in $\operatorname{Re} s > 0$*; at* $s = 1$ *the right side is* $(\log n)/n$*.*

Proof. The left side is holomorphic in $\operatorname{Re} s > 0$ because g_n is bounded and supported in $(0,1]$. The right side is holomorphic there because the factor $n^{-s} - n^{-1}$ vanishes at the pole of ζ. It suffices to prove the identity for $0 < \operatorname{Re} s < 1$. The substitution $y = 1/(nx)$ gives

$$\int_0^\infty \Big\{\frac{1}{nx}\Big\}x^{s-1}\,\mathrm{d}x = n^{-s}\int_0^\infty \{y\}y^{-s-1}\,\mathrm{d}y.$$

On $0 < y < 1$ the integral is $1/(1-s)$. On $y > 1$ it is $1/(s-1) - \zeta(s)/s$, by the formula $\zeta(s) = s/(s-1) - s\int_1^\infty \{y\}y^{-s-1}\,\mathrm{d}y$, valid for $\operatorname{Re} s > 0$ [36, §2.1]. The sum is $-\zeta(s)/s$. Subtracting n^{-1} times the case $n = 1$ gives the lemma. □

Consequently, if $c_n = -\mu(n)w_n$ and $D(s) = \sum_{n\le M}\mu(n)w_n n^{-s}$, the residual $u = 1 - \sum_{n\le M} c_n g_n$ has the Mellin transform

$$\widehat{u}(s) = \int_0^1 u(x)x^{s-1}\,\mathrm{d}x = \frac{1 - \zeta(s)\big(D(s) - D(1)\big)}{s} \qquad (\operatorname{Re} s > 0). \tag{13.2}$$

If $|w_n| \le 1$ we shall use the trivial bound

$$|D(1)| \le \sum_{n\le M}\frac{1}{n} \le 1 + \log M. \tag{13.3}$$

Lemma 13.2 (Plancherel and evaluation)**.** *Let* $\sigma \ge 1/2$ *and let* u *be bounded and measurable on* $(0,1)$*. Then*

$$\|u\|_\sigma^2 = \frac{1}{2\pi}\int_{-\infty}^\infty |\widehat{u}(\sigma + it)|^2\,\mathrm{d}t,$$

and for every s_0 with $\operatorname{Re} s_0 > \sigma$,

$$|\widehat{u}(s_0)| \le \frac{\|u\|_\sigma}{\sqrt{2(\operatorname{Re} s_0 - \sigma)}}.$$

Proof. Put $v(y) = \mathrm{e}^{-\sigma y} u(\mathrm{e}^{-y})$ for $y > 0$ and $v(y) = 0$ for $y \le 0$. Then $v \in L^1 \cap L^2(\mathbb{R})$, $\int |v|^2 = \|u\|_\sigma^2$, and $\int v(y)\mathrm{e}^{-ity}\,\mathrm{d}y = \widehat{u}(\sigma + it)$; the first assertion is Plancherel's theorem. The second is the Cauchy–Schwarz inequality applied to $u(x)x^{\sigma-1/2} \cdot x^{s_0-\sigma-1/2}$, since $\int_0^1 x^{2(\operatorname{Re} s_0-\sigma)-1}\,\mathrm{d}x = 1/(2(\operatorname{Re} s_0 - \sigma))$. □

Upper bounds for the residuals need the size of ζ and $1/\zeta$ to the right of a zero-free line. We state the result for Dirichlet L-functions, uniformly in the character, because Section 14 uses it in that form. For $q = 1$ the bound for the reciprocal is Lemma A.2.

Lemma 13.3 (Subpower growth in a zero-free half-plane)**.** *Let $1/2 \le \theta < 1$, $0 < \delta \le 1/4$ and $\kappa > 0$. Let χ be a primitive Dirichlet character of conductor $q \ge 1$ and suppose that $L(s,\chi)$ has no zero in $\operatorname{Re} s > \theta$. Then, for $\theta + \delta \le \sigma \le 2$ and all real t, with the restriction $|t| \ge 4$ when $q = 1$,*

$$\big(q(2+|t|)\big)^{-\kappa} \ll_{\theta,\delta,\kappa} |L(\sigma+it,\chi)| \ll_{\theta,\delta,\kappa} \big(q(2+|t|)\big)^{\kappa}.$$

The implied constants do not depend on χ or q. When $q = 1$, the lower bound for $|\zeta(\sigma+it)|$ holds for all real t, the reciprocal being extended by 0 at $s = 1$.

The proof, by the Borel–Carathéodory inequality and the three-circles theorem as in [36, §14.2], is in Appendix C.

13.2. **Convergence in shifted norms.** The Riesz-weighted Möbius coefficients give the residuals

$$u_M = 1 + \sum_{n \le M} \mu(n)\Big(1 - \frac{n}{M}\Big) g_n, \qquad D_M(s) = \sum_{n \le M} \mu(n)\Big(1 - \frac{n}{M}\Big) n^{-s}. \tag{13.4}$$

Theorem 13.4 (Convergence in shifted norms)**.** *Suppose $1/2 \le \theta < \sigma < 1$ and $\zeta(s) \ne 0$ for $\operatorname{Re} s > \theta$. Then for every $0 < \delta < \sigma - \theta$,*

$$d_{M,\sigma} \le \|u_M\|_\sigma \ll_{\sigma,\theta,\delta} M^{-\delta}.$$

Conversely, let $\sigma \ge 1/2$ and let ρ be a zero of ζ with $\operatorname{Re}\rho > \sigma$. Then, for every M and every choice of coefficients,

$$\Big\|1 - \sum_{n \le M} c_n g_n\Big\|_\sigma^2 \ge \frac{2(\operatorname{Re}\rho - \sigma)}{|\rho|^2}, \qquad \text{so} \qquad d_{M,\sigma}^2 \ge \frac{2(\operatorname{Re}\rho - \sigma)}{|\rho|^2}. \tag{13.5}$$

Proof. For $y > 0$ one has $\frac{1}{2\pi i}\int_{\operatorname{Re} w = 2} y^w \frac{\mathrm{d}w}{w(w+1)} = (1 - 1/y)_+$: for $y > 1$ close the contour to the left, with residues 1 and $-1/y$; for $y < 1$ close it to the right; the case $y = 1$ follows by continuity. Hence, for $\operatorname{Re} s \ge 1/2$,

$$D_M(s) = \frac{1}{2\pi i}\int_{\operatorname{Re} w = 2} \frac{M^w}{w(w+1)}\,\frac{\mathrm{d}w}{\zeta(s+w)}. \tag{13.6}$$

Fix $s = \sigma + it$ and $\kappa \in (0, 1/4)$, and move the line to $\operatorname{Re} w = -\delta$. On and to the right of the new line $\operatorname{Re}(s+w) \ge \sigma - \delta > \theta$, so Lemma A.2, applied with the margin $\min(\sigma - \delta - \theta, 1/5)$, gives $1/\zeta(s+w) \ll (1 + |t| + |\operatorname{Im} w|)^\kappa$ there; for $\operatorname{Re}(s+w) \ge 2$ the reciprocal is bounded. The horizontal integrals at height $\pm V$ are $O(M^2 V^{\kappa-2})$ for fixed t and tend to zero. The only pole crossed is $w = 0$, with residue $1/\zeta(s)$; the pole at $w = -1$ is not reached because $\delta < 1$. Therefore

$$\Big|D_M(s) - \frac{1}{\zeta(s)}\Big| \ll M^{-\delta}\int_{-\infty}^{\infty} \frac{(1 + |t| + |v|)^\kappa}{|-\delta + iv|\,|1 - \delta + iv|}\,\mathrm{d}v \ll_{\sigma,\theta,\delta,\kappa} M^{-\delta}(1+|t|)^\kappa.$$

The same computation at $s = 1$, where the integrand is holomorphic at $w = 0$ because $1/\zeta(1+w)$ vanishes there, gives $D_M(1) \ll M^{-\delta}$.

By (13.2),

$$\widehat{u}_M(s) = \frac{-\zeta(s)\big(D_M(s) - 1/\zeta(s)\big) + \zeta(s)D_M(1)}{s}.$$

On $\operatorname{Re} s = \sigma$ we have $\zeta(s) \ll (1+|t|)^{\kappa}$: for $|t| \ge 4$ this is Lemma 13.3, and for $|t| \le 4$ the function is bounded because $\sigma < 1$. Thus $\widehat{u}_M(\sigma + it) \ll M^{-\delta}(1+|t|)^{2\kappa-1}$, which is square integrable since $\kappa < 1/4$. Lemma 13.2 gives the upper bound.

For the converse, let $u = 1 - \sum c_n g_n$. By Lemma 13.1, $\widehat{u}(\rho) = 1/\rho$, because the transform of each g_n vanishes at a zero of ζ. The evaluation bound of Lemma 13.2 gives $|\rho|^{-1} \le \|u\|_\sigma (2(\operatorname{Re}\rho - \sigma))^{-1/2}$. □

A point on the line $\operatorname{Re} s = \sigma$ itself is not a bounded evaluation in the weighted space, so (13.5) says nothing about zeros on that line. Theorem 13.4 leaves open the endpoint $\sigma = \theta$, where zeros may lie on the line of integration. The next theorem settles it.

13.3. **The endpoint.**

Theorem 13.5 (Endpoint equivalence)**.** *Let $1/2 \le \sigma < 1$. Then*

$$\lim_{M\to\infty} d_{M,\sigma} = 0 \quad\Longleftrightarrow\quad \zeta(s) \ne 0 \ \text{ for } \operatorname{Re} s > \sigma.$$

Hence $\{\sigma \in [1/2, 1) : d_{M,\sigma} \to 0\} = [\Theta, 1)$, and $d_{M,\sigma} \ll_{\sigma,\delta} M^{-\delta}$ for every $\delta < \sigma - \Theta$.

At $\sigma = 1/2$ this is the theorem of Báez-Duarte [3], which sharpened the criterion of Nyman and Beurling. Beurling [7] treated the closure problem in $L^p(0,1)$ and proved that it is solvable exactly when ζ has no zero in $\operatorname{Re} s > 1/p$; the weighted norms $\|\cdot\|_\sigma$ are the Hilbert-space counterpart of his scale. The argument below is Báez-Duarte's, carried over to the weighted norms.[18]

Proof. If $d_{M,\sigma} \to 0$, then (13.5) shows that no zero has real part greater than σ.

Assume $\zeta(s) \ne 0$ for $\operatorname{Re} s > \sigma$. For $0 < \varepsilon < 1/2$ consider

$$u_{M,\varepsilon} = 1 + \sum_{n \le M} \mu(n) n^{-\varepsilon}\Big(1 - \frac{n}{M}\Big) g_n.$$

By Lemma 13.1, its transform is

$$U_{M,\varepsilon}(s) = \frac{1 - \zeta(s)\big(D_M(s+\varepsilon) - D_M(1+\varepsilon)\big)}{s}.$$

In (13.6) with $s+\varepsilon$ in place of s, move the line to $\operatorname{Re} w = -\varepsilon/2$. Then $\operatorname{Re}(s+\varepsilon+w) \ge \sigma + \varepsilon/2$, and Lemma A.2 applies in the zero-free half-plane $\operatorname{Re} s > \sigma$ with the margin $\varepsilon/2 < 1/4$. As in the proof of Theorem 13.4, for $\operatorname{Re} s = \sigma$ and any $\kappa > 0$,

$$D_M(s+\varepsilon) = \frac{1}{\zeta(s+\varepsilon)} + O_{\varepsilon,\kappa}\big(M^{-\varepsilon/2}(1+|t|)^{\kappa}\big),$$
$$D_M(1+\varepsilon) = \frac{1}{\zeta(1+\varepsilon)} + O_{\varepsilon}\big(M^{-\varepsilon/2}\big).$$

Put

$$U_\varepsilon(s) = \frac{1}{s}\Big(1 - \frac{\zeta(s)}{\zeta(s+\varepsilon)}\Big) + \frac{\zeta(s)}{s\,\zeta(1+\varepsilon)}.$$

Then $U_{M,\varepsilon}(s) - U_\varepsilon(s) \ll_{\varepsilon,\kappa} M^{-\varepsilon/2}|\zeta(s)|(1+|t|)^{\kappa}/|s|$ on $\operatorname{Re} s = \sigma$. The convexity bound $\zeta(\sigma+it) \ll_\kappa (1+|t|)^{1/4+\kappa}$ for $\sigma \ge 1/2$, $|t| \ge 1$ [36, Ch. V] makes the square of this error at

[18] Burnol [13], sharpening [4], proved that the distance in $L^2(0,1)$ from 1 to the span of the functions $\{\vartheta/x\} - \vartheta\{1/x\}$ with $1/M \le \vartheta \le 1$ is at least $(C + o(1))/\sqrt{\log M}$, where $C > 0$ is given by a sum over the zeros on the critical line. The functions g_n are the cases $\vartheta = 1/n$, so the same lower bound holds for $d_{M,1/2}$. No power rate is therefore possible at $\sigma = 1/2$, in contrast with the interior of $[\Theta, 1)$.

most a constant times $M^{-\varepsilon}(1+|t|)^{-3/2+4\kappa}$. This is integrable provided $\kappa < 1/8$, and we fix such a κ. By Lemma 13.2,

$$\limsup_{M\to\infty} d_{M,\sigma} \le \limsup_{M\to\infty} \|u_{M,\varepsilon}\|_\sigma \le \Big(\frac{1}{2\pi}\int_{-\infty}^{\infty} |U_\varepsilon(\sigma+it)|^2\,\mathrm{d}t\Big)^{1/2}. \tag{13.7}$$

It remains to show that the right side tends to zero with ε.

The second term of U_ε has L^2 norm $O(1/\zeta(1+\varepsilon)) = O(\varepsilon)$, because $\zeta(s)/s$ is square integrable on the line by the convexity bound and $\zeta(1+\varepsilon) > 1/\varepsilon$.

For the first term, let $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. Hadamard's factorisation [36, §2.12] is $\xi(s) = \mathrm{e}^{b_0+b_1 s}\prod_\rho (1-s/\rho)\mathrm{e}^{s/\rho}$, the product over the nontrivial zeros converging absolutely. Since $\sum_\rho |\rho|^{-2} < \infty$, the series $\sum_\rho \operatorname{Re}(1/\rho)$ converges absolutely, and

$$\left|\frac{\xi(s)}{\xi(s+\varepsilon)}\right| = \exp\Big(-\varepsilon\Big(\operatorname{Re} b_1 + \sum_\rho \operatorname{Re}\frac{1}{\rho}\Big)\Big)\prod_\rho \frac{|s-\rho|}{|s+\varepsilon-\rho|}.$$

The exponential factor is bounded by an absolute constant C_0 for $0 < \varepsilon \le 1/2$.[19] Every zero $\rho = \beta + i\gamma$ has $\beta \le \sigma = \operatorname{Re} s$, so

$$|s+\varepsilon-\rho|^2 = (\sigma-\beta+\varepsilon)^2 + (t-\gamma)^2 \ge (\sigma-\beta)^2 + (t-\gamma)^2 = |s-\rho|^2.$$

Each factor of the product thus has modulus at most 1, and $|\xi(s)| \le C_0|\xi(s+\varepsilon)|$. Consequently

$$\begin{aligned}\left|\frac{\zeta(s)}{\zeta(s+\varepsilon)}\right| &\le C_0\left|\frac{(s+\varepsilon)(s+\varepsilon-1)}{s(s-1)}\right|\left|\frac{\Gamma((s+\varepsilon)/2)}{\Gamma(s/2)}\right| \\ &\le C_\sigma(1+|t|)^{\varepsilon/2} \le C_\sigma(1+|t|)^{1/4}.\end{aligned}$$

Here the rational factor is at most $2\,(1+1/(2(1-\sigma)))$ because $|s| \ge 1/2$ and $|s-1| \ge 1-\sigma$, and the Gamma quotient is $O((1+|t|)^{\varepsilon/2})$ uniformly for $0 < \varepsilon \le 1/2$ by Stirling's formula [36, §4.12]. So the first term of U_ε is dominated by $(1 + C_\sigma(1+|t|)^{1/4})/|s|$, which is square integrable and independent of ε. As $\varepsilon \downarrow 0$ it tends to zero at every point of the line where $\zeta(s) \ne 0$, hence almost everywhere. Dominated convergence shows that its L^2 norm tends to zero. Since the left side of (13.7) does not depend on ε, it is zero.

The description of the set of exponents follows from the definition of Θ, and the rate is Theorem 13.4 with $\theta = \Theta$. □

Remark 13.6. The theorem identifies the statement "$d_{M,\sigma} \to 0$" for one $\sigma < 1$, with or without a rate, with the statement that ζ has no zero in $\operatorname{Re} s > \sigma$. No zero-free half-plane $\operatorname{Re} s > \theta$ with $\theta < 1$ has been established, so no exponent $\sigma < 1$ is at present known to belong to the convergence set. The exponent $\sigma = 1$ is different.

13.4. **The endpoint rate.** At $\sigma = 1/2$ Theorem 13.5 gives no rate. The rate is known for the distance of Báez-Duarte, and it transfers to $d_{M,1/2}$. Write $\varphi_n(x) = \{1/(nx)\}$ and $\chi = \mathbf{1}_{(0,1]}$ on $(0,\infty)$, let $\|\cdot\|$ be the norm of $L^2(0,\infty)$, let d_M^{BD} be the distance from χ to the span of $\varphi_1,\dots,\varphi_M$, and let γ_0 be Euler's constant. In this notation $g_n = \varphi_n - \varphi_1/n$.

Lemma 13.7 (Comparison with the Báez-Duarte distance)**.** *Let* $\kappa = \|\varphi_1\|$. *Then* $\kappa^2 = \log 2\pi - \gamma_0 = 1.26066\ldots$, *and*

$$d_M^{BD} \le d_{M,1/2} \le (1+\kappa)\, d_M^{BD}.$$

Proof. For $c \in \mathbb{C}^M$ put $r_c = \chi - \sum_{n\le M} c_n\varphi_n$ and $A(c) = \sum_{n\le M} c_n/n$. For $x > 1$ we have $0 < 1/(nx) < 1$, so $r_c(x) = -A(c)/x$. Since $\sum c_n g_n = \sum c_n\varphi_n - A(c)\varphi_1$, the function $1 - \sum c_n g_n$ on $(0,1]$, extended by 0, equals $r_c + A(c)\varphi_1$ on $(0,\infty)$. Hence $\|1-\sum c_n g_n\|_{1/2} \le \|r_c\| + \kappa|A(c)|$, and $|A(c)|^2 = \int_1^\infty |r_c|^2 \le \|r_c\|^2$. Taking c optimal for d_M^{BD} gives the upper bound. Conversely,

[19] It equals 1, since $b_1 = -\sum_\rho \operatorname{Re}(1/\rho)$; the bound is all that is used.

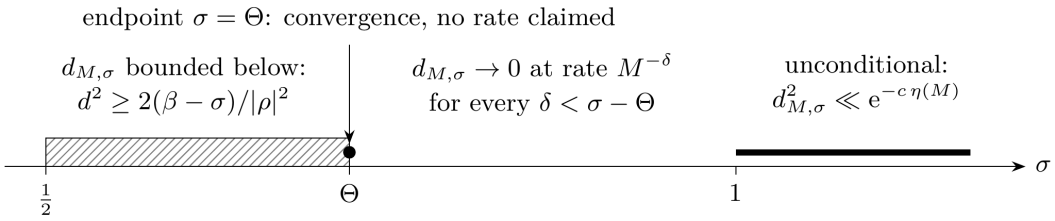


FIGURE 10. The convergence set of the reciprocal-dilate problem in the weighted norms $\|\cdot\|_\sigma$. For $\frac{1}{2} \leq \sigma < \Theta$ a zero $\rho = \beta + i\gamma$ with $\beta > \sigma$ keeps the distance $d_{M,\sigma}$ bounded below by (13.5); for $\Theta < \sigma < 1$ the distance tends to 0 at a power rate (Theorem 13.4); at $\sigma = \Theta$ there is convergence and no rate is claimed (Theorem 13.5); for $\sigma \geq 1$ the rate is unconditional (Theorem 13.11).

given $a_2, \dots, a_M$, put $c_n = a_n$ for $n \geq 2$ and $c_1 = -\sum_{n\geq 2} a_n/n$; then $A(c) = 0$ and $\|1 - \sum a_n g_n\|_{1/2} = \|r_c\| \geq d_M^{BD}$. Finally $\kappa^2 = 1 + \int_1^\infty \{y\}^2 y^{-2}\,\mathrm{d}y = \log 2\pi - \gamma_0$. □

For $0 < \operatorname{Re} s < 1$ one has $\int_0^\infty \varphi_n(x) x^{s-1}\,\mathrm{d}x = -n^{-s}\zeta(s)/s$ (proof of Lemma 13.1), so by Mellin–Plancherel on $\operatorname{Re} s = \frac{1}{2}$,

$$\|r_c\|^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left|1 - \zeta(s) D_c(s)\right|^2 \frac{\mathrm{d}t}{|s|^2}, \qquad D_c(s) = -\sum_{n\leq M} c_n n^{-s}, \quad s = \tfrac{1}{2} + it. \tag{13.8}$$

Theorem 13.8 (Endpoint constant)**.** *Assume the Riemann hypothesis and* $\sum_{|\gamma|\leq T} |\zeta'(\rho)|^{-2} \ll T^{3/2-\delta}$ *for some* $\delta > 0$*. Then, with* $u_M^{\log} = 1 + \sum_{n\leq M} \mu(n)(1 - \frac{\log n}{\log M}) g_n$*,*

$$\lim_{M\to\infty} (\log M)\, d^2_{M,1/2} = \lim_{M\to\infty} (\log M)\, \|u_M^{\log}\|^2_{1/2} = 2 + \gamma_0 - \log 4\pi = 0.046191\ldots.$$

Proof. Write $C_0 = 2 + \gamma_0 - \log 4\pi$. Take $c_n = -\mu(n)(1 - \log n/\log M)$. Then $D_c = V_M$, the polynomial of Bettin, Conrey and Farmer, and [6, Theorem 1] together with (13.8) gives $\|r_c\|^2 \sim C_0/\log M$. Next $A(c) = -\sum_{n\leq M} \mu(n)/n + (\log M)^{-1} \sum_{n\leq M} \mu(n) \log n/n$. Under the Riemann hypothesis $\sum_{n\leq y} \mu(n) \ll y^{1/2+\eta}$ [36, §14.25], so by partial summation $\sum \mu(n) n^{-s}$ and $\sum \mu(n)(\log n) n^{-s}$ converge at $s = 1$ to 0 and to $\lim_{s\to 1} \zeta'(s)/\zeta(s)^2 = -1$, with tails $\ll M^{-1/2+2\eta}$. Hence $A(c) = -1/\log M + O(M^{-1/2+2\eta})$. By the proof of Lemma 13.7, $\|u_M^{\log}\|_{1/2} \leq \|r_c\| + \kappa|A(c)| = (C_0/\log M)^{1/2}(1 + o(1))$, so $\limsup (\log M) d^2_{M,1/2} \leq \limsup (\log M) \|u_M^{\log}\|^2_{1/2} \leq C_0$.

For the lower bound, the functions φ_n, $n \leq M$, are among Burnol's functions $x \mapsto \{\vartheta/x\}$ with $1/M \leq \vartheta \leq 1$. By Lemma 13.7 and [13, Theorem 1.3], $\liminf (\log M) d^2_{M,1/2} \geq \sum_\rho m_\rho^2/|\rho|^2$, summed over distinct zeros with multiplicities m_ρ. The hypothesis forces $\zeta'(\rho) \neq 0$, so $m_\rho = 1$. Under the Riemann hypothesis $1/|\rho|^2 = 1/\rho + 1/\bar\rho$ summed symmetrically, and $\sum_\rho \operatorname{Re}(1/\rho) = 1 + \frac{1}{2}\gamma_0 - \frac{1}{2}\log 4\pi$ [15, Ch. 12]. So the sum is C_0. □

Theorem 13.9 (Endpoint rate under the Riemann hypothesis)**.** *Assume the Riemann hypothesis, fix* $0 < \delta \leq \frac{1}{2}$*, and put* $\varepsilon_M = 50(\log\log M)^{5/2+\delta}(\log M)^{-1/2}$*. For the functions* $u_{M,\varepsilon}$ *of the proof of Theorem 13.5,*

$$d^2_{M,1/2} \leq \|u_{M,\varepsilon_M}\|^2_{1/2} \ll_\delta (\log\log M)^{5/2+\delta}(\log M)^{-1/2} \qquad (M \geq M_0(\delta)).$$

Proof. Write $\varepsilon = \varepsilon_M$, $s = \frac{1}{2} + it$, $M_x(s) = \sum_{n\leq x} \mu(n) n^{-s}$ and $D_M(s) = \sum_{n\leq M} \mu(n)(1 - n/M) n^{-s}$. Under the Riemann hypothesis $\zeta(s+\varepsilon) \neq 0$, and by (13.2) the transform of $u_{M,\varepsilon}$ on the line is

$$\frac{1 - \zeta(s)/\zeta(s+\varepsilon)}{s} - \frac{\zeta(s)(D_M(s+\varepsilon) - 1/\zeta(s+\varepsilon))}{s} + \frac{\zeta(s) D_M(1+\varepsilon)}{s}.$$

By Lemma 13.2, $\|u_{M,\varepsilon}\|_{1/2} \le J_\varepsilon^{1/2} + I^{1/2} + \kappa|D_M(1+\varepsilon)|$, where $J_\varepsilon = \frac{1}{2\pi}\int |1-\zeta(s)/\zeta(s+\varepsilon)|^2 \frac{\mathrm{d}t}{|s|^2}$ and $I = \frac{1}{2\pi}\int |\zeta(s)|^2 |D_M(s+\varepsilon) - 1/\zeta(s+\varepsilon)|^2 \frac{\mathrm{d}t}{|s|^2}$. By [5, Proposition 2], $J_\varepsilon \ll \varepsilon$.

Since $\sum_{n\le M} a_n(1-n/M) = M^{-1}\int_0^M \sum_{n\le x} a_n \,\mathrm{d}x$, the triangle inequality in L^2 gives $I^{1/2} \le M^{-1}\int_0^M I_x^{1/2}\,\mathrm{d}x$, where I_x is I with M_x in place of D_M. For $\sqrt{M} \le x \le M$, $N = \lfloor x \rfloor$ satisfies $\log N \ge \frac14 \log M$ and $\log\log N \le \log\log M$, so $\varepsilon \ge 25(\log\log N)^{5/2+\delta}(\log N)^{-1/2}$, and [5, Proposition 3] gives $I_x \ll_\delta N^{-\varepsilon/2} \le M^{-\varepsilon/8}$. For $x < \sqrt{M}$, $|M_x(s+\varepsilon)| \le 2\sqrt{x}$, $\frac{1}{2\pi}\int |\zeta(s)|^2 \frac{\mathrm{d}t}{|s|^2} = \kappa^2 \le 2$ and $|\zeta(s)/\zeta(s+\varepsilon)| \le 1 + |1-\zeta(s)/\zeta(s+\varepsilon)|$, so $I_x^{1/2} \le 2\sqrt{2}\sqrt{x} + 1 + J_\varepsilon^{1/2}$. Hence $I^{1/2} \ll_\delta M^{-1/4} + M^{-\varepsilon/16}$.

Finally $\zeta(1+\varepsilon) > 1/\varepsilon$, so $0 < 1/\zeta(1+\varepsilon) < \varepsilon$, and for $x \ge 1$, $\sum_{n>x} \mu(n) n^{-1-\varepsilon} \ll_\eta x^{-1/2+\eta}$ uniformly in ε, by partial summation. Averaging over x as above gives $|D_M(1+\varepsilon)| \le 2\varepsilon + O(M^{-1/2+\eta})$. Since $M^{-\varepsilon_M/16} = \exp\bigl(-\frac{25}{8}(\log\log M)^{5/2+\delta}(\log M)^{1/2}\bigr) \ll \varepsilon_M$, all terms are $\ll_\delta \varepsilon_M^{1/2}$. □

Remark 13.10. By Lemma 13.7, the bound of Theorem 13.9 also follows, with the factor $(1+\kappa)^2$, from the theorem of Balazard and de Roton [5] for d_M^{BD}; Theorem 13.9 shows that the construction of Theorem 13.5 attains it. Whether $d_{M,1/2}^2 \ll 1/\log M$ follows from the Riemann hypothesis alone, or with simple zeros, is not known.

13.5. **The exponent one.** Write $\mathcal{M}(x) = \sum_{n\le x}\mu(n)$ for the summatory function of the Möbius function; the letter M is reserved for the length of the approximant. The classical estimate of Walfisz [37] is

$$\mathcal{M}(x) \ll x\exp\bigl(-c_0'\,\eta(x)\bigr), \qquad \eta(x) = (\log x)^{3/5}(\log\log x)^{-1/5} \qquad (x \ge 3), \tag{13.9}$$

with an absolute constant $c_0' > 0$.

Theorem 13.11 (Unconditional rate at $\sigma = 1$)**.** *Let $u_M^\sharp = 1 + \sum_{n\le M}\mu(n) g_n$. There is an absolute constant $c > 0$ such that, for $M \ge 3$ and every $\sigma \ge 1$,*

$$d_{M,\sigma}^2 \le \|u_M^\sharp\|_\sigma^2 \le \|u_M^\sharp\|_1^2 \ll \exp\bigl(-c\,\eta(M)\bigr).$$

Proof. The first two inequalities are immediate. Put $R_M(s) = \sum_{n>M}\mu(n)n^{-s}$ for $\operatorname{Re} s \ge 1$. By (13.9) and partial summation, this series and the series of its termwise derivative converge uniformly on compact subsets of the line $\operatorname{Re} s = 1$, and

$$|R_M(1+it)| + |R_M'(1+it)| \ll (1+|t|)\,\varepsilon_1(M), \qquad \varepsilon_1(M) \ll \exp\bigl(-c_1'\eta(M)\bigr), \tag{13.10}$$

for some absolute $c_1' > 0$. Moreover $\sum_{n\ge1}\mu(n)n^{-1-it} = 1/\zeta(1+it)$ for all real t, with the value 0 at $t = 0$. By (13.2) the transform of $u_M^\sharp$ on the line $\operatorname{Re} s = 1$ is therefore

$$U_M(s) = \frac{\zeta(s)\bigl(R_M(s) - R_M(1)\bigr)}{s}.$$

Three ranges of t are treated separately. For $|t| \le 1$, the mean value theorem and (13.10) give $|R_M(s) - R_M(1)| \ll |t|\varepsilon_1$, and $t\zeta(1+it)$ is bounded, so $U_M \ll \varepsilon_1$. For $1 \le |t| \le T$, the bound $\zeta(1+it) \ll \log(2+|t|)$ [36, Theorem 3.5] and (13.10) give $U_M \ll \varepsilon_1\log(2+|t|)$. For $|t| > T$, the trivial bound $|\sum_{n\le M}\mu(n)n^{-1-it}| \le 1 + \log M$ gives $U_M \ll (\log M)\log(2+|t|)/|t|$. By Lemma 13.2,

$$\|u_M^\sharp\|_1^2 \ll \varepsilon_1^2\, T\log^2(2+T) + \frac{(\log M)^2\log^2(2+T)}{T}.$$

The choice $T = 1/\varepsilon_1$ gives $\|u_M^\sharp\|_1^2 \ll \varepsilon_1(\log M)^2\log^2(2/\varepsilon_1)$, and the logarithmic factors are absorbed by taking $c < c_1'$. The details are in Appendix F. □

The rate is that of the prime number theorem with the Vinogradov–Korobov error term. A power rate $d_{M,1} \ll M^{-\delta}$ is not excluded by any lower bound in this paper, but our proof of any such rate would pass through a zero-free half-plane.

13.6. **Fixed tapers at the critical norm.** By Theorem 13.5, under the Riemann hypothesis the residuals u_M of (13.4) tend to zero in $\|\cdot\|_\sigma$ for every $\sigma > 1/2$. They do not tend to zero in the critical norm, and neither do the residuals of any other fixed taper, whether or not the Riemann hypothesis holds.

Theorem 13.12 (Every fixed taper of bounded variation)**.** *Let $w : [0,1] \to \mathbb{C}$ have bounded variation and satisfy $w(0+) = 1$. Put*

$$U_M = 1 + \sum_{n \le M} \mu(n)\, w(n/M)\, g_n.$$

For every nontrivial zero ρ of ζ with $\beta = \operatorname{Re}\rho \ge 1/2$,

$$\limsup_{M\to\infty} M^{1/2-\beta}\|U_M\|_{1/2} > 0,$$

and the limit superior is infinite if ρ is a multiple zero. In particular $\|U_M\|_{1/2}$ does not tend to zero. This applies to the Riesz tapers $w(t) = (1-t)^k$ of every real order $k > 0$ and to the sharp cut-off $w = 1$.

The proof is in Appendix G. Its steps are these. The identity $\sum_{n\le y}\mu(n)\lfloor y/n\rfloor = 1$ turns $U_M(1/y)$ into a sum of $\mu(n)$ against a dilated kernel. Integration against a test function W of vanishing first moment removes the term that carries $D(1)$ and leaves a sum $H(X) = \sum_n \mu(n) k_W(n/X)$ with

$$|H(X)| \le \sqrt{X}\,\|vW\|_{L^2}\|U_X\|_{1/2} + O(\log X),$$

whose Mellin transform is $\widehat{k}_W(s)/\zeta(s)$. The test function can be chosen with $\widehat{k}_W(\rho) \ne 0$; this is where the bounded variation of w and the normalisation $w(0+) = 1$ are used. The pole at ρ then prevents $H(X) = o(X^\beta)$.

For the Riesz family the exact rate of growth can be read off.

Corollary 13.13 (Exact growth of the linear residual)**.** *For the residuals u_M of* (13.4)*,*

$$\limsup_{M\to\infty}\|u_M\|_{1/2} > 0 \qquad \text{and} \qquad \limsup_{M\to\infty}\frac{\log(1+\|u_M\|_{1/2})}{\log M} = \Theta - \frac{1}{2}.$$

Thus the Riemann hypothesis is equivalent to $\|u_M\|_{1/2} \ll_\varepsilon M^\varepsilon$ for every $\varepsilon > 0$, although these norms do not tend to zero. For every taper w as in Theorem 13.12 the second limit superior is at least $\Theta - 1/2$.

Proof. The first assertion is Theorem 13.12 at a zero on the critical line. The lower bound in the second follows from the same theorem applied to zeros with real part approaching Θ.

For the upper bound take $s = 1/2 + it$ in (13.6) and move the line to $\operatorname{Re} w = \Theta - 1/2 + \delta$ with $0 < \delta < 1/4$. This line lies to the right of $w = 0$, so no pole is crossed. On it $\operatorname{Re}(s+w) = \Theta + \delta$. If $\Theta < 1$, Lemma A.2 gives $1/\zeta(s+w) \ll (1 + |t| + |\operatorname{Im} w|)^\kappa$ for any $\kappa > 0$; if $\Theta = 1$, the lemma is not available, and we use instead $|1/\zeta(s+w)| \le \zeta(1+\delta)$. In both cases

$$|D_M(1/2+it)| \ll_{\delta,\kappa} M^{\Theta-1/2+\delta}(1+|t|)^\kappa.$$

With (13.3) and the convexity bound $\zeta(1/2+it) \ll (1+|t|)^{1/4+\kappa}$, the transform (13.2) is square integrable on the critical line with norm $O_\delta(M^{\Theta-1/2+\delta})$, provided $\kappa < 1/8$. Lemma 13.2 gives $\|u_M\|_{1/2} \ll_\delta M^{\Theta-1/2+\delta}$. Let $\delta \downarrow 0$. □

13.7. **The harmonic cost of the coefficients.** Theorem 13.12 concerns a profile held fixed as M grows. The following necessary condition applies to every finite choice of coefficients.

Theorem 13.14 (Harmonic coefficient cost)**.** *Let $U = 1 + \sum_{n=2}^{M} a_n g_n$, put $d = \|U\|_{1/2}$ and $a_n = 0$ for $n > M$. For every integer $K \ge 2$,*

$$\Big(\sum_{n\ge 2}\frac{|a_n|^2}{n}\Big)^{1/2} \ge \Big(\sum_{2\le n\le K}\frac{\mu(n)^2}{n}\Big)^{1/2} - 2d\sqrt{K}\sum_{m\le K}\frac{1}{m}.$$

Consequently, along any family of approximants with $d \to 0$,

$$\sum_{n\ge 2} \frac{|a_n|^2}{n} \ge \Big(\frac{12}{\pi^2} - o(1)\Big) \log \frac{1}{d},$$

and for $0 < d \le \mathrm{e}^{-4}$,

$$\sum_{n\ge 2} \frac{|a_n|^2}{n} \ge \frac{12}{\pi^2} \log \frac{1}{d} - \frac{24}{\pi^2} \log\log \frac{1}{d} - 12.4. \tag{13.11}$$

The proof, in Appendix G, expresses $a_n - \mu(n)$ by Möbius inversion through the increments of the step function $y \mapsto U(1/y)$, whose weighted mean square is controlled by d.

A bounded harmonic norm $\sum |a_n|^2/n$ therefore excludes convergence at the critical exponent, whatever the shape of the coefficients. The Möbius coefficients themselves have $\sum_{n\le M} \mu(n)^2/n \sim (6/\pi^2)\log M$. The theorem is a constraint on constructions and gives no lower bound for $d_{M,1/2}$ by itself.

13.8. **The logarithmic taper.** A taper that changes shape with M lies outside Theorem 13.12. The simplest is logarithmic:

$$u_M^{\log} = 1 + \sum_{n\le M} \mu(n)\Big(1 - \frac{\log n}{\log M}\Big) g_n, \qquad D_M^{\log}(s) = \sum_{n\le M} \mu(n)\Big(1 - \frac{\log n}{\log M}\Big) n^{-s}.$$

Let $\psi(y) = \sum_{n\le y} \Lambda(n)$ and $E(y) = \lfloor y \rfloor - \psi(y)$. Expanding the floors in (13.1) and using $\sum_{n\mid m} \mu(n) = \mathbf{1}_{m=1}$ and $\sum_{n\mid m} \mu(n)\log n = -\Lambda(m)$ gives the exact formula

$$(\log M)\, u_M^{\log}(1/y) = (\log M) D_M^{\log}(1)\, \lfloor y \rfloor - \psi(y) \qquad (1 \le y < M). \tag{13.12}$$

Proposition 13.15 (Logarithmic taper in shifted norms)**.** *Suppose $1/2 \le \theta < \sigma < 1$ and $\zeta(s) \ne 0$ for $\operatorname{Re} s > \theta$. Then for every $0 < \delta < \sigma - \theta$,*

$$\big\| (\log M)\, u_M^{\log} - E(1/\,\cdot\,) \big\|_\sigma \ll_{\sigma,\theta,\delta} M^{-\delta},$$

and the Mellin transform of $x \mapsto E(1/x)$ on $\operatorname{Re} s = \sigma$ is $(\zeta(s) + \zeta'(s)/\zeta(s))/s$. In particular $\|u_M^{\log}\|_\sigma \sim \|E(1/\,\cdot\,)\|_\sigma / \log M$.

The proof is in Appendix G; it repeats that of Theorem 13.4 with the kernel $M^w/(w^2 \log M)$, whose double pole at $w = 0$ produces the derivative of $1/\zeta$.

The logarithmic taper thus converges in the shifted norms only at the rate $1/\log M$, and its scaled limit has a pole of its transform at every zero on the line $\operatorname{Re} s = \sigma$. At the critical exponent the relevant quantity is the interior mean square of E. For integer $M \ge 2$ put

$$\widetilde{u}_M = u_M^{\log} + M\Big(\frac{1}{\log M} - D_M^{\log}(1)\Big) g_M, \qquad I(Y) = \int_1^Y |E(y)|^2 \frac{\mathrm{d}y}{y^2}.$$

Since $g_M(1/y) = \lfloor y \rfloor / M$ for $1 \le y < M$, formula (13.12) gives $(\log M)\widetilde{u}_M(1/y) = E(y)$ on that range, and so

$$\|\widetilde{u}_M\|^2_{L^2(1/M,1)} = \frac{I(M)}{(\log M)^2}. \tag{13.13}$$

Proposition 13.16 (Growth forced by an off-line zero)**.** *Let ρ be a zero of ζ of order m with $\beta = \operatorname{Re}\rho > 1/2$. Then*

$$\limsup_{Y\to\infty} \frac{I(Y)}{Y^{2\beta-1}} \ge \frac{m^2}{(2\beta-1)|\rho|^2}.$$

Consequently the Riemann hypothesis holds if and only if $\|\widetilde{u}_M\|_{L^2(1/M,1)}$ is bounded, and in that case its square is $O(1/\log M)$.

The proof is in Appendix G.

Since I is nondecreasing, a zero with $\beta > 1/2$ gives

$$\limsup_{M\to\infty} M^{1-2\beta}(\log M)^2 \,\|\widetilde{u}_M\|^2_{L^2(1/M,1)} > 0.$$

Table 16 compares the families.

TABLE 16. Explicit coefficient families $c_n = -\mu(n)w_n$, $n \le M$. The second column assumes $\zeta \neq 0$ in $\operatorname{Re} s > \theta$ and $\theta < \sigma < 1$, $\delta < \sigma - \theta$, except in the fifth row, which is unconditional. The last column gives the number of the theorem, proposition or corollary.

Taper w_n	Norm $\|\cdot\|_\sigma$, $\sigma > \theta$	Critical norm	Reference
$1 - n/M$	$\ll M^{-\delta}$	exponent $\Theta - \frac{1}{2}$; no decay	13.4, 13.13
$w(n/M)$, w fixed	—	$\limsup M^{1/2-\beta}\|U_M\| > 0$	13.12
$n^{-\varepsilon}(1 - n/M)$	$\to 0$ also at $\sigma = \theta$ $(\varepsilon \to 0)$	—	13.5
$1 - \log n/\log M$	$\asymp 1/\log M$	interior norm $O(1)$ iff RH	13.15, 13.16
1, with $\sigma \ge 1$	$\ll \exp(-c\,\eta(M))$	no decay	13.11, 13.12
arbitrary a_n	—	$\sum \frac{\lvert a_n\rvert^2}{n} \ge (\frac{12}{\pi^2} - o(1))\log\frac{1}{d}$	13.14

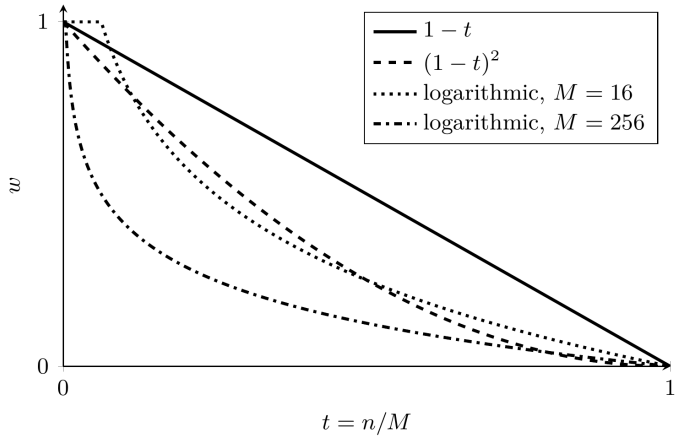


FIGURE 11. Fixed taper profiles $1 - t$ and $(1 - t)^2$, and the logarithmic profile $\min(1, -\log t/\log M)$ for $M = 16$ and $M = 256$. The logarithmic profile depends on M and tends to 0 at each fixed $t > 0$ as $M \to \infty$. It therefore lies outside Theorem 13.12 and is treated separately in Proposition 13.15.

13.9. **The ordinary-dilate model.** The balanced reciprocal dilates should not be confused with the ordinary dilates $\{kx\}$. For those the approximation problem has an exact solution that does not involve the zeros.

Proposition 13.17 (Ordinary dilates)**.** *Let*

$$d_M^{\mathrm{ord}} = \inf_{c_1,\dots,c_M} \Big\| 1 - \sum_{k\le M} c_k\{kx\} \Big\|_{L^2(0,1)},$$

let φ be Euler's function and $J_2(d) = d^2\prod_{p\mid d}(1 - p^{-2})$. Then

$$\big(d_M^{\mathrm{ord}}\big)^2 = \frac{1}{1 + 3S_M}, \qquad S_M = \sum_{d\le M} \frac{\varphi(d)^2}{J_2(d)} \sim M\prod_p \Big(1 - \frac{2}{p(p+1)}\Big).$$

The proof is a diagonalisation of the Gram matrix $\int_0^1 \{jx\}\{kx\}\,\mathrm{d}x = \frac{1}{4} + \gcd(j,k)^2/(12jk)$ and is given in Appendix G.

So $d_M^{\mathrm{ord}} \asymp M^{-1/2}$ whatever the zeros of ζ do. The difference is visible on the Mellin side:

$$\int_0^1 \{kx\}x^{s-1}\,\mathrm{d}x = \frac{k}{s+1} - \frac{k-1}{s} + \frac{k^{-s}}{s}\sum_{r=1}^{k-1} r^s \qquad (\mathrm{Re}\, s > 0),$$

which is not a multiple of $k^{-s}\zeta(s)/s$; at $k=1$ it is $1/(s+1)$. The transforms of the ordinary dilates have no common zero at the zeros of ζ, and the obstruction (13.5) has no analogue for them.

14. Dirichlet characters

The criterion of Theorem 3.1 extends to twists by Dirichlet characters and then characterises the generalised Riemann hypothesis. Averaging over characters does not make the problem easier: the character of conductor one is always present.

14.1. **The signed energy of a twist.** For a Dirichlet character χ, including the trivial character of conductor 1, put

$$L_\chi(x) = \sum_{n\le x} \lambda(n)\chi(n), \qquad Q_\chi(N) = \frac{1}{N}\sum_{t\in\mathbb{Z}} \Big| \sum_{\substack{t<n\le t+N \\ 1\le n\le N}} \lambda(n)\chi(n)\Big|^2.$$

The window count in the proof of Proposition 2.1 does not use that the sequence is real, and gives

$$Q_\chi(N) = \frac{1}{N}\Big(|L_\chi(N)|^2 + \sum_{j=1}^{N-1}\big(|L_\chi(j)|^2 + |L_\chi(N)-L_\chi(j)|^2\big)\Big).$$

Since $|z|^2 + |w-z|^2 \ge |w|^2/2$,

$$\tfrac{1}{2}|L_\chi(N)|^2 \le Q_\chi(N) \le 5\max_{j\le N}|L_\chi(j)|^2. \tag{14.1}$$

Theorem 14.1 (Signed energy for Dirichlet characters)**.** *Let $1/2 \le \theta \le 1$. The following are equivalent.*

(1) No Dirichlet L-function has a zero in $\mathrm{Re}\, s > \theta$.
(2) For every character χ and every $\varepsilon > 0$, $L_\chi(N) \ll_{\chi,\varepsilon} N^{\theta+\varepsilon}$.
(3) For every character χ and every $\varepsilon > 0$, $Q_\chi(N) \ll_{\chi,\varepsilon} N^{2\theta+\varepsilon}$.

The characters in the second and third conditions may be restricted to primitive ones. At $\theta = 1/2$ each condition is equivalent to the generalised Riemann hypothesis for all Dirichlet L-functions.

Proof. The second and third conditions are equivalent by (14.1).

Assume the second condition for a character χ. For $\mathrm{Re}\, s > 1$ the Euler factor $(1+\chi(p)p^{-s})^{-1} = (1-\chi(p)p^{-s})/(1-\chi(p)^2p^{-2s})$ gives

$$F_\chi(s) = \sum_{n\ge1}\lambda(n)\chi(n)n^{-s} = \frac{L(2s,\chi^2)}{L(s,\chi)},$$

and by partial summation $F_\chi(s) = s\int_1^\infty L_\chi(x)x^{-s-1}\,\mathrm{d}x$ is holomorphic in $\mathrm{Re}\, s > \theta$. The identity $L(s,\chi)F_\chi(s) = L(2s,\chi^2)$ persists there. At a zero ρ of $L(s,\chi)$ with $\mathrm{Re}\,\rho > \theta \ge 1/2$ the left side would vanish, while $L(2\rho,\chi^2) \ne 0$ by the Euler product, since $\mathrm{Re}(2\rho) > 1$. Hence $L(s,\chi)$ has no zero in $\mathrm{Re}\, s > \theta$. A character and the primitive character inducing it have the same zeros in $\mathrm{Re}\, s > 0$, so primitive characters suffice.

Assume the first condition and fix χ modulo q, induced by the primitive character χ^*. If $\theta = 1$ the second condition is trivial. Let $\theta < 1$. Since $L(s,\chi) = L(s,\chi^*)\prod_{p\mid q}(1-\chi^*(p)p^{-s})$

and each factor of the finite product has modulus between $1-2^{-1/2}$ and 2 on $\operatorname{Re} s \ge 1/2$, Lemma 13.3 gives, for $0<\delta\le 1/4$ and $\kappa>0$,

$$\frac{1}{L(\sigma+it,\chi)} \ll_{\chi,\delta,\kappa} (2+|t|)^{\kappa} \qquad (\theta+\delta\le\sigma\le 2),$$

where for χ^* trivial the reciprocal vanishes at $s=1$. The numerator $L(2s,\chi^2)$ is bounded by $\zeta(2\theta+2\delta)$ in this strip. So F_χ is holomorphic for $\operatorname{Re} s\ge\theta+\delta$ and is $O((2+|t|)^{\kappa})$ there. The proof of Proposition A.3 uses only this bound and $|\lambda(n)\chi(n)|\le 1$; it gives $L_\chi(x)\ll_{\chi,\varepsilon} x^{\theta+\varepsilon}$.

For the last assertion, the functional equation of a primitive L-function maps its zeros in the critical strip to their reflections in the line $\operatorname{Re} s=1/2$, so the absence of zeros in $\operatorname{Re} s>1/2$ puts all nontrivial zeros on that line. □

The constants are allowed to depend on χ. For the generalised Riemann hypothesis the bounds must nevertheless hold for every character.

14.2. **Large conductors.** For $X>0$ define the smoothed twists and their weighted mean square over primitive characters,

$$B_\chi(X)=\sum_{n\ge1}\lambda(n)\chi(n)\mathrm{e}^{-n/X}, \qquad A_*(X)=\sum_{q\ge1}\frac{1}{q^2\varphi(q)}\sideset{}{^*}\sum_{\chi \bmod q}|B_\chi(X)|^2.$$

The star restricts the sum to primitive characters, and $q=1$ contributes $|B(X)|^2$. By (12.1), $|B_\chi(X)|\le X$, so $A_*(X)\le\zeta(2)X^2$.

Theorem 14.2 (Large conductors)**.** *For $X>0$ and $Q\ge1$,*

$$\sum_{q\ge Q}\frac{1}{q^2\varphi(q)}\sideset{}{^*}\sum_{\chi \bmod q}|B_\chi(X)|^2\ll\frac{X^2}{Q^3}+\frac{X}{Q},$$

with an absolute implied constant. In particular the conductors $q\ge X^{1/3}$ contribute $O(X)$ for $X\ge1$. The same bound holds with λ replaced by any sequence of modulus at most 1*.*

Proof. Let (a_n) have $|a_n|\le1$ and put $T_\chi(y)=\sum_{n\le y}a_n\chi(n)$. On a block $R\le q<2R$ we have $1/(q^2\varphi(q))\le R^{-3}q/\varphi(q)$, and the multiplicative large sieve inequality [27], [22, Theorem 7.13] gives

$$\sum_{R\le q<2R}\frac{1}{q^2\varphi(q)}\sideset{}{^*}\sum_{\chi \bmod q}|T_\chi(y)|^2\le R^{-3}(y+4R^2)\,y.$$

Partial summation gives $\sum_n a_n\chi(n)\mathrm{e}^{-n/X}=X^{-1}\int_0^\infty T_\chi(y)\mathrm{e}^{-y/X}\,\mathrm{d}y$. By Minkowski's integral inequality in the weighted ℓ^2 norm over the block, the square root of the block's contribution is at most

$$\frac{1}{XR^{3/2}}\int_0^\infty (y+2R\sqrt{y})\mathrm{e}^{-y/X}\,\mathrm{d}y=\frac{X}{R^{3/2}}+\sqrt{\frac{\pi X}{R}}.$$

Squaring and summing over $R=2^jQ$, $j\ge0$, gives the result. □

For general completely multiplicative signs the term X^2/Q^3 cannot be removed. Take an odd prime $r\in[Q,3Q]$ and let f be the completely multiplicative function with $f(p)=\chi_r(p)$ for $p\ne r$ and $f(r)=1$, where χ_r is the Legendre symbol. Then $f\chi_r$ is the indicator of the integers prime to r, and $\sum_{(n,r)=1}\mathrm{e}^{-n/X}\ge X/3$ for $X\ge2$. This one character contributes at least $X^2/(9r^3)\gg X^2/Q^3$.

14.3. **The exact exponent across conductors.**

Theorem 14.3 (Exact exponent across conductors)**.** *Let Θ_D be the supremum of the real parts of the zeros of all primitive Dirichlet L-functions, ζ included. Then*

$$\limsup_{X\to\infty} \frac{\log(1+A_*(X))}{\log X} = 2\Theta_D.$$

If $\Theta_D < 1$, then for every $\sigma \in (\Theta_D, 1)$ and $\kappa > 0$,

$$|B_\chi(X)| \le C(\sigma,\kappa)\, q^\kappa X^\sigma \qquad (X \ge 1) \tag{14.2}$$

for every primitive character χ of every conductor q, with $C(\sigma,\kappa)$ independent of χ and q.

The proof is in Appendix E. The lower bound uses one character at a time. The upper bound moves the contour in the Mellin inversion formula for B_χ to $\operatorname{Re} s = \sigma$ and uses Lemma 13.3; the uniformity of that lemma in χ and q is essential, because the resulting bound (14.2) is summed over all conductors.

By the functional equations, the generalised Riemann hypothesis for Dirichlet L-functions is the statement $\Theta_D = 1/2$. Theorems 14.2 and 14.3 reduce it to the bound

$$\sum_{q<X^{1/3}} \frac{1}{q^2\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(X)|^2 \ll_\varepsilon X^{1+\varepsilon} \qquad (\varepsilon > 0),$$

which is open. The term $q = 1$ is $|B(X)|^2$, so the positive average contains the original problem.

14.4. **Orthogonality and square divisors.** Let $A(X)$ be defined as $A_*(X)$ but with the sum over all characters modulo q. Orthogonality of characters gives

$$A(X) = \sum_{n,m\ge1} \lambda(n)\lambda(m)\mathrm{e}^{-(n+m)/X} \sum_{\substack{q\mid n-m\\(q,n)=1}} \frac{1}{q^2},$$

where every q divides 0; the inner sum is at most $\zeta(2)$, which justifies the rearrangement. Since $\sum_{(q,n)=1} q^{-2} = \zeta(2)\sum_{d\mid n}\mu(d)d^{-2}$, the diagonal $n = m$ equals

$$\zeta(2)\sum_{d\ge1}\frac{\mu(d)}{d^2\,(\mathrm{e}^{2d/X}-1)} = \frac{\zeta(2)}{2\zeta(3)}X + O(1),$$

by $(\mathrm{e}^y-1)^{-1} = y^{-1} + O(1)$ for $y > 0$. The diagonal is thus of exact order X, and all further information lies in the signed off-diagonal terms, which include $|B(X)|^2 - \sum_n \mathrm{e}^{-2n/X}$ from $q = 1$.

The identities $\lambda(n) = \sum_{d^2\mid n}\mu(n/d^2)$ and $\mu(n) = \sum_{d^2\mid n}\mu(d)\lambda(n/d^2)$ relate B_χ to $M_\chi(X) = \sum_n \mu(n)\chi(n)\mathrm{e}^{-n/X}$:

$$B_\chi(X) = \sum_{d\ge1}\chi(d)^2 M_\chi(X/d^2), \qquad M_\chi(X) = \sum_{d\ge1}\mu(d)\chi(d)^2 B_\chi(X/d^2).$$

Multiplication by $\chi(d)^2$ is a contraction in the weighted ℓ^2 norm over characters, and the terms are bounded by X/d^2. By Minkowski's inequality and $\sum_d d^{-1-2\varepsilon} < \infty$, a bound $O_\varepsilon(X^{1/2+\varepsilon})$ for the weighted norm of either family, valid for all $X > 0$, implies the same bound for the other. The Liouville and Möbius forms of the averaged problem are thus equivalent, and in both the cancellation at small conductors sits in the term $d = 1$.

14.5. **Why increasing the modulus retains conductor one.** Theorem 14.2 is indexed by the primitive conductor, not by the modulus. For the principal character $\chi_{0,p}$ modulo a prime p, removing the multiples of p gives

$$B_{\chi_{0,p}}(X) = B(X) + B(X/p), \tag{14.3}$$

since $\lambda(pm) = -\lambda(m)$. If $X \geq 2$ and $p \geq 2X \log X$, then by (12.1)

$$|B_{\chi_{0,p}}(X) - B(X)| \leq \frac{1}{\mathrm{e}^{p/X} - 1} \leq \frac{1}{X^2 - 1} \leq \frac{2}{X^2}.$$

A twist of arbitrarily large modulus can thus reproduce $B(X)$ almost exactly. It has conductor one and is not part of the large-conductor tail. On the Mellin side (14.3) multiplies $\Phi(s)$ by $1 + p^{-s}$, which has no zero in $\operatorname{Re} s > 0$ and removes no pole.

Nor can the tail be used to reconstruct the conductor-one term by an inequality that is uniform in the coefficients. For a sequence a with $|a_n| \leq 1$ let $\mathrm{Tail}_a(X, Q)$ denote the left side of Theorem 14.2 with a in place of λ; then $\mathrm{Tail}_a(X, X^{1/3}) \ll X$. Suppose that

$$\Big|\sum_{n\geq 1} a_n \mathrm{e}^{-n/X}\Big|^2 \leq K(X)\, \mathrm{Tail}_a(X, X^{1/3})$$

held for all such a. The constant sequence $a_n = 1$ has untwisted sum $(\mathrm{e}^{1/X} - 1)^{-1} \sim X$, so $K(X) \gg X$, and the inequality would return only the trivial bound $O(X^2)$. Any reconstruction that does better must use arithmetic cancellation special to λ.

15. What is unconditional

The equivalences of the preceding sections have the form "a bound with exponent θ holds if and only if ζ has no zero in $\operatorname{Re} s > \theta$". This section records what can be proved with no hypothesis, and then lists the conditional statements in one table. The unconditional results are transfers of known theorems; we state them with their sources.

15.1. **The full window.**

Theorem 15.1 (Unconditional bounds for L and Q_λ)**.** *There is an absolute constant $c > 0$ such that, for $x \geq 3$ and $N \geq 3$,*

$$L(x) \ll x \exp\bigl(-c\,\eta(x)\bigr), \qquad Q_\lambda(N) \ll N^2 \exp\bigl(-2c\,\eta(N)\bigr),$$

where $\eta(x) = (\log x)^{3/5}(\log\log x)^{-1/5}$. Explicitly, for all $x \geq 1$ and all integers $N \geq 1$,

$$|L(x)| \leq \frac{x}{2641} + 8.3\sqrt{x}, \qquad Q_\lambda(N) \leq 5\Bigl(\frac{N}{2641} + 8.3\sqrt{N}\Bigr)^2.$$

Proof. Since $\lambda = \mu * \mathbf{1}_\square$, where $\mathbf{1}_\square$ is the indicator of the squares,

$$L(x) = \sum_{d \leq \sqrt{x}} \mathcal{M}\Bigl(\frac{x}{d^2}\Bigr). \tag{15.1}$$

The rate. For $3 \leq x < 25$ the bound is trivial. Let $x \geq 25$. The function η is increasing on $[5, \infty)$. For $d \leq x^{1/4}$ we have $x/d^2 \geq \sqrt{x} \geq 5$ and $\eta(x/d^2) \geq \eta(\sqrt{x}) \geq 2^{-3/5}\eta(x)$, so by (13.9) these terms contribute at most a constant times $\zeta(2)\, x \exp(-2^{-3/5} c_0' \eta(x))$. For $d > x^{1/4}$ the trivial bound $|\mathcal{M}(y)| \leq y$ gives a contribution of at most $x \sum_{d > x^{1/4}} d^{-2} \leq 2x^{3/4}$. This proves the bound for L with $c = 2^{-3/5} c_0'$. The bound for Q_λ follows from $Q_\lambda(N) \leq 5 \max_{j \leq N} |L(j)|^2$, which is a consequence of (2.6), because $y \exp(-c\eta(y))$ is increasing for large y and tends to infinity.

Explicit constants. Cohen, Dress and El Marraki [14] proved

$$|\mathcal{M}(y)| \leq \frac{y}{4345} \qquad (y \geq y_0 = 2\,160\,535). \tag{15.2}$$

For $1 \leq y \leq 3 \cdot 10^6$ one has $|\mathcal{M}(y)| \leq \sqrt{y}$; this is a finite verification, summarised in Table 26 of Appendix I. In (15.1) the terms with $x/d^2 \geq y_0$ contribute at most $\zeta(2)x/4345 <$

$x/2641$, since $\zeta(2)/4345 = 1/2641.44\ldots$ The remaining terms have $\sqrt{x/y_0} < d \le \sqrt{x}$ and $|\mathcal{M}(x/d^2)| \le \sqrt{x}/d$, and the sum of $1/d$ over this range is at most $1 + \frac{1}{2}\log y_0 = 8.2929\ldots <$ 8.3. The bound for Q_λ follows as before, the majorant being increasing. Appendix I gives the details. □

The rate for $\mathcal{M}$ is classical, and the transfer to L and Q_λ is routine; only the explicit constants for Q_λ seem not to have been recorded. Inserting the bound $|\mathcal{M}(y)| \le 0.013\, y/\log y$ for $y \ge 1.1 \cdot 10^6$ of Ramaré [34] in (15.1) gives in the same way an explicit saving of one logarithm. The explicit inequality is far from the truth in the range where L can be computed: for $x \le 3 \cdot 10^6$ the ratio $|L(x)|/(x/2641 + 8.3\sqrt{x})$ does not exceed 0.159.

15.2. **Short windows.** For windows of length $H < N$ the theorem of Matomäki and Radziwiłł on multiplicative functions in short intervals gives a saving for almost all windows, and hence for the energy. The following statement is their theorem in the notation of this paper; the saving is theirs, and our contribution is the bookkeeping of the boundary windows.

Theorem 15.2 (Short windows, after Matomäki and Radziwiłł)**.** *For integers* $16 \le H \le N$*,*

$$\mathcal{Q}_\lambda(N, H) \ll NH\left(\Big(\frac{\log\log H}{\log H}\Big)^2 + (\log N)^{-1/100}\right).$$

The trivial bound is $\mathcal{Q}_\lambda(N, H) \le 2NH$*.*

The proof is in Appendix H. Their Theorem 1 controls, outside an exceptional set, all windows that lie inside $[\max(H, \sqrt{N}), N]$, and the windows near the two ends of $[1, N]$ are estimated by Theorem 15.1.

The saving does not persist up to $H = N$: there the first term is of size $(\log\log N/\log N)^2$, and Theorem 15.1 is stronger.

15.3. **The barrier.**

Proposition 15.3 (Any power saving is a zero-free half-plane)**.** *Let* $\delta > 0$ *and let* $H(N)$ *be integers with* $1 \le H(N) \le N$*. If*

$$\mathcal{Q}_\lambda(N, H(N)) \ll N^{1-\delta} H(N) \qquad \text{for all } N,$$

then $\zeta(s) \ne 0$ *for* $\operatorname{Re} s > 1 - \delta/2$*.*

Proof. Each $n \le N$ lies in H windows and at most $N + H - 1$ windows meet $[1, N]$, so $HL(N) = \sum_t S(t)$ and the Cauchy–Schwarz inequality give $|L(N)|^2 \le (N + H - 1)\mathcal{Q}_\lambda(N, H)/H$; this is the lower bound in Proposition 2.1. With $H = H(N)$ it yields $|L(N)|^2 \le 2N\,\mathcal{Q}_\lambda(N, H)/H \ll N^{2-\delta}$, and Theorem 3.1 with $\theta = 1 - \delta/2$ applies. □

Theorems 15.1 and 15.2 save less than any power of N, and by Proposition 15.3 this is the limit of what can be proved without a zero-free half-plane. The same holds for the other objects of this paper; Table 17 lists the unconditional statements.

15.4. **Conditional inputs and what they give.** Table 18 assembles the consequences of a zero-free half-plane. In every row the conclusion, required for all $\varepsilon > 0$ or for the stated range of exponents, is equivalent to the hypothesis. No value $\theta < 1$ is known for which the hypothesis holds, and $\theta = 1/2$ is the Riemann hypothesis.

Two kinds of published information about the zeros bear on these tables, and neither moves an exponent. The first is numerical: by Platt and Trudgian [33], all zeros with $0 < \operatorname{Im} s \le 3 \cdot 10^{12}$ lie on the critical line. A zero $\rho = \beta + i\gamma$ with $\beta > 1/2$ therefore has $|\gamma| > 3 \cdot 10^{12}$. The lower bounds it forces are correspondingly small in their constants, with $|\Gamma(\rho)|$ of order $\mathrm{e}^{-\pi|\gamma|/2}$ in Theorem 12.7 and $|\rho|^{-2} < 10^{-24}$ in (13.5), but the exponents β and 2β are unchanged, and a verification to a finite height cannot bound Θ.

The second is the zero-density theory. Guth and Maynard [18] proved that the number $N(\sigma, T)$ of zeros with real part at least σ and height at most T is at most $T^{30(1-\sigma)/13+o(1)}$.

TABLE 17. Unconditional bounds. Here $\eta(x) = (\log x)^{3/5}(\log\log x)^{-1/5}$ and $\ell(t) = (\log t)^{2/3}(\log\log t)^{1/3}$.

Quantity	Bound	Input	Theorem
$L(x)$	$\le x/2641 + 8.3\sqrt{x}$	[14]	15.1
$L(x)$	$\ll x\exp(-c\,\eta(x))$	[37]	15.1
$Q_\lambda(N)$	$\le 5(N/2641 + 8.3\sqrt{N})^2$	[14]	15.1
$Q_\lambda(N)$	$\ll N^2\exp(-2c\,\eta(N))$	[37]	15.1
$\mathcal{Q}_\lambda(N,H)$	$\ll NH\big((\frac{\log\log H}{\log H})^2 + (\log N)^{-1/100}\big)$	[25]	15.2
$B(X)$	$\le X\exp(-c\log X/\ell(\log X))$	[17], Lemma 12.3	12.4
$d^2_{M,\sigma}$, $\sigma \ge 1$	$\ll \exp(-c\,\eta(M))$	[37]	13.11
$A_*(X)$, $q \ge X^{1/3}$	$\ll X$	[27]	14.2

TABLE 18. Conditional inputs and what they give. Hypothesis for the first six rows: $\zeta(s) \ne 0$ for $\operatorname{Re} s > \theta$, where $1/2 \le \theta < 1$. Hypothesis for the last two rows: no Dirichlet L-function has a zero in $\operatorname{Re} s > \theta$.

Object	Conclusion	Reference
Liouville sum	$L(x) \ll x^{\theta+\varepsilon}$	Thm. 3.1
Signed energy	$Q_\lambda(N) \ll N^{2\theta+\varepsilon}$	Thm. 3.1
Smoothed sum	$B(X) \ll X^{\theta+\varepsilon}$	Prop. 12.1
Reciprocal dilates	$d_{M,\sigma} \to 0$ for $\theta \le \sigma < 1$	Thm. 13.5
Reciprocal dilates	$d_{M,\sigma} \ll M^{-\delta}$ for $\delta < \sigma - \theta$	Thm. 13.4
Riesz residual	$\|u_M\|_{1/2} \ll M^{\theta-1/2+\varepsilon}$	Cor. 13.13
Twisted sums	$L_\chi(N) \ll_\chi N^{\theta+\varepsilon}$, $Q_\chi(N) \ll_\chi N^{2\theta+\varepsilon}$	Thm. 14.1
Conductor average	$A_*(X) \ll X^{2\theta+\varepsilon}$	Thm. 14.3

This bounds the number of zeros to the right of a line. It does not exclude any one of them, and a single zero ρ with $\beta > \theta$ already forces $B(X) = \Omega_\pm(X^\beta)$ by Theorem 12.7, $d^2_{M,\sigma} \ge 2(\beta-\sigma)/|\rho|^2$ for $\sigma < \beta$ by (13.5), and $Q_\lambda(N) > N^{2\sigma}$ for all large N and each $\sigma \in [1/2, \beta)$ by Theorem 4.2. Removing a sparse set of scales or averaging over them does not help. The exponents in Table 18 respond only to a zero-free half-plane.

16. THE CRITICAL BOUND

The results above fix the shape of the remaining problem. The Riemann hypothesis is equivalent to

$$Q(N) \ll_\varepsilon N^{1+\varepsilon} \qquad \text{for every } \varepsilon > 0, \tag{16.1}$$

and by Theorem 4.2 it is equally equivalent to (16.1) along any one infinite sequence of N. The bound (16.1) is an upper bound for a positive definite Toeplitz form at the single vector λ (Proposition 2.3), and the three ways of attacking it that the paper examines each meet a precise obstruction.

Correlations. Fejér's identity writes $Q(N)$ through the shifted correlations of λ. Bounding these one shift at a time, uniformly in the endpoint, needs the collar hypothesis of Section 7, which implies the generalized Riemann hypothesis for every Dirichlet character, is false at its endpoint for λ (Section 9), and in its sharp form without the factor N^ε fails along every infinite set of dyadic scales for every sign sequence (Theorem 8.5). Any successful use of correlations must therefore keep the cancellation between different shifts.

Properties stable under perturbation. By Theorem C the exponent of the energy becomes at least 2β when λ is altered on a suitable set of x^β primes, for every $\beta \in (1/2, 1)$, while every property in Table 6 is kept. A proof of (16.1) must use information about λ that such

a change destroys, that is, information that distinguishes λ from the functions λ_S of that theorem, with S of size x^β and $\beta > 1/2$.

Approximation. The reciprocal-dilate problem of Section 13 has the same convergence set as the energy (Theorem 13.5). At $\sigma = 1/2$ no fixed taper of bounded variation converges at all (Theorem 13.12), convergence forces the harmonic norm of the coefficients to grow like $(12/\pi^2)\log(1/d)$ (Theorem 13.14), and the interior norm of the logarithmic taper stays bounded if and only if the Riemann hypothesis holds (Proposition 13.16). The approximation side offers an equivalent target, not an easier one.

Three questions are left open by the paper.

(1) *Boundedness of $Q(N)/N$.* By Theorem 5.6 the set $B_{1/2}$ is infinite, so under the Riemann hypothesis B_Θ is infinite; whether its complement is infinite is not known. By Proposition 5.8, $Q(N)/N$ is bounded exactly when $L(x) \ll \sqrt{x}$. It is unbounded if the positive ordinates are linearly independent over $\mathbb{Q}$ (Corollary 5.10), while under the Riemann hypothesis and $J_{-1}(T) \ll T$ its logarithmic mean is the finite number S of Theorem 5.13, so averaging cannot decide the question. For $N \le 3 \cdot 10^7$ the largest value is $Q(N)/N = 1.1534$ at $N = 12\,897\,104$ (Table 30).

(2) *The endpoint rate.* Under the Riemann hypothesis $d_{M,1/2}^2 \ll (\log\log M)^{5/2+\delta}(\log M)^{-1/2}$ (Theorem 13.9), and under the additional hypothesis of Bettin, Conrey and Farmer, $(\log M)\, d_{M,1/2}^2 \to 2 + \gamma_0 - \log 4\pi$ (Theorem 13.8), so the lower bound of Burnol [13] has the right order. An upper bound of order $(\log M)^{-1/2}$ for $d_{M,1/2}$ from the Riemann hypothesis alone, or with simple zeros, is open.

(3) *Explicit constants.* Theorem 15.1 gives $|L(x)| \le x/2641 + 8.3\sqrt{x}$ from the bound of Cohen, Dress and El Marraki for the Mertens function. Any improvement of the constant 4345 in that bound, or of the range in which $|M(y)| \le \sqrt{y}$ has been checked, improves the constants for L and Q at once.

None of these questions, if answered, would prove (16.1). The bound itself is a statement about the rightmost zero of ζ, and Theorems A–C show that it cannot be reached through the energy without, at some stage, proving a zero-free half-plane.

Appendix A. Calculations for the Liouville energy

This appendix contains the calculations used in Sections 2–5: the window counts behind the Fejér identity, the algebra of Lemma 2.4, the Perron calculation for Theorem 3.1, the extremal values of the weight K, and the kernel estimates for Proposition 4.1.

A.1. **Window counts.** A *window* is a set $(t, t+H] \cap \mathbb{Z} = \{t+1, \dots, t+H\}$ with $t \in \mathbb{Z}$.

Lemma A.1. *Let $N, H \ge 1$ be integers.*

(a) For integers m, n, the number of windows containing both m and n is $(H - |m-n|)_+$. In particular each integer lies in exactly H windows, and the windows meeting $[1, N]$ are those with $1 - H \le t \le N - 1$; there are $N + H - 1$ of them.

(b) Let $H \ge N$. The windows meeting $[1, N]$ are of three kinds:

- *$N - H \le t \le 0$: the window contains $[1, N]$; there are $H - N + 1$ such t;*
- *$1 - H \le t \le N - H - 1$: the window meets $[1, N]$ in the prefix $[1, j]$ with $j = t + H$, and j takes each value in $\{1, \dots, N-1\}$ exactly once;*
- *$1 \le t \le N - 1$: the window meets $[1, N]$ in the suffix $[t+1, N]$, and $j = t$ takes each value in $\{1, \dots, N-1\}$ exactly once.*

Proof. (a) The window $(t, t+H]$ contains m and n if and only if $t < \min(m, n)$ and $\max(m, n) \le t + H$, that is, $\max(m, n) - H \le t \le \min(m, n) - 1$. The number of such integers t is $\min(m, n) - \max(m, n) + H = H - |m - n|$ if this is positive, and zero otherwise. With $m = n$ the count is H. The window meets $[1, N]$ if and only if $t + 1 \le N$ and $t + H \ge 1$.

(b) The window contains $[1,N]$ if and only if $t\le 0$ and $t+H\ge N$. If $t+H\le N-1$, then $t\le N-H-1\le -1$, so the window starts to the left of 1 and its intersection with $[1,N]$ is $[1,t+H]$; as t runs over $[1-H,N-H-1]$, $j=t+H$ runs over $[1,N-1]$. If $t\ge 1$, then $t+H\ge 1+N>N$, and the intersection is $[t+1,N]$. The three ranges of t are disjoint and their union is $[1-H,N-1]$. □

The shift form. By Lemma A.1(a),

$$H\,\mathcal{Q}_a(N,H)=\sum_{t\in\mathbb{Z}}\Big(\sum_{\substack{n\le N\\ t<n\le t+H}}a_n\Big)^2=\sum_{m,n\le N}a_m a_n\,(H-|m-n|)_+.$$

The pairs with $m=n$ contribute $H\sum_{n\le N}a_n^2$. The pairs with $|m-n|=h\ge 1$ occur only for $h\le N-1$, have a nonzero weight only for $h\le H-1$, and contribute $2(H-h)\sum_{n\le N-h}a_n a_{n+h}=2(H-h)C_{a,h}(N-h)$. Division by H gives (2.3).

The Fourier form. Expanding the square,

$$\Big|\sum_{j=1}^{H}\mathrm{e}(j\alpha)\Big|^2=\sum_{j,k=1}^{H}\mathrm{e}((j-k)\alpha)=\sum_{|h|<H}(H-|h|)\,\mathrm{e}(h\alpha),$$

since $j-k=h$ has $H-|h|$ solutions. Hence $F_H(\alpha)=\sum_{|h|<H}(1-|h|/H)\mathrm{e}(h\alpha)$ and $0\le F_H(\alpha)\le F_H(0)=H$. Since $|A_N(\alpha)|^2=\sum_{m,n\le N}a_m a_n\mathrm{e}((m-n)\alpha)$ and $\int_0^1\mathrm{e}(k\alpha)\,\mathrm{d}\alpha$ is 1 for $k=0$ and 0 for other integers k,

$$\int_0^1|A_N(\alpha)|^2F_H(\alpha)\,\mathrm{d}\alpha=\sum_{m,n\le N}a_m a_n\Big(1-\frac{|m-n|}{H}\Big)_+=\mathcal{Q}_a(N,H).$$

The boundary form. For $H\ge N$, Lemma A.1(b) gives

$$H\,\mathcal{Q}_a(N,H)=(H-N+1)L_a(N)^2+\sum_{j=1}^{N-1}L_a(j)^2+\sum_{j=1}^{N-1}\big(L_a(N)-L_a(j)\big)^2,$$

which is (2.5). At $H=N$, write $x_j=L_a(j)$ and $y=L_a(N)$. Then $x_j^2+(y-x_j)^2=2x_j^2-2x_jy+y^2=2(x_j-\frac12 y)^2+\frac12 y^2$, so that

$$N\,Q_a(N)=y^2+\frac{N-1}{2}\,y^2+2\sum_{j=1}^{N-1}\big(x_j-\tfrac12 y\big)^2=\frac{N+1}{2}\,y^2+2\sum_{j=1}^{N-1}\big(x_j-\tfrac12 y\big)^2,$$

which is (2.6).

A.2. **The algebra of Lemma 2.4.** Let a be real, $x_j=L_a(j)$, $y=L_a(N)$, $Q=Q_a(N)$, $V=V_a(N)=\sum_{j\le N}x_j^2$.

Proof of (2.11). Both terms of (2.6) are nonnegative, so

$$y^2\le\frac{2N}{N+1}\,Q\le 2Q,\qquad\sum_{j=1}^{N-1}\big(x_j-\tfrac12 y\big)^2\le\frac{N}{2}\,Q.$$

From $x_j=(x_j-\frac12 y)+\frac12 y$ and $(p+q)^2\le 2p^2+2q^2$ we get $x_j^2\le 2(x_j-\frac12 y)^2+\frac12 y^2$. Therefore

$$V=\sum_{j=1}^{N-1}x_j^2+y^2\le 2\cdot\frac{N}{2}\,Q+\frac{N-1}{2}\,y^2+y^2\le NQ+(N-1)Q+2Q=(2N+1)Q.$$

Proof of (2.12). Assume $|a_n|\le 1$. In (2.5) at $H=N$ use $(y-x_j)^2\le 2y^2+2x_j^2$:

$$N\,Q\le y^2+\sum_{j=1}^{N-1}(3x_j^2+2y^2)=(2N-1)y^2+3\sum_{j=1}^{N-1}x_j^2\le 2N\,y^2+3V,$$

so $Q\le 2y^2+3V/N$. Let $m=|y|$; we may assume $m>0$. Since $|x_N-x_j|\le N-j$, we have $|x_j|\ge m-(N-j)\ge m/2$ whenever $N-m/2\le j\le N$. The number of

integers in $[N - m/2, N]$ is $\lfloor m/2 \rfloor + 1 \geq m/2$, and they are positive because $m \leq N$ gives $N - m/2 \geq N/2 > 0$. Hence

$$V \geq \frac{m}{2} \cdot \Big(\frac{m}{2}\Big)^2 = \frac{m^3}{8}, \qquad \text{that is,} \qquad y^2 = m^2 \leq 4V^{2/3}.$$

Also $|x_j| \leq j$ gives $V \leq \sum_{j \leq N} j^2 \leq N^3$, so $N \geq V^{1/3}$ and $V/N \leq V^{2/3}$. Altogether $Q \leq 2 \cdot 4V^{2/3} + 3V^{2/3} = 11V^{2/3}$.

Table 19 records the extreme values of the two ratios for $a = \lambda$ in a finite range; they show that neither constant is close to being attained by the Liouville function.

TABLE 19. The two inequalities of Lemma 2.4 for $a = \lambda$: largest observed ratios for $1 \leq N \leq 3 \cdot 10^6$. Computed values, given as a check; the lemma is proved above for all N.

inequality	proved	largest ratio found (computed)
$V(N) \leq (2N+1)Q_\lambda(N)$	$V/((2N+1)Q_\lambda) \leq 1$	0.4125
$Q_\lambda(N) \leq 11V(N)^{2/3}$	$Q_\lambda/(11V^{2/3}) \leq 1$	0.1336

A.3. **From a zero-free half-plane to cancellation.** This subsection proves the implication (i)⇒(ii) of Theorem 3.1. Throughout, $\frac{1}{2} \leq \theta < 1$ and $\zeta(s) \neq 0$ for $\operatorname{Re} s > \theta$. The argument is the classical one for the consequences of the Riemann hypothesis [36, Sections 14.2 and 14.25], carried out to the right of the line $\operatorname{Re} s = \theta$. We write $s = \sigma + it$.

Two classical facts are used. The first is the Borel–Carathéodory inequality [28, Section 6.1]: if g is holomorphic on a neighbourhood of the disc $|s - s_0| \leq R$ and $0 < r < R$, then

$$\max_{|s-s_0| \leq r} |g(s)| \leq \frac{2r}{R-r} \max_{|s-s_0|=R} \operatorname{Re} g(s) + \frac{R+r}{R-r} |g(s_0)|. \tag{A.1}$$

The second is Hadamard's three-circles theorem: if g is holomorphic on a neighbourhood of $|s - s_0| \leq r_2$, $0 < r_0 < r_1 < r_2$, and $\mathcal{M}(r) = \max_{|s-s_0|=r} |g(s)|$, then

$$\mathcal{M}(r_1) \leq \mathcal{M}(r_0)^{1-\alpha} \mathcal{M}(r_2)^{\alpha}, \qquad \alpha = \frac{\log(r_1/r_0)}{\log(r_2/r_0)} \in (0,1). \tag{A.2}$$

Lemma A.2 (Subpower growth of the reciprocal)**.** *Let $\frac{1}{2} \leq \theta < 1$ and suppose $\zeta(s) \neq 0$ for $\operatorname{Re} s > \theta$. For $0 < \delta < \frac{1}{4}$ and $\eta > 0$,*

$$\Big|\frac{1}{\zeta(\sigma + it)}\Big| \ll_{\theta,\delta,\eta} (1 + |t|)^{\eta} \qquad (\theta + \delta \leq \sigma \leq 2),$$

where $1/\zeta$ takes the value 0 at $s = 1$.

Proof. Bounded t. The function $1/\zeta$ is continuous on the compact set $\theta + \delta \leq \sigma \leq 2$, $|t| \leq 4$, because ζ has no zero there and its pole at $s = 1$ is a zero of $1/\zeta$. So $1/\zeta$ is bounded on this set.

Large t. Let $|t| \geq 4$, $s_0 = 2 + it$, and

$$R = 2 - \theta - \tfrac{1}{4}\delta, \qquad r_2 = 2 - \theta - \tfrac{1}{2}\delta, \qquad r_1 = 2 - \theta - \delta, \qquad r_0 = \tfrac{1}{2}.$$

Then $r_0 < r_1 < r_2 < R < \frac{3}{2}$. The closed disc $D_R = \{|s - s_0| \leq R\}$ lies in the half-plane $\sigma \geq \theta + \frac{1}{4}\delta$, where ζ has no zero, and it does not contain $s = 1$, since $|s_0 - 1| \geq |t| \geq 4$. The same holds for a slightly larger open disc. On it there is a holomorphic branch $g = \log \zeta$, which we normalize by $g(s_0) = \sum_p \sum_{k \geq 1} k^{-1} p^{-k s_0}$. On the half-plane $\sigma \geq \frac{3}{2}$, which contains the disc $|s - s_0| \leq r_0$, this series represents g, and

$$|g(s)| \leq \sum_p \sum_{k \geq 1} \frac{p^{-3k/2}}{k} = \log \zeta(\tfrac{3}{2}) \qquad (|s - s_0| \leq r_0). \tag{A.3}$$

For $\sigma > 0$, $s \neq 1$, partial summation gives [36, Section 2.1]

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\}x^{-s-1}\,\mathrm{d}x,$$

so that $|\zeta(s)| \le |s|/|s-1| + |s|/\sigma$. On D_R we have $\sigma \ge \frac12$, $|s-1| \ge 1$ and $|s| \le 4 + |t|$, hence $|\zeta(s)| \le 3(4+|t|)$ and

$$\max_{s\in D_R} \operatorname{Re} g(s) = \max_{s\in D_R} \log|\zeta(s)| \le \log(12+3|t|).$$

By (A.1) with $r = r_2$, $R - r_2 = \frac14\delta$, and (A.3) at s_0,

$$\mathcal{M}(r_2) \le \frac{8r_2}{\delta}\log(12+3|t|) + \frac{4(R+r_2)}{\delta}\log\zeta(\tfrac32) \le C_1\log(2+|t|),$$

with $C_1 = C_1(\delta)$. By (A.2) and (A.3),

$$\mathcal{M}(r_1) \le \big(\log\zeta(\tfrac32)\big)^{1-\alpha}\big(C_1\log(2+|t|)\big)^{\alpha} \le C_2\big(\log(2+|t|)\big)^{\alpha},$$

where $\alpha = \log(2r_1)/\log(2r_2) < 1$ depends only on θ and δ. By the maximum principle the same bound holds on the whole disc $|s - s_0| \le r_1$, which contains the segment $\theta + \delta \le \sigma \le 2$ at height t. There

$$\Big|\frac{1}{\zeta(s)}\Big| = e^{-\operatorname{Re} g(s)} \le \exp\big(C_2(\log(2+|t|))^{\alpha}\big) \le C_3(1+|t|)^{\eta},$$

because $C_2u^{\alpha} \le \eta u + C_4$ for all $u \ge 0$, with $C_4 = C_4(C_2,\alpha,\eta)$. □

The truncated Perron formula is used in the following form [28, Section 5.1]; compare [36, Lemma 3.12]. Let $\sum_n a_n n^{-s}$ converge absolutely for $\sigma > 1$ to $D(s)$. For $c > 1$, $T \ge 1$ and $x \ge 1$ not an integer,

$$\sum_{n\le x} a_n = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} D(s)\frac{x^s}{s}\,\mathrm{d}s + O\Big(\sum_{n\ge1}|a_n|\Big(\frac{x}{n}\Big)^c \min\Big\{1, \frac{1}{T|\log(x/n)|}\Big\}\Big), \tag{A.4}$$

with an absolute implied constant.

Proposition A.3 (Perron calculation)**.** *Let $\frac12 \le \theta < 1$ and suppose $\zeta(s) \neq 0$ for $\operatorname{Re} s > \theta$. Then $L(x) \ll_{\theta,\varepsilon} x^{\theta+\varepsilon}$ for every $\varepsilon > 0$ and all $x \ge 1$.*

Proof. Fix $0 < \varepsilon < 1$, and choose

$$0 < \delta < \min(\tfrac14, \tfrac12\varepsilon), \qquad \sigma_0 = \theta + \delta, \qquad 0 < \eta < \tfrac14\varepsilon.$$

Let $x = k + \frac12$ with an integer $k \ge 3$, and put $c = 1 + 1/\log x$ and $T = x^2$. Then $1 < c < 2$ and $x^c = ex$. Let $D(s) = \zeta(2s)/\zeta(s)$.

The integrand. On the half-plane $\sigma \ge \sigma_0$, $|\zeta(2s)| \le \zeta(2\sigma_0)$, a constant depending on θ and δ, because $2\sigma_0 > 1$. The function D is holomorphic on $\sigma > \theta$: the only pole of $\zeta(2s)$ is at $s = \frac12 < \sigma_0$, and ζ has no zero. By Lemma A.2,

$$|D(\sigma+it)| \le C_5(1+|t|)^{\eta} \qquad (\sigma_0 \le \sigma \le 2), \tag{A.5}$$

with $C_5 = C_5(\theta,\delta,\eta)$.

The Perron error. In (A.4) with $a_n = \lambda(n)$ we have $|a_n| = 1$. For $x/2 < n < 2x$, $(x/n)^c \le 2^c \le 4$ and

$$\Big|\log\frac{x}{n}\Big| = \Big|\int_n^x \frac{\mathrm{d}v}{v}\Big| \ge \frac{|x-n|}{\max(x,n)} \ge \frac{|x-n|}{2x}.$$

Since x is a half-integer, the numbers $|x-n|$ for n in this range are of the form $j + \frac12$ with $0 \le j \le x$, each value occurring at most twice. These n contribute at most

$$4 \cdot \frac{2x}{T} \cdot 2\sum_{0\le j\le x}\frac{1}{j+1/2} \le \frac{16x}{T}(3+\log x).$$

For $n \le x/2$ or $n \ge 2x$ we have $|\log(x/n)| \ge \log 2$, and these n contribute at most

$$\frac{x^c}{T\log 2}\sum_{n\ge 1} n^{-c} = \frac{ex\,\zeta(c)}{T\log 2} \le \frac{ex(1+\log x)}{T\log 2},$$

since $\zeta(c) \le 1 + 1/(c-1)$. With $T = x^2$ the error term in (A.4) is $O(x^{-1}\log x) = O(1)$.

The contour shift. The function $D(s)x^s/s$ is holomorphic on a neighbourhood of the rectangle with vertices $\sigma_0 \pm iT$, $c \pm iT$. By Cauchy's theorem the integral over $[c-iT, c+iT]$ equals the integral over $[\sigma_0 - iT, \sigma_0 + iT]$ plus the integrals over the two horizontal sides. On the horizontal sides $|s| \ge T$, and by (A.5)

$$\Big|\int_{\sigma_0\pm iT}^{c\pm iT} D(s)\frac{x^s}{s}\,\mathrm{d}s\Big| \le 2\,C_5(1+T)^{\eta}\frac{x^c}{T} \le C_6\,x^{1+2\eta-2} \le C_6.$$

On the left side $|s| \ge \max(\frac12, |t|) \ge \frac14(1+|t|)$, so

$$\begin{aligned}\Big|\int_{\sigma_0-iT}^{\sigma_0+iT} D(s)\frac{x^s}{s}\,\mathrm{d}s\Big| &\le 4\,C_5\,x^{\sigma_0}\int_{-T}^{T}(1+|t|)^{\eta-1}\,\mathrm{d}t\\ &\le \frac{8C_5}{\eta}\,x^{\sigma_0}(1+T)^{\eta} \le C_7\,x^{\theta+\delta+2\eta}.\end{aligned}$$

Since $\delta + 2\eta < \varepsilon$, we obtain $|L(x)| \le C_8\,x^{\theta+\varepsilon}$ for half-integers $x \ge \frac72$, with $C_8 = C_8(\theta,\varepsilon)$.

All x. For real $x \ge 3$, $L(x) = L(\lfloor x\rfloor + \frac12)$ and $\lfloor x\rfloor + \frac12 \le 2x$, so $|L(x)| \le 2C_8 x^{\theta+\varepsilon}$. For $1 \le x < 3$, $|L(x)| \le 3$. For $\varepsilon \ge 1$ the bound is trivial. □

A.4. **The weight K and the Gram formulas.** Let $\operatorname{Re} z, \operatorname{Re} z' > -\frac12$. From $w_z(u) = (u^z, 1-u^z)$ and $\int_0^1 u^p\,\mathrm{d}u = 1/(1+p)$ for $\operatorname{Re} p > -1$,

$$\begin{aligned}\langle w_z, w_{z'}\rangle &= \int_0^1 \Big(u^{z+\overline{z'}} + (1-u^z)(1-u^{\overline{z'}})\Big)\,\mathrm{d}u\\ &= 1 - \frac{1}{1+z} - \frac{1}{1+\overline{z'}} + \frac{2}{1+z+\overline{z'}}.\end{aligned} \tag{A.6}$$

With $z = z' = s = \sigma + i\tau$ this gives

$$K(s) = 1 - 2\,\operatorname{Re}\frac{1}{1+s} + \frac{2}{1+2\sigma} = 1 + \frac{2}{1+2\sigma} - \frac{2(1+\sigma)}{(1+\sigma)^2+\tau^2},$$

which is (3.5). Since $\overline{w_z} = w_{\bar z}$, we have $\operatorname{Re} w_z = \frac12(w_z + w_{\bar z})$, and (A.6) applied to the four pairs gives

$$\begin{aligned}\langle \operatorname{Re} w_z, \operatorname{Re} w_{z'}\rangle &= \frac14 \sum_{\zeta_1\in\{z,\bar z\}}\sum_{\zeta_2\in\{z',\overline{z'}\}}\Big(1 - \frac{1}{1+\zeta_1} - \frac{1}{1+\overline{\zeta_2}} + \frac{2}{1+\zeta_1+\overline{\zeta_2}}\Big)\\ &= 1 - \operatorname{Re}\frac{1}{1+z} - \operatorname{Re}\frac{1}{1+z'} + \operatorname{Re}\frac{1}{1+z+z'} + \operatorname{Re}\frac{1}{1+z+\overline{z'}},\end{aligned}$$

which is (5.9). Two special cases are used in Corollary 5.4 with $J = 1$ and $\rho = \frac12 + i\gamma$:

$$\begin{aligned}\|\operatorname{Re} w_\rho\|^2 &= \frac32 + \operatorname{Re}\frac{1}{1+2\rho} - 2\,\operatorname{Re}\frac{1}{1+\rho},\\ \langle w_{1/2}, \operatorname{Re} w_\rho\rangle &= \frac13 - \operatorname{Re}\frac{1}{1+\rho} + 2\,\operatorname{Re}\frac{1}{3/2+\rho}.\end{aligned}$$

Proof of (3.6). Let $\sigma \ge \frac12$. The subtracted term in (3.5) is positive and $2/(1+2\sigma) \le 1$, so $K(s) < 2$. Since $K(\frac12 + i\tau) \to 2$ as $|\tau| \to \infty$, the value 2 is the supremum, and it is not attained. For fixed σ the subtracted term is largest at $\tau = 0$, where

$$K(\sigma) = 1 + \frac{2}{1+2\sigma} - \frac{2}{1+\sigma} = 1 - \varphi(\sigma), \qquad \varphi(\sigma) = \frac{2\sigma}{(1+2\sigma)(1+\sigma)}.$$

Dividing numerator and denominator by σ, $\varphi(\sigma) = 2/(\sigma^{-1} + 3 + 2\sigma)$. By the inequality of the arithmetic and geometric means, $\sigma^{-1} + 2\sigma \geq 2\sqrt{2}$, with equality only at $\sigma = 1/\sqrt{2}$. Hence

$$\varphi(\sigma) \leq \frac{2}{3 + 2\sqrt{2}} = 6 - 4\sqrt{2}, \qquad K(s) \geq K(\sigma) \geq 4\sqrt{2} - 5 = 0.65685\ldots,$$

with equality only for $\tau = 0$ and $\sigma = 1/\sqrt{2}$. Finally $K(\frac{1}{2}) = 1 + 1 - \frac{4}{3} = \frac{2}{3}$, and for $\sigma = \frac{1}{2}$ the formula (3.5) reads $K(\frac{1}{2} + i\tau) = 2 - 2\,\mathrm{Re}(\frac{3}{2} + i\tau)^{-1}$.

A.5. **Kernel estimates for Proposition 4.1.** We use the following standard bound [36, Section 5.1]: for all real $\kappa_1 < \kappa_2$ there is $A_0 = A_0(\kappa_1, \kappa_2)$ such that

$$|(s-1)\zeta(s)| \ll_{\kappa_1,\kappa_2} (1 + |t|)^{A_0} \qquad (\kappa_1 \leq \sigma \leq \kappa_2). \tag{A.7}$$

Let $\rho_0 = \beta_0 + i\gamma_0$, ν, W, w, G and $I(Y)$ be as in Proposition 4.1 and its proof. When $\rho_0 = \frac{1}{2}$ we put $\nu = 0$ in the formulas for W, so that in both cases

$$W(s) = \frac{s(s-1)\zeta(s)}{(s-\rho_0)^\nu}\, e^{(s-\rho_0)^2}.$$

Lemma A.4. *The statements* (K1)–(K4) *in the proof of Proposition 4.1 hold.*

Proof. The function W. The function $(s-1)\zeta(s)$ is entire, and when $\zeta(\rho_0) = 0$ it has a zero of order ν at $\rho_0 \neq 1$. So W is entire. For $s = \sigma + it$,

$$\bigl|e^{(s-\rho_0)^2}\bigr| = e^{(\sigma-\beta_0)^2 - (t-\gamma_0)^2}.$$

Let $\kappa_1 < \kappa_2$. For $|s - \rho_0| \geq 1$ the denominator $(s-\rho_0)^\nu$ has modulus at least 1, and on $|s - \rho_0| \leq 1$ the entire function W is bounded. With (A.7) this gives

$$|W(\sigma + it)| \leq C(\kappa_1, \kappa_2)\, e^{-t^2/2} \qquad (\kappa_1 \leq \sigma \leq \kappa_2,\ t \in \mathbb{R}), \tag{A.8}$$

because $(1 + |t|)^{A_0+1} e^{-(t-\gamma_0)^2} \ll e^{-t^2/2}$.

(K1). By (A.8) the integral defining $w(u)$ converges absolutely for every κ. For $\kappa_1 < \kappa_2$, Cauchy's theorem on the rectangle with vertices $\kappa_1 \pm iT$, $\kappa_2 \pm iT$, together with (A.8) on the horizontal sides as $T \to \infty$, shows that the integrals over $\mathrm{Re}\, s = \kappa_1$ and $\mathrm{Re}\, s = \kappa_2$ agree. On the line $\mathrm{Re}\, s = \kappa$ we have $|u^{-s}| = u^{-\kappa}$, hence

$$|w(u)| \leq \frac{u^{-\kappa}}{2\pi} \int_{-\infty}^{\infty} |W(\kappa + it)|\, \mathrm{d}t.$$

Taking $\kappa = A$ and $\kappa = -A$ gives $|w(u)| \leq C_A \min(u^A, u^{-A})$. With $A = 1$, $|w(u)|^2 u^{-2} \leq C_1^2 \min(1, u^{-4})$, which is integrable on $(0, \infty)$; so $C_w < \infty$.

(K2). Take $\kappa = 2$ in the definition of w. Since $|L(x)| \leq x$,

$$\int_1^\infty \int_{-\infty}^\infty |L(x)|\, |W(2+it)| \Big(\frac{x}{Y}\Big)^{-2} \mathrm{d}t\, \frac{\mathrm{d}x}{x} \leq Y^2 \int_1^\infty x^{-2}\, \mathrm{d}x \int_{-\infty}^\infty |W(2+it)|\, \mathrm{d}t < \infty.$$

By Fubini's theorem and (3.1),

$$\begin{aligned} I(Y) &= \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} W(s)\, Y^s \Big(\int_1^\infty L(x) x^{-s-1}\, \mathrm{d}x \Big)\, \mathrm{d}s \\ &= \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} W(s) M(s)\, Y^s\, \mathrm{d}s. \end{aligned}$$

(K3). By the definitions,

$$G(s) = W(s)M(s) = \frac{s(s-1)\zeta(s)}{(s-\rho_0)^\nu}\, e^{(s-\rho_0)^2} \cdot \frac{\zeta(2s)}{s\,\zeta(s)} = \frac{(s-1)\zeta(2s)}{(s-\rho_0)^\nu}\, e^{(s-\rho_0)^2},$$

first where $\zeta(s) \neq 0$ and then as an identity of meromorphic functions. The function $(s-1)\zeta(2s)$ is holomorphic on $\mathbb{C}$ except for a simple pole at $s = \frac{1}{2}$. If $\zeta(\rho_0) = 0$ with $\beta_0 > \frac{1}{2}$, the only pole of G in $\operatorname{Re} s > \frac{1}{2}$ is at ρ_0, and it has order exactly ν, since the numerator

$$h(s) = (s-1)\zeta(2s)e^{(s-\rho_0)^2}$$

satisfies $h(\rho_0) = (\rho_0 - 1)\zeta(2\rho_0) \neq 0$: indeed $\operatorname{Re}(2\rho_0) > 1$ and $\rho_0 \neq 1$. If $\rho_0 = \frac{1}{2}$, then $G = h$ has the single simple pole $s = \frac{1}{2}$, with

$$\lim_{s \to 1/2} (s - \tfrac{1}{2})\, G(s) = (\tfrac{1}{2} - 1) \cdot \tfrac{1}{2} \cdot e^0 = -\tfrac{1}{4}, \tag{A.9}$$

because $\zeta(2s) \sim 1/(2s-1)$ as $s \to \frac{1}{2}$.

(K4), the case $\zeta(\rho_0) = 0$. Let $b = \frac{1}{2}(\frac{1}{2} + \beta_0)$, so that $\frac{1}{2} < b < \beta_0$. On the strip $b \leq \sigma \leq 2$ we have $|\zeta(2s)| \leq \zeta(2b)$, and therefore, for $|s - \rho_0| \geq \frac{1}{2}(\beta_0 - b)$,

$$|G(\sigma + it)| \leq C(1 + |t|)\, e^{(2-\beta_0)^2 + \beta_0^2}\, e^{-(t-\gamma_0)^2} \qquad (b \leq \sigma \leq 2). \tag{A.10}$$

By the residue theorem on the rectangle with vertices $b \pm iT$, $2 \pm iT$, with $T \to \infty$, and by (K2),

$$I(Y) = \operatorname*{Res}_{s=\rho_0} \bigl(G(s)Y^s\bigr) + \frac{1}{2\pi i} \int_{b-i\infty}^{b+i\infty} G(s)Y^s \, \mathrm{d}s.$$

By (A.10) the last integral is at most $C_b Y^b$ in modulus. Since $G(s)Y^s = h(s)Y^s(s-\rho_0)^{-\nu}$ and $\frac{\mathrm{d}^k}{\mathrm{d}s^k} Y^s = (\log Y)^k Y^s$, Leibniz's rule gives

$$\begin{aligned} \operatorname*{Res}_{s=\rho_0} \bigl(G(s)Y^s\bigr) &= \frac{1}{(\nu-1)!} \frac{\mathrm{d}^{\nu-1}}{\mathrm{d}s^{\nu-1}} \bigl[h(s)Y^s\bigr]_{s=\rho_0} \\ &= Y^{\rho_0} \sum_{k=0}^{\nu-1} \frac{h^{(\nu-1-k)}(\rho_0)}{k!\,(\nu-1-k)!} (\log Y)^k. \end{aligned}$$

The term $k = \nu - 1$ is $h(\rho_0)(\log Y)^{\nu-1} Y^{\rho_0}/(\nu-1)!$, and the other terms are smaller by a factor $O(1/\log Y)$. Hence there is Y_1 such that, for $Y \geq Y_1$,

$$\Bigl|\operatorname*{Res}_{s=\rho_0} \bigl(G(s)Y^s\bigr)\Bigr| \geq \frac{|h(\rho_0)|}{2\,(\nu-1)!}\, (\log Y)^{\nu-1}\, Y^{\beta_0} \geq \frac{|h(\rho_0)|}{2\,(\nu-1)!}\, Y^{\beta_0},$$

the last step for $Y \geq e$. Since $b < \beta_0$, the term $C_b Y^b$ is at most half of this for large Y. This gives (K4) with $\kappa_0 = |h(\rho_0)|/(4(\nu-1)!)$.

(K4), the case $\rho_0 = \frac{1}{2}$. Here $G(s) = (s-1)\zeta(2s)e^{(s-1/2)^2}$. By (A.7) applied to $\zeta(2s)$, the function $(2s-1)\zeta(2s)$ is bounded by a power of $1 + |t|$ on the strip $\frac{1}{4} \leq \sigma \leq 2$, so $|G(\sigma + it)| \ll e^{-t^2/2}$ there for $|s - \frac{1}{2}| \geq \frac{1}{8}$. The residue theorem on the rectangle with vertices $\frac{1}{4} \pm iT$, $2 \pm iT$, and (A.9), give

$$I(Y) = -\frac{Y^{1/2}}{4} + \frac{1}{2\pi i} \int_{1/4-i\infty}^{1/4+i\infty} G(s)Y^s \, \mathrm{d}s = -\frac{Y^{1/2}}{4} + O\bigl(Y^{1/4}\bigr).$$

So $|I(Y)| \geq \frac{1}{8} Y^{1/2}$ for large Y, which is (K4) with $\beta_0 = \frac{1}{2}$ and $\kappa_0 = \frac{1}{8}$. □

Remark A.5. The Gaussian factor in W serves only to make all contour integrals absolutely convergent; any entire factor of rapid decay in vertical strips that does not vanish at ρ_0 would do. The factor $s(s-1)\zeta(s)(s-\rho_0)^{-\nu}$ is the essential one. Table 20 lists what each factor of W removes from $M(s) = \zeta(2s)/(s\zeta(s))$.

Appendix B. Calculations for the obstructions

This appendix contains the computations referred to in Sections 6–11. Numerical values appear only in tables and serve as checks; no proof depends on them.

TABLE 20. The factors of the kernel W in (4.2) and their effect on $G = WM$.

factor of $W(s)$	effect on $G(s) = W(s)\,\zeta(2s)/(s\,\zeta(s))$
s	cancels the factor $1/s$ of M
$(s-1)\zeta(s)$	entire; cancels $1/\zeta(s)$, hence all poles of M at zeros of ζ
$(s-\rho_0)^{-\nu}$	restores the pole at the chosen zero ρ_0, with its full order ν
$e^{(s-\rho_0)^2}$	decay $e^{-(t-\gamma_0)^2}$ on vertical lines; equal to 1 at ρ_0

B.1. **Rounding controls.** *The discrepancy identity* (10.2). List the primes above P as $p_1 < p_2 < \cdots$ and put $S_0 = 0$, $S_i = S(p_i) = S_{i-1} + q_{p_i}$. By (10.1), $I_{p_i} = \lfloor S_i \rfloor - \lfloor S_{i-1} \rfloor$, and since $0 < q_{p_i} < 1$ this difference is 0 or 1. Summing over $i \le n$ gives $\sum_{i\le n} I_{p_i} = \lfloor S_n \rfloor$, hence

$$\sum_{i\le n} (I_{p_i} - q_{p_i}) = \lfloor S_n \rfloor - S_n \in (-1, 0].$$

Both sides of (10.2) are constant between consecutive primes, which gives the identity for all real $t \ge P$.

The centred series (10.4). For $T > P$ and $\operatorname{Re} s > 0$, Abel summation gives

$$\sum_{P<p\le T} (I_p - q_p) p^{-s} = E(T)T^{-s} + s\int_P^T E(t)t^{-s-1}\,\mathrm{d}t.$$

Since $|E| \le 1$, the first term tends to zero and the integral converges absolutely as $T \to \infty$, uniformly for $\operatorname{Re} s \ge \sigma_1 > 0$ and $|s|$ bounded. This proves (10.4), the holomorphy of A on $\operatorname{Re} s > 0$, and the bound $|A(s)| \le |s| P^{-\operatorname{Re} s}/\operatorname{Re} s$.

The logarithmic factorization (10.5). For $\operatorname{Re} s > 1$ all the following series converge absolutely. The Euler products give

$$\log G_P(s) = \sum_{p\in\mathcal{S}_P} \big(\log(1+p^{-s}) - \log(1-p^{-s})\big) = 2\sum_{p>P} I_p \sum_{k\ \text{odd}} \frac{p^{-ks}}{k},$$

$$\log\zeta(s+1-\beta) = \sum_p \sum_{k\ge1} \frac{p^{-k(s+1-\beta)}}{k}.$$

In the difference, the terms with $k = 1$ are

$$2\sum_{p>P} I_p p^{-s} - \sum_p p^{-(s+1-\beta)} = 2\sum_{p>P} (I_p - q_p)p^{-s} - \sum_{p\le P} p^{-(s+1-\beta)},$$

because $2q_p p^{-s} = p^{-(s+1-\beta)}$. The first sum is $2A(s)$. The terms with $k \ge 3$ in $\log G_P$ and $k \ge 2$ in $\log\zeta(s+1-\beta)$ are the two double series of (10.5). For $\operatorname{Re} s \ge \sigma > 1/2$ they are bounded in absolute value by

$$\frac{2}{3}\sum_p \frac{p^{-3\sigma}}{1-p^{-2\sigma}} \le \frac{4}{3}\sum_p p^{-3\sigma}, \qquad \frac{1}{2}\sum_p \frac{p^{-2(\sigma+1-\beta)}}{1-p^{-(\sigma+1-\beta)}} \le 2\sum_p p^{-2(\sigma+1-\beta)},$$

using $p^{-2\sigma} \le 1/2$ and $p^{-(\sigma+1-\beta)} \le 2^{-1/2}$. Both majorants converge, since $3\sigma > 3/2$ and $2(\sigma+1-\beta) > 3 - 2\beta > 1$.

Table 21 shows the first steps of the rounding and compares the counting function of $\mathcal{S}_P$ with $S(x)$ and with the leading term $x^\beta/(2\beta\log x)$; the convergence to the latter is slow, as for the prime number theorem itself.

B.2. **Euler-factor bounds.** Three elementary inequalities enter Lemma 10.3, Theorem 10.4 and Table 10.

Exponential bounds. For complex w, $|e^w - 1| \le \sum_{j\ge1} |w|^j/j! = e^{|w|} - 1$. For $0 \le x < 1$, comparing $\sum_{j\ge1} x^j/j!$ with $\sum_{j\ge1} x^j$ gives $e^x - 1 \le x/(1-x)$.

TABLE 21. The rounding (10.1) for $\beta = 3/4$, $P = 2$. Left: the first fourteen primes. Right: the counting function of $\mathcal{S}_P$.

p	q_p	$S(p)$	I_p	$E(p)$
3	0.3799	0.3799	0	−0.3799
5	0.3344	0.7143	0	−0.7143
7	0.3074	1.0217	1	−0.0217
11	0.2746	1.2962	0	−0.2962
13	0.2633	1.5596	0	−0.5596
17	0.2462	1.8058	0	−0.8058
19	0.2395	2.0453	1	−0.0453
23	0.2283	2.2736	0	−0.2736
29	0.2155	2.4891	0	−0.4891
31	0.2119	2.7010	0	−0.7010
37	0.2027	2.9037	0	−0.9037
41	0.1976	3.1013	1	−0.1013
43	0.1953	3.2965	0	−0.2965
47	0.1910	3.4875	0	−0.4875

x	$\pi_{\mathcal{S}_P}(x)$	$S(x)$	$\dfrac{x^\beta}{2\beta \log x}$
10^3	20	20.95	17.16
10^4	86	86.26	72.38
10^5	375	375.12	325.63
10^6	1711	1711.82	1525.96

A rational bound for C_r. For $0 < r < 1/2$, comparison with the integral gives $\zeta(2-2r) \le 1 + \int_1^\infty t^{-(2-2r)}\,\mathrm{d}t = (2-2r)/(1-2r)$, hence

$$C_r = \frac{\zeta(2-2r)}{(1-2r)^2} \le \frac{2-2r}{(1-2r)^3}.$$

The rows of Table 10. Let $d = 1-r-\beta$ and $P = 2^e$ with ed an integer. Then $\Delta(P, 1-r) \le 2P^{-d}/d = 2^{1-ed}/d$, and by (10.7) the largest of the differences $|\mathcal{J}_P^{(k)}(1) - \mathcal{J}_\lambda^{(k)}(1)|$, $k \le K$, is at most $M\Delta/(1-\Delta)$, where $M = C_r \max_{k\le K} k!\, r^{-k}$. All quantities in Table 22 are rational numbers, and the last column is rounded upwards.

TABLE 22. Rational data for Table 10.

β	r	d	K	$C_r \le$	$\max\limits_{k\le K} k!\, r^{-k}$	$M \le$	$\Delta \le$	bound
3/4	1/8	1/8	3	112/27	3072	114688/9	2^{-36}	$1.9 \cdot 10^{-7}$
3/4	1/8	1/8	0	112/27	1	112/27	2^{-16}	$6.4 \cdot 10^{-5}$
3/4	1/8	1/8	6	112/27	188743680	2348810240/3	2^{-56}	$1.1 \cdot 10^{-8}$
5/8	1/4	1/8	3	12	384	4608	2^{-36}	$6.8 \cdot 10^{-8}$
7/8	1/16	1/16	3	960/343	24576	23592960/343	2^{-55}	$2.0 \cdot 10^{-12}$

For instance, in the first row $P = 2^{320}$, $P^{-d} = 2^{-40}$, $\Delta \le 2 \cdot 8 \cdot 2^{-40} = 2^{-36}$, and $(114688/9) \cdot 2^{-36}/(1-2^{-36}) < 1.9 \cdot 10^{-7}$.

B.3. **The Vandermonde system.** In the proof of Theorem 10.6 the moments satisfy

$$|t_j| \le \sum_{n\ge 2} n^{\beta-2}(\log n)^j < \infty, \qquad \int_1^\infty x^{\beta-2}(\log x)^j\,\mathrm{d}x = \frac{j!}{(1-\beta)^{j+1}},$$

the integral indicating their size. Write $x_m = b_m 2^{-m}$ and $s_j = -t_j/(-\log 2)^j$. The system (10.9) becomes

$$\sum_{m=0}^{k-1} x_m\, m^j = s_j \qquad (0 \le j < k),$$

with the convention $0^0 = 1$. Its matrix $(m^j)_{0\le j,m<k}$ is a Vandermonde matrix with nodes $0, 1, \dots, k-1$ and determinant $\prod_{0\le m<n<k}(n-m) = \prod_{n=1}^{k-1} n! \neq 0$. To invert it, let

$$\ell_m(x) = \prod_{\substack{0\le n<k\\ n\neq m}} \frac{x-n}{m-n} = \sum_{j=0}^{k-1} \ell_{m,j} x^j$$

be the Lagrange polynomials of the nodes. Since $\sum_m m^i \ell_m(x)$ is the polynomial of degree less than k that interpolates x^i at the nodes, it equals x^i for $0 \le i < k$; comparing coefficients, $\sum_m m^i \ell_{m,j} = \delta_{ij}$. Hence

$$x_m = \sum_{j=0}^{k-1} \ell_{m,j} s_j, \qquad b_m = 2^m \sum_{j=0}^{k-1} \ell_{m,j} \frac{-t_j}{(-\log 2)^j} \tag{B.1}$$

is the unique solution. The coefficients $\ell_{m,j}$ are rational; Table 23 lists them for $k \le 4$. For $k = 1$ the system reads $b_0 = -t_0$, and the sequence is $a_1 = -\varepsilon_0 t_0$, $a_n = \varepsilon_0 n^{\beta-1}\cos(\gamma \log n)$ for $n \ge 2$, with $\varepsilon_0 = 1/(1+|t_0|)$. For $k = 2$, $b_0 = -t_0 - t_1/\log 2$ and $b_1 = 2t_1/\log 2$.

TABLE 23. Coefficients $\ell_{m,j}$ of the Lagrange polynomials on the nodes $0, \dots, k-1$.

k	m	$\ell_{m,0}$	$\ell_{m,1}$	$\ell_{m,2}$	$\ell_{m,3}$
2	0	1	-1		
2	1	0	1		
3	0	1	$-3/2$	$1/2$	
3	1	0	2	-1	
3	2	0	$-1/2$	$1/2$	
4	0	1	$-11/6$	1	$-1/6$
4	1	0	3	$-5/2$	$1/2$
4	2	0	$-3/2$	2	$-1/2$
4	3	0	$1/3$	$-1/2$	$1/6$

B.4. **Rudin–Shapiro blocks.** Let $p^{(m)}, q^{(m)} \in \{\pm 1\}^{2^m}$ be the coefficient vectors of P_m, Q_m. The recursion (8.1) reads

$$p^{(m+1)} = (p^{(m)}, q^{(m)}), \qquad q^{(m+1)} = (p^{(m)}, -q^{(m)}) \tag{B.2}$$

as concatenations. An induction on $m - j$ shows that every aligned block of length 2^j inside $p^{(m)}$ or $q^{(m)}$ is one of $\pm p^{(j)}, \pm q^{(j)}$, and the same holds in the infinite sequence r, whose first 2^m terms are $p^{(m)}$.

The complementary identity. For complex u, v, $|u+v|^2 + |u-v|^2 = 2|u|^2 + 2|v|^2$. With $u = P_m(z)$, $v = z^{2^m} Q_m(z)$ and $|z| = 1$ this gives $|P_{m+1}|^2 + |Q_{m+1}|^2 = 2(|P_m|^2 + |Q_m|^2)$, and (8.2) follows from $|P_0|^2 + |Q_0|^2 = 2$. In particular $|P_m(z)|, |Q_m(z)| \le 2^{(m+1)/2}$ on the unit circle. Writing $P_m(z)\overline{P_m(z)} = \sum_{|h|<2^m} c_P(h) z^h$ with $c_P(h) = \sum_t p_t^{(m)} p_{t+h}^{(m)}$ for $h \ge 0$, and similarly for Q_m, the identity (8.2) is equivalent to

$$c_P(h) + c_Q(h) = 0 \qquad (1 \le h < 2^m). \tag{B.3}$$

The sup norm. Let $N \ge 1$ and $2^J \le N < 2^{J+1}$. The binary expansion $N = \sum_{j\in\mathcal{B}} 2^j$, read from the largest exponent downwards, cuts $[1, N]$ into consecutive aligned blocks, one of each length 2^j, $j \in \mathcal{B}$. The exponential sum over an aligned block of length 2^j is a unimodular multiple of $\pm P_j(\mathrm{e}(\alpha))$ or $\pm Q_j(\mathrm{e}(\alpha))$. Therefore

$$\Big|\sum_{n\le N} r_n \mathrm{e}(n\alpha)\Big| \le \sum_{j=0}^{J} 2^{(j+1)/2} < \frac{2\cdot 2^{J/2}}{\sqrt{2}-1} = (2+2\sqrt{2})\, 2^{J/2} \le (2+2\sqrt{2})\sqrt{N}.$$

Fixed shifts. Let $h, Y \geq 1$. If $h > Y$ then $|C_{r,h}(Y)| \leq Y < \sqrt{hY}$. Let $h \leq Y$, choose k with $\sqrt{hY} \leq 2^k < 2\sqrt{hY}$ and put $D = 2^k$, so that $h \leq D$. Let $g = \lfloor Y/(2D) \rfloor$ and cut $[1, 2Dg]$ into the aligned blocks $\Lambda_i = (2D(i-1), 2Di]$, $1 \leq i \leq g$. The remaining starting points $n \in (2Dg, Y]$ number fewer than $2D$. Fix i and let Λ_i' and Λ_i'' be the two halves of Λ_i. By (B.2) and the block structure, r restricted to Λ_i' is $\pm p^{(k)}$ and r restricted to Λ_i'' is $\pm q^{(k)}$. Split the $n \in \Lambda_i$ into four classes:

- $n, n+h \in \Lambda_i'$: these contribute $c_P(h)$, the signs cancelling in the product;
- $n, n+h \in \Lambda_i''$: these contribute $c_Q(h)$;
- $n \in \Lambda_i'$, $n+h \in \Lambda_i''$: exactly h values of n;
- $n \in \Lambda_i''$, $n+h \notin \Lambda_i$: exactly h values of n.

Since $h \leq D$ there are no other cases. By (B.3) with $m = k$ the first two classes cancel when $h < D$, and both are empty when $h = D$. So $|\sum_{n \in \Lambda_i} r_n r_{n+h}| \leq 2h$, and

$$|C_{r,h}(Y)| \leq 2hg + 2D \leq \frac{hY}{D} + 2D \leq \sqrt{hY} + 4\sqrt{hY} = 5\sqrt{hY}.$$

The repeated block. Applying (B.2) twice,

$$p^{(m+2)} = (p^{(m)},\, q^{(m)},\, p^{(m)},\, -q^{(m)}).$$

For $H = 2^m$ and $1 \leq n \leq H$ this gives $r_{n+2H} = r_n$ and $r_{n+3H} = -r_{n+H}$. Hence

$$C_{r,2H}(H) = \sum_{n=1}^{H} r_n^2 = H, \qquad C_{r,2H}(2H) = H + \sum_{n=H+1}^{2H} r_n r_{n+2H} = 0.$$

The endpoint $Y = H$ satisfies $Y \leq N - h$ for $N = 3H$, $h = 2H$, so $C^*_{r,2H}(3H) \geq H$ and $\mathbb{E}_r(3H, H) \geq H^2$.

Table 24 lists the first terms and numerical checks of the three estimates.

TABLE 24. Rudin–Shapiro data. Top: $r_1, \dots, r_{16}$, grouped into the blocks $p^{(2)}, q^{(2)}, p^{(2)}, -q^{(2)}$. Bottom: computed values of $\sup_\alpha |\sum_{n \leq N} r_n \mathrm{e}(n\alpha)|/\sqrt{N}$ (bound $2 + 2\sqrt{2} < 4.83$) and of $\max |C_{r,h}(Y)|/\sqrt{hY}$ over $1 \leq h, Y \leq 2048$ (bound 5).

$p^{(2)}$	$q^{(2)}$	$p^{(2)}$	$-q^{(2)}$
$+\,+\,+\,-$	$+\,+\,-\,+$	$+\,+\,+\,-$	$-\,-\,+\,-$

N	10	100	1000	4096	5000
$\sup_\alpha \lvert A_N(\alpha)\rvert/\sqrt{N}$	1.90	1.92	1.63	1.41	2.06

$\max_{h,Y \leq 2048} \lvert C_{r,h}(Y)\rvert/\sqrt{hY}$	1.414
attained at (h, Y)	$(1, 2)$

B.5. **The Legendre kernel.** Define Λ_j by Rodrigues' formula

$$\Lambda_j(t) = \frac{1}{2^j j!} \frac{\mathrm{d}^j}{\mathrm{d}t^j} (t^2 - 1)^j.$$

It is a polynomial of degree j with leading coefficient $\ell_j = (2j)!/(2^j (j!)^2)$, and it has the parity of j.

Orthogonality and norm. The function $(t^2 - 1)^j$ vanishes to order j at ± 1. For a polynomial g, integrating by parts j times therefore gives

$$\int_{-1}^{1} \Lambda_j(t) g(t)\, \mathrm{d}t = \frac{(-1)^j}{2^j j!} \int_{-1}^{1} (t^2 - 1)^j g^{(j)}(t)\, \mathrm{d}t,$$

which vanishes if $\deg g < j$. Taking $g = \Lambda_j$, whose j-th derivative is $j!\, \ell_j$,

$$\int_{-1}^{1} \Lambda_j^2 = \frac{\ell_j}{2^j} \int_{-1}^{1} (1 - t^2)^j\, \mathrm{d}t.$$

Put $W_j = \int_{-1}^{1}(1-t^2)^j\,\mathrm{d}t$. One integration by parts gives $W_j = 2j\int_{-1}^{1} t^2(1-t^2)^{j-1}\,\mathrm{d}t = 2j(W_{j-1}-W_j)$, so $W_j = \frac{2j}{2j+1}W_{j-1}$ and $W_j = 2^{2j+1}(j!)^2/(2j+1)!$. Hence $\int_{-1}^{1}\Lambda_j^2 = 2/(2j+1)$. By Leibniz' rule the j-th derivative of $(t-1)^j(t+1)^j$ at $t=1$ is $j!\,2^j$, so $\Lambda_j(1)=1$.

Values at the origin. By parity $\Lambda_j(0)=0$ for odd j. For $j=2i$ the coefficient of t^{2i} in $(t^2-1)^{2i}$ is $(-1)^i\binom{2i}{i}$, so

$$\Lambda_{2i}(0) = \frac{(2i)!\,(-1)^i\binom{2i}{i}}{2^{2i}(2i)!} = (-1)^i\binom{2i}{i}4^{-i}.$$

Let $w_i = (\binom{2i}{i}4^{-i})^2$. Then $w_0=1$ and $w_{i+1} = w_i(2i+1)^2/(2i+2)^2$. We claim $w_i \le 1/(i+1)$. This holds for $i=0$, and if it holds for i then

$$w_{i+1} \le \frac{(2i+1)^2}{4(i+1)^3} \le \frac{1}{i+2},$$

because $(2i+1)^2(i+2) = 4i^3+12i^2+9i+2 \le 4i^3+12i^2+12i+4 = 4(i+1)^3$.

The kernel. By the substitution $t=4u$, the functions

$$\phi_j(u) = \sqrt{2(2j+1)}\,\Lambda_j(4u) \qquad (0\le j\le m)$$

form an orthonormal basis of the space of polynomials of degree at most m in $L^2[-1/4,1/4]$, and $K_m(v,u) = \sum_{j\le m}\phi_j(v)\phi_j(u)$. For $p=\sum_j c_j\phi_j$,

$$\int_{-1/4}^{1/4} K_m(0,u)p(u)\,\mathrm{d}u = \sum_j c_j\phi_j(0) = p(0),$$

$$|p(0)|^2 \le \sum_j |c_j|^2 \sum_j \phi_j(0)^2 = K_m(0,0)\int_{-1/4}^{1/4}|p|^2,$$

and $\int_{-1/4}^{1/4}K_m(0,u)^2\,\mathrm{d}u = K_m(0,0)$. This is (11.2). Finally

$$K_m(0,0) = 2\sum_{i\le m/2}(4i+1)w_i \le 2\sum_{i=0}^{\lfloor m/2\rfloor}\frac{4i+1}{i+1} \le 8\Big(\Big\lfloor\frac{m}{2}\Big\rfloor+1\Big) \le 4(m+2),$$

which is (11.3). Table 25 compares the two sides; the true growth is $K_m(0,0)\sim 4m/\pi$.

TABLE 25. The diagonal value $K_m(0,0)$ and the bound (11.3).

m	0	2	4	8	16	32	64
$K_m(0,0)$	2.000	4.500	7.031	12.112	22.291	42.658	83.400
$4(m+2)$	8	16	24	40	72	136	264

B.6. **Elementary prime sums.** We use Chebyshev's bounds in the form $\pi(t) \le 6t/\log t$ for $t\ge 2$ and $\vartheta(t) = \sum_{p\le t}\log p \le \pi(t)\log t \le 6t$ [1].

The sum $\sum_{p\le y}p^{-1/2}$. For $y\ge 4$, partial summation gives

$$\sum_{p\le y}p^{-1/2} = \frac{\pi(y)}{\sqrt{y}} + \frac{1}{2}\int_2^y \frac{\pi(t)}{t^{3/2}}\,\mathrm{d}t \le \frac{6\sqrt{y}}{\log y} + 3\int_2^y \frac{\mathrm{d}t}{\sqrt{t}\,\log t}.$$

On $[2,\sqrt{y}]$ use $\log t\ge\log 2$, and on $[\sqrt{y},y]$ use $\log t\ge\frac12\log y$:

$$\int_2^y \frac{\mathrm{d}t}{\sqrt{t}\log t} \le \frac{2y^{1/4}}{\log 2} + \frac{4\sqrt{y}}{\log y}.$$

Hence $\sum_{p\le y} p^{-1/2} \le 18\sqrt{y}/\log y + 9y^{1/4}$. For $0 < u \le 2^{-1/2}$, $\log(1+u) \le u$ and $-\log(1-u) \le u/(1-u)$ give $\log\frac{1+u}{1-u} \le \frac{2u}{1-u} < 7u$. Therefore

$$-\log c_y(1/2) = \sum_{p\le y} \log\frac{1+p^{-1/2}}{1-p^{-1/2}} \le \frac{126\sqrt{y}}{\log y} + 63y^{1/4},$$

and for $y = (\log X)^2$ the right side is $63\log X/\log\log X + 63\sqrt{\log X} = o(\log X)$.

The sum $P_u = \sum_{p\le u}(\log p)p^{-\sigma}$, $0 < \sigma < 1$. Partial summation with $\vartheta(t) \le 6t$ gives

$$P_u = \vartheta(u)u^{-\sigma} + \sigma\int_2^u \vartheta(t)t^{-\sigma-1}\,\mathrm{d}t \le 6u^{1-\sigma} + \frac{6\sigma}{1-\sigma}u^{1-\sigma} = \frac{6u^{1-\sigma}}{1-\sigma}.$$

This is the bound used in Proposition 11.6, where also

$$\sum_{p\le R}(\log p)p^{-\sigma}\log\frac{R}{p} = \sum_{p\le R}(\log p)p^{-\sigma}\int_p^R \frac{\mathrm{d}u}{u} = \int_2^R P_u\frac{\mathrm{d}u}{u} \le \frac{6R^{1-\sigma}}{(1-\sigma)^2}.$$

With the prime number theorem $\vartheta(t) \sim t$ in place of Chebyshev's bound the same partial summation gives $P_u \sim u^{1-\sigma}/(1-\sigma)$, as used in Theorem 11.5.

Appendix C. Growth in a zero-free half-plane

This appendix proves Lemma 13.3. The method is that of [36, §14.2]; the point to check is the uniformity in the character.

Let χ be primitive of conductor $q \ge 1$ and suppose $L(s,\chi)$ has no zero in $\operatorname{Re} s > \theta$, where $1/2 \le \theta < 1$. Let $0 < \delta \le 1/4$ and let t be real, with $|t| \ge 4$ if $q = 1$. Put $s_0 = 2 + it$ and

$$r_0 = \tfrac12, \qquad r_1 = 2-\theta-\delta, \qquad r_2 = 2-\theta-\tfrac{\delta}{2}, \qquad R = 2-\theta-\tfrac{\delta}{4},$$

so that $r_0 < r_1 < r_2 < R \le 3/2$. Every point $\sigma + it$ with $\theta + \delta \le \sigma \le 2$ lies in the closed disc of radius r_1 about s_0.

The logarithm. The closed disc $|s - s_0| \le R$ lies in $\operatorname{Re} s \ge \theta + \delta/4$, where $L(s,\chi)$ has no zero. It does not contain the point $s = 1$ when $q = 1$, because then $|\operatorname{Im} s| \ge 4 - R > 1$. So there is a holomorphic branch $G(s) = \log L(s,\chi)$ on a neighbourhood of this disc, equal to $\sum_p\sum_{k\ge1}\chi(p)^k p^{-ks}/k$ for $\operatorname{Re} s > 1$. On the disc of radius r_0 we have $\operatorname{Re} s \ge 3/2$ and therefore $|G(s)| \le \log\zeta(3/2) < 1$.

An upper bound for $\operatorname{Re} G$. If $q > 1$, the sums of χ over intervals are bounded by q in modulus, and partial summation gives, for $\operatorname{Re} s > 0$,

$$L(s,\chi) = s\int_1^\infty \Big(\sum_{n\le x}\chi(n)\Big)x^{-s-1}\,\mathrm{d}x, \qquad |L(s,\chi)| \le \frac{q|s|}{\operatorname{Re} s}.$$

If $q = 1$, the formula $\zeta(s) = s/(s-1) - s\int_1^\infty \{x\}x^{-s-1}\,\mathrm{d}x$ gives $|\zeta(s)| \le |s|/|s-1| + |s|/\operatorname{Re} s$. On the disc of radius R we have $|s| \le 4 + |t|$, $\operatorname{Re} s \ge 1/2$, and $|s-1| \ge 1$ when $q = 1$. In both cases

$$\operatorname{Re} G(s) = \log|L(s,\chi)| \le \Lambda := \log\big(6q(2+|t|)\big) \qquad (|s-s_0| \le R),$$

and $\Lambda \ge \log 12 > 1$.

Borel–Carathéodory. For a function G holomorphic on $|s - s_0| \le R$ and $0 < r < R$,

$$\max_{|s-s_0|\le r}|G(s)| \le \frac{2r}{R-r}\max_{|s-s_0|=R}\operatorname{Re} G(s) + \frac{R+r}{R-r}|G(s_0)|$$

[28, §6.1]. With $r = r_2$, $R - r_2 = \delta/4$ and $|G(s_0)| < 1 < \Lambda$, this gives $\max_{|s-s_0|\le r_2}|G| \le (24/\delta)\Lambda$.

Three circles. Let $\mathsf{M}(r) = \max_{|s-s_0|=r}|G(s)|$. Hadamard's theorem states that $\log\mathsf{M}(r)$ is a convex function of $\log r$; hence

$$\mathsf{M}(r_1) \le \mathsf{M}(r_0)^{1-\tau}\mathsf{M}(r_2)^{\tau}, \qquad \tau = \frac{\log(r_1/r_0)}{\log(r_2/r_0)} \in (0,1),$$

where τ depends only on θ and δ. With $\mathsf{M}(r_0) < 1$ we get $|G(s)| \le (24\Lambda/\delta)^\tau$ on the disc of radius r_1, by the maximum principle. Since $\tau < 1$, for every $\kappa > 0$ there is $C(\theta, \delta, \kappa)$ with $(24\Lambda/\delta)^\tau \le \kappa\Lambda + C(\theta, \delta, \kappa)$. Therefore, for $\theta + \delta \le \sigma \le 2$,

$$\bigl|\log|L(\sigma + it, \chi)|\bigr| \le |G(\sigma + it)| \le \kappa \log(6q(2 + |t|)) + C(\theta, \delta, \kappa),$$

which is the two-sided bound of the lemma. No constant depends on χ or q.

Small $|t|$ for $q = 1$. The reciprocal $1/\zeta$, extended by 0 at $s = 1$, is continuous on the compact set $\theta + \delta \le \sigma \le 2$, $|t| \le 4$, which contains no zero of ζ. It is therefore bounded there. This completes the proof.

Appendix D. Contour estimates for the smoothed sum

This appendix proves the estimates (12.5), (12.6) and (12.7) used for Theorem 12.4. The notation is that of the proof: $T = \log X \ge 3$, $\omega = c_1/\ell(T) \le \omega_0 \le 1/4$, and $\Phi(s) = \Gamma(s)\zeta(2s)/\zeta(s)$.

D.1. **The Gamma function on vertical lines.**

Lemma D.1 (Elementary Gamma bounds)**.** *For real t and $0 < \sigma \le 2$,*

$$|\Gamma(\sigma + it)| \le \Gamma(\sigma)\,|\Gamma(2 + it)|, \qquad |\Gamma(2 + it)|^2 = \frac{(1 + t^2)\,\pi t}{\sinh \pi t} \le 2\pi(1 + |t|)^3 \mathrm{e}^{-\pi|t|}.$$

Consequently, for $1/2 \le \sigma \le 2$ and all real t,

$$|\Gamma(\sigma + it)| \le 5\,(1 + |t|)^{3/2} \mathrm{e}^{-\pi|t|/2}.$$

Proof. The Weierstrass product $1/\Gamma(s) = s\mathrm{e}^{\gamma s} \prod_{n\ge1}(1 + s/n)\mathrm{e}^{-s/n}$ [28, Appendix C] gives, for $\sigma > 0$,

$$\left|\frac{\Gamma(\sigma)}{\Gamma(\sigma + it)}\right|^2 = \prod_{k\ge0}\Bigl(1 + \frac{t^2}{(\sigma + k)^2}\Bigr).$$

Each factor decreases as σ increases, so $|\Gamma(\sigma + it)|/\Gamma(\sigma)$ is nondecreasing in σ; comparing with $\sigma = 2$, where $\Gamma(2) = 1$, gives the first inequality. The reflection formula gives $|\Gamma(1 + it)|^2 = \Gamma(1 + it)\Gamma(1 - it) = \pi t/\sinh \pi t$, and $|\Gamma(2 + it)|^2 = (1 + t^2)|\Gamma(1 + it)|^2$. For $x \ge 0$ the inequality $\mathrm{e}^{2x} \ge 1 + x$ is equivalent to $x/\sinh x \le 2(1 + x)\mathrm{e}^{-x}$. With $x = \pi|t|$, and $1 + t^2 \le (1 + |t|)^2$, $1 + \pi|t| \le \pi(1 + |t|)$, this gives the bound for $|\Gamma(2 + it)|^2$. Finally $\Gamma(\sigma) \le \Gamma(1/2) = \sqrt{\pi}$ on $[1/2, 2]$, and $\sqrt{\pi} \cdot \sqrt{2\pi} = 4.44\ldots < 5$. □

The exponent $3/2$ is not optimal for $\sigma < 2$ but is harmless here. The decay $\mathrm{e}^{-\pi|t|/2}$ is the source of the saving: with $T = \log X$,

$$\mathrm{e}^{-\pi T/2} = X^{-\pi/2}, \qquad 2 - \frac{\pi}{2} = 0.42920\ldots < \frac{1}{2}.$$

D.2. **The region $|t| \le 3$.** The rectangle $\{1 - \omega_0 \le \sigma \le 2,\ |t| \le 3\}$ is compact and contains no zero of ζ, by the choice of ω_0. The function $1/\zeta$, with the value 0 at $s = 1$, is continuous on it. Let K_0 be its maximum there. For $|t| \ge 3$ and $\sigma \ge 1 - c_1/\ell(|t|)$, Lemma 12.3 and $\ell(|t|) \le \log|t|$ give $|1/\zeta(\sigma + it)| \le C_1 \log|t|$. Since $\omega \le \omega_0$ and $1 - \omega \ge 1 - c_1/\ell(|t|)$ for $3 \le |t| \le T$, we have, with $C_2 = \max(K_0, C_1)$,

$$\left|\frac{1}{\zeta(\sigma + it)}\right| \le C_2 \log(3 + |t|) \qquad (1 - \omega \le \sigma \le 2,\ |t| \le T). \tag{D.1}$$

In the same region $\mathrm{Re}(2s) \ge 2 - 2\omega \ge 3/2$, so $|\zeta(2s)| \le \zeta(3/2)$.

D.3. **The three estimates.** *The half-lines.* On $\operatorname{Re} s = 2$ we have $|\zeta(4+2it)| \le \zeta(4)$ and $|1/\zeta(2+it)| \le \sum_n \mu(n)^2 n^{-2} = \zeta(2)/\zeta(4)$, so $|\zeta(2s)/\zeta(s)| \le \zeta(2)$. By Lemma D.1,

$$\int_{\operatorname{Re} s=2,\ |t|\ge T} |\Phi(s)X^s|\,|\,\mathrm{d}s| \le 2\sqrt{2\pi}\,\zeta(2)X^2 \int_T^\infty (1+t)^{3/2}\mathrm{e}^{-\pi t/2}\,\mathrm{d}t.$$

Writing $t = T+u$ and using $1+T+u \le (1+T)(1+u)$,

$$\int_T^\infty (1+t)^{3/2}\mathrm{e}^{-\pi t/2}\,\mathrm{d}t \le (1+T)^{3/2}\mathrm{e}^{-\pi T/2}\int_0^\infty (1+u)^{3/2}\mathrm{e}^{-\pi u/2}\,\mathrm{d}u.$$

The last integral is an absolute constant, and $\mathrm{e}^{-\pi T/2} = X^{-\pi/2}$. This proves (12.5).

The horizontal segments. For $s = \sigma \pm iT$ with $1-\omega \le \sigma \le 2$, Lemma D.1, the bound $|\zeta(2s)| \le \zeta(3/2)$ and Lemma 12.3 at height T give

$$|\Phi(\sigma \pm iT)| \le 5(1+T)^{3/2}X^{-\pi/2}\,\zeta(3/2)\,C_1\ell(T).$$

The segment has length at most $5/4$ and $X^\sigma \le X^2$ on it. Since $(1+T)^{3/2}\ell(T) \ll T^{3/2}\log T \ll T^2$, this proves (12.6).

The left side. For $s = 1-\omega+it$ with $|t| \le T$, Lemma D.1 and (D.1) give

$$|\Phi(1-\omega+it)| \le 5\,\zeta(3/2)\,C_2\,(1+|t|)^{3/2}\log(3+|t|)\,\mathrm{e}^{-\pi|t|/2},$$

whose integral over the real line is an absolute constant. As $|X^s| = X^{1-\omega}$, this proves (12.7).

The three estimates give $B(X) \ll X^{1-\omega} + X^{2-\pi/2}(\log X)^2$, and

$$X^{-\omega} = \exp\Bigl(-\frac{c_1\log X}{(\log\log X)^{2/3}(\log\log\log X)^{1/3}}\Bigr),$$

which tends to zero more slowly than any negative power of X. The second term is smaller than the first for large X, and Theorem 12.4 follows.

Appendix E. Further proofs for the smoothed sum and its twists

Proof of Proposition 12.9. If the limit superior is infinite there is nothing to prove. Otherwise let C exceed it, so that $\mathcal{H}_B(X) \le CX$ for $X \ge X_1$. For $\kappa > 0$, Tonelli's theorem and the substitution $X = Y/t$ give

$$\int_0^\infty \mathcal{H}_B(X)X^{-1-\kappa}\frac{\mathrm{d}X}{X} = r(1+\kappa)P(\kappa),$$

where

$$r(s) = \int_{1/2}^2 t^s\frac{\mathrm{d}t}{t} = \frac{2^s-2^{-s}}{s}, \qquad P(\kappa) = \int_0^\infty |B(Y)|^2 Y^{-1-\kappa}\frac{\mathrm{d}Y}{Y}.$$

The part of the left side over $X \le X_1$ is bounded uniformly for $0 < \kappa \le 1$, by (12.1), and the part over $X \ge X_1$ is at most C/κ. Hence $P(\kappa) \le (C/\kappa + O(1))/r(1+\kappa)$. For $\delta > 0$ the Cauchy–Schwarz inequality gives

$$\Bigl|\int_1^\infty B(Y)Y^{-1/2-\delta}\frac{\mathrm{d}Y}{Y}\Bigr|^2 \le \int_1^\infty |B(Y)|^2Y^{-1-\delta}\frac{\mathrm{d}Y}{Y}\int_1^\infty Y^{-\delta}\frac{\mathrm{d}Y}{Y} \le \frac{P(\delta)}{\delta}.$$

The same inequality with absolute values inside shows that $\int_1^\infty B(Y)Y^{-s}\,\mathrm{d}Y/Y$ converges absolutely for $\operatorname{Re} s > 1/2$. It is therefore holomorphic there and, by (12.2), equals $\Phi(s)$ minus an entire function. Multiplying by δ^2 and letting $\delta \downarrow 0$, the residue (12.3) gives

$$c_0^2 \le \liminf_{\delta\downarrow 0}\,\delta\,P(\delta) \le \frac{C}{r(1)} = \frac{2C}{3}.$$

Let C decrease to the limit superior. □

Proof of Theorem 14.3. Let r be the limit superior. Every primitive χ carries a positive weight, so $B_\chi(X) \ll_{\chi,\varepsilon} X^{r/2+\varepsilon}$. As in (12.2),

$$\int_0^\infty B_\chi(X) X^{-s} \frac{\mathrm{d}X}{X} = \Gamma(s) \frac{L(2s,\chi^2)}{L(s,\chi)} \qquad (\operatorname{Re} s > 1),$$

and the integral is holomorphic in $\operatorname{Re} s > r/2$. A zero ρ of $L(s,\chi)$ with $\operatorname{Re}\rho > 1/2$ is a pole of the right side, because $\Gamma(\rho)L(2\rho,\chi^2) \neq 0$; hence $\operatorname{Re}\rho \le r/2$. For $q = 1$ the pole at $s = 1/2$ gives $r \ge 1$. Thus $r \ge 2\Theta_D$.

If $\Theta_D = 1$ the bound $A_*(X) \le \zeta(2)X^2$ gives $r \le 2$. Let $\Theta_D < 1$ and fix $\sigma \in (\Theta_D, 1)$ and $0 < \kappa < 1/2$. Lemma 13.3, with $\theta = \Theta_D$ and $\delta = \min(\sigma - \Theta_D, 1/4)$, gives

$$\left| \frac{1}{L(u+it,\chi)} \right| \le C'(\sigma,\kappa)\big(q(2+|t|)\big)^\kappa \qquad (\sigma \le u \le 2,\ t \in \mathbb{R})$$

for every primitive χ, and the constant is the same for all characters and all conductors. This uniformity is essential, since the bound is summed over q below. The analogue of (12.4) is

$$B_\chi(X) = \frac{1}{2\pi i} \int_{\operatorname{Re} s = 2} \Gamma(s) \frac{L(2s,\chi^2)}{L(s,\chi)} X^s \,\mathrm{d}s.$$

The integrand is holomorphic in the strip $\sigma \le \operatorname{Re} s \le 2$, where $|L(2s,\chi^2)| \le \zeta(2\sigma)$. By Lemma D.1, $|\Gamma(u+it)| \le 5(1+|t|)^{3/2}\mathrm{e}^{-\pi|t|/2}$ in the strip, so the horizontal integrals tend to zero and the line may be moved to $\operatorname{Re} s = \sigma$:

$$|B_\chi(X)| \le \frac{5\,\zeta(2\sigma)C'(\sigma,\kappa)}{2\pi}\, q^\kappa X^\sigma \int_{-\infty}^{\infty} (1+|t|)^{3/2}(2+|t|)^\kappa \mathrm{e}^{-\pi|t|/2}\,\mathrm{d}t.$$

This is (14.2). Since there are at most $\varphi(q)$ primitive characters modulo q,

$$A_*(X) \le C(\sigma,\kappa)^2 X^{2\sigma} \sum_{q \ge 1} q^{-2+2\kappa} \ll_{\sigma,\kappa} X^{2\sigma}.$$

Letting $\sigma \downarrow \Theta_D$ gives $r \le 2\Theta_D$. □

Appendix F. The Plancherel computation at the exponent one

This appendix supplies the details of the proof of Theorem 13.11. Write $\varepsilon(u) = \exp(-c_0'\eta(u))$ for $u \ge 3$, so that $|\mathcal{M}(u)| \le C_W u\,\varepsilon(u)$ by (13.9), and let $h(v) = v^{3/5}(\log v)^{-1/5}$, so that $\eta(u) = h(\log u)$. The function h is increasing for $v \ge \mathrm{e}^{1/3}$, and $h(v) \ge \sqrt{v}$ for all $v > 1$, because $\sqrt{v} \ge \log v$.

F.1. **The tail of the Möbius series on the line.** For a continuously differentiable ϕ and $3 \le M < Y$, partial summation gives

$$\sum_{M < n \le Y} \mu(n)\phi(n) = \mathcal{M}(Y)\phi(Y) - \mathcal{M}(M)\phi(M) - \int_M^Y \mathcal{M}(u)\phi'(u)\,\mathrm{d}u.$$

Take $\phi(u) = u^{-s}$ with $\operatorname{Re} s \ge 1$. Then $|\phi(u)| \le 1/u$ and $|\phi'(u)| \le |s|u^{-2}$, and letting $Y \to \infty$,

$$|R_M(s)| \le C_W\,\varepsilon(M) + C_W|s| \int_M^\infty \varepsilon(u) \frac{\mathrm{d}u}{u} \qquad (\operatorname{Re} s \ge 1). \tag{F.1}$$

The integral converges, as shown below, so the series $\sum_n \mu(n)n^{-s}$ converges uniformly on compact subsets of $\operatorname{Re} s \ge 1$. Its sum is continuous there and equals $1/\zeta(s)$ for $\operatorname{Re} s > 1$; hence it equals $1/\zeta(s)$ on the line $\operatorname{Re} s = 1$ as well, with the value 0 at $s = 1$. With $\phi(u) = u^{-s}\log u$, for which $|\phi'(u)| \le (|s|\log u + 1)u^{-2}$, the same computation bounds the termwise derivative:

$$|R_M'(s)| \le C_W\,\varepsilon(M)\log M + C_W \int_M^\infty \varepsilon(u)\big(|s|\log u + 1\big)\frac{\mathrm{d}u}{u} \qquad (\operatorname{Re} s \ge 1).$$

The differentiated series converges uniformly on compact subsets of the line, so $t \mapsto R_M(1+it)$ is continuously differentiable with derivative $iR'_M(1+it)$. Put

$$\varepsilon_1(M) = \varepsilon(M)\log M + \int_M^\infty \varepsilon(u)(1+\log u)\frac{\mathrm{d}u}{u}.$$

Then $|R_M(1+it)| + |R'_M(1+it)| \le 4C_W(1+|t|)\varepsilon_1(M)$, which is the first part of (13.10).

For the second part let $M \ge 5$, so that $\log M \ge \mathrm{e}^{1/3}$. The substitution $v = \log u$ and the inequalities $h(v) \ge h(\log M)$ and $h(v) \ge \sqrt{v}$ for $v \ge \log M$ give

$$\begin{aligned}\int_M^\infty \varepsilon(u)(1+\log u)\frac{\mathrm{d}u}{u} &= \int_{\log M}^\infty \mathrm{e}^{-c_0' h(v)}(1+v)\,\mathrm{d}v \\ &\le \mathrm{e}^{-c_0'\eta(M)/2}\int_0^\infty \mathrm{e}^{-c_0'\sqrt{v}/2}(1+v)\,\mathrm{d}v.\end{aligned}$$

Also $\varepsilon(M)\log M = \exp(-c_0'\eta(M) + \log\log M) \ll \exp(-c_0'\eta(M)/2)$, because $\eta(M) \ge \sqrt{\log M}$. Hence $\varepsilon_1(M) \ll \exp(-c_1'\eta(M))$ with $c_1' = c_0'/2$.

F.2. **The three ranges.** On $\operatorname{Re} s = 1$, $s \ne 1$, we have $\sum_{n\le M}\mu(n)n^{-s} = 1/\zeta(s) - R_M(s)$, and at $s = 1$ the sum equals $-R_M(1)$. By (13.2) with $w_n = 1$,

$$U_M(s) = \frac{1 - \zeta(s)\bigl(1/\zeta(s) - R_M(s) + R_M(1)\bigr)}{s} = \frac{\zeta(s)\bigl(R_M(s) - R_M(1)\bigr)}{s}.$$

Write $s = 1 + it$ and $\varepsilon_1 = \varepsilon_1(M)$.

For $0 < |t| \le 1$: $|R_M(1+it) - R_M(1)| \le |t| \max_{|\tau|\le 1}|R'_M(1+i\tau)| \le 8C_W|t|\varepsilon_1$, and $|t\zeta(1+it)|$ is bounded on this range, so $|U_M(1+it)| \ll \varepsilon_1$.

For $1 \le |t| \le T$: $|R_M(1+it) - R_M(1)| \ll (1+|t|)\varepsilon_1$, $|\zeta(1+it)| \ll \log(2+|t|)$ and $|s| \ge |t|$, so $|U_M(1+it)| \ll \varepsilon_1\log(2+|t|)$.

For $|t| > T \ge 1$: from the first form of U_M and $|\sum_{n\le M}\mu(n)(n^{-s} - n^{-1})| \le 2(1+\log M)$,

$$|U_M(1+it)| \le \frac{1 + 2(1+\log M)|\zeta(1+it)|}{|t|} \ll \frac{(\log M)\log(2+|t|)}{|t|}.$$

The function $u_M^\sharp$ is bounded by $1 + M$, so Lemma 13.2 applies with $\sigma = 1$:

$$\begin{aligned}\|u_M^\sharp\|_1^2 &= \frac{1}{2\pi}\int_{-\infty}^\infty |U_M(1+it)|^2\,\mathrm{d}t \\ &\ll \varepsilon_1^2 + \varepsilon_1^2\int_1^T \log^2(2+t)\,\mathrm{d}t + (\log M)^2\int_T^\infty \frac{\log^2(2+t)}{t^2}\,\mathrm{d}t.\end{aligned}$$

The two integrals are $O(T\log^2(2+T))$ and $O(\log^2(2+T)/T)$. With $T = 1/\varepsilon_1$,

$$\|u_M^\sharp\|_1^2 \ll \varepsilon_1\,(\log M)^2\log^2(2/\varepsilon_1) \ll \exp\bigl(-c_1'\eta(M)\bigr)(\log M)^4,$$

since $\log(2/\varepsilon_1) \ll \eta(M) \le \log M$. As $(\log M)^4 = \exp(4\log\log M)$ and $\eta(M) \ge \sqrt{\log M}$, the right side is $O(\exp(-c\,\eta(M)))$ for any fixed $c < c_1'$. This assumes $M \ge 5$ and $\varepsilon_1(M) \le 1$, which holds for $M \ge M_0$ with an absolute M_0; for $3 \le M < M_0$ the bound holds after adjusting the implied constant, since $\|u_M^\sharp\|_1 \le 1 + M$.

Appendix G. Proofs for the reciprocal-dilate problem

This appendix contains the proofs deferred from Section 13. The notation is that of the corresponding statements.

G.1. **Fixed tapers.**

Proof of Theorem 13.12. Extend w by 0 to $t > 1$ and put $f = 1 - w$ on $(0, \infty)$. Then f is bounded and of bounded variation, $f(0+) = 0$ and $f(t) = 1$ for $t > 1$. For real $X \geq 1$ define U_X by the same formula as U_M, and let $D_{X,w}(1) = \sum_{n \leq X} \mu(n) w(n/X)/n$, so that $|D_{X,w}(1)| \ll_w \log(2X)$.

Rescaling. By (13.1) and the identity $\sum_{n \leq y} \mu(n) \lfloor y/n \rfloor = 1$ for $y \geq 1$,

$$U_X(1/y) = \lfloor y \rfloor D_{X,w}(1) + \sum_{n \leq y} \mu(n) f(n/X) \Big\lfloor \frac{y}{n} \Big\rfloor \qquad (y \geq 1). \tag{G.1}$$

Real parameters. Let $m = \lfloor X \rfloor$. For $n \leq m$ the point n/X lies in $[(n-1)/m, n/m]$, so $\sum_{n \leq m} |w(n/X) - w(n/m)| \leq \operatorname{Var}(w)$ and $|U_X - U_m| \leq \operatorname{Var}(w)$ on $(0, 1)$. At a fixed $x = 1/y \in (0, 1)$ and for $K > y$, the terms with $n \leq K$ tend to zero as $m \to \infty$ because $w(0+)$ exists, and the terms with $n > K$ contribute at most $y \operatorname{Var}(w)/K$, since $g_n(1/y) = \lfloor y \rfloor / n$ for $n > y$. Hence $U_X - U_m \to 0$ pointwise, and by dominated convergence $\|U_X - U_{\lfloor X \rfloor}\|_{1/2} \to 0$. Because $\beta \geq 1/2$, this gives $X^{1/2-\beta} \|U_X\|_{1/2} \leq m^{1/2-\beta}(\|U_m\|_{1/2} + o(1))$, so the limit superior over real X does not exceed the limit superior over integers. It suffices to prove the assertions for real $X \to \infty$.

A test function. Let $W \in C_c^\infty(0, \infty)$ be supported in $[a, b]$ with $\int v W(v) \, \mathrm{d}v = 0$, and put

$$k_W(t) = f(t) \int_0^\infty W(v) \Big\lfloor \frac{v}{t} \Big\rfloor \, \mathrm{d}v = -f(t) \int_0^\infty W(v) \Big\{ \frac{v}{t} \Big\} \, \mathrm{d}v,$$
$$H(X) = \sum_{n \geq 1} \mu(n) k_W(n/X).$$

The function k_W is bounded, vanishes for $t > b$, and tends to 0 as $t \to 0$; so $\widehat{k}_W(s) = \int_0^\infty k_W(t) t^{s-1} \, \mathrm{d}t$ is holomorphic in $\operatorname{Re} s > 0$, and $H(X)$ is a finite sum that vanishes for $X < 1/b$. For $X \geq 1/a$, integrate (G.1) with $y = Xv$ against $W(v)$. The first term contributes $D_{X,w}(1) \int \lfloor Xv \rfloor W(v) \, \mathrm{d}v = -D_{X,w}(1) \int \{Xv\} W(v) \, \mathrm{d}v = O_W(\log 2X)$, and the second contributes $H(X)$. On the other side, the substitution $x = 1/(Xv)$ and the Cauchy–Schwarz inequality give

$$\Big| \int_0^\infty U_X\Big(\frac{1}{Xv}\Big) W(v) \, \mathrm{d}v \Big| \leq \sqrt{X} \, \|vW\|_{L^2} \, \|U_X\|_{1/2}.$$

Therefore

$$|H(X)| \leq \sqrt{X} \, \|vW\|_{L^2} \, \|U_X\|_{1/2} + O_W(\log 2X). \tag{G.2}$$

For $\operatorname{Re} s > 1$, termwise integration gives

$$\int_0^\infty H(X) X^{-s-1} \, \mathrm{d}X = \frac{\widehat{k}_W(s)}{\zeta(s)}. \tag{G.3}$$

Detection of the zero. We claim that W can be chosen with $\widehat{k}_W(\rho) \neq 0$. By Fubini's theorem,

$$\widehat{k}_W(\rho) = \int_0^\infty W(v) J_\rho(v) \, \mathrm{d}v, \qquad J_\rho(v) = -\int_0^\infty f(t) \Big\{ \frac{v}{t} \Big\} t^{\rho - 1} \, \mathrm{d}t.$$

The integral for J_ρ converges absolutely because $0 < \beta < 1$; the substitution $t = v/u$ shows $|J_\rho(v)| \ll v^\beta$; and J_ρ is continuous by dominated convergence. If every admissible W annihilated J_ρ, then $J_\rho(v) = \kappa v$ for a constant κ, and the bound v^β with $\beta < 1$ would force $J_\rho \equiv 0$. Put $F(r) = f(\mathrm{e}^r)$ and $k_\rho(u) = \{\mathrm{e}^u\} \mathrm{e}^{-\rho u}$. Then $J_\rho(\mathrm{e}^r) = -\mathrm{e}^{\rho r}(k_\rho * F)(r)$. The kernel k_ρ is integrable, and by the computation in the proof of Lemma 13.1 its Fourier transform is

$$\int_{-\infty}^\infty k_\rho(u) \mathrm{e}^{-i\xi u} \, \mathrm{d}u = -\frac{\zeta(\rho + i\xi)}{\rho + i\xi},$$

whose real zeros are isolated. The function F is bounded and of bounded variation, with $F(-\infty) = 0$ and $F(+\infty) = 1$; its derivative is a finite complex measure ν of total mass 1.

If $k_\rho * F = 0$, then $k_\rho * \nu = 0$, and taking Fourier transforms gives $\widehat{k}_\rho\,\widehat{\nu} = 0$.[20] But $\widehat{\nu}$ is continuous with $\widehat{\nu}(0) = 1$, so $\widehat{k}_\rho$ would vanish on a neighbourhood of 0. This contradiction proves the claim.

Conclusion. Fix such a W. The right side of (G.3) is meromorphic in $\operatorname{Re} s > 0$ and has at ρ a pole of the same order as the zero. Suppose $\limsup X^{1/2-\beta}\|U_X\|_{1/2} = 0$. By (G.2), $H(X) = o(X^\beta)$, so the integral in (G.3) is holomorphic in $\operatorname{Re} s > \beta$, the identity persists there, and

$$\varepsilon \int_0^\infty |H(X)| X^{-\beta-\varepsilon-1}\,\mathrm{d}X \longrightarrow 0 \qquad (\varepsilon \downarrow 0),$$

as one sees by splitting the integral at a large X_1. This is incompatible with a pole at ρ. If the limit superior is finite, then $H(X) = O(X^\beta)$ and the integral at $\rho + \varepsilon$ is $O(1/\varepsilon)$, which excludes a pole of order two or more. □

G.2. **The harmonic cost.**

Proof of Theorem 13.14. By (13.1), $U(1/y)$ is constant on $k \le y < k+1$ for each $k \ge 1$; call the value V_k and set $V_0 = 0$. With $C = \sum_{n\ge2} a_n/n$,

$$V_k = 1 + kC - \sum_{n\ge2} a_n \Big\lfloor \frac{k}{n} \Big\rfloor \qquad (k \ge 1),$$

$$\Delta V_r := V_r - V_{r-1} = C + \mathbf{1}_{r=1} - \sum_{n\mid r,\ n\ge2} a_n \qquad (r \ge 1).$$

Möbius inversion gives $a_n - \mu(n) = -\sum_{r\mid n} \mu(n/r)\Delta V_r$ for every $n \ge 2$. Also

$$d^2 = \int_1^\infty |U(1/y)|^2 \frac{\mathrm{d}y}{y^2} = \sum_{k\ge1} \frac{|V_k|^2}{k(k+1)},$$

whence, for $R \ge 1$,

$$\sum_{r\le R} \frac{|\Delta V_r|^2}{r} \le 2\sum_{r\le R} \frac{|V_r|^2}{r} + 2\sum_{r\le R} \frac{|V_{r-1}|^2}{r} \le 2(R+1)d^2 + 2(R-1)d^2 = 4Rd^2.$$

Writing $n = mr$ and applying Minkowski's inequality to the sum over m,

$$\Big(\sum_{2\le n\le K} \frac{|a_n - \mu(n)|^2}{n}\Big)^{1/2} \le \sum_{m\le K} \frac{1}{\sqrt{m}} \Big(\sum_{r\le K/m} \frac{|\Delta V_r|^2}{r}\Big)^{1/2} \le 2d\sqrt{K} \sum_{m\le K} \frac{1}{m}.$$

The triangle inequality in the norm $(\sum_{2\le n\le K} |\cdot|^2/n)^{1/2}$ gives the first assertion.

Let $c = 6/\pi^2$ and $Q_2(t) = \sum_{n\le t} \mu(n)^2$. From $\mu(n)^2 = \sum_{r^2\mid n} \mu(r)$ one gets $Q_2(t) = \sum_{r\le\sqrt{t}} \mu(r)\lfloor t/r^2 \rfloor$, and bounding the fractional parts by $\sqrt{t}$ and the tail $t\sum_{r>\sqrt{t}} r^{-2}$ by $2\sqrt{t}$ gives $|Q_2(t) - ct| \le 3\sqrt{t}$ for $t \ge 1$. Partial summation then yields

$$\sum_{2\le n\le K} \frac{\mu(n)^2}{n} = \frac{Q_2(K)}{K} - 1 + \int_1^K \frac{Q_2(t)}{t^2}\,\mathrm{d}t \ge c\log K - 7.$$

Let $\Lambda = \log(1/d) \ge 4$ and $K = \lfloor \mathrm{e}^{2\Lambda}/\Lambda^4 \rfloor \ge 11$. Then $\log K \ge 2\Lambda - 4\log\Lambda - \log 2$ and

$$e := 2d\sqrt{K} \sum_{m\le K} \frac{1}{m} \le \frac{2(1+2\Lambda)}{\Lambda^2} \le \frac{4.5}{\Lambda}.$$

With $A = (c\log K - 7)_+ \le 2c\Lambda$, the first assertion and $(\sqrt{A} - e)_+^2 \ge A - 2e\sqrt{A}$ give

$$\sum_{n\ge2} \frac{|a_n|^2}{n} \ge 2c\Lambda - 4c\log\Lambda - c\log 2 - 7 - \frac{9\sqrt{2c}}{\sqrt{\Lambda}}.$$

[20] The differentiation is carried out after pairing with a smooth compactly supported test function; Fubini's theorem applies because $k_\rho \in L^1$, F is bounded and ν is finite.

Since $c\log 2 < 0.43$ and $9\sqrt{2c}/\sqrt{\Lambda} < 4.97$ for $\Lambda \ge 4$, the constant terms sum to less than 12.4, which proves (13.11); the asymptotic statement follows. □

G.3. **The logarithmic taper.**

Proof of Proposition 13.15. For $y > 0$, $\frac{1}{2\pi i}\int_{\operatorname{Re} w=2} y^w \,\mathrm{d}w/w^2 = (\log y)_+$, so with $F = 1/\zeta$,

$$D_M^{\log}(s) = \frac{1}{2\pi i \,\log M}\int_{\operatorname{Re} w=2} \frac{M^w}{w^2} F(s+w)\,\mathrm{d}w.$$

Moving the line to $\operatorname{Re} w = -\delta$ as in the proof of Theorem 13.4, the double pole at $w = 0$ contributes $F(s)\log M + F'(s)$, and for $\operatorname{Re} s = \sigma$

$$D_M^{\log}(s) = F(s) + \frac{F'(s)}{\log M} + O_{\sigma,\theta,\delta,\kappa}\Big(\frac{M^{-\delta}(1+|t|)^{\kappa}}{\log M}\Big).$$

At $s = 1$, where $F(1) = 0$ and $F'(1) = 1$, the same computation gives $D_M^{\log}(1) = (1 + O(M^{-\delta}))/\log M$. Inserting both in (13.2) and using $-\zeta F' = \zeta'/\zeta$, the transform of $(\log M)u_M^{\log}$ on $\operatorname{Re} s = \sigma$ equals $(\zeta(s)+\zeta'(s)/\zeta(s))/s$ up to an error $O(M^{-\delta}|\zeta(s)|(1+|t|)^{\kappa}/|s|)$. By Lemma 13.3 and Lemma 13.2 with $\kappa < 1/4$, the functions $(\log M)u_M^{\log}$ converge in $\|\cdot\|_\sigma$, with error $O(M^{-\delta})$, to a function with that transform. By (13.12) and $(\log M)D_M^{\log}(1) \to 1$ they converge pointwise on $(0,1)$ to $E(1/x)$. A norm limit and a pointwise limit agree almost everywhere. □

G.4. **Growth forced by an off-line zero.**

Proof of Proposition 13.16. Put $\alpha = 2\beta - 1$ and suppose $I(y) \le cy^{\alpha}$ for $y \ge y_0$. For $0 < \varepsilon \le 1$, integration by parts gives

$$\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y) = (\alpha+\varepsilon)\int_1^\infty I(y)y^{-\alpha-\varepsilon-1}\,\mathrm{d}y \le O(1) + \frac{c(\alpha+\varepsilon)}{\varepsilon}.$$

Let $F_E(s) = \int_1^\infty E(y)y^{-s-1}\,\mathrm{d}y$. For $\operatorname{Re} s > 1$ it equals $(\zeta(s) + \zeta'(s)/\zeta(s))/s$. By the Cauchy–Schwarz inequality,

$$\begin{aligned}\Big(\int_1^\infty |E(y)|y^{-\beta-\varepsilon-1}\,\mathrm{d}y\Big)^2 &\le \int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y)\int_1^\infty y^{-1-\varepsilon}\,\mathrm{d}y\\ &= \frac{1}{\varepsilon}\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y).\end{aligned}$$

Hence F_E is holomorphic in $\operatorname{Re} s > \beta$, where it coincides with its continuation, and $|F_E(\rho+\varepsilon)|^2 \le \varepsilon^{-1}(O(1)+c(\alpha+\varepsilon)/\varepsilon)$. The residue of ζ'/ζ at ρ is m, so $F_E(\rho+\varepsilon) \sim m/(\rho\varepsilon)$. Letting $\varepsilon \downarrow 0$ gives $m^2/|\rho|^2 \le c\alpha$. This holds for every c exceeding the limit superior, which proves the inequality.

If the interior norms are bounded, then $I(M) \ll (\log M)^2$ by (13.13), and by monotonicity $I(Y) \ll \log^2(2Y)$; the lower bound excludes a zero with $\beta > 1/2$. Under the Riemann hypothesis, Cramér's estimate $\int_Y^{2Y} |\psi(y)-y|^2\,\mathrm{d}y \ll Y^2$ [28, Ch. 13] gives $I(M) \ll \log M$. □

G.5. **Ordinary dilates.**

Proof of Proposition 13.17. The function $\{x\} - 1/2$ has Fourier coefficients $-1/(2\pi i\ell)$ for $\ell \ne 0$, and Parseval's identity gives

$$\int_0^1 \{jx\}\{kx\}\,\mathrm{d}x = \frac{1}{4} + \frac{\gcd(j,k)^2}{12\,jk}.$$

Put $t = \sum_k c_k$ and $u_d = \sum_{d\mid k\le M} c_k/k$; the map $(c_k) \mapsto (u_d)$ is triangular and invertible. Since $\sum_{d\mid k}\varphi(d) = k$ and $\sum_{d\mid k} J_2(d) = k^2$, we have $t = \sum_d \varphi(d)u_d$ and

$$\Big\|1 - \sum_{k\le M} c_k\{kx\}\Big\|^2 = \Big|1 - \frac{t}{2}\Big|^2 + \frac{1}{12}\sum_{d\le M} J_2(d)|u_d|^2.$$

For fixed t, the Cauchy–Schwarz inequality shows that the minimum of the last sum is $|t|^2/S_M$, attained at $u_d = t\varphi(d)/(S_M J_2(d))$. The resulting quadratic in t is minimal at $t = 6S_M/(1+3S_M)$, with value $1/(1+3S_M)$. Finally $a(d) = \varphi(d)^2/J_2(d)$ is multiplicative with $a(p^k) = (p-1)/(p+1)$ for $k \ge 1$, so $a = 1 * b$ with b supported on squarefree integers and $b(p) = -2/(p+1)$. Since $\sum_d |b(d)|/d < \infty$, $S_M = \sum_d b(d)\lfloor M/d \rfloor \sim M \sum_d b(d)/d$, which is the stated product. □

Appendix H. Short windows

Proof of Theorem 15.2. We use [25, Theorem 1] for the multiplicative function λ: there are absolute constants $C, C' > 1$ such that, for $2 \le h \le X$ and $\delta > 0$,

$$\Big|\frac{1}{h}\sum_{x\le n\le x+h}\lambda(n) - m_X\Big| \le \delta + C'\,\frac{\log\log h}{\log h}, \qquad m_X = \frac{1}{X}\sum_{X\le n\le 2X}\lambda(n), \tag{H.1}$$

for all integers $x \in [X, 2X]$ outside a set of cardinality at most

$$CX\left(\frac{(\log h)^{1/3}}{\delta^2 h^{\delta/25}} + \frac{1}{\delta^2(\log X)^{1/50}}\right).$$

Let $16 \le h \le X$ and take $\delta = \max\big(125\log\log h/\log h,\ (\log X)^{-1/200}\big)$. Then $h^{-\delta/25} \le (\log h)^{-5}$ and $\delta^{-2} \le \min\big((\log h)^2, (\log X)^{1/100}\big)$, so the exceptional set has at most $CX((\log h)^{-8/3} + (\log X)^{-1/100})$ elements. On it the left side of (H.1) is at most 4. Hence

$$\frac{1}{X}\sum_{X\le x\le 2X}\Big|\frac{1}{h}\sum_{x\le n\le x+h}\lambda(n) - m_X\Big|^2 \ll \Big(\frac{\log\log h}{\log h}\Big)^2 + (\log X)^{-1/100}. \tag{H.2}$$

By Theorem 15.1, $m_X \ll 1/\log X$, so (H.2) holds with m_X removed.

Recall that $\mathcal{Q}_\lambda(N,H) = H^{-1}\sum_t |S(t)|^2$ with $S(t) = \sum_{t<n\le t+H,\ 1\le n\le N}\lambda(n)$. Put $T_0 = \max(H, \lceil\sqrt{N}\,\rceil)$. For $T_0 \le t \le N-H$ the window lies inside $[1,N]$, and with $x = t+1$ the sum in (H.1) with $h = H$ equals $S(t) + \lambda(t+H+1)$. Cover the range of x by dyadic blocks $[X, 2X]$ with $X = 2^k T_0 \le N$. Each has $H \le X$ and $\log X \ge \frac12 \log N$, and the blocks have total length at most $2N$. From (H.2),

$$\frac{1}{H}\sum_{T_0\le t\le N-H}|S(t)|^2 \ll NH\left(\Big(\frac{\log\log H}{\log H}\Big)^2 + (\log N)^{-1/100}\right) + \frac{N}{H},$$

and the last term is smaller than the first.

The remaining windows are those with $-H < t < T_0$ or $N-H < t < N$; there are at most $3H + \sqrt{N} + 1$ of them. Each satisfies $|S(t)| \le \min(H, 2\max_{j\le N}|L(j)|) \le \min(H, C''N/\log N)$ by Theorem 15.1. If $H \le \sqrt{N}$ their contribution to $\mathcal{Q}_\lambda(N,H)$ is at most $H^{-1}\cdot 5\sqrt{N}\cdot H^2 = 5\sqrt{N}\,H$. If $H > \sqrt{N}$ it is at most $5\min(H, C''N/\log N)^2 \le 5C''NH/\log N$. Both are within the stated bound.

The trivial bound follows from $|S(t)| \le H$ and the count of at most $N + H - 1 \le 2N$ windows. □

Appendix I. The explicit constants

This appendix gives the details of the explicit part of Theorem 15.1.

I.1. **The decomposition.** Every n is uniquely $n = d^2 m$ with m squarefree, and $\lambda(n) = \mu(m)$ is then one term of $\sum_{d^2\mid n}\mu(n/d^2)$; the other terms vanish because n/d'^2 is squarefree only for $d' = d$. Thus $\lambda(n) = \sum_{d^2\mid n}\mu(n/d^2)$, and summing over $n \le x$ gives (15.1).

Let $y_0 = 2\,160\,535$ and split the sum at $d = \sqrt{x/y_0}$.

Large arguments. If $x/d^2 \ge y_0$, then (15.2) gives $|\mathcal{M}(x/d^2)| \le x/(4345\,d^2)$. The sum over all such d is at most

$$\frac{\zeta(2)}{4345}\,x = \frac{x}{2641.44\ldots} < \frac{x}{2641}.$$

Small arguments. If $1 \le x/d^2 < y_0$, the finite verification $|\mathcal{M}(y)| \le \sqrt{y}$ for $1 \le y \le 3 \cdot 10^6$ gives $|\mathcal{M}(x/d^2)| \le \sqrt{x}/d$. Let $z = \sqrt{x/y_0}$ and $w = \sqrt{x}$, and let d_1 be the least integer greater than z. If $z \ge 1$, then

$$\sum_{z<d\le w} \frac{1}{d} \le \frac{1}{d_1} + \int_{d_1}^{w} \frac{\mathrm{d}u}{u} \le 1 + \log \frac{w}{z} = 1 + \tfrac{1}{2}\log y_0.$$

If $z < 1$, then $x < y_0$ and $\sum_{d\le w} 1/d \le 1 + \log w \le 1 + \frac{1}{2}\log y_0$. Numerically $1 + \frac{1}{2}\log y_0 = 8.29293\ldots < 8.3$.

Adding the two parts, $|L(x)| \le x/2641 + 8.3\sqrt{x}$ for every $x \ge 1$. By (2.6), $Q_\lambda(N) \le 5\max_{j\le N} |L(j)|^2$, and the majorant is increasing in j, which gives the bound for $Q_\lambda(N)$.

I.2. **The finite verification.** Table 26 summarises a sieve computation of μ and λ up to $3 \cdot 10^6$. It confirms $|\mathcal{M}(y)| \le \sqrt{y}$ on $[1, 3 \cdot 10^6]$, with equality only at $y = 1$. It also shows that the threshold in (15.2) cannot be lowered: $\mathcal{M}(y_0 - 1) = -498$, whereas $(y_0-1)/4345 = 497.246\ldots$; and on $[y_0, 3\cdot 10^6]$ the largest value of $4345\,|\mathcal{M}(y)|/y$ is $0.9995\ldots$

Table 26. Finite checks for Theorem 15.1. Left: the maximum of $|\mathcal{M}(y)|/\sqrt{y}$ over the integers y in each range, and the point where it is attained. Right: $L(x)$ against the explicit majorant $x/2641 + 8.3\sqrt{x}$.

Range of y	Maximum	at y	x	$L(x)$	Majorant
$[2, 10^3]$	0.8944	5	10^2	-2	83.0
$[10^3, 10^4]$	0.4722	2803	10^3	-14	262.8
$[10^4, 10^5]$	0.4630	24 185	10^4	-94	833.8
$[10^5, 10^6]$	0.4378	300 551	10^5	-288	2662.6
$[10^6, 3 \cdot 10^6]$	0.4182	1 066 854	10^6	-530	8678.6
			$3 \cdot 10^6$	-942	15 512.0

Over $1 \le x \le 3 \cdot 10^6$ the largest value of $|L(x)|/(x/2641 + 8.3\sqrt{x})$ is $0.1580\ldots$, attained at $x = 96\,862$, and $\max_{x\le 3\cdot 10^6} |L(x)| = 1803$. The explicit inequality is thus a safe majorant and not an approximation. Its only inputs are (15.2) and the finite range of $\mathcal{M}$; no zero-free region enters.

I.3. **Constants used in Part C.** Table 27 lists the numerical constants that occur in Sections 12–15.

Table 27. Numerical constants.

Constant	Value	Where used
$\zeta(1/2)$	$-1.46035\ldots$	(12.3)
$c_0 = -\sqrt{\pi}/(2\zeta(1/2))$	$0.60685\ldots$	Proposition 12.8
$\frac{3}{2}c_0^2$	$0.55241\ldots$	Proposition 12.9
$2 - \pi/2$	$0.42920\ldots$	Theorem 12.4
$\sqrt{\pi} \cdot \sqrt{2\pi}$	$4.44288\ldots$	Lemma D.1
$12/\pi^2$	$1.21585\ldots$	Theorem 13.14
$\zeta(2)/4345$	$1/2641.44\ldots$	Theorem 15.1
$1 + \frac{1}{2}\log 2\,160\,535$	$8.29293\ldots$	Theorem 15.1
$\zeta(2)/(2\zeta(3))$	$0.68421\ldots$	Section 14, diagonal

Appendix J. Elementary prime bounds and exact checks

J.1. **Chebyshev-type bounds.** Sums over primes in this paper are estimated with the following elementary bounds, which need neither the prime number theorem nor any zero-free region. Let $\vartheta(t) = \sum_{p\le t} \log p$ and let $\pi(t)$ count the primes up to t.

Lemma J.1. *For $t \geq 2$ and $y \geq 2$,*

$$\vartheta(t) \leq 4(\log 2)\, t, \qquad \pi(t) \leq \frac{7t}{\log t},$$
$$\sum_{p \leq y} \frac{1}{\sqrt{p}} \leq \frac{36\sqrt{y}}{\log y}, \qquad \sum_{p \leq y} \frac{\log p}{p} \leq 3(1 + \log y).$$

Proof. The product of the primes in $(n, 2n]$ divides $\binom{2n}{n} \leq 4^n$, so $\vartheta(2n) - \vartheta(n) \leq 2n \log 2$. Summing over $n = 2^j$ gives $\vartheta(2^k) \leq 2^{k+1} \log 2$, and for $2^{k-1} < t \leq 2^k$ this yields $\vartheta(t) \leq 4(\log 2)t$.

Primes in $(\sqrt{t}, t]$ have $\log p \geq \frac{1}{2} \log t$, so $\pi(t) \leq \sqrt{t} + 2\vartheta(t)/\log t$. Since $\log t \leq \sqrt{t}$, the first term is at most $t/\log t$, and $1 + 8 \log 2 < 7$.

By partial summation,

$$\sum_{p \leq y} \frac{1}{\sqrt{p}} = \frac{\pi(y)}{\sqrt{y}} + \frac{1}{2} \int_2^y \frac{\pi(t)}{t^{3/2}}\, \mathrm{d}t \leq \frac{7\sqrt{y}}{\log y} + \frac{7}{2} \int_2^y \frac{\mathrm{d}t}{\sqrt{t}\, \log t}.$$

On $[2, \sqrt{y}]$ the integrand is at most $1/(\sqrt{t} \log 2)$, and on $[\sqrt{y}, y]$ it is at most $2/(\sqrt{t} \log y)$; the integral is at most $2y^{1/4}/\log 2 + 4\sqrt{y}/\log y$. Since $y^{1/4} \log y \leq (4/\mathrm{e})\sqrt{y}$, the total is at most $(7 + 14 + 7 \cdot 4/(\mathrm{e} \log 2))\sqrt{y}/\log y < 36\sqrt{y}/\log y$.

Finally $\sum_{p \leq y} (\log p)/p = \vartheta(y)/y + \int_2^y \vartheta(t) t^{-2}\, \mathrm{d}t \leq 4(\log 2)(1 + \log y)$, and $4 \log 2 < 3$. □

J.2. **Exact checks of the sign of B.** The values $B(1) > 0$ and $B(3) < 0$ quoted after Proposition 12.8 follow from a finite sum and a geometric tail. For every $K \geq 1$ and $X > 0$,

$$\Big| B(X) - \sum_{n \leq K} \lambda(n) \mathrm{e}^{-n/X} \Big| \leq \frac{\mathrm{e}^{-(K+1)/X}}{1 - \mathrm{e}^{-1/X}}.$$

With $\lambda(1), \ldots, \lambda(9) = 1, -1, -1, 1, -1, 1, -1, -1, 1$,

$$\sum_{n \leq 9} \lambda(n) \mathrm{e}^{-n} = 0.195689\ldots, \qquad \frac{\mathrm{e}^{-10}}{1 - \mathrm{e}^{-1}} < 7.2 \cdot 10^{-5},$$

so $0.1956 < B(1) < 0.1958$. Similarly

$$\sum_{n \leq 40} \lambda(n) \mathrm{e}^{-n/3} = -0.079206\ldots, \qquad \frac{\mathrm{e}^{-41/3}}{1 - \mathrm{e}^{-1/3}} < 4.1 \cdot 10^{-6},$$

so $-0.0793 < B(3) < -0.0792$. Table 28 records these values. The sum B therefore takes both signs at small scales, and any statement about its sign can only be an eventual one.

Table 28. Certified values of B at two small scales.

X	Terms K	Partial sum	Tail bound
1	9	0.195689	$7.2 \cdot 10^{-5}$
3	40	−0.079206	$4.1 \cdot 10^{-6}$

Appendix K. Machine verification

The finite identities and the numerical constants of the paper are checked twice, once by a Lean 4 project over Mathlib and once by three Julia programs. Both are archived with the paper at doi:10.5281/zenodo.24197589. No proof in the text uses them; they guard against slips in algebra and in arithmetic.

K.1. **The Lean project.** The project builds with Lean 4.35 and the Mathlib release pinned in its manifest. It contains no unproved statement and adds no axiom: for every theorem in Table 29 the command `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The statements are about real numbers and finite sums, so that each one can be compared with the displayed formula it formalises.[21]

TABLE 29. Theorems of the Lean project. The left column gives the Lean name, the middle column the statement in the notation of the paper, and the right column the place where it is used. All twenty-four theorems build without `sorry`.

Lean name	Statement	Used in
File `Energy.lean`		
`boundary_term`	$x^2+(y-x)^2 = 2(x-\frac{1}{2}y)^2+\frac{1}{2}y^2$	(2.6)
`completion`	the completion identity at $H=N$	(2.6)
`half_sq_le_energy`	$\frac{1}{2}L_a(N)^2 \le Q_a(N)$	(2.7)
`energy_nonneg`	$Q_a(N)\ge 0$, with equality only if all $L_a(j)=0$	Prop. 2.3
`gram_counting`	$K^2M \le KM^2+16BK(K-1)$, $K=2M \Rightarrow M<32B$	Thm. 8.5
`plancherel_weight_lower`	$K(\sigma+i\tau)>0.65$ for $\sigma\ge 0$	(3.6)
`plancherel_weight_min`	$K(\sigma+i\tau)\ge 4\sqrt{2}-5$ for $\sigma\ge 0$	(3.6)
`zeta_two_constant`	$\zeta(2)/4345 < 1/2641$	Thm. 15.1
`log_threshold`	$1+\frac{1}{2}\log 2160535 < 8.3$	Thm. 15.1
`shift_correlation_le`	$\left\lvert\sum_{n\le Y} a_n a_{n+h}\right\rvert \le Y$ for $\lvert a_n\rvert\le 1$	Sec. 7
`oscillation_constant`	$2.3723^2/2 > 2.8139$	Thm. 5.6
`ten_thirds_count`	$N(N-1)+N^2+\frac{4}{3}N^2 \le \frac{10}{3}N^2$	Prop. 5.8
`positions_exponents`	the two exponent comparisons in the proof	Prop. 4.7
File `GramCore.lean`		
`diagonal_sq_le_frobenius`, `trace_sq_le_dimension_mul_frobenius`	$(\operatorname{tr}A)^2 \le m\operatorname{tr}(A^{\mathsf T}A)$ for $m\times m$ real A	Thm. 8.5
`frobenius_sum`, `family_rank_lower`, `family_energy_upper`	expansion of $\operatorname{tr}((\sum_i B_i)^2)$ and its two bounds	Thm. 8.5
`gram_rank_obstruction`, `twice_dimension_bound`	the counting inequality for K nearly orthogonal Gram matrices	Thm. 8.5
`averagedGram_diagonal`, `averagedGram_entry_sq_le_one`, `averagedGram_trace_and_self`, `averaged_sign_rank_obstruction`	averaged Gram matrices of sign vectors: unit diagonal, entries in $[-1,1]$, trace M, and the rank obstruction	Thm. 8.5

K.2. **The Julia programs.** The programs use exact integer and rational arithmetic, and for the transcendental constants the ball arithmetic of Arb through the package `Arblib.jl`, so that every reported value comes with a rigorous enclosure. Table 30 lists the checks; all of them pass.

K.3. **Numerical illustration.** Table 31 and Figure 12 show the quantities of the paper for N and x up to 10^7 and $3\cdot 10^6$, and X up to $6\cdot 10^4$. For $N\le 3\cdot 10^7$ the ratio $Q(N)/N$ has largest value 1.1534, attained at $N=12\,897\,104$, where $L(N)/\sqrt{N}=-1.3583$; this is consistent with the order $N^{1+\varepsilon}$ that the Riemann hypothesis predicts, and its logarithmic

[21]The analytic arguments (contour shifts, Mellin inversion, the kernel of Pintz) are not formalised. The project checks every step that reduces to a finite identity or a numerical inequality.

TABLE 30. Checks made by the Julia programs. Ranges are complete: every integer in the stated range is tested. Enclosures are certified intervals, written with their midpoints rounded to five digits.

Program	Check	Result
`finite_identities`	window energy equals the prefix form, all 8190 sign strings of lengths 1 to 12	exact equality
	completion identity, 2000 random prefix vectors	exact equality
	$L(x) = \sum_{d \le \sqrt{x}} M(x/d^2)$, $1 \le x \le 1024$	exact equality
	Abel-jet constants 112/27 and 114688/9 of Table 22	exact
	Rudin–Shapiro complementarity, $m \le 10$, and the block pattern	exact
`energy_checks`	$\frac{1}{2}L(N)^2 \le Q(N) \le 5 \max_{j \le N} L(j)^2$, all $N \le 3 \cdot 10^7$	no violation
	$\lvert M(y)\rvert \le \sqrt{y}$, $1 \le y \le 3 \cdot 10^7$	no violation
	$\lvert M(y)\rvert \le y/4345$, $2160535 \le y \le 3 \cdot 10^7$	max ratio 0.9995
	$\lvert L(x)\rvert \le x/2641 + 8.3\sqrt{x}$, $1 \le x \le 3 \cdot 10^7$	max ratio 0.158
	monotone core, (2.11) and (2.12), $N \le 3 \cdot 10^7$	max ratios 0.41, 0.13
	$\min K = 4\sqrt{2} - 5$ on a grid in $\sigma \ge \frac{1}{2}$	0.65685
	largest value of $Q(N)/N$, $N \le 3 \cdot 10^7$	1.1534 at $N = 12\,897\,104$
	largest value of $\lvert L(x)\rvert/\sqrt{x}$, $x \le 3 \cdot 10^7$	1.3583 at the same point
	mean of $Q(N)/N^2$ against $1/\log$, $10^4 < N \le 3 \cdot 10^7$	0.4540
`pole_certificates`	1000 disjoint boxes isolating the first zeros on $\operatorname{Re} s = \frac{1}{2}$	certified
	right side of (5.5), $n = 1000$	$[0.45770 \pm 10^{-34}]$
	right side of (5.8), $J = 1$, $m = 100$	$[0.65142 \pm 10^{-35}]$
	right side of (5.8), $J = 2$, $m = 30$ (3721 frequencies)	$[0.72446 \pm 10^{-35}]$

mean 0.4540 is close to the constant $S > 0.4579$ of Theorem 5.13. The modified function f that is $+1$ on about $x^{3/4}$ primes has $\log\lvert L_f(x)\rvert/\log x$ close to 3/4, as Theorem 6.3 requires; and $B(X)/\sqrt{X}$ approaches the value $-c_0 = -0.60685$ of the pole at $s = 1/2$ (Proposition 12.8).[22]

TABLE 31. Values of L, of the energy Q and of the sparse flip f with $+1$ on $\lfloor 2^{3j/4} \rfloor$ primes of each interval $(2^j, 2^{j+1}]$, and of the smoothed sum B. Computed exactly, except B, which is rounded to four digits.

$N = x$	$L(N)$	$Q(N)$	$Q(N)/N$	$\dfrac{\log Q(N)}{\log N}$	$L_f(x)$	$\dfrac{\log\lvert L_f(x)\rvert}{\log x}$
10^2	-2	26.12	0.2612	0.7085	—	—
10^3	-14	261.35	0.2614	0.8057	52	0.575
10^4	-94	5783.54	0.5784	0.9405	594	0.694
10^5	-288	56316.88	0.5632	0.9501	5126	0.742
10^6	-530	382792.54	0.3828	0.9305	30750	0.748
10^7	-842	3835484.17	0.3835	0.9405	252510	0.772
X	10^3	10^4	$3 \cdot 10^4$	$6 \cdot 10^4$		
$B(X)/\sqrt{X}$	-0.5752	-0.5969	-0.6011	-0.6028		

[22] Finite data of this kind cannot decide the Riemann hypothesis: by Platt and Trudgian [33] a zero off the line would have height above $3 \cdot 10^{12}$, and its effect in this range would be invisible.

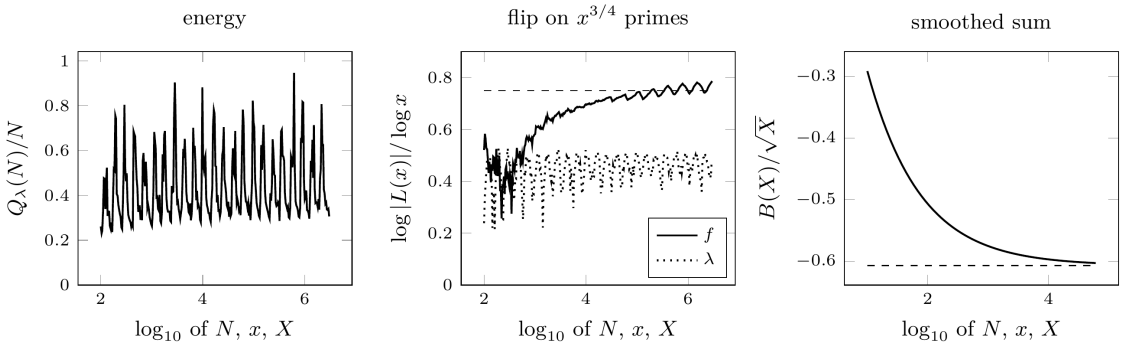


FIGURE 12. Left: the ratio $Q(N)/N$ at 300 logarithmically spaced $N \le 3{\cdot}10^6$. Middle: the exponent $\log|L(x)|/\log x$ for the sparse flip f on about $x^{3/4}$ primes (solid) and for λ (dotted); the dashed line is 3/4. Right: $B(X)/\sqrt{X}$, with the dashed line at $-c_0 = -0.60685$.

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The energy identities rest on Fejér's kernel, the passage between summatory functions and Dirichlet series on Landau's theorem and on Perron's formula, and the zero-one law on the mean-value kernel of Pintz. The modified Liouville functions of Section 6 are those of Borwein, Choi and Coons, and part (i) of the sensitivity theorem is a form of a theorem of Aymone. The approximation results continue the work of Nyman, Beurling and Báez-Duarte and of Báez-Duarte, Balazard, Landreau and Saias, and the explicit constants use the bound of Cohen, Dress and El Marraki for the Mertens function.

AUTHOR CONTRIBUTIONS

Deep Bhattacharjee, who leads the project, posed the problem in the form treated here, carried out the research and wrote the paper. The proofs are his: the energy identities and the equivalence theorem, the zero-one law, the sensitivity theorem and its threshold, the Mellin and Nyman–Beurling results, the explicit bounds and the appendices. He made all the computations: the Lean 4 project of Table 29, the Julia programs of Table 30 with their exact and ball-arithmetic evaluations, the numerical tables and the figures; and he brought the work to its present state. Ushashi Bhattacharya read the manuscript and verified its arguments, its constants and its references. Both authors approved the final version.

OPEN GLUON AND ANAXIA

Two tools built by D. B. stand behind the way this project was organised. Open Gluon is a workspace he assembled and maintains, which keeps pinned copies of the mathematical libraries, the source record of the project and the checks in one reproducible place. Anaxia, the engine of the eBLLM project, is his construction engine: it stores each construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact rational arithmetic. He wrote both and developed them from the constructions of his earlier papers [8, 9, 10]. Three of their constructions were used here. The first is the reduction of every target bound to a typed statement about the rightmost zero, which showed that each route proposed for the critical energy bound was equivalent to a zero-free half-plane and led to the form of Theorem 3.1 and of the zero-one law. The second is the perturbation test on the signs of λ, which flips the values on a prescribed set of primes and records whether the energy exponent moves; it produced the experiment in the middle panel of Figure 12 and the threshold of Corollary 6.5. The third is exact certification: the Julia programs recompute the constants of the paper in exact rational arithmetic wherever

the value is rational, and in certified ball arithmetic otherwise. The engine proposes routes and tests them; every statement of this paper is proved in the text.

Declarations

Data availability. No data were generated. The paper, the Lean 4 project and the Julia programs are archived at doi:10.5281/zenodo.24197589. The Lean project builds over Mathlib without unproved statements and without added axioms; every named result depends only on the three standard axioms of Lean (propositional extensionality, choice and quotient soundness). It covers the finite identities and the numerical constants of Table 29; the analytic arguments are not formalised. The Julia programs reproduce the tables and the certificates of Table 30, and all their checks pass. Nothing in the paper rests on either: every statement has its analytical proof in the text.

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